Lectures on Algebra I

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Contents

1 Fields and Galois Theory

1.1 Basic notions

Let F be a field. We always assume that $0 \neq 1$ in F. In F, there exists a smallest field, called the *prime field* in F, which is simply the field generated by 1 under the operations $+, -, \times, \div$. It is either $\mathbb{Q}$ or $\mathbb{F}_p$ ($\mathbb{Z}/p\mathbb{Z}$), depending on whether $p.1$ is zero or not in F. We say that F is of characteristic 0 or p accordingly.

It is an easy, but useful fact that a field has a unique maximal ideal: (0). So if $F \to A$ is a ring homomorphism, where F is a field, then it is either injective, or zero. In practice, we don't consider the latter case.

If we have two fields F, K and F is contained in K, we say that K is an extension of F. Sometimes write K/F is an extension of fields. We can view K as a vector space over F, so we can talk about the dimension of K over F, called the degree of the extension K/F, denoted by $[K : F]$. This degree could be well infinite. In the case it is finite, we call that K/F is a finite extension, otherwise, an infinite extension.

Proposition 1.1.1. *Let $F \subset E \subset K$ be a tower of finite extensions, then*

$$[K : F] = [K : E][E : F].$$

If $\{u_1, \dots, u_m\}$ is a basis for K/E and $\{v_1, \dots, v_n\}$ is a basis for E/F, then

$$\{u_i v_j : 1 \leq i \leq m; 1 \leq j \leq n\}$$

is a basis for K/F.

Proof. This is an easy exercise. □

Suppose $F \subset K$ are fields and S is a subset of K. Let

$$F[S] := \cap\{A : A \text{ is a subring of } K \text{ containing } F \cup S\}$$

$$F(S) := \cap\{A : A \text{ is a subfield of } K \text{ containing } F \cup S\}$$

i.e. the smallest subring (resp. subfield) containing $F \cup S$. $F(S)$ is called the subfield of E obtained from F by adjoining S, or generated by S over F. It is clear that $F[S]$ is an integral domain with fraction field $F(S)$. When S is a finite set, say $S = \{u_1, \dots, u_n\}$, we write

$$F[S] = F[u_1, \dots, u_n], \quad F(S) = F(u_1, \dots, u_n).$$

In the case $S = \{u\}$, $F(u)$ is called a *simple extension* of F.. Then, explicitly, $F[u] = \{P(u) : P \in F[X]\}$, and $F(u)$ is the set of $P(u)/Q(u)$, where $P, Q \in F[X]$ and $Q(u) \neq 0$.

1.2 Algebraic extension

Let K/F be a field extension and $u \in K$.

Definition 1.2.1. *We say that u is algebraic over F if there exists a non-zero polynomial $P \in F[X]$ such that $P(u) = 0$. Otherwise, u is called* transcendental.

An extension K/F is called algebraic *if every element of K is algebraic over F.*

Proposition 1.2.2. (i) *Consider the ring homomorphism $\theta_u : F[X] \to F[u]$ sending X to u. Then u is algebraic if and only if $\ker(\theta_u)$ is non-zero.*

(ii) *If u is algebraic, $\ker(\theta_u) = (P)$ for some unique monic polynomial $P \in F[X]$. It is called the minimal polynomial of u.*

(iii) *If u is algebraic, then $F[u]$ is in fact a field so that $F[u] = F(u)$ and we have*

$$[F[u] : F] = \deg P.$$

Proof. (i) If u is algebraic, then $P(u) = 0$ for some P, so that $P \in \ker(\theta_u)$. The converse is also clear.

(ii) By (i), $\ker(\theta_u)$ is non-zero, then use that $F[X]$ is a PID.

(iii) Any non-zero prime ideal of $F[X]$ is maximal, so that $F[u] \cong F[X]/\ker(\theta_u)$ is already a field. It is easy to see that $\{1, u, \ldots, u^{n-1}\}$, where $n = \deg P$, is a basis of $F[u]$ over F. □

Corollary 1.2.3. *Let $F \subset K$ be fields and $u \in K$. Then u is algebraic over F if and only if $[F(u) : F]$ is finite, if and only if $F(u)$ is algebraic over F.*

Proof. It follows from Proposition 1.2.2. □

Corollary 1.2.4. *Let $F \subset K$ be fields and $u, v \in K$. If u, v are algebraic, so are $u + v$, $u.v$ and u/v if $v \neq 0$.*

Proof. Consider the extensions $F \subset F(u) \subset F(u, v)$. The elements listed above all belong to $F(u, v)$. Since v is algebraic over F, it is also algebraic over $F(u)$, so that $F(u, v) = F(u)(v)$ is a finite extension over $F(u)$, hence it is a finite extension over F. The assertion then follows from Corollary 1.2.3. □

Corollary 1.2.5. *Let $F \subset E \subset K$ be a tower of extensions where both K/E and E/F are algebraic. Then K/F is also algebraic.*

Proof. Let $u \in K$. Since it is algebraic over E, there exists $P \in E[X]$, say $P(X) = a_n X^n + \cdots + a_0$, such that $P(u) = 0$. Set $E' = F(a_0, \ldots, a_n)$, then u is in fact algebraic over E', hence $E'(u)$ is a finite extension over E'. On the other hand, all a_i are algebraic over F, so E' is a finite extension over F by Corollary 1.2.3. Therefore, $E'(u)$ is finite over F. But $F(u) \subset E'(u)$, we get $F(u)$ is finite over F, and therefore u is algebraic over F. □

Corollary 1.2.6. *If $F \subset E, E' \subset K$, and $E/F, E'/F$ are algebraic extensions, then so is EE'/F.*

Here EE' is called the *compositum* of E, E' inside K, it is just the smallest subfield of K containing E, E'. (1)

Proof. An element $\gamma \in EE'$ can be written of the form

$$\gamma = \frac{u_1 v_1 + \cdots + u_n v_n}{u_1' v_1' + \cdots + u_m' v_m'}, \quad u_i, u_i' \in E, \; v_i, v_i' \in E'.$$

The result follows from Corollary 1.2.4. $\square$

Conversely, here is a general method to construct field extensions.

Lemma 1.2.7. *Let F be a field and $P \in F[X]$ of degree n. Then there is a simple extension $K = F(u)$ such that $P(u) = 0$ and $[K : F] \leq n$.*

Proof. Choose an irreducible factor of P, say Q. Then $F[X]/(Q(X))$ is a field extension of F. Its degree is that of Q, hence $\leq n$. $\square$

Next is a general lemma which will be used frequently later.

Lemma 1.2.8. *Let $F(u)/F$ be a simple extension with u algebraic of minimal polynomial P. If E/F is an extension and $v \in E$ satisfies $P(v) = 0$, then there exists a unique F-embedding $\iota : F(u) \to E$ such that $\iota(u) = v$.*

Proof. The uniqueness is clear. Prove the existence. We know that $F(u) = F[u] \cong F[X]/(P)$. Now, the universal property of $F[X]$ implies a ring homomorphism $F[X] \to F[v] \subset E$, sending X to v. Since $P(v) = 0$, this morphism factors through $F[X]/(P)$. We thus get a morphism of rings $F[u] \to E$ between two fields. $\square$

1.3 Splitting fields and normal extensions

Definition 1.3.1. *If K/F be an extension and $P \in F[X]$, then we say P splits over K if P factors as*

$$P(X) = c(X - u_1) \cdots (X - u_n), \quad c \in F, \; u_i \in K.$$

For example, $P(X) = X^2 - 3 \in \mathbb{Q}[X]$ splits over the field $\mathbb{Q}[\sqrt{3}]$.

Definition 1.3.2. *Let $P \in F[X]$ and K be an extension of F. We say that K is a splitting field of P if K is a minimal field over which P splits.*

Theorem 1.3.3. *(Existence) Let F be a field and $P \in F[X]$ with $\deg(P) = n$. Then there is a field K containing F such that K is a splitting field of P over F and $[K : F] \leq n!$.*

(1) Attention: when we talk about this, we need to precise the larger field in which we are taking compositum.

Proof. We do induction on n, and we allow F. If P splits over F, then we take $K = F$. This proves the case $n = 1$. Suppose $n > 1$, and the result is true for all polynomials of degree $< n$.

By Lemma 1.2.7, there is a finite extension F_1 over F of degree $\leq n$ such that $P(u) = 0$ for some $u \in F_1$. Namely, viewing P as an element in $F_1[X]$, it decomposes as

$$P(X) = (X - u)Q(X).$$

Clearly $\deg(Q) = n - 1$. By the inductive hypothesis applied to $Q \in F_1[X]$ (and to the new field F_1), there is a splitting field K of Q over $F_1[X]$ of degree $\leq (n-1)!$. Then it is clear that K is a splitting field of P over F and has degree $\leq n!$. $\square$

Proposition 1.3.4. *(Uniqueness of splitting field). Let $\alpha : F \to F'$ be an isomorphism of fields and $P \in F[X]$. If K and K' are splitting fields of F of $P \in F[X]$, and of $\alpha P \in F'[X]$ respectively, then there is a field isomorphism $\tau : K \to K'$ extending α.*

Proof. We do induction on $[K : F] =: n$ (not $\deg P$!); note that *a priori* we don't assume $[K' : F] = n$. If $n = 1$, this is ok. Suppose that $n > 1$ and the result is true for all splitting fields which are extensions of dimension strictly less than n.

We may assume that P does not split over F. Choose an irreducible factor P_1 of P (it could be P) with $m = \deg P_1 > 1$ (possible because P does not split). Then, since P, hence also P_1, splits over K, P_1 has a root in K, say u. But αP_1 also has a root in K', say v. Lemma 1.2.8 says that there exists a field embedding $K \supset F(u) \overset{\beta}{\hookrightarrow} F(v) \subset K'$; β is also surjective, hence is an isomorphism extending α. Using inductive hypothesis, we get an F-isomorphism $K \to K'$ extending β. $\square$

Algebraic closure

Definition 1.3.5. *A field F is said to be algebraically closed if it has no proper algebraic extension.*

Proposition 1.3.6. *Let F be a field. The following statements are equivalent.*

(i) *F is algebraically closed.*

(ii) *Every non-constant $P \in F[X]$ factors as a (finite) product of linear factors in $F[X]$.*

(iii) *Every non-constant $P \in F[X]$ has a root in F.*

Proof. (i)$\Longrightarrow$(ii) Let $P \in F[X]$ and choose an irreducible factor of P, say P'. consider $K := F[X]/(P')$, then K is a finite extension of F of degree $\deg P'$. But F is algebraically closed, so $K = F$ and $\deg P' = 1$ by Proposition 1.2.2(iii), i.e. $P' = X - u$ for some $u \in F$. This being true for every irreducible factor of P, we get the assertion.

(ii)$\Longrightarrow$(iii) Trivial.

(iii)$\Longrightarrow$(i) Otherwise, let K be a non-trivial algebraic extension of F. By choosing $u \in K \backslash F$, we may assume $K = F(u)$ is a simple extension. Let P be the minimal polynomial of u. By assumption, P has a root in F, but this contradicts the fact that P is irreducible over F. $\square$

The fundamental theorem of algebra says:

Theorem 1.3.7. *The field $\mathbb{C}$ of complex numbers is algebraically closed.*

Definition 1.3.8. *An extension K of F is said to be an algebraic closure of F if K is algebraic over F and K is algebraically closed.*

Remark that $\mathbb{C}$ is *not* an algebraic closure of $\mathbb{Q}$, since $\mathbb{C}$ contains numbers, like e, π which are not algebraic over $\mathbb{Q}$. We denote by $\overline{\mathbb{Q}}$ the algebraic closure contained in $\mathbb{C}$.

Theorem 1.3.9. *If F is a field, then F has an algebraic closure and any two algebraic closures of F are F-isomorphic.*

Proof. (Artin) We first construct an extension E of F in which every polynomial in $F[X]$ of degree ≥ 1 has a root. Let $S = \{P_\lambda : \lambda \in \Lambda\}$ be the set of all monic irreducible polynomials in $F[X]$. For each $P_\lambda \in S$, introduce new variables $y_{\lambda,1}, \ldots, y_{\lambda,d_\lambda}$ where $d_\lambda = \deg P_\lambda$. Let $R = F[y_{\lambda,i}]$ be the polynomial ring over F generated by $y_{\lambda,i}$ for all $\lambda \in \Lambda$ and all $i \leq \deg P_\lambda$. Write

$$P_\lambda - \prod_{i=1}^{d_\lambda}(X - y_{\lambda,i}) = \sum_{j=0}^{d_\lambda - 1} r_{\lambda,j} X^j \in R[X] \tag{1}$$

with $r_{\lambda,j} \in F[y_{\lambda,i}]$. Let I be the ideal in R generated by the $r_{\lambda,j}$. We claim that $I \subset R$ is a proper ideal of R. If not, I is the unit ideal, then there exists finite $\{r_{\lambda,j}\}$ which generate 1. So only a finite number of polynomials are involved, say $\{P_\lambda : \ \lambda \in \Lambda_1\}$ where Λ_1 is a finite subset of Λ, and let K be a finite extension in which every P_λ splits, using Lemma 1.2.7. Numerate the roots of P_λ as $u_{\lambda,1}, \ldots, u_{\lambda,d_\lambda}$. Now evaluate $y_{\lambda,i}$ at $u_{\lambda,i}$ (for $\lambda \in \Lambda_1$) and set $y_{\lambda,i} = 0$ (or any value) otherwise, we deduce $r_{\lambda,j}(u_{\lambda,j}) = 0$ from (1). Since 1 can be generated by these $\{r_{\lambda,j} : \ \lambda \in \Lambda_1, 0 \leq j \leq d_\lambda - 1\}$, we get $0 = 1$ in K, contradiction.

The claim implies that I is contained in a maximal ideal $\mathfrak{m}$ in R (this uses Zorn's lemma). Set $K := R/\mathfrak{m}$, which is an extension of F.

It is clear that K is algebraic over F, because K is generated by $y_{\lambda,i} \pmod{\mathfrak{m}}$ and every $y_{\lambda,i} \pmod{\mathfrak{m}}$ is an algebraic element over F, with P_λ being its minimal polynomial by (1).

It is left to prove that K itself is algebraically closed. Let K'/K be an algebraic extension and $x \in K'$. Then x is also algebraic over F. By construction, its minimal polynomial splits over K, thus $x \in K$. $\square$

Now we show that algebraic closures are unique up to (non-canonical) isomorphisms.

Lemma 1.3.10. *(Extension lemma) Let F be a field and L an algebraically closed field. Let $\alpha : F \hookrightarrow L$ be an embedding. If K is an algebraic extension of F, then there exists an extension of α to an embedding $K \hookrightarrow L$.*

In particular, if K is algebraic closure of F, then get $K \hookrightarrow L$. If moreover, L is algebraic over F, then $K \cong L$.

Proof. Let S be the set of (E, τ), where $F \subset E \subset K$ and $\tau : E \hookrightarrow L$ extending α. If (E, τ), $(E', \tau') \in S$, then we say $(E, \tau) \leq (E', \tau')$ iff $E \subset E'$ and $\tau'|_E = \tau$. We get a partial order on S, and one checks that any chain (totally ordered subset) has an upper bound. By Zorn's lemma, there exists a maximal element in S, say (E, τ). We must have $E = K$. Otherwise, there is $u \in K \backslash E$, and since u is algebraic over E, we get an extension of τ: $E(u) \hookrightarrow L$, a contradiction. $\square$

We cite the theorem of Artin-Schreier.

Theorem 1.3.11. *If F is a field which is of finite index in $\overline{F}$ and $[\overline{F} : F] > 1$. Then F is a real closed field: $[\overline{F} : F] = 2$ and -1 is not a square in F, and $\overline{F} = F(\sqrt{-1})$.*

To study splitting field, it is often convenient to choose an algebraic closure $\overline{F}/F$.

Normal extensions

Let F be a field and fix an algebraic closure $\overline{F}$. All algebraic extensions of F considered below will be taken in $\overline{F}$.

Definition 1.3.12. *An algebraic extension K/F is called* normal *if for every irreducible polynomial $P \in F[X]$, if it has one root in K, then it splits over K.*

Remark 1.3.13. *Here it is necessary to consider irreducible polynomials.*

Let $u, v \in \overline{F}$, we say that u and v are *conjugates* over F if they have the same minimal polynomial over F. We thus have the following: *$F \subset K$ is normal if and only if K is closed under conjugates*, i.e. if $u \in K$, then all conjugates of u also belong to K.

We show that splitting fields are normal extensions.

Theorem 1.3.14. *Let K/F be an algebraic extension. The following statements are equivalent.*

(i) *K is a normal extension of F.*

(ii) *K is a splitting field over F of a family of polynomials $\{P_i \in F[X] : i \in I\}$. (In other words, K is the minimal field of $\overline{F}$ over which all P_i splits.)*

(iii) *Every F-embedding $\iota : K \hookrightarrow \overline{F}$ satisfies $\iota(K) = K$.*

Proof. (i)$\Longrightarrow$(ii) We choose a set $S = \{u_i : \ i \in I\}$ such that $K = F(S)$; this is always possible. Let P_i be the minimal polynomial of u_i over F for each $i \in I$. Since u_i is a root of P_i and K/F is normal, all roots of P_i are contained in K. Thus each P_i splits over K. Because K is generated by the roots of P_i, K is a splitting field of $\{P_i : i \in I\}$.

(ii)$\Longrightarrow$(iii) By assumption, K is generated by the roots of $\{P_i : i \in I\}$. The F-embedding $\iota : E \hookrightarrow \overline{F}$ is determined by the images of roots of the P_i's. It is clear that if u_i is a root of P_i, then $\iota(u_i)$ is also a root of P_i , hence $\iota(u_i) \in K$.

(iii)$\Longrightarrow$(i) Let Q be an (arbitray) irreducible polynomial in $F[X]$ with a root $u \in K$. We need show that Q splits in K. Let $v \in \overline{F}$ be another root of Q. We then get an F-embedding $F(u) \hookrightarrow \overline{F}$ sending u to v by Lemma 1.2.7. This can be extended to an F-embedding $\iota : K \hookrightarrow \overline{F}$. By assumption $\iota(K) = K$, in particular $v = \iota(u) \in K$. □

Corollary 1.3.15. (i) *Let $F \subset E \subset K$ be a tower of algebraic (not necessarily finite) extensions. If K/F is normal, then K/E is normal (but E/F not necessarily).*

(ii) *If E/F and E'/F are normal extensions, then so is EE'/F.*

Proof. Exercise. □

Remark 1.3.16. *In general, even if K/E and E/F are both normal, K/F need not be normal. For example, $F = \mathbb{Q}$, $E = \mathbb{Q}(\sqrt{2})$ and $K = \mathbb{Q}(\sqrt[4]{2})$.*

1.4 Separable and purely inseparable extensions

Definition 1.4.1. *A polynomial $P \in F[X]$ is called* separable *if it has no multiple roots. Otherwise it is called* inseparable.

For $P \in F[X]$, we have the notion of 'derivate': if $P = \sum_{k=0}^n a_k X^k$, then

$$P' := \sum_{k=1}^{n} k a_k X^{k-1}.$$

Lemma 1.4.2. *Let $P \in F[X]$. Then P has multiple roots in $\overline{F}$ if and only if $(P, P') \neq 1$. When P is irreducible, the latter condition is equivalent to $P' = 0$.*

Proof. Write $P(X) = \prod_{k=1}^n (X - a_k)$ with $a_k \in \overline{F}$. Then

$$P' = \sum_{i=1}^{n} (X - a_1) \cdots (X - a_{i-1})(X - a_{i+1}) \cdots (X - a_n).$$

If it has a multiple root, say $a_1 = a_2$, then $X - a_1$ still divides P', so $(P, P') \neq 1$.[(2)] Conversely, if P has no multiple roots, then $P'(a_i) \neq 0$ for any $1 \leq i \leq n$, i.e. $(P, P') = 1$. □

Corollary 1.4.3. *If $\mathrm{char}(F) = 0$, then any irreducible polynomial has only simple roots in $\overline{F}$.*

For an irreducible polynomial $P \in F[X]$, the following statements are equivalent:

[(2)] Strictly speaking, here we only get $(P, P') \neq 1$ in $\overline{F}[X]$, but we can deduce $(P, P') \neq 1$ in $F[X]$.

(1) P is inseparable;

(2) $P' = 0$;

(3) $\mathrm{char}(F) = p$, and there exist $P_1 \in F[X]$ such that $P(X) = P_1(X^p)$ (since $ka_k = 0$ for all $0 \leq k \leq d$, so that either $a_k = 0$ or $p|k$).

We can continue the procedure in (3): if P_1 is again inseparable, we get P_2 such that $P_1(X) = P_2(X^p)$, etc. The degree of P being finite, we finally get a *separable* polynomial $P_e \in F[X]$ such that

$$P(X) = P_e(X^{p^e}).$$

If we write $n = \deg P$, $n_s = \deg P_e$, then $n = p^e n_s$. We call n_s the *separable degree* of P, and p^e the *inseparable degree.*

Definition 1.4.4. (i) *Let K/F be an algebraic extension. An element $u \in K$ is called* separable*, if its minimal polynomial is separable. Otherwise, u is called* inseparable.

(ii) *The extension K/F is called* separable*, if every element of K is separable. Otherwise, K/F is called* inseparable *extension.*

Example 1.4.5. *Consider the function field $F = \mathbb{F}_p(t)$, where p is a prime number. Show that the polynomial $P(X) = X^p - t \in F[X]$ is irreducible and inseparable.*

Lemma 1.4.6. *Assume* $\mathrm{char}(F) = p$ *and let $u \in \overline{F}$. Then u is separable if and only if $F(u) = F(u^p)$.*

Proof. $\Longrightarrow$ we have $F' := F(u^p) \subset F(u)$. Over $F(u^p)$, the polynomial $X^p - u^p$ annihilates u, so the minimal polynomial of u over F' divides $X^p - u^p = (X - u)^p$ (as $\mathrm{char}(F) = p$). But u is separable over F', so its minimal polynomial has no multiple root, hence is just $X - u$, i.e. $u \in F'$.

$\Longleftarrow$ Assume that $F(u) = F(u^p)$. We show that u is separable, i.e. its minimal polynomial P is separable. If not, we write $P = P_1(X^p)$ for some P_1, and set $n_1 := n/p$. Since u^p has minimal polynomial P_1, we have $[F(u^p) : F] = n_1$. But $[F(u) : F] = n$, a contradiction. $\square$

Proposition 1.4.7. *Assume* $\mathrm{char}(F) = p$ *and let K/F be a finite extension degree d. The following statement are equivalent:*

(i) *K/F is a separable extension;*

(ii) *We have $K = FK^p$. (Note that $K^p := \{x^p : x \in K\}$ is a sub-field of K, but not necessarily contains F.)*

(iii) *There exists a basis $\{e_1, \ldots, e_d\}$ of K/F such that $\{e_1^p, \ldots, e_d^p\}$ are still a basis.*

(iii)' *For any basis $\{f_1, \ldots, f_d\}$ of K/F, $\{f_1^p, \ldots, f_d^p\}$ is still a basis.*

(iii)” *For any elements* $f_1, \dots, f_r$ *of* K *which are linearly independent over* F, $\{f_1^p, \dots, f_r^p\}$ *are still linearly independent over* F.

Remark 1.4.8. *Proposition 1.4.7 is not true for infinite extensions, e.g.* $K = \mathbb{F}_p(t^{1/p^\infty})$.

Proof. We first show that (iii), (iii)’, (iii)” are equivalent.

(iii)$\Longleftrightarrow$(iii)’ there is an invertible matrix $M \in M_n(F)$ such that

$$\begin{pmatrix} f_1 \\ \vdots \\ f_d \end{pmatrix} = M \cdot \begin{pmatrix} e_1 \\ \vdots \\ e_d \end{pmatrix}.$$

Taking p-power, get

$$\begin{pmatrix} f_1^p \\ \vdots \\ f_d^p \end{pmatrix} = M^{(p)} \cdot \begin{pmatrix} e_1^p \\ \vdots \\ e_d^p \end{pmatrix}.$$

Here $M^{(p)}$ denotes (a_{ij}^p) if $M = (a_{ij})$; since $\mathrm{char}(F) = p$, we have $\det M^{(p)} = (\det M)^p \neq 0$. In particular, $M^{(p)}$ is still invertible, hence $\{f_i^p\}$ is a basis.

(iii)’$\Longleftrightarrow$(iii)” It suffices to extend the set $\{f_1, \dots, f_r\}$ to a basis $\{f_1, \dots, f_d\}$.

Now consider other implications.

(i)$\Longrightarrow$(ii) For any $u \in K$, $F(u) = F(u^p)$, hence $FK^p = K$.

(ii)$\Longrightarrow$(iii) Let $\{e_1, \dots, e_d\}$ be a basis of K/F, i.e. $K = Fe_1 + Fe_2 + \cdots + Fe_d$. Passing to the p-th power and noting that $\mathrm{char}(K) = p$, we get

$$K^p = F^p e_1^p + \cdots + F^p e_d^p$$

hence(3)

$$FK^p = Fe_1^p + \cdots + Fe_d^p.$$

But $[FK^p : F] = [K : F] = d$, so $e_1^p, \dots, e_d^p$ must be a basis.

(iii)”$\Longrightarrow$(i) Fix $u \in K$, we need to show u is separable over F. Take the canonical basis of $F(u)$ over F which gives a basis of $F(u^p)$ over F by passing to the p-th power by (iii)”. In particular, $[F(u^p) : F] = [F(u) : F]$, hence $F(u^p) = F(u)$ and we conclude by Lemma 1.4.6. □

Corollary 1.4.9. *Assume* $\mathrm{char}(F) = p$ *and let* $u \in \overline{F}$. *Then* u *is separable over* F *if and only if* $F(u)/F$ *is a separable extension.*

Proof. $\Longleftarrow$ is obvious.

$\Longrightarrow$ Write $K = F(u)$. It is direct to see that $K^p = F(u^p)$. On the other hand, we have $K = F(u^p)$ by Lemma 1.4.6 (as u is separable). Thus the equality $K = FK^p$ holds and we conclude by Proposition 1.4.7. □

(3)Note that $[FK^p : F]$ is finite, so FK^p is equal to the minimal *subring* containing F and K^p, using Proposition 1.2.2(iii). Denote $L = Fe_1^p + \cdots + Fe_d^p$. Then L contains F and K^p, and L is a ring as $e_i^p e_j^p = (e_i e_j)^p \in K^p \subset L$. Thus $FK^p \subseteq L$; the inclusion $\supseteq$ is obvious.

Corollary 1.4.10. *Let $F \subset E \subset K$ be a tower of algebraic extensions. Then K/F is separable if and only if both K/E and E/F are separable.*

Proof. If they are finite extensions, use (ii): $K = EK^p = FE^pK^p = FK^p$. The general case can be reduced to this. Precisely let $u \in K$ and $P(X) = \sum_k a_k X^k \in E[X]$ the minimal polynomial. Replacing E by $F(a_0, \ldots, a_n)$; clearly E is separable over F. Moreover u is separable over E, we are thus reduced to the case of finite extensions. $\square$

Corollary 1.4.11. *Let $F \subset E, E' \subset \overline{F}$ and assume $E/F, E'/F$ are separable extensions, then so is EE'/F.*

Proof. Again we reduced the proof to the case of finite extensions. We need to check $F(EE')^p$ contains both E and E', this is true since $FE^p = E$ and $FE'^p = E'$. $\square$

Definition 1.4.12. *Let $F \subset K$ be an algebraic extension. Let K_s be the set of all separable elements in K. Then K_s is an intermediate field of K/F (by Corollary 1.4.9 and Corollary 1.4.11), called maximal separable extension of F in K. When it is finite, $[K : F]_s := [K_s : F]$ is called the separable degree.*

When $K = \overline{F}$, we write $F^{\mathrm{sep}} := K_s$, called a separable closure of F.

Purely inseparable extensions

Definition 1.4.13. *An element $u \in \overline{F}$ is* purely inseparable *over F if $u^{p^m} \in F$ for some $m \in \mathbb{N}$. By convention, if $u \in F$, then u is* purely inseparable*: take $m = 0$.*

An algebraic extension K/F is purely inseparable extension if every element of K is purely inseparable over F.

Corollary 1.4.14. (i) *If K/F is a finite purely inseparable extension, then $[K : F]$ is a power of p.*

(ii) *If K/E and E/F are both purely inseparable extensions, so is K/F.*

(iii) *A compositum of purely inseparable extensions is still purely inseparable.*

Proof. Left as an exercise. $\square$

Proposition 1.4.15. *Let K/F be an algebraic extension, with K_s the maximal separable subextension in K. Then K/K_s is purely inseparable.*

Definition 1.4.16. *We define the inseparable degree of K/F to be $[K : K_s]$, denoted by $[K : F]_i$, which makes*

$$[K : F] = [K : F]_s [K : F]_i.$$

The link between separable extensions and the number of embeddings from K to $\overline{F}$ is as follows. Let K/F be an algebraic extension. Write $\mathrm{Hom}_F(K, \overline{F})$ for the set of F-embeddings $K \hookrightarrow \overline{F}$.

Proposition 1.4.17. *Let K/F be a finite extension. Then*

$$[K : F]_s = |\mathrm{Hom}_F(K, \overline{F})|.$$

Proof. Let K_s be the maximal separable subextension. We first observe that if $\tau : K \hookrightarrow \overline{F}$ is an F-embedding, then τ is determined by its restriction to K_s. In fact, K/K_s is purely inseparable, i.e. for any $u \in K$, $u^{p^m} \in K_s$ for some $m \geq 0$, hence if

$$v := \tau(u^{p^m}) = (\tau(u))^{p^m} \in \overline{F}.$$

But any element in $\overline{F}$ has a unique p^m-th power root, namely the equation $X^{p^m} - v = 0$ has a unique solution in $\overline{F}$ for any $v \in \overline{F}$, denoted by $v^{p^{-m}}$. We must have

$$\tau(u) = v^{p^{-m}}.$$

As a consequence we get

$$\mathrm{Hom}_F(K, \overline{F}) = \mathrm{Hom}_F(K_s, \overline{F}),$$

i.e. we may assume that K is separable over F.

We first treat the case when $K = F(u)$ is a simple extension. Then the minimal polynomial P of u has degree $n = [K : F]$. Since P is separable, it has n distinct roots in $\overline{F}$, say $u_1 = u, u_2, \ldots, u_n$. For each i, we get an embedding $K \hookrightarrow \overline{F}$ by sending u to u_i. Of course, these are different, so we get (at least) n embedding. Conversely, any embedding must send u to some u_i, hence we have the equality $|\mathrm{Hom}_F(K, \overline{F})| = n$, proving the assertion in this case.

In general, since K/F is finite, K is generated over F by finite elements, then do an easy induction. □

1.5 Galois extensions

Let K/F be an extension, we write $\mathrm{Aut}_F(K)$ for the group of F-automorphisms of K: it becomes a group, the binary operation being the composition: $(\sigma, \tau) \mapsto \sigma \circ \tau$. Note that, in general, $\mathrm{Aut}_F(K) \neq \mathrm{Hom}_F(K, \overline{F})$; this is true only when K is normal.

We have two basic operations.

1. If H is a subgroup of $\mathrm{Aut}_F(K)$, we can attach the *fixed field*:

$$K^H := \{u \in K : \forall \tau \in H, \tau(u) = u\}.$$

Obviously K^H is an intermediate field of K/F.

2. If E/F is a subextension of K/F, then we have $\mathrm{Aut}_E(K)$: the group of E-automorphisms of K. It is a subgroup of $\mathrm{Aut}_F(K)$.

These operations are both order-reversing:

$$H_1 \subset H_2 \Rightarrow K^{H_1} \supset K^{H_2}$$

$$E_1 \subset E_2 \Rightarrow \mathrm{Aut}_{E_1}(K) \supset \mathrm{Aut}_{E_2}(K).$$

Definition 1.5.1. *A* Galois extension *of F is a normal and separable algebraic extension of F. If K/F is Galois, we define the Galois group* $\mathrm{Gal}(K/F)$ *to be* $\mathrm{Aut}_F(K)$.[4]

Definition 1.5.2. *Let K/F be a separable extension. The* Galois closure *of K/F (inside $\overline{F}$) is the smallest Galois extension of F containing K.*

Lemma 1.5.3. *Let K/F be a finite Galois extension. Then*

(i) $|\mathrm{Gal}(K/F)| = [K : F]$;

(ii) $K^{\mathrm{Gal}(K/F)} = F$.

Proof. (1) Since K/F is separable, we have $|\mathrm{Hom}_F(K, \overline{F})| = [K : F]$ by Proposition 1.4.17. But K is normal, every F-embedding $K \hookrightarrow \overline{F}$ has image K, so $\mathrm{Hom}_F(K, \overline{F}) = \mathrm{Aut}_F(K) = \mathrm{Gal}(K/F)$.

(2) It is clear that $F \subset K^{\mathrm{Gal}(K/F)}$. For any $u \in K^{\mathrm{Gal}(K/F)}$, denote by P its minimal polynomial in $F[X]$ which is separable. If $v \in \overline{F}$ is a root of P, we get an F-embedding $F(u) \hookrightarrow \overline{F}$ mapping u to v; it extends to an embedding $\tau : K \hookrightarrow \overline{F}$ by Lemma 1.3.10. Since K is normal, τ belongs to $\mathrm{Gal}(K/F)$. By assumption, u is fixed by $\mathrm{Gal}(K/F)$, i.e. $u = \tau(u) = v$. Therefore P has only one root in $\overline{F}$. Since P has no multiple roots, we must have $\deg P = 1$ and $u \in F$. □

The next lemma is quite general !

Lemma 1.5.4 (Artin)**.** *Let K be a field. Let H be a finite subgroup of* $\mathrm{Aut}(K)$ *and $E = K^H$. Then K/E is a Galois extension of degree* [5]

$$[K : E] \leq |H|.$$

Proof. First, we prove that the extension K/E is Galois. Let $u \in K$, we want to write down its minimal polynomial over E. Let O be the H-orbit of u, namely $O = \{\tau(u) : \tau \in H\}$ (without multiplicities), and consider

$$P_u(X) := \prod_{\alpha \in O} (X - \alpha).$$

More precisely, if H_1 denotes the stabilizer of u and if $H = \coprod_i g_i H_1$ is the left coset decomposition of H over H_1, then $P_u(X) = \prod_i (X - g_i u)$. Since $\tau(P_u) = P_u$ for any $\tau \in H$, the coefficients of P_i lie in $K^H = E$, i.e. $P_u \in E[X]$. This implies that P, the minimal polynomial of u over E, divides P_u. Since P_u has no multiple roots, so neither does P. A conjugate of u is one of $\alpha \in O$, so lies in K, i.e. K/E is normal.

Next, we prove that $[K : E] \leq |H| =: n$. Let $v_1, \dots, v_{n+1}$ be arbitrary $n + 1$ elements of K, we have to show that they are linearly dependent over E. We may assume $v_i \neq 0$

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[4] In some references, the notion $\mathrm{Gal}(K/F) := \mathrm{Aut}_F(K)$ is used for any field extension K/F (even not algebraic). But we prefer to write $\mathrm{Gal}(K/F)$ only for Galois extensions.

[5] We in fact have an equality.

for any i. Let $\tau_1, \ldots, \tau_n$ be the elements of H. In the matrix $(\tau_i(v_j))_{n\times(n+1)}$, the $n+1$ columns are linearly dependent over K, so we can find a set of $n+1$ elements c_j in K, *not all zero*, such that the following system of n homogeneous equations holds:

$$\sum_{j=1}^{n+1} c_j(\tau_i(v_j)) = 0, \quad i = 1, 2, \ldots, n.$$

Among all such sets $\{c_1, c_2, \ldots, c_{n+1}\}$, we choose one with the smallest number of non-zeros. We assume that $\{c_1, c_2, \ldots, c_{n+1}\}$ has already been chosen in this fashion. Let, say $c_1, c_2, \ldots, c_r \neq 0$, $c_{r+1} = \cdots = c_{n+1} = 0$. Then $r \geq 2$, because if $r = 1$ then $\tau_i(v_1) = 0$ and hence $v_1 = 0$ (in fact one of τ_i is the identity.) We have then

$$\sum_{j=1}^{r} c_j \cdot \tau_i(v_j) = 0, \quad i = 1, \ldots, n \tag{2}$$

and, in particular, taking for τ_i the identity of H we have

$$\sum_{j=1}^{r} c_j v_j = 0. \tag{3}$$

We may assume $c_1 = 1$. **Claim**: $c_2, \ldots, c_r$ belong to E, whence by (3), $v_1, \ldots, v_{n+1}$ are indeed linearly dependent over E, as wanted.

Proof of Claim. Since $E = K^H$, we have to prove that $\tau_i(c_j) = c_j$ for $i = 1, 2, \ldots, n$. Fix $\tau \in H$ (i.e. one of $\{\tau_1, \ldots, \tau_n\}$). Apply to (2) the automorphism τ we get

$$\sum_{j=1}^{r} \tau(c_j) \cdot \tau\tau_i(v_j) = 0, \quad i = 1, 2, \ldots, n.$$

The elements $\{\tau\tau_i\}$ give again all the elements of H. Hence we have

$$\sum_{j=1}^{r} \tau(c_j)\tau_i(v_j) = 0, \quad \forall i.$$

Noting that $c_1 = 1 = \tau(1)$, we find

$$\sum_{j=2}^{r} (\tau(c_j) - c_j)(\tau_i(v_j)) = 0, \quad \forall i.$$

Here we have a set of n relations similar to (2), but the number of terms in each of these relations is less than r. Hence, by the choice of the set $\{c_1, \ldots, c_r, 0, \ldots, 0\}$, we must have

$$\tau(c_j) = c_j, \quad j = 2, 3, \ldots, r.$$

This being true for any $\tau \in H$, we obtain $c_j \in K^H = E$. $\square$

1.6 The Galois correspondence

The fundamental theorem of Galois theory is the following

Theorem 1.6.1. *Let K/F be a finite Galois extension. There is a one-to-one correspondence between the subgroups H of $G = \mathrm{Gal}(K/F)$ and the intermediate fields $F \subset E \subset K$. The correspondence is given by $E = K^H$ and $H = \mathrm{Gal}(K/E)$.*

Proof. By Lemma 1.5.3, we have $K^{\mathrm{Gal}(K/E)} = E$ for any $E \subset K$. We need to prove $\mathrm{Gal}(K/K^H) = H$. Clearly, $H \subset \mathrm{Gal}(K/K^H)$, both viewed as subgroups of G, so we are left to prove

$$|\mathrm{Gal}(K/K^H)| \leq |H|,$$

which follows from Artin's Lemma 1.5.4. □

Corollary 1.6.2. *Let K/F be a finite Galois extension. If $F \subset E \subset K$, then E/F is Galois if and only if $\mathrm{Gal}(K/E)$ is a normal subgroup of $\mathrm{Gal}(K/F)$. When this holds, the Galois group $\mathrm{Gal}(E/F)$ is isomorphic to $\mathrm{Gal}(K/F)/\mathrm{Gal}(K/E)$.*

Proof. Let $G = \mathrm{Gal}(K/F)$ and $H = \mathrm{Gal}(K/E)$. For any $\tau \in G$, $\{\tau(x) : x \in E\}$ is still an intermediate field of K/F, denoted by $\tau(E)$. One checks that $\tau H \tau^{-1} = \mathrm{Gal}(K/\tau(E))$.

If E/F is normal, then $\tau(E) = E$ for any τ, hence $\tau H \tau^{-1} = H$ and H is normal in G. Conversely, if H is normal in G, we have

$$\mathrm{Gal}(K/E) = H = \tau H \tau^{-1} = \mathrm{Gal}(K/\tau(E)).$$

This implies $E = K^H = K^{\tau H \tau^{-1}} = \tau(E)$, hence E/F is a normal extension.

Finally, we define a homomorphism of groups

$$\mathrm{Gal}(K/F) \to \mathrm{Gal}(E/F), \quad \tau \mapsto \tau|_E.$$

Its kernel is clearly $\mathrm{Gal}(K/E)$. It is surjective because any F-embedding $E \hookrightarrow \overline{F}$ can be extended to an F-embedding $K \hookrightarrow \overline{F}$ by Lemma 1.3.10 (alternatively, one can argue by comparing their orders). □

Proposition 1.6.3. *Let E/F, K/F be two algebraic extensions and assume E/F is finite Galois.*

(i) *The extension EK/K is also finite Galois.*

(ii) *There is an isomorphism $\mathrm{Gal}(EK/K) \cong \mathrm{Gal}(E/E \cap K)$, hence it can be viewed as a subgroup of $\mathrm{Gal}(E/F)$.*

Remark 1.6.4. *As a consequence of Proposition 1.6.3 (ii), we have $[E : E \cap K] = [EK : K]$. However, be attention that this need not be true if E/F is not Galois, e.g. $E = \mathbb{Q}(\sqrt[3]{2})$, $K = \mathbb{Q}(\omega\sqrt[3]{2})$. So correspondingly, the restriction map $\mathrm{Hom}_K(EK, \overline{F}) \to \mathrm{Hom}_{E \cap K}(E, \overline{F})$ need not be surjective in general.*

Proof. (i) Since E/F is normal, E is a splitting field of some polynomial $P \in F[X]$. Hence EK is the splitting field of P over K, and that EK/K is normal. The extension is separable by Proposition 1.4.7: assume $\mathrm{char}(F) = p$, then $EK = E^p K = (EK)^p K$ as $E = FE^p$.

(2) Consider the map

$$\iota : \mathrm{Gal}(EK/K) \to \mathrm{Gal}(E/(E \cap K)), \quad \sigma \mapsto \sigma|_E.$$

This is clearly a group homomorphism. If $\sigma \in \mathrm{Gal}(EK/K)$ is such that $\sigma|_E = \mathrm{Id}_E$, then it is identity on EK (as σ fixes K). This proves the injectivity of ι.

We prove the surjectivity. Let H be the image of ι, which is a subgroup of $\mathrm{Gal}(E/E \cap K)$. By the Galois correspondence, $H = \mathrm{Gal}(E/E \cap K)$ if and only if $E^H = E \cap K$. The inclusion $E \cap K \supset E^H$ is obvious. Conversely, let $u \in E^H$. Then $u \in EK$ is fixed by $\mathrm{Gal}(EK/K)$, hence $u \in K$, and $u \in E \cap K$. So we get the equality $E \cap K = E^H$ and concludes the proof. □

Proposition 1.6.5. *Let $K/F, K'/F$ be two finite Galois extensions. Then KK'/F is still finite Galois and there is an injection* $\mathrm{Gal}(KK'/F) \hookrightarrow \mathrm{Gal}(K/F) \times \mathrm{Gal}(K'/F)$.

Proof. It is clear that KK'/F is Galois, by Corollary 1.3.15 and Corollary 1.4.11. The injection is given by $\sigma \mapsto (\sigma|_K, \sigma|_{K'})$. It is an isomorphism if and only if $K \cap K' = F$ comparing the orders (using $[KK' : F] = [KK' : K][K : F] = [K' : F][K' : F]$). □

Definition 1.6.6. *A Galois extension K/F is called* abelian *if* $\mathrm{Gal}(K/F)$ *is an abelian group.*

Corollary 1.6.7. *A compositum of abelian extensions is still abelian.*

Remark 1.6.8. *The largest abelian extension inside $\overline{F}$ is denoted by F^{ab}. The structure of* $\mathrm{Gal}(F^{\mathrm{ab}}/F)$ *is the main object of class field theory.*

The following direct corollary is, however, non trivial.

Corollary 1.6.9. *Let K/F be a finite separable extension, then there are only finitely many subfields with $F \subset E \subset K$.*

Proof. By replacing K by its Galois closure, which is still finite over F, we may assume K is Galois: an intermediate field between F and K is also an intermediate field between F and K^{Gal}. Then use Galois correspondence to conclude, since $\mathrm{Gal}(K/F)$ has only finitely many subgroups. □

Primitive element theorem

Theorem 1.6.10. *If K/F is a finite separable extension, then $K = F(u)$ for some $u \in K$, i.e. K is a simple extension of F.*

Proof. If F is a finite field (hence so is K), then it follows from the fact that $K^\times$ is a cyclic group.

Suppose F is infinite for the rest of the proof. We know that there are only finitely many subfields $F \subset E \subset K$. Write $K = F(u_1, \dots, u_n)$ for suitable $u_i \in K$. An induction shows that we may assume $n = 2$.

For each $r \in F$, $F(u_1 + ru_2)$ is an intermediate field of K/F. The above observation, using that F is infinite, shows that there exist $r \neq s \in F$ such that

$$F(u_1 + ru_2) = F(u_1 + su_2).$$

But then $(r - s)u_2 \in F(u_1 + ru_2)$, which implies that u_2, and also u_1, lie in $F(u_1 + ru_2)$. Thus $K(u_1, u_2) = F(u_1 + ru_2)$ is a simple extension. $\square$

Remark 1.6.11. *Theorem 1.6.10 is false if K/F is not separable.* ***Example****: take $F = \mathbb{F}_p(x, y)$, $K = \mathbb{F}_p(x^{1/p}, y^{1/p})$, then $[K : F] = p^2$, and $F_r := F(x^{1/p} + ry^{1/p})$ is an intermediate field of K/F for any $r \in F$ of degree p. Thus $F_r \subsetneq K$ for any $r \in F$, and consequently $F_r \neq F_{r'}$ if $r \neq r'$.*

Examples of Galois correspondence

Let $K = \mathbb{Q}(\sqrt[3]{2}, \xi_3)$. It is a Galois extension over $\mathbb{Q}$ of degree 6, and $F := \mathbb{Q}(\xi_3)$ is a Galois subextension of degree 2. Note that $G = \mathrm{Gal}(K/\mathbb{Q})$ is not abelian, since $\mathbb{Q}(\sqrt[3]{2})$ is an intermediate field which is not Galois over F. Thus, G is isomorphic to D_3 (or S_3), the dihedral group of order 6. It has a presentation

$$\{r, a \mid r^3 = 1, a^2 = 1, ara = r^{-1}\}.$$

To explicitly construct an isomorphism $G \cong D_3$, we define

$$r \in \mathrm{Gal}(K/\mathbb{Q}(\xi_3)) : \sqrt[3]{2} \mapsto \sqrt[3]{2}\xi_3$$

$$a \in \mathrm{Gal}(K/\mathbb{Q}(\sqrt[3]{2})) : \xi_3 \mapsto \xi_3^2.$$

It is easy to check the relation $ara = r^{-1}$. The lattice of subgroups of G is as follows:

$$\begin{tikzcd}
 & \{\mathrm{Id}\} \arrow[dl, dash] \arrow[d, dash] \arrow[dr, dash] \arrow[drr, dash] & & \\
\langle r \rangle \arrow[dr, dash] & \langle a \rangle \arrow[d, dash] & \langle ar \rangle \arrow[dl, dash] & \langle ar^2 \rangle \arrow[dll, dash] \\
 & D_3 & &
\end{tikzcd}$$

and the corresponding intermediate fields of $K/\mathbb{Q}$ are:

$$\begin{tikzcd}
& K \arrow[dl, dash] \arrow[d, dash] \arrow[dr, dash] \arrow[drr, dash] & & \\
\mathbb{Q}(\xi_3) \arrow[dr, dash] & \mathbb{Q}(\sqrt[3]{2}) \arrow[d, dash] & \mathbb{Q}(\sqrt[3]{2}\xi_3) \arrow[dl, dash] & \mathbb{Q}(\sqrt[3]{2}\xi_3^2) \arrow[dll, dash] \\
& \mathbb{Q} & &
\end{tikzcd}$$

1.7 Galois groups of polynomials

Definition 1.7.1. *Let $f \in F[X]$ be a separable polynomial and K be its splitting field. The Galois group* $\mathrm{Gal}(K/F)$ *is called the Galois group of f and denoted by G_f.*

In most of the time, we will assume $f \in F[X]$ is moreover irreducible.

Lemma 1.7.2. *Let $f \in F[X]$ be a separable irreducible polynomial of degree n. Then G_f is isomorphic to a transitive subgroup of the symmetric group S_n. Moreover, the order $o(G_f)$ is divided by n.*

Recall that a subgroup $H < S_n$ is called *transitive* if for all i, j, there exists $\sigma \in H$ such that $\sigma(i) = j$ (this does not mean $(ij) \in H$).

Proof. Let $\alpha_1, \ldots, \alpha_n$ be the distinct roots of f. Let S_n be the group of permutations of the set $T := \{\alpha_1, \ldots, \alpha_n\}$. Since $\sigma \in G_f$ is determined by the images $\sigma(\alpha_i)$, which still lie in T, σ defines a permutation on T, i.e. $\sigma \in S_n$. For any $1 \leq i, j \leq n$, there exists an F-isomorphism $K \xrightarrow{\sim} K$ sending α_i to α_j, so G_f is transitive. The last statement follows from the facts that $o(G_f) = [K : F] = [K : F(\alpha_1)][F(\alpha_1) : F]$ (where K is the splitting field of f) and $[F(\alpha_1) : F] = n$. $\square$

Let $\alpha_1, \ldots, \alpha_n \in \overline{F}$ be the roots of f. Set

$$\Delta(f) = \prod_{1 \leq i < j \leq n} (\alpha_i - \alpha_j), \quad D(f) = \Delta(f)^2 = \prod_{1 \leq i < j \leq n} (\alpha_i - \alpha_j)^2.$$

Here $D(f)$ is called the *discriminant* of f. Note that $\Delta(f) \neq 0$ as f is separable.

Example 1.7.3. *We have*

$$D(X^2 + bX + c) = b^2 - 4c, \quad D(X^3 + px + q) = -4p^3 - 27q^2.$$ (6)

Let $\sigma \in S_n$; it naturally acts on the set $\{\alpha_1, \ldots, \alpha_n\}$. It is easy to see that

$$\sigma\Delta(f) = \mathrm{sign}(\sigma)\Delta(f),$$

where $\mathrm{sign}(\sigma) \in \{\pm 1\}$ denotes the *signature* of σ. Hence $\sigma D(f) = D(f)$. This implies that $D(f) \in F[X]$, since the Galois group G_f is in particular a subgroup of S_n. Also we see that $\Delta(f)$ is fixed by σ if and only if $\mathrm{sign}(\sigma) = 1$, if and only if $\sigma \in A_n$. Therefore, we get the following:

(6) In general, $D(X^3 + bX^2 + cX + d) = b^2c^2 - 4c^3 - 4b^3d - 27d^2 + 18bcd$

Lemma 1.7.4. *Assume* $\mathrm{char}(F) \neq 2$. *Then*

$$G_f \subset A_n \Longleftrightarrow \Delta(f) \in F^\times \Longleftrightarrow D(f) \in F^{\times 2}.$$

From now on, we assume $\mathrm{char}(F) \neq 2$.

1.7.1 $\deg f = 3$

We know that G_f is a subgroup of S_3 and $3|o(G_f)$, so $o(G_f) \in \{3, 6\}$ and there are two possibilities for G_f: A_3 (the unique subgroup of order 3) and S_3. Lemma 1.7.4 implies that $G_f = A_3$ if and only if $D(f)$ has a square root in $F^\times$.

Example 1.7.5. *Take* $f = X^3 - 3X + 1 \in \mathbb{Q}[X]$. *It is irreducible, with discriminant* $-4(-3)^3 - 27 = 81 = 9^2$, *so* $G_f = A_3$. *But for* $f = X^3 + 3X + 1 \in \mathbb{Q}[X]$, *we have* $G_f = S_3$.

1.7.2 $\deg f = 4$

We first list the subgroups of S_4, up to conjugation (except for 1 and S_4 itself):

- S_2, A_3 or S_3: order 2, 3 or 6 respectively, not transitive;
- $\{\mathrm{Id}, (12)(34)\}$: order 2, not transitive;
- $\langle (12), (34) \rangle$: not normal, not transitive;
- $\langle (1234) \rangle = \{\mathrm{Id}, (1234), (13)(24), (1432)\}$: order 4, cyclic group, transitive, not normal;
- $V_4 := \{\mathrm{Id}, (12)(34), (13)(24), (14)(23)\}$: order 4, normal abelian subgroup, transitive;
- $\langle (1234), (13) \rangle \cong D_4$ (dihedral group): order 8 and Sylow 2-subgroup, transitive;
- A_4; normal subgroup, order 12, transitive.

In summary, transitive subgroups of S_4 whose order is divided by 4 are as follows:

- C_4, conjugates of $\langle (1234) \rangle$;
- $V_4 = \{\mathrm{Id}, (12)(34), (13)(24), (14)(23)\}$;
- D_4, conjugates of $\langle (1234), (13) \rangle$;
- A_4;
- S_4.

Moreover, if H contains a transposition (ij), then $H = D_4, A_4, S_4$.

The subgroup V_4 will play a crucial role. Let $\alpha_1, \alpha_2, \alpha_3, \alpha_4$ be the roots of f. Consider the following elements

$$\gamma_1 = \alpha_1\alpha_2 + \alpha_3\alpha_4$$

$$\gamma_2 = \alpha_1\alpha_3 + \alpha_2\alpha_4$$

$$\gamma_3 = \alpha_1\alpha_4 + \alpha_2\alpha_3.$$

They are pairwise distinct, e.g. $\gamma_1 - \gamma_2 = (\alpha_1 - \alpha_4)(\alpha_2 - \alpha_3) \neq 0$. It is clear that S_4 permutes the set $\{\gamma_1, \gamma_2, \gamma_3\}$.

Lemma 1.7.6. *The stabilizer of each γ_i is a Sylow 2-subgroup, e.g.*

$$\mathrm{Stab}(\gamma_2) = \langle (1234), (13) \rangle.$$

The group V_4 fixes all $\gamma_1, \gamma_2, \gamma_3$. The fixed field of $G_f \cap V_4$ is $F(\gamma_1, \gamma_2, \gamma_3)$, and $F(\gamma_1, \gamma_2, \gamma_3)$ is Galois over F with Galois group $G_f/(G_f \cap V_4)$.

Proof. Obvious. $\square$

We consider the polynomial

$$g(X) := (X - \gamma_1)(X - \gamma_2)(X - \gamma_3),$$

which has coefficients in F; it is called the *cubic resolvent* of f.

Lemma 1.7.7. *If $f = X^4 + bX^3 + cX^2 + dX + e$, then*

$$g = X^3 - cX^2 + (bd - 4e)X - b^2e + 4ce - d^2.$$

Moreover, f and g have the same discriminants.

Proof. The first statement is a direct computation. For example, from $f(X) = \prod_{i=1}^4 (X - \alpha_i)$ we get

$$c = \sum_{1 \le i < j \le 4} \alpha_i\alpha_j = \sum_{i=1}^3 \gamma_i$$

which explains the coefficient of X^2 in g. The last statement is also clear by using $\gamma_1 - \gamma_2 = (\alpha_1 - \alpha_4)(\alpha_2 - \alpha_3)$ and similar decompositions for $\gamma_1 - \gamma_3$ and $\gamma_2 - \gamma_3$. $\square$

In particular, when $b = c = 0$, $g = X^3 - 4eX - d^2$. When $b = d = 0$, $g = X^3 - cX^2 - 4eX + 4ce$.

Theorem 1.7.8. *Let $M := F(\gamma_1, \gamma_2, \gamma_3)$ be the splitting field of g over F and $m := [M : F]$. Then*

- *if $m = 1$, then $G_f = V_4$;*

- *if $m = 2$, then G_f is isomorphic to C_4 or D_4; moreover, G_f is isomorphic to D_4 if and only if f is irreducible in $M[X]$;*

- *if $m = 3$, then $G_f = A_4$;*

- *if $m = 6$, then $G_f = S_4$.*

Proof. This is because that $[M : F] = |\mathrm{Gal}(M/F)| = |G_f/G_f \cap V_4|$. It is easy to check that $[C_4 : C_4 \cap V_4] = 2$ and (since $V_4 \subset D_4$)

$$[D_4 : V_4] = 2, \quad [A_4 : V_4] = 3, \quad [S_4 : V_4] = 6.$$

This proves all the statements except the last one in the case $m = 2$.

Assume $m = 2$. The extension K/M is Galois with Galois group $G_f \cap V_4$, which has order 2 (resp. 4) if G_f is isomorphic to C_4 (resp. D_4). If f is irreducible in $M[X]$, then $G_f \cap V_4$ acts transitively on the four roots of f, so has order 4. Conversely, if f is reducible in $M[X]$, then f must decompose as $f_1 f_2$, with $\deg f_i = 2$. Therefore $G_f \cap V_4$ can not act transitively on $\{\alpha_i\}$, so $G_f \cap V_4$ is not transitive, equivalently $G_f = C_4$. $\square$

Proposition 1.7.9. *Assume that* $\mathrm{char}(F) \neq 2$. *Let $f = X^4 + bX^3 + cX^2 + dX + e \in F[X]$ be a separable irreducible polynomial. Assume $m = [M : F] = 2$, and let γ be the unique root of the cubic resolvent of f. Then G_f is isomorphic to D_4 if and only if either $D(f)(b^2 - 4c + 4\gamma) \in F^\times \backslash F^{\times 2}$, or $b^2 - 4c + 4\gamma = 0$ and $D(f)(\gamma^2 - 4e) \in F^\times \backslash F^{\times 2}$.*

Proof. We may assume $\gamma = \gamma_2$, so that $G_f < \langle (1234), (13) \rangle$ (see Lemma 1.7.6), with equality if and only if $G_f \cong D_4$. Since both (1234) and (13) are odd transformations, we have

$$(1234)\Delta(f) = -\Delta(f), \quad (13)\Delta(f) = -\Delta(f).$$

We want to find a suitable element $0 \neq \beta \in K$ satisfying

$$(1234)\beta = -\beta, \quad (13)\beta = \beta, \tag{4}$$

so that $\Delta(f)\beta$ is fixed by (1234) but not by (13). Consequently, if $G_f = \langle (1234) \rangle \cong C_4$, then $\Delta(f)\beta \in F^\times$ and so $D(f)\beta^2 \in F^{\times 2}$, while if $G_f = \langle (1234), (13) \rangle \cong D_4$ then $D(f)\beta^2 \notin F^{\times 2}$.

First assume $\alpha_1 + \alpha_3 \neq \alpha_2 + \alpha_4$, and set

$$\beta := (\alpha_1 + \alpha_3) - (\alpha_2 + \alpha_4).$$

It is easy to see that (4) is satisfied. Since $\beta^2 = b^2 - 4c + 4\gamma$ (recall $\gamma = \gamma_2$), we obtain that $G_f \cong D_4$ if and only if $D(f)(b^2 - 4c + 4\gamma) \notin F^{\times 2}$ (when $b^2 - 4c + 4\gamma \neq 0$).

If $\beta = 0$, we set

$$\beta' := \alpha_1\alpha_3 - \alpha_2\alpha_4.$$

Then $\beta' \neq 0$, otherwise we would get $\{\alpha_1, \alpha_3\} = \{\alpha_2, \alpha_4\}$, which is impossible. As before, one checks that (4) is satisfied for β' and that $\beta'^2 = \gamma^2 - 4e$. We conclude that if $b^2 - 4c + 4\gamma = 0$, then $G_f \cong D_4$ if and only if $D(f)(\gamma^2 - 4e) \notin F^{\times 2}$. $\square$

1.8 Examples

We recall some facts about the irreducibility of polynomials in $\mathbb{Q}[X]$. The first one is Gauss's lemma.

Proposition 1.8.1. *Let $f(X) = a_n X^n + a_{n-1}X^{n-1} + \cdots + a_0 \in \mathbb{Z}[X]$ be primitive, i.e. $\mathrm{GCD}\{a_i\} = 1$. Then $f(X)$ is irreducible in $\mathbb{Z}[X]$ if and only if it is irreducible in $\mathbb{Q}[X]$.*

For example, f is primitive if it is monic.

Proposition 1.8.2 (Eisenstein's criterion)**.** *Let $f = a_n X^n + a_{n-1}X^{n-1} + \cdots + a_0 \in \mathbb{Z}[X]$. Suppose that there is a prime p such that*

- *p does not divide a_n (ok if $a_n = 1$);*
- *p divides $a_{n-1}, \ldots, a_0$;*
- *p^2 does not divide a_0.*

Then f is irreducible in $\mathbb{Q}[X]$.

We also have the following simple fact.

Lemma 1.8.3. *Suppose $r \in \mathbb{Q}$ is a root of $f(X) = a_n X^n + a_{n-1}X^{n-1} + \cdots + a_0 \in \mathbb{Z}[X]$. Let $r = c/d$ with $(c, d) = 1$. Then $c|a_0$ and $d|a_n$. In particular, if $a_n = 1$, we get $r \in \mathbb{Z}$.*

This is easy to prove. In general, it is not enough for the irreducibility of f. But when $\deg f = 3$, f is reducible if and only if f has a root in $\mathbb{Q}$, so Lemma 1.8.3 is applicable.

Example 1.8.4. *Let $f = X^3 - 8X + 4 \in \mathbb{Q}[X]$. It is irreducible, because the only possible roots are ± 1, ± 2, ± 4, and we check that this is not the case.*

Remark 1.8.5. *Another way to show irreducibility of monic polynomial $f \in \mathbb{Z}[X]$ is to show $f \pmod p$ is irreducible in $\mathbb{F}_p[X]$ for some prime p.*

For example, $f(X) = X^4 - X^2 - 1$ is irreducible mod 3, so f is itself irreducible. Indeed, it is easy to see that $f(0), f(1), f(2) \neq 0$ in $\mathbb{F}_3$, and that $f \pmod p$ is not divisible by X^2+1, X^2+X+2 or X^2+2X+2 in $\mathbb{F}_3[X]$ (these are the only irreducible polynomials of degree 2 in $\mathbb{F}_3[X]$).

Next, we give some examples of irreducible polynomials and study their Galois groups.

Example 1.8.6. *Consider $f(X) = X^4 + 4X^2 + 2 \in \mathbb{Q}[X]$. It is irreducible by Eisenstein's criterion. Its cubic resolvent is*

$$g(X) = X^3 - 4X^2 - 8X + 32 = (X - 4)(X^2 - 8).$$

Therefore $M = \mathbb{Q}[\sqrt{2}]$. So $m = 2$ and G_f is either C_4 or D_4. Over M, f decomposes as

$$(X^2 + 2 - \sqrt{2})(X^2 + 2 + \sqrt{2}),$$

so $G_f \cong C_4$ by Theorem 1.7.8. One can also verify the conclusion using Proposition 1.7.9: $b = 0$, $c = 4$, $\gamma = 4$ so $b^2 - 4c + 4\gamma = 0$, but $D(f)(\gamma^2 - 4e) = (32^2 \times 2) \times 8 \in (\mathbb{Q}^\times)^2$.

Example 1.8.7. *Consider $f(X) = X^4 + 2X + 2 \in \mathbb{Q}[X]$. It is irreducible by Eisenstein's criterion for $p = 2$. Its cubic resolvent is*

$$g(X) = X^3 - 8X - 4.$$

One checks that g is also irreducible over $\mathbb{Q}$, so $m \geq 3$. As in §1.7.1, we compute the discriminant:

$$D(g) = -4(-8)^3 - 27(-4)^2 = 16 \times 101$$

which is not a square in $\mathbb{Q}$. So $m = 6$ and $G_f \cong S_4$.

Example 1.8.8. *Consider $f(X) = X^4 - X^2 - 1 \in \mathbb{Q}[X]$; it is irreducible as seen in Remark 1.8.5. Let K be the splitting field of f. It is easy to see that $\alpha = \sqrt{(1+\sqrt{5})/2}$ is a root of f, and the four conjugates of α are*

$$\alpha_1 = \alpha, \quad \alpha_2 = i/\alpha, \quad \alpha_3 = -\alpha, \quad \alpha_4 = -i/\alpha.$$

As a consequence, $K = \mathbb{Q}(\alpha, i)$ and $[K : \mathbb{Q}] = 8$ (note that $i \notin \mathbb{Q}(\alpha)$ as $\alpha \in \mathbb{R}$).

The cubic resolvent of f is

$$g = X^3 + X^2 + 4X + 4 = (X^2 + 4)(X + 1).$$

Thus $m = [M : \mathbb{Q}] = 2$. From Theorem 1.7.8 we deduce that $\mathrm{Gal}(K/\mathbb{Q}) \cong D_4$ (as $[K : \mathbb{Q}] = 8$). Note that $\gamma_2 = \alpha_1\alpha_3 + \alpha_2\alpha_4 = -1$ is the unique root of g in $\mathbb{Q}$, so $G_f = \langle (1234), (13) \rangle$ (when viewed as subgroups of S_4) by Lemma 1.7.6.

For simplicity, write $r = (1234)$ and $s = (13)$. The following table tells the explicit action of $\sigma \in G_f$:

σ	id	r	r^2	r^3	s	rs	r^2s	r^3s
$\sigma(\alpha)$	α	i/α	$-\alpha$	$-i/\alpha$	$-\alpha$	$-i/\alpha$	α	i/α
$\sigma(i)$	i	$-i$	i	$-i$	$-i$	i	$-i$	i

Below is the lattice of subgroups of D_4:

$$\begin{tikzcd}
 & & \{\mathrm{Id}\} \arrow[dll, dash] \arrow[dl, dash] \arrow[d, dash] \arrow[dr, dash] \arrow[drr, dash] & & \\
\langle s \rangle \arrow[dr, dash] & \langle r^2 s \rangle \arrow[d, dash] & \langle r^2 \rangle \arrow[dl, dash] \arrow[d, dash] \arrow[dr, dash] & \langle rs \rangle \arrow[d, dash] & \langle r^3 s \rangle \arrow[dl, dash] \\
 & \langle r^2, s \rangle \arrow[dr, dash] & \langle r \rangle \arrow[d, dash] & \langle r^2, rs \rangle \arrow[dl, dash] & \\
 & & D_4 & &
\end{tikzcd}$$

and the corresponding intermediate fields of $K/\mathbb{Q}$ are:

$$\begin{tikzcd}
& & \mathbb{Q}(\alpha,i) \arrow[dll, dash] \arrow[dl, dash] \arrow[d, dash] \arrow[dr, dash] \arrow[drr, dash] & & \\
\mathbb{Q}(\alpha i) \arrow[dr, dash] & \mathbb{Q}(\alpha) \arrow[d, dash] & \mathbb{Q}(\sqrt{5},i) \arrow[dl, dash] \arrow[d, dash] \arrow[dr, dash] & \mathbb{Q}(\sqrt{1-2i}) \arrow[d, dash] & \mathbb{Q}(\sqrt{1+2i}) \arrow[dl, dash] \\
& \mathbb{Q}(\sqrt{5}) \arrow[dr, dash] & \mathbb{Q}(\sqrt{-5}) \arrow[d, dash] & \mathbb{Q}(i) \arrow[dl, dash] & \\
& & \mathbb{Q} & &
\end{tikzcd}$$

For example, to determine $E_1 := K^{\langle s\rangle}$, we note that αi is fixed by s, so $\mathbb{Q}(\alpha i) \subset E_1$; this must be an equality as $[\mathbb{Q}(\alpha i) : \mathbb{Q}] = 4 = [E_1 : \mathbb{Q}]$. To determine $E_2 := K^{\langle rs\rangle}$, we note that

$$\alpha + (rs)\alpha = \alpha - i/\alpha$$

is fixed by rs; since $(\alpha - i/\alpha)^2 = 1 - 2i$, we obtain $E_2 = \mathbb{Q}(\sqrt{1-2i})$ (as $[E_2 : \mathbb{Q}] = 4 = [\mathbb{Q}(\sqrt{1-2i}) : \mathbb{Q}]$). To determine $E_3 := K^{\langle r^2,s\rangle}$, we note that it is a subfield of $\mathbb{Q}(\alpha i)$, and $(\alpha i)^2 = -(1+\sqrt{5})/2 \in E_3$, so $E_3 = \mathbb{Q}(\sqrt{5})$ (as $[E_3 : \mathbb{Q}] = 2$).

1.9 Polynomials with Galois group S_p over $\mathbb{Q}$

Theorem 1.9.1. *If p is a prime number and $f \in \mathbb{Q}[x]$ an irreducible polynomial of degree p with exactly two non-real roots in $\mathbb{C}$, then the Galois group of f over $\mathbb{Q}$ is isomorphic to the symmetric group S_p.*

Recall Cauchy's theorem: if G is a finite group, p is a prime and $p|o(G)$, then G has an element of order p. (7)

Proof. Let K be the splitting field of f and G_f its Galois group. By Cauchy's theorem, there is an element σ of order p, thus G contains a p-cycle (as p is a prime !).

Let $\tau : \mathbb{C} \to \mathbb{C}$ denote the complex conjugation $a + bi \mapsto a - bi$. The restriction $\tau|_K$ defines an element of $\mathrm{Gal}(K/\mathbb{Q})$, which is not trivial. Moreover, it just exchanges the two non-real roots and fixes the other real roots, so it is a 2-cycle (i.e. a transposition). We then conclude by Lemma 1.9.2 below. □

Lemma 1.9.2. *If p is a prime number, S_p is generated by any transposition and any p-cycle.*

Proof. We may assume the transposition is $\tau = (12)$ (up to renumbering). Also we may write the p-cycle σ as $(1 i_2 \cdots i_p)$. If $i_r = 2$ then σ^{r-1} maps 1 to 2, and is still a p-cycle (here we use that p is a prime). So we may assume $\sigma = (12 j_3 \cdots j_p)$, and after

(7) In fact, by Sylow's theorem, G has a subgroup H of order p^n, where $p^n \| o(G)$; since a p-group is solvable, H has subgroups of order p^r for any $r < n$.

renumbering $\{j_3, \dots, j_p\}$ we may assume $\sigma = (123\cdots p)$ (without changing τ). Namely we are left to prove that S_p is generated by (12) and $(123\cdots p)$. For this, one checks that for $i = 1, \dots, p-2$,

$$\sigma^i(12)\sigma^{-i} = (i+1\ i+2),$$

and that these transpositions generate S_p. $\square$

Remark 1.9.3. *In Lemma 1.9.2, it is crucial to work with S_p for a prime number p. For example, (13) and (1234) generate D_4 inside S_4.*

Example 1.9.4. *The polynomial $f(x) = x^5 - 5x + 2$ has Galois group S_5. First, it is irreducible, since putting $x = y - 2$, $f(y-2) = (y-2)^5 - 5(y-2) + 2$ with constant coefficient being $-32 + 10 + 2 = -20$, so that we can apply Eisenstein's criterion. Next, we check that f has 3 real roots, using $f'(x) = 15x^4 - 15 = 0$ has 2 real roots ± 1.*(8)

Dedekind's theorem

Theorem 1.9.5. *(Dedekind) Let $f(X) \in \mathbb{Z}[X]$ be a monic irreducible polynomial of degree m, and let p be a prime number such that $(f \mod p)$ has only simple roots.*(9) *Suppose that $f = \prod_i f_i$ with f_i irreducible of degree m_i in $\mathbb{F}_p[X]$. Then G_f contains an element whose cycle decomposition is of type*

$$m = m_1 + \cdots + m_r.$$

Proof. See Jacobson, Basic Algebra I, §4.16. $\square$

Example 1.9.6. *Consider $f(X) = X^5 - X - 1 \in \mathbb{Z}[X]$. It is irreducible as it is modulo 3. Modulo 2, this factors as $(X^2+X+1)(X^3+X^2+1)$. Hence G_f contains (12345) and a cycle of the form $\sigma = (ab)(cde)$. We check that $\sigma^3 = (ab)^3 = (ab)$ is a transposition. Therefore $G_f = S_5$ by Lemma 1.9.2.*

1.10 Finite fields

A field F is called *finite* if $|F|$ is finite. They are also called *Galois fields.* A basis example is $\mathbb{F}_p := \mathbb{Z}/(p)$ where p is a prime number; it has p elements.

Lemma 1.10.1. *A finite field has a positive characteristic, say p. Its cardinality is then a power of p.*

Proof. The first assertion is obvious. Since F is an $\mathbb{F}_p$-vector space, its cardinality equals to p^n where $n = [F : \mathbb{F}_p]$. $\square$

(8) we have $f(+\infty) > 0$, $f(1) < 0$, $f(-1) > 0$, $f(-\infty) < 0$.

(9) This is equivalent to that $D(f)$ is not divisible by p.

Theorem 1.10.2. *For every finite p-power q, there exists a finite field of cardinality q. Moreover, such a field is unique up to isomorphism.*

Proof. Given $q = p^m$, let $F/\mathbb{F}_p$ be the splitting field of the polynomial $X^q - X \in \mathbb{F}_p[X]$. Since $(u+v)^q = u^q + v^q$ in F, the roots of $X^q - X$ in F form a subfield. On the other hand, $X^q - X$ is separable as $(X^q - X)' = -1$ is prime to $X^q - X$ (see Lemma 1.4.2). Therefore, it has q distinct roots in F. By construction, F is exactly generated by the roots, hence $F = \{u \in F : u^q = u\}$ and has cardinality q.

Next we show any finite field of cardinality q, say F', is isomorphic to F. Since $|F'^{\times}| = q-1$, we have $u^{q-1} = 1$ for any $u \in F'^{\times}$, hence $u^q = u$ for any $u \in F'$. This shows that F' is a splitting field of $X^q - X$. We then conclude by the uniqueness of splitting fields (see Proposition 1.3.4). □

We denote by $\mathbb{F}_q$ the unique finite field of cardinality q. To give a more explicit structure of $\mathbb{F}_q$, especially for the multiplicative structure, we need another construction of $\mathbb{F}_q$. Write $q = p^n$. Take an irreducible polynomial $f \in \mathbb{F}_p[X]$ of degree n, and let α be a root of f in $\overline{\mathbb{F}}_p$. Since $\mathbb{F}_p[\alpha]$ is a field extension over $\mathbb{F}_p$ of degree n, it must be isomorphic to $\mathbb{F}_q$ (for any such f!) by Theorem 1.10.2. As a consequence, any element in $\mathbb{F}_q$ can be uniquely written as

$$a_{d-1}\alpha^{n-1} + \cdots + a_1\alpha + a_0, \quad a_i \in \mathbb{F}_p$$

and the addition and multiplication in $\mathbb{F}_q$ can be made explicit via f.

Being a splitting field of a separable polynomial, $\mathbb{F}_q$ is a Galois extension over $\mathbb{F}_p$. More generally, we have the following result.

Theorem 1.10.3. *(i) For $m, n \in \mathbb{N}$, there exists a field embedding $\mathbb{F}_{p^m} \hookrightarrow \mathbb{F}_{p^n}$ if and only if $m|n$.*

(ii) The group $\mathrm{Gal}(\mathbb{F}_{p^n}/\mathbb{F}_p)$ *is isomorphic to $\mathbb{Z}/n\mathbb{Z}$, and is generated by $x \mapsto x^p$, which is called* Frobenius.

(iii) An algebraic closure of $\mathbb{F}_p$ is isomorphic to $\varinjlim_n \mathbb{F}_{p^n}$.

Proof. (i) The condition is clearly sufficient. Conversely, if $\mathbb{F}_{p^m} \hookrightarrow \mathbb{F}_{p^n}$, then $\mathbb{F}_{p^n}$ becomes a vector space over $\mathbb{F}_{p^m}$. If d denotes the dimension, then $p^n = (p^m)^d$, in particular $m|n$.

(ii) We know that $|\mathrm{Gal}(\mathbb{F}_{p^n}/\mathbb{F}_p)| = n = [\mathbb{F}_{p^n} : \mathbb{F}_p]$. On the other hand, $\sigma : x \mapsto x^p$ is an element of $\mathrm{Gal}(\mathbb{F}_{p^n}/\mathbb{F}_p)$, so it suffices to prove that σ has order n. If d denotes the order of σ, then $x^{p^d} = x$ for any $x \in \mathbb{F}_{p^n}$, so $\mathbb{F}_{p^n} \hookrightarrow \mathbb{F}_{p^d}$ and $n|d$ by (i). But, we also have $d|n$ (as $\sigma^n = \mathrm{Id}$), hence $d = n$.

(iii) This is because any irreducible polynomial splits in $\overline{\mathbb{F}}_p$. □

Remark 1.10.4. *As an analog of Theorem 1.10.3(ii), for any p-power q and any $n \geq 1$, $\mathbb{F}_{q^n}/\mathbb{F}_q$ is Galois of group $\mathbb{Z}/n\mathbb{Z}$ with a generator given by $x \mapsto x^q$.*

The following result proves the primitive element theorem 1.6.10 in finite field case.

Proposition 1.10.5. *Let F be a finite field, then $F^{\times}$ forms a cyclic group.*

More generally, we prove the following result:

Proposition 1.10.6. *If F is a field, and G is a finite subgroup of $F^\times$, then G is cyclic.*

Proof. First note that G is an abelian group, hence is a finite direct product of cyclic groups:

$$G \cong H_1 \times H_2 \times \cdots \times H_r,$$

where $H_i = \langle x_i \rangle$ is a cyclic group of order m_i.

We claim that $(m_i, m_j) \neq 1$ for any $i \neq j$; this will imply that G is cyclic. Suppose this is not true, and let k be the LCM (least common multiple) of all m_i. Then $k < o(G) = m_1 \cdots m_r$. However, every element of G satisfies $f(x) = x^k - 1 \in F[x]$, while f has at most k roots in F (even in $\overline{F}$). Contradiction. $\square$

Counting irreducible polynomials of given degree

Let q be a finite p-power. Denote by $S_q(n)$ the set of monic, irreducible polynomials in $\mathbb{F}_q[X]$ of degree n and by $N_q(n)$ its cardinality. Note the following facts:

- $\mathbb{F}_{q^n}$ is the splitting field of the separable polynomial $g_n := X^{q^n} - X$, so $f|g_n$ for any $f \in S_q(n)$; actually for any $d|n$, $\mathbb{F}_{q^d}$ is a subfield of $\mathbb{F}_{q^n}$ by Theorem 1.10.3(i), so $f|g_n$ for any $f \in S_q(d)$.

- Any root of g_n generates a subfield of $\mathbb{F}_{q^n}$, hence its minimal polynomial lies in $S_q(d)$ with $d|n$.

Thus we obtain

$$X^{q^n} - X = \prod_{d|n} \prod_{f \in S_q(d)} f$$

for all $n \geq 1$, and in particular

$$q^n = \sum_{d|n} d N_q(d).$$

This implies the formula

$$N_q(n) = \frac{1}{n} \sum_{d|n} \mu(d) q^{n/d} \tag{5}$$

by the Möbius inversion formula. Here $\mu(d)$ denotes the Möbius function

$$\mu(d) = \begin{cases} 1 & d = 1 \\ 0 & p^2 | d \text{ for some prime } p \\ (-1)^r & d = p_1 \cdots p_r \text{ for distinct primes } p_i. \end{cases}$$

Example 1.10.7. *We have $N_p(3) = \frac{1}{3}(p^3 - p)$, and $N_p(4) = \frac{1}{4}(p^4 - p^2)$.*

1.11 Cyclotomic extensions

Let ξ_n be a *primitive* n-th root of unity in $\mathbb{C}$, i.e. $\xi_n^n = 1$ and $\xi_n^k \neq 1$ if $k < n$. Recall the following basic facts:

(i) the set of n-th roots of unity is just $\{\zeta_n^k : 0 \leq k \leq n-1\}$;

(ii) the root ζ_n^k is primitive if and only if $(k,n) = 1$; hence the number of primitive n-th roots of 1 is $\phi(n)$, the Euler's function. Explicitly we have $\phi(n) = n\prod_{p|n}(1-1/p)$.

Lemma 1.11.1. *The extension $\mathbb{Q}(\xi_n)/\mathbb{Q}$ is Galois.*

Proof. We only need to show $\mathbb{Q}(\xi_n)/\mathbb{Q}$ is normal. This is clear, since a conjugate of ξ_n is again a (primitive) n-th root of unity, hence is a power of ξ_n and contained in $\mathbb{Q}(\xi_n)$. □

We want to determine the degree $[\mathbb{Q}(\xi_n) : \mathbb{Q}]$, which amounts to determining the degree of the minimal polynomial of ξ_n over $\mathbb{Q}$. Clearly, $X^n - 1$ annihilates ξ_n, but it is not minimal. It is natural de define

$$\Phi_n(X) := \prod_{\xi:\ \text{primitive } n\text{-th root of } 1} (X - \xi).$$

By Galois correspondence, Φ_n has coefficients in $\mathbb{Q}$, because it is fixed by $\mathrm{Gal}(\mathbb{Q}(\xi_n)/\mathbb{Q})$ by construction.

For example, we have $\Phi_1 = X - 1$, $\Phi_2 = X + 1$, $\Phi_3 = X^2 + X + 1$, $\Phi_4 = X^2 + 1$, $\Phi_5 = X^4 + X^3 + X^2 + X + 1$, $\Phi_6 = X^2 - X + 1$, etc. By definition, we have

$$X^n - 1 = \prod_{d|n} \Phi_d.$$

Let P denote the minimal polynomial of ξ_n over $\mathbb{Q}$. Then $P|\Phi_n$. Our goal is to show $P = \Phi_n$.

Lemma 1.11.2. *Let p be a prime not dividing n. If u is a root of $P(X)$, then so is u^p.*

Proof. We have $P|(X^n - 1)$; so we may decompose in $\mathbb{Q}[X]$

$$X^n - 1 = P(X) \cdot Q(X).$$

Gauss lemma implies that $P, Q \in \mathbb{Z}[X]$ (being monic). Assume that u^p is not a root of P. Then u^p is a root of Q. Equivalently, u is a root of $Q(X^p)$ (as u^p is a root of $X^n - 1$). But P, being irreducible, is the minimal polynomial of u, hence $P(X)$ divides $Q(X^p)$. Taking reduction mod p, we see that $\overline{P(X)}$ divides $\overline{Q(X^p)}$ in $\mathbb{F}_p[X]$. On the other hand, since $a^p = a$ for $a \in \mathbb{F}_p$ we have

$$\overline{Q(X^p)} = \overline{Q(X)}^p,$$

so $\overline{Q(X^p)}$ has the same roots as $\overline{Q(X)}$ (although with different multiplicities). This implies that any root of $\overline{P}$ is also a root of $\overline{Q}$. However, we know by construction

$$\overline{P} \cdot \overline{Q} = \overline{X^n - 1} \in \mathbb{F}_p[X],$$

a contradiction with the separability of $\overline{X^n - 1}$ (as $(n,p) = 1$). □

Theorem 1.11.3. *We have $P = \Phi_n$; in particular Φ_n is irreducible. The Galois group* $\mathrm{Gal}(\mathbb{Q}(\xi_n)/\mathbb{Q})$ *is isomorphic to* $(\mathbb{Z}/n\mathbb{Z})^\times$.

Proof. We show $P = \Phi_n$, i.e. $\Phi_n | P$. It amounts to showing that every *primitive* n-th root of unity is a root of P. Fix such a ξ and write $\xi = \xi_n^k$, with $(k, n) = 1$. By decomposing k as a product of primes p_i, with $(p_i, n) = 1$, we see that ξ_n^k is a root of P by Lemma 1.11.2, .

For $\sigma \in \mathrm{Gal}(\mathbb{Q}(\xi_n)/\mathbb{Q})$, it sends ξ_n to ξ_n^k with $(k, n) = 1$. Then $\sigma \mapsto (k \mod n)$ gives the desired map to $(\mathbb{Z}/n\mathbb{Z})^\times$. □

Since $\mathbb{Q}(\xi_n)$ is an abelian extension over $\mathbb{Q}$, any subfield is also a Galois extension of $\mathbb{Q}$. On the other hand, for any field extension F of $\mathbb{Q}$, $F(\xi_n)/F$ is also Galois with $\mathrm{Gal}(F(\xi_n)/F)$ being a subgroup of $(\mathbb{Z}/n\mathbb{Z})^\times$.

Remark 1.11.4. *An important theorem, named after Kronecker-Weber, says that $\mathbb{Q}^{\mathrm{ab}}$ (the maximal abelian extension of $\mathbb{Q}$ contained in $\mathbb{C}$) is obtained by adjoining to $\mathbb{Q}$ all n-th roots of unity for all $n \in \mathbb{N}$.*

We recall the group structure of $(\mathbb{Z}/n\mathbb{Z})^\times$. If $n = p_1^{\alpha_1} \cdots p_r^{\alpha_r}$ is the prime decomposition, then

$$(\mathbb{Z}/n\mathbb{Z})^\times \cong (\mathbb{Z}/p_1^{\alpha_1}\mathbb{Z})^\times \times \cdots \times (\mathbb{Z}/p_r^{\alpha_r}\mathbb{Z})^\times,$$

For a prime $p \geq 3$, $(\mathbb{Z}/p^\alpha\mathbb{Z})^\times$ is a cyclic group isomorphic to

$$\mathbb{Z}/(p-1)\mathbb{Z} \times \mathbb{Z}/p^{\alpha-1}\mathbb{Z}.$$

For $p = 2$, $(\mathbb{Z}/2^\alpha\mathbb{Z})^\times$ is not cyclic when $\alpha \geq 3$ and is isomorphic to $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2^{\alpha-2}\mathbb{Z}$.

We have the following beautiful result.

Theorem 1.11.5. *Let $q = p^d$. Let $n \geq 1$ be prime to p. Then the polynomial $\Phi_n(X)$ is irreducible in $\mathbb{F}_q[X]$ if and only if the class of q in the group $(\mathbb{Z}/n\mathbb{Z})^\times$ is a generator.*

Proof. Left as an exercise. □

1.12 Solvable groups and radical extensions

Let G be a finite group.

Definition 1.12.1. *We say that G is* solvable *if G has a series of subgroups*

$$G = G_0 \supset G_1 \supset \cdots \supset G_n = \{e\}$$

such that where G_i is normal in G_{i-1} and G_{i-1}/G_i is an abelian group.

Example 1.12.2. *The symmetric groups S_3, S_4 are solvable, but S_5 is not.*

We recall the following standard facts about solvable groups:

(i) If G is solvable and $H < G$ is a subgroup, then H is solvable.

(ii) If G is solvable and $H < G$ is a normal subgroup, then G/H is solvable.

(iii) If $1 \to G_1 \to G \to G_2 \to 1$ is an exact sequence, then G is solvable if and only if G_1, G_2 are solvable.

(iv) If G is solvable, then G has a normal subgroup H of prime index (i.e. $[G : H]$ is a prime). Indeed, let H be a maximal normal subgroup of G, then the quotient group G/H is simple and also solvable, so it is abelian and of prime order (being simple).

Let F be a field of characteristic 0.

Proposition 1.12.3. *Let $a \in F$. If K is the splitting field of $X^n - a$ over F then the Galois group* $\mathrm{Gal}(K/F)$ *is solvable.*

Proof. Let α be a root of $X^n - a$. Then the roots of $X^n - a$ are exactly $\{\alpha\xi_n^i; 0 \leq i \leq n-1\}$, where ξ_n is a (fixed) primitive root of unity. Thus, the splitting field K is equal to $F(\alpha, \xi_n)$.

Letting $E = F(\xi_n)$ we obtain a tower of extensions

$$\begin{tikzcd}[arrows=dash]
 & F(\alpha, \xi_n) \arrow[dl] \arrow[dr] & \\
F(\alpha) \arrow[dr] & & E = F(\xi_n) \arrow[dl] \\
 & F &
\end{tikzcd}$$

It is clear that E/F is a Galois extension and the group $\mathrm{Gal}(E/F)$ is a subgroup of $(\mathbb{Z}/n\mathbb{Z})^\times$, hence is abelian. It suffices to prove $\mathrm{Gal}(K/E)$ is also abelian. If $\sigma \in \mathrm{Gal}(K/E)$ sends u to $u\xi_n^k$ and τ sends u to $u\xi_n^l$, then $\sigma\tau$ and $\tau\sigma$ both send u to $u\xi_n^{k+l}$, since $\xi_n \in E$. This proves the claim and that sending σ to k induces an embedding $\mathrm{Gal}(K/E) \hookrightarrow \mathbb{Z}/n\mathbb{Z}$. $\square$

Definition 1.12.4. *A finite extension K/F is said to be* radical *if there exist intermediate fields*

$$F = K_0 \subset K_1 \subset \cdots \subset K_n = K$$

such that for each i, $K_i = K_{i-1}(u_i)$ with $u_i \in K_i$ satisfying $u_i^{m_i} \in K_{i-1}$ for some $m_i \geq 0$.

Roughly, a radical extension of F is an extension of F which can be obtained by adjoining a sequence of m-th roots of elements. Note that a radical extension need not be Galois (see Proposition 1.12.3).

Below are some basic properties about radical extensions.

(i) If $F \subset E \subset K$ is a tower of field extensions and K/F is radical, then K/E is radical (but E/F need not be radical).

Example. Consider $F = \mathbb{Q}$ and $K = \mathbb{Q}(\xi_7)$ which is clearly a radical extension of F. Since $\mathrm{Gal}(\mathbb{Q}(\xi_7)/\mathbb{Q}) = \mathbb{Z}/6\mathbb{Z}$, the Galois correspondence implies that $\mathbb{Q}(\xi_7)$ has a subfield E which has degree 3 over $\mathbb{Q}$. Note that $E/\mathbb{Q}$ is Galois, and there is no proper intermediate field of $E/\mathbb{Q}$.

We claim that $E/\mathbb{Q}$ is not radical. Indeed, if $E = \mathbb{Q}(u)$ with $u^m \in \mathbb{Q}$ and we choose m to be minimal, then $m \geq 3$; moreover, if τ denotes a generator of $\mathrm{Gal}(E/\mathbb{Q}) \cong \mathbb{Z}/3\mathbb{Z}$, then $\tau(u) = u\xi_m$ where ξ_m is a primitive m-th root of unity.(10) Since $E/\mathbb{Q}$ is Galois, we deduce that E contains ξ_m, hence $E = \mathbb{Q}(\xi_m)$. But this is not possible because there is no solution for the equation $\phi(m) = 3$.

(ii) If K/E and E/F are finite radical extensions, then K/F is radical.

(iii) If $E/F, E'/F$ are radical, then EE'/F is also radical.

Proposition 1.12.5. *Let E/F be a (finite) radical extension, and K/F be the Galois closure of E. Then K/F is also radical and* $\mathrm{Gal}(K/F)$ *is solvable.*

Proof. The first assertion follows from the property (iii) above, as K is the compositum of the conjugates of E. By an easy induction, the second assertion follows from Proposition 1.12.3 (which treats the case of a simple radical extension). □

Lemma 1.12.6. *Let K/F be a Galois extension with $[K : F] = p$ is prime. Assume that F contains ξ_p, primitive p-th root of unity. Then K/F is a simple radical extension, i.e. there exists $u \in K$ such that $u^p \in F$ and $F(u) = K$.*

Proof. By assumption $\mathrm{Gal}(K/F)$ is a cyclic group of order p, say generated by τ. We will construct a non-zero element $u \in K$ satisfying $\tau(u) = \xi_p^{-1}u$. Then $\tau(u) \neq u$ implies $u \notin F$ and $\tau(u^p) = (\xi_p^{-1}u)^p = u^p$ implies $u^p \in F$. Since $[K : F]$ is prime, there is no proper intermediate field of K/F, so we must have $K = F(u)$.

By linear independence of characters(11), we may choose $v \in K$ such that $u := \sum_{i=0}^{p-1} \xi_p^i \tau^i(v)$ is nonzero. Then (as $\xi_p \in F$ is fixed by τ)

$$\tau(u) = \textstyle\sum_{i=0}^{p-1} \xi_p^i \tau^{i+1}(v) = \xi_p^{-1}(\sum_{i=0}^{p-1} \xi_p^{i+1}\tau^{i+1}(v)) = \xi_p^{-1}u$$

as required. □

(10) Proof: if $\tau(u) = u\xi_d$ for some $d < m$, then we would have $\tau(u^d) = (u\xi_d)^d = u^d$, so $u^d \in \mathbb{Q}$ which contradicts the choice of m.

(11) Let G be an abelian group and L be a field. Let $\chi_1, \dots, \chi_n$ be distinct characters $G \to L^\times$. Then they are linearly independent: if $c_1, \dots, c_n \in L$ satisfy

$$c_1\chi_1(g) + \cdots + c_n\chi_n(g) = 0, \quad \forall g \in G$$

then $c_i = 0$ for $1 \leq i \leq n$.

Theorem 1.12.7. *Let K/F be a finite Galois extension such that* $\mathrm{Gal}(K/F)$ *is solvable. Then K is contained in a radical extension of F.*

Proof. We prove by induction on $n := [K : F]$ (the base field F is allowed to change). The case $n = 1$ is trivial.

Since $\mathrm{Gal}(K/F)$ is solvable, it has a normal subgroup H of prime index. We fix such a prime p and adjoin the p-th roots of unity to K and F. It is clear that $[K(\xi_p) : F(\xi_p)] \leq [K : F]$.

Case 1. If $[K(\xi_p) : F(\xi_p)] < n$, then by inductive hypothesis $K(\xi_p)$ is contained in a radical extension $L/F(\xi_p)$, as illustrated by the diagram below:

$$\begin{tikzcd}
K \arrow[r, dash] \arrow[d, dash] & K(\xi_p) \arrow[r, dash] \arrow[d, dash] & L \arrow[dl, dash, "\text{radical}"] \\
F \arrow[r, dash] & F(\xi_p)
\end{tikzcd}$$

Since $F(\xi_p)/F$ is radical, L/F is also radical.

Case 2. If $[K(\xi_p) : F(\xi_p)] = n$, then $\mathrm{Gal}(K(\xi_p)/F(\xi_p)) \cong \mathrm{Gal}(K/F)$ and we may view H as a subgroup of $\mathrm{Gal}(K(\xi_p)/F(\xi_p))$. Let $E = K(\xi_p)^H$ which is a Galois extension over $F(\xi_p)$ of degree p. Lemma 1.12.6 implies that $E/F(\xi_p)$ is a simple radical extension. Since $[K(\xi_p) : E] = \frac{n}{p} < n$, by induction $K(\xi_p)$ is contained in a radical extension L/E. As before, L/F is also radical and the proof is complete. $\square$

1.13 Solvability of algebraic equations

Let F be a field of characteristic 0.

Definition 1.13.1. *We say that $f \in F[X]$ is solvable by radicals over F if f splits in some radical extension of F, i.e. the splitting field of f is contained in some radical extension of F. Equivalently, the roots of f can be obtained by the algebraic operations of $+, -, \times, \div$ and taking m-th roots.*

Remark 1.13.2. *In general, for a polynomial which is solvable by radicals, its splitting field need not be a radical extension.*

Theorem 1.13.3. *Let $f \in F[X]$. Then f is solvable by radicals if and only if G_f is solvable.*

Proof. Let K_f/F be the splitting field of f. Assume that K_f is contained in a radical extension K; we may assume K/F is Galois using Proposition 1.12.5. Then $\mathrm{Gal}(K/F)$ is solvable by Proposition 1.12.5 again. Being a quotient group of $\mathrm{Gal}(K/F)$, G_f is also solvable. The converse follows from Theorem 1.12.7. $\square$

Therefore, we have a necessary and sufficient condition for a polynomial to be solvable by radicals: its Galois group is a solvable group. Since S_n is not solvable for $n \geq 5$, we obtain the following important result.

Corollary 1.13.4. *The general equation of degree n can not be solved by radicals for $n \geq 5$.*

1.14 Applications: constructibility

We are allowed to do two basic types of operations:

(a) Given two points on the plane, we can construct the line joining them.

(b) Given a point A, and a radius $r > 0$, we can draw the circle with centre A and radius r.

In this way, starting from points $O = (0,0)$ and $(1,0)$, we get new points, called *constructible points*, by

- intersecting two lines
- intersecting a line with a circle
- intersecting two circles.

Remark that we can do the following, less obvious, operations:

(a') Given a segment S joining A and B, can construct a circle with diameter AB.

Proof: Choose r large enough and construct two circles around respectively A and B of radius r. Denote the two intersections by C and C'. Then the line CC' intersects with AB at the mid-point of AB, then the circle follows. Note that CC' is perpendicular to AB, i.e. $CC' \perp AB$.

(b') Given any line ℓ and any point P, we can construct a line passing P and perpendicular to ℓ; if P is not on ℓ, then we can construct a line passing P and parallel to ℓ.

Proof: Choose r large enough such that the circle around P of radius r intersects with ℓ at A and B. Construct C, C' as in (a'); then the line CC' passes P and is perpendicular to AB. Now assume $P \notin \ell$. It suffices to construct ℓ' such that $P \in \ell'$ and $\ell' \perp CC'$; then $\ell' \parallel \ell$.

Using (b'), we easily see that a point (a,b) is constructible, if and only if $(a,0)$ and $(0,b)$ are both constructible. So it suffices to study the constructible points of the form $(a,0)$. For this purpose, we make the following definition.

Definition 1.14.1. *A real number $c \in \mathbb{R}$ is said to be constructible if and only if the point $(c,0)$ is constructible. We let*

$$\mathfrak{C} := \{c \in \mathbb{R} : c \text{ is constructible}\}.$$

The first fact is the following.

Proposition 1.14.2. *$\mathfrak{C}$ is a subfield of $\mathbb{R}$ (hence contains $\mathbb{Q}$).*

Proof. Of course, if $c \in \mathfrak{C}$, then $-c \in \mathfrak{C}$ by doing a circle of radius c. Let $a, b \in \mathfrak{C}$, we want $a+b \in \mathfrak{C}$: we may assume $a, b > 0$, then construct a circle around $(a, 0)$ of radius b, which will intersect with the x-axis at $(a+b, 0)$.

Next we show $a^{-1} \in \mathfrak{C}$: construct the line ℓ passing $(a, 0)$ and $(0, 1)$, and construct the line parallel to ℓ and passing $(1, 0)$, the intersection with y-axis gives $(0, 1/a)$, in this way we get $(1/a, 0)$.

Finally we show $ab \in \mathfrak{C}$ in a similar way. Construct first the line ℓ which passes $(1, 0)$ and $(0, a)$, and do the line parallel to ℓ and passing $(b, 0)$, the intersection with y-axis gives $(0, ab)$. □

Another observation is

Lemma 1.14.3. *If $c \in \mathfrak{C}$ and $c > 0$, then $\sqrt{c} \in \mathfrak{C}$.*

Proof. Construct the circle $\mathcal{D}$ with diameter the segment joining $(0, 0)$ and $(1+c, 0)$. The line parallel to y-axis passing $(1, 0)$ will intersect with $\mathcal{D}$ on some point $(1, c')$ (choose the one in the first quadrant). Then we check that $c' = \sqrt{c}$. □

Theorem 1.14.4. *Let K/F (where $F, K \subset \mathbb{R}$) be a normal extension of degree 2^n for some $n \in \mathbb{N}$. Then $F \subset \mathfrak{C}$ implies $K \subset \mathfrak{C}$.*

Proof. We do induction on n. If $n = 1$, then $K = F(u)$ for some u. The minimal polynomial of u is $X^2 + bX + c$ (degree 2). By changing u by $u + b/2$, we may assume $u^2 \in F$. Lemma 1.14.3 implies that $u \in \mathfrak{C}$.

Suppose that $n > 1$ and the result is true for all $m < n$. Let $G = \mathrm{Gal}(K/F)$, whose order is 2^n. By a standard result from group theory, G has a normal subgroup H of index 2.[12] Let $K' = K^H$. Then $[K' : F] = 2$ and we saw that $K' \subset \mathfrak{C}$. Again by the inductive hypothesis, we get $K \subset \mathfrak{C}$. □

Corollary 1.14.5. *Let $\mathbb{Q} \subset F \subset \mathbb{R}$ with $F/\mathbb{Q}$ normal and $[F : \mathbb{Q}] = 2^n$. Then $F \subset \mathfrak{C}$.*

Remark 1.14.6. *The condition that $F/\mathbb{Q}$ being "normal" in Corollary 1.14.5 is necessary. To see this, let $f \in \mathbb{Q}[X]$ be an irreducible polynomial of degree 4 with $G_f \cong S_4$. Let α_i ($1 \leq i \leq 4$) be the four roots of f. Then $[\mathbb{Q}(\alpha_i) : \mathbb{Q}] = 4$. If $\mathbb{Q}(\alpha_i) \subset \mathfrak{C}$ for all i, then $K_f \subset \mathfrak{C}$ too. But $K_f/\mathbb{Q}$ contains a subfield of degree 3 over $\mathbb{Q}$, which cannot be contained in $\mathfrak{C}$ (by Theorem 1.14.8 below). Contradiction.*

Our next task is to show the converse of Corollary 1.14.5: if $c \in \mathfrak{C}$, then $[\mathbb{Q}(c) : \mathbb{Q}] = 2^n$ for some $n \in \mathbb{N}$. For this, we need to analyse the three types of intersections.

Let F be a subfield of $\mathbb{R}$. The plane of F means the subset $F \times F$ of the plane $\mathbb{R}^2$. By a *line* in F, we mean a line joining two points P, Q of the plane of F. By a *circle* in F, we mean a circle whose centre lies in the plane of F and the radius lies in F.

[12] Proof: the center of G is non-trivial; by induction $G/Z(G)$ has a normal subgroup of index 2 whose preimage gives the required subgroup $H \lhd G$.

Lemma 1.14.7. *(1) If ℓ_1 and ℓ_2 are distinct non-parallel lines in F, then $\ell_1 \cap \ell_2$ is a point in $F \times F$.*

(2) If ℓ is a line in F and $\mathcal{D}$ is a circle in F with $\ell \cap \mathcal{D} \neq \emptyset$, then $\ell \cap \mathcal{D}$ consists of one or two points in the plane of $F(u)$ for some $u \in \mathbb{R}$ with $u^2 \in F$ (the case $u \in F$ is allowed).

(3) If $\mathcal{D}_1$ and $\mathcal{D}_2$ are distinct circles in F and $\mathcal{D}_1 \cap \mathcal{D}_2 \neq \emptyset$, then $\mathcal{D}_1 \cap \mathcal{D}_2$ consists of one or two points in the plane of $F(u)$ for some $u \in \mathbb{R}$ with $u^2 \in F$.

Proof. Recall that a line in F has equation $ax + by + c = 0$ with $a, b, c \in F$ and a, b not both 0, and a circle in F has equation

$$x^2 + y^2 + ax + by + c = 0, \quad a, b, c \in F.$$

One easily deduces the results by solving the equations $\ell = 0$ and $\mathcal{D} = 0$. $\square$

Theorem 1.14.8. *If $c \in \mathfrak{C}$, i.e. c is a constructible number, then c is algebraic over $\mathbb{Q}$ and $[\mathbb{Q}(c) : \mathbb{Q}] = 2^n$ for some $n \geq 0$.*

Proof. By assumption, there is a finite sequence of ruler and compass constructions giving the point $(c, 0)$. At the first stage, we produce a point (a_0, b_0) from intersections of the form (1)-(3) using the subfield $F_0 = \mathbb{Q}$. By Lemma 1.14.7, there is an element $u_0 \in \mathbb{R}$ with $u_0^2 \in F_0$ such that $F_0(a_0, b_0) \in F_0(u_0)$; here $u_0 \in F$ is allowed. Letting $F_1 = F_0(u_0)$, we continue to produce a point (a_1, b_1) with F_1 as the base field. We then get $u_1 \in \mathbb{R}$ with $u_1^2 \in F_1$ such that $F_1(a_1, b_1) \subset F_1(u_1)$. Continuing this procedure, we get a sequence of fields F_k such that $F_k \subset F_{k+1}$ and $[F_{k+1} : F_k] = 1$ or 2. Now the procedure stops after finitely many steps, and $Q(c) \subset F_n$. Since $[F_n : \mathbb{Q}] = 2^n$, $[\mathbb{Q}(c) : \mathbb{Q}]$ is also a power of 2. $\square$

The above discussion can be generalized as follows. Choose a subfield $F \subset \mathbb{R}$ and let

$$\mathfrak{C}_F := \{c \in \mathbb{R} : c \text{ is constructible from } F\}.$$

By a similar discussion as above, especially by Theorem 1.14.4, we get the following result.

Theorem 1.14.9. *If $K \subset \mathbb{R}$ is a normal extension of F and $[K : F] = 2^n$ for some $n \geq 0$, then $F \subset \mathfrak{C}_F$. Conversely, for any $c \in \mathfrak{C}_F$, c is algebraic over F and $[\mathbb{Q}(c) : \mathbb{Q}] = 2^n$ for some $n \geq 0$.*

Below we will apply the above results to the three classical problems in Greek mathematics, namely, *squaring the circle*, *doubling the cube* and *trisecting an angle*.

1. **Squaring the circle**: construct a square whose area is equal to that of a circle.

This amounts to constructing $\sqrt{\pi}$. As π is not algebraic,[13] this is not possible by Theorem 1.14.8.

[13] This was proved by Lindemann in 1882. His proof is based on the transcendence of e proved by Hermite in 1873 and Euler's identity $e^{i\pi} + 1 = 0$.

2. **Doubling the cube**: given the length of an edge of a cube, construct a second cube having double the volume of the first.

This amounts to constructing $\sqrt[3]{2}$. Since $[\mathbb{Q}(\sqrt[3]{2}) : \mathbb{Q}] = 3$, this is not possible by Theorem 1.14.8.

3. **Trisecting an arbitrary angle**: construct $\theta/3$ for a given angle θ.

Here, we say an angle θ is *constructible* if we can construct two intersecting lines with angle θ, equivalently, if the point $(\cos(\theta), \sin(\theta))$ is constructible.

Lemma 1.14.10. *Let $\theta \in \mathbb{R}$. The following statements are equivalent:*

(i) *The angle θ is constructible.*

(ii) *The number $\cos(\theta)$ is constructible.*

(iii) *The number $\sin(\theta)$ is constructible.*

Proof. This is obvious. □

As a consequence, if θ, θ' are two constructible angles, then the angle $m\theta + n\theta'$ is also constructible for $m, n \in \mathbb{Z}$, using trigonometric identities.

Example 1.14.11. *If $\theta = \pi/3 = 60°$, since $\cos(\theta) = 1/2$, θ is a constructible angle. Now let $\alpha = \theta/3 = \pi/9$. Then we have*

$$1/2 = \cos(3\alpha) = 4\cos^3(\alpha) - 3\cos(\alpha),$$

so that $\cos(\alpha)$ satisfies the equation

$$f(X) = 4X^3 - 3X - \frac{1}{2} \in \mathbb{Q}[X].$$

One checks that f is irreducible, hence $[\mathbb{Q}(\cos(\alpha)) : \mathbb{Q})] = 3$ and α is not constructible.

Example 1.14.12. *If $\theta = \pi/4 = 45°$, then $\cos(15°)$ is constructible. Indeed, as $\cos(45°) = \frac{\sqrt{2}}{2}$, $\cos(15°)$ is a root of*

$$f(x) = 4X^3 - 3X - \frac{\sqrt{2}}{2} = 0.$$

Over $\mathbb{Q}(\sqrt{2})[X]$, this polynomial is reducible and decomposes as

$$f = \big(X + \frac{\sqrt{2}}{2}\big)(4X^2 - 2\sqrt{2}X - 1) = 0.$$

One computes that $\cos(15°) = \frac{\sqrt{6}+\sqrt{2}}{4} \in \mathbb{Q}(\sqrt{2}, \sqrt{3})$, hence it is constructible by Corollary 1.14.5.

4. **Constucting regular polynons**. Next we turn to the problem of constructing a regular polygon with n sides where $n \in \mathbb{N}$ (i.e. a polygon whose sides have equal length). For each $n \in \mathbb{N}$, let θ_n be the angle with radian measure $2\pi/n$. We have the following observation:

Fact: *The problem of constructing a regular polygon with n sides is equivalent to that of constructing the angle θ_n, for we may inscribe the regular polygon in the unit circle.*

By Lemma 1.14.10, this amonts to constructing the number $\cos(\theta_n)$. A more important observation is the following theorem.

Theorem 1.14.13. *Let ξ_n be a primitive n-th root of unity. A regular polygon with n sides is constructible if and only if $[\mathbb{Q}(\xi_n) : \mathbb{Q}]$ is a power of 2.*

Proof. We may take $\xi_n = e^{i\theta_n}$. The Euler formula says

$$\xi_n = \cos(\theta_n) + i\sin(\theta_n), \quad \xi_n^{-1} = cos(\theta_n) - i\sin(\theta_n)$$

which give the relation

$$\xi_n + \xi_n^{-1} = 2\cos(\theta_n).$$

In particular $\cos(\theta_n) \in \mathbb{Q}(\xi_n)$ and $[\mathbb{Q}(\xi_n) : \mathbb{Q}(\cos(\theta_n))] = 1$ or 2, because

$$\xi_n^2 + 1 = 2\cos(\theta_n)\xi_n.$$

Assume $\cos(\theta_n)$ is constructible, then $[\mathbb{Q}(\cos(\theta_n)) : \mathbb{Q}]$ is a power of 2, so is $[\mathbb{Q}(\xi_n) : \mathbb{Q}]$.

Conversely, assume $[\mathbb{Q}(\xi_n) : \mathbb{Q}]$ is a power of 2. Then, $[]\mathbb{Q}(\cos(\theta_n)) : \mathbb{Q}]$ is also a power of 2. Since $\mathbb{Q}(\xi_n)/\mathbb{Q}$ is an abelian Galois extension, $\mathbb{Q}(\cos(\theta_n))$ is also a Galois extension of $\mathbb{Q}$. Hence $\mathbb{Q}(\cos(\theta_n)) \subset \mathfrak{C}$ and $\cos(\theta_n)$ is constructible. □

Combined with results in §1.11, we deduce the following sufficient and necessary condition for a regular polygon with n sides to be constructible.

Corollary 1.14.14. *A regular polygon with n sides is constructible if and only if n has prime decomposition*

$$n = 2^k p_1 \cdots p_s$$

where $k, s \geq 0$, and p_i is a Fermat prime for each i.

Recall that a *Fermat prime* is a prime number p of the form $1 + 2^{2^m}$ for some m, like $p = 3, 5, 17, \ldots$ This is equivalent to requiring $p = 1 + 2^m$ for some m by the next Lemma.

Lemma 1.14.15. *If $1 + 2^m$ is a prime, then m is a 2-power.*

Proof. For odd m, we have $(x + y)|(x^m + y^m)$. Thus if $m \geq 3$ is odd, then $3|(1 + 2^m)$ and so $1 + 2^m$ can not be a prime. If $m = kl$ where k is even and $l \geq 3$ is odd, then similarly $(1 + 2^k)|(1 + 2^m)$. Therefore m can not have any odd prime factor. □

1.15 Infinite Galois theory

Topological groups

Definition 1.15.1. *A group G equipped with a topology is a* topological group *if the maps*

$$G \times G \to G, \quad (g,h) \mapsto gh; \qquad G \to G, \quad g \mapsto g^{-1}$$

are both continuous.

Example 1.15.2. • $(\mathbb{R}, +)$ *and* $(\mathbb{R}^\times, \times)$ *are topological groups.*

• *If G is a finite group, it becomes a topological group when equipped with the discrete topology.*

It follows from the definition that left multiplication $L_a : G \to G, g \mapsto ag$ is continuous, since it is the composition of

$$G \overset{g\mapsto(a,g)}{\longrightarrow} G \times G \overset{(g,h)\mapsto gh}{\longrightarrow} G.$$

Moreover, L_a is a homeomorphism with inverse $L_{a^{-1}}$.

Lemma 1.15.3. *Let G be a topological group and H be a subgroup of G.*

(i) *Equipped with the subspace topology, H is a topological goup.*

(ii) *The closure of H is still a subgroup. Moreover, if H is normal, then so is its closure.*

(iii) *G is Hausdorff if and only if $\{e\}$ is closed in G.*

Proof. Exercise. □

Lemma 1.15.4. *Let $H \subset G$ be a subgroup.*

(i) *If H is open, then H is also closed.*

(ii) *If H is closed of finite index, then H is open. The converse also holds when G is compact.*

Proof. (i) Let S be a set of representatives of cosets G/H, then $G \backslash H = \cup_{\sigma \in S \backslash \{e\}} \sigma H$. Note that each σH is open, and any union of open sets is still open, hence H is closed.

(ii) If $[G : H]$ is finite, then S is finite and H is open because it is the complementary of a *finite* union of closed sets, which is again closed. When G is compact and H is open, the coset decomposition in (i) gives a *disjoint* open cover of G, hence the cover must be finite. □

As a consequence, topological groups are nice topological spaces: topological properties can be just tested around the identity!

Let $\{G_i \mid i \in I\}$ be a family of groups indexed by a *directed*(14) partially ordered set I and suppose we have a family of morphisms

$$p_{ij} : G_j \to G_i, \ \forall i, j \in I, i \leq j,$$

called *transition maps*, with the following properties:

- p_{ii} is the identity map;
- $p_{ik} = p_{ij} \circ p_{jk}$ for all $i \leq j \leq k$.

We call the data (G_i, G_{ij}) an inverse system of groups indexed by I.

The *inverse limit* $\varprojlim G_i$ is defined to be a group G with group homomorphisms $p_i : G \to G_i$ such that $p_{ij} \circ p_j = p_i$, $\forall i, j \in I$, $i \leq j$, satisfying the following universal property: if (H, f_i) is another solution such that $p_{ij} \circ f_j = f_i$, then there is a unique homomorphism $\theta : H \to G$ such that $p_i \circ \theta = f_i$ for all $i \in I$.

From the universal property it is easy to see that the inverse limit, if exists, is unique up to a unique isomorphism. In the category of groups, the inverse limit indeed exists and has the following explicit description

$$\varprojlim G_i = \Big\{(g_i) \in \prod_{i \in I} G_i \mid p_{ij}(g_j) = g_i \ \forall i \leq j\Big\}.$$

In particular, $\varprojlim G_i$ is a subset of $\prod_i G_i$.

Example 1.15.5. *Assume $G_i = \mathbb{Z}/p^i\mathbb{Z}$ for $i \geq 1$, with natural maps $G_i = \mathbb{Z}/p^i\mathbb{Z} \to \mathbb{Z}/p^{i-1}\mathbb{Z} = G_{i-1}$, $x \mapsto x$ mod p^{i-1}. Then*

$$\begin{aligned} \varprojlim_i \mathbb{Z}/p^i\mathbb{Z} &= \{(x_i)_{i \geq 1} | x_{i-1} \equiv x_i \bmod p^{i-1}\} \\ &= \{(a_0, a_1, \dots) | 0 \leq a_i \leq p-1\} \\ &= \Big\{\sum_{i \geq 0} a_i p^i | 0 \leq a_i \leq p-1\Big\}. \end{aligned}$$

*It is denoted by $\mathbb{Z}_p$, and called the p-*adic integers.

If we consider inverse limits in the category of topological groups, then equipped with the subspace topology of $\prod_{i \in I} G_i$, $\varprojlim G_i$ becomes a topological group and is the inverse limit of G_i in the category of topological groups.

Definition 1.15.6. *An inverse limit of finite (discrete) topological groups is called a* profinite group.

Proposition 1.15.7. *A profinite group is compact, Hausdorff and totally disconnected.*

(14) i.e. for any $i, j \in I$, there exists $k \in I$ such that $i, j \leq k$.

Proof. By Tychonoff's theorem, a product of compact topological spaces is compact. Thus $\prod_i G_i$ is compact. Since $\varprojlim G_i \subset \prod_i G_i$, it's enough to prove $\varprojlim G_i$ is closed. Given $(g_i)_i \in \prod_i G_i$ but $(g_i)_i \notin \varprojlim G_i$, there exists some p_{ij} such that $p_{ij}(g_j) \neq g_i$. Define

$$U = \{g_i\} \times \{g_j\} \times \prod_{k \neq i,j} G_k$$

which is open in $\prod_i G_i$ since G_i's are discrete. Then $(g_i)_i \in U$, but $U \cap \varprojlim G_i = \emptyset$, namely the complement of $\varprojlim G_i$ in $\prod_i G_i$ is open.

Any product of Hausdorff topological spaces is Hausdorff and any subspace of a Hausdorff space is Hausdorff. The result follows.

Recall that a space X is called *totally disconnected* if for every $x \in X$ the connected component containing x is $\{x\}$ itself.[15] We know that any product of totally disconnected spaces is totally disconnected and any subspace of a totally disconnected space is totally disconnected. Clearly, each G_i is disconnected, thus $\varprojlim G_i$ is totally disconnected. $\square$

Theorem 1.15.8. *If the transition morphisms $G_i \to G_j$ are all surjective, then the morphisms $G \to G_i$ are also surjective.*

Proof. Exercise. $\square$

Krull topology

Let K/F be a (possibly infinite) Galois extension. For any intermediate field E, K/E is still Galois, whose group $\mathrm{Gal}(K/E)$ is a subgroup of $\mathrm{Gal}(K/F)$. Set $G = \mathrm{Gal}(K/F)$.

Remark 1.15.9. *For any F-embedding $\tau : K \to K$, it is automatically an automorphism. Indeed, let $x \in K$ and take a finite Galois subextension E/F which contains x. Then $\tau|_E : E \to K$ must have image equal to E, in particular, $x \in \mathrm{Im}(\tau)$.*

Definition 1.15.10. *We take*

$$\mathcal{N} = \{\mathrm{Gal}(K/E) : E/F \text{ is finite and Galois}\}$$

as a neighbourhood basis [16] *of $\mathrm{Id}_K \in G$. For any $\sigma \in G$, let $\sigma\mathcal{N}$ be its neighbourhood basis. The topology thus defined is called the* Krull topology *on G.*

It is direct to check that G becomes a topological group under the Krull topology.

Remark 1.15.11. *(i) Equally, we can take*

$$\mathcal{N}' = \{\mathrm{Gal}(K/E) : E/F \text{ is finite}\}$$

[15] With the subspace topology from $\mathbb{R}$, $\mathbb{Q}$ is totally disconnected but not discrete.

[16] A neighbourhood basis for a point x in a topological space X is a family of neighbourhoods $\mathcal{N}$ such that for any open set U containing x, there exists $N \in \mathcal{N}$ such that $N \subset U$.

as a neighbourhood basis of Id_K*. In fact, if* E/F *is finite and if* E' *denotes its Galois closure, then* E'/F *is finite and Galois. Since* $\mathrm{Gal}(K/E)$ *contains a finite index subgroup* $\mathrm{Gal}(K/E')$ *which is open, it is open too.*

(ii) For any intermediate field E*,* $\mathrm{Gal}(K/E)$ *is closed. In fact, Let* $F \subset E' \subset E$ *be all* finite *subextensions of* F*, then* E *is the compositum of all such* E'*, and*

$$\mathrm{Gal}(K/E) = \bigcap_{E'} \mathrm{Gal}(K/E')$$

is the intersection of closed subsets (as $\mathrm{Gal}(K/E')$ *is open).*

Lemma 1.15.12. *For any intermediate field* E *which is finite Galois over* F*, the map*

$$G \to \mathrm{Gal}(E/F), \quad \sigma \mapsto \sigma|_E$$

is a continuous surjection. Here $\mathrm{Gal}(E/F)$ *is equipped with the discrete topology.*

Proof. For any $\iota : E \to E$, we can extend it to an F-isomorphism $K \to K$, see Lemma 1.3.10 (and Remark 1.15.9). Hence the morphism is surjective. The kernel of the morphism is $\mathrm{Gal}(K/E)$ which is open, so it is continuous. $\square$

We index all finite Galois extensions E/F contained in K by I, in such a way that $E \leq E'$ iff $E \subseteq E'$. Then I is a partially order set and is directed as $E, E' \leq EE'$. This is an inverse system, so we can define the inverse limit $\varprojlim_E \mathrm{Gal}(E/F)$. Using Lemma 1.15.12, we obtain a continuous morphism of topological groups

$$\iota : G \to \varprojlim_E \mathrm{Gal}(E/F).$$

Proposition 1.15.13. *The map* ι *is an homeomorphism.*

Proof. If $\sigma \in G$ is such that $\sigma|_E = \mathrm{Id}_E$ for all finite Galois subextensions E/F, then σ must be identity on K since every element of K is contained in some finite subextension E. This proves the injectivity of ι.

Conversely, if $(\sigma_E)_E \in \varprojlim_E \mathrm{Gal}(E/F)$, then we define an F-automorphism σ of K as follows: for $x \in K$, choose E large enough containing x and let $\sigma(x) := \sigma_E(x) \in E \subset K$. This is well defined: if we choose another E', then, since EE'/F is also finite and Galois, we have

$$\sigma_{E'}(x) = \sigma_{EE'}(x) = \sigma_E(x).$$

Since ι is continuous, it suffices to prove that ι is open. Let E'/F be a finite Galois extension, so that $H = \mathrm{Gal}(K/E')$ is open in G for the Krull topology. We need to show that it is also open for the inverse limit topology on $\varprojlim_E \mathrm{Gal}(E/F)$, which is clear because

$$\Big(\prod_{E \neq E'} \mathrm{Gal}(E/F) \times \{\mathrm{Id}_{E'}\} \Big) \cap G = \mathrm{Gal}(K/E').$$

This proves the openness of ι. $\square$

Corollary 1.15.14. *Under the Krull topology, G is compact Hausdorff and totally disconnected.*

Proof. This follows directly from Proposition 1.15.7 and Proposition 1.15.13. $\square$

Proposition 1.15.15. *(1) For any Galois extension K/F, $K^{\mathrm{Gal}(K/F)} = F$.*

(2) Let H be any subgroup of G, then $\mathrm{Gal}(K/K^H)$ is the closure of H.

Proof. (1) Every element in K/F lies in a finite Galois extension of F, so this follows from the finite case.

(2) Since $\mathrm{Gal}(K/K^H)$ is closed and contains H, it contains $\overline{H}$. Now take $\sigma \notin \overline{H}$, we want to show that σ moves some element in K^H, i.e. $\sigma \notin \mathrm{Gal}(K/K^H)$. Since $\sigma \notin \overline{H}$, there exists an open set $\mathrm{Gal}(K/E)$ (where E/F is finite Galois) such that

$$\sigma \mathrm{Gal}(K/E) \cap H = \emptyset.$$

Consider $\phi : G \to \mathrm{Gal}(E/F)$ which is surjective with kernel $\mathrm{Gal}(K/E)$; let $\phi(H)$ be the image of H. The assumption implies that $\phi(\sigma) = \sigma|_E \notin \phi(H)$. Therefore, by Galois correspondence in finite case, i.e. $\phi(H) = \mathrm{Gal}(E/E^{\phi(H)})$, σ moves some element in $E^{\phi(H)}$. But one checks that $E^{\phi(H)} \subset K^H$, the result follows. $\square$

We deduce the main theorem of infinite Galois theory.

Theorem 1.15.16. *Let K/F be Galois with Galois group G. There is one-to-one bijection between the set of closed subgroups of G and the set of intermediate fields between K and F, given by*

$$H \mapsto K^H, \quad E \mapsto \mathrm{Gal}(K/E).$$

Moreover

(i) *a closed subgroup H is normal if and only if K^H is Galois over F and in this case,* $\mathrm{Gal}(K^H/F) \cong G/H$.

(ii) *H is open if and only if K^H is finite over F, and in this case $[G : H] = [K^H : F]$.*

Proof. The one-to-one correspondence is easily derived from the Proposition 1.15.15.

(i) The same proof of Corollary 1.6.2 is also valid here.

(ii) If H is open, then it contains $\mathrm{Gal}(K/E)$ for some finite Galois extension E/F. We have $K^H \subset K^{\mathrm{Gal}(K/E)} = E$ where the equality follows from Proposition 1.15.15. Since E/F is finite, K^H/F is also finite. Conversely, since H is assumed to be closed, we have $\mathrm{Gal}(K/K^H) = H$ by Proposition 1.15.15 again. If K^H/F is finite, then H is open by definition.

To see the equality $[G : H] = [E : F]$ where $E = K^H$, we note that when H is normal the assertion follows from (i). In general, let E' be the Galois closure of E and $H' = \mathrm{Gal}(K/E') < H$, then $[G : H'] = [E' : F]$ as just noted. Since $[H : H'] = [E' : E]$ by (i) applied to K/E, the result follows. $\square$

Example 1.15.17. *Consider the finite field* $\mathbb{F}_p$ *where* p *is a prime. There exists a unique extension of degree* n *for each* n*, namely* $\mathbb{F}_{p^n}$*. We know* $\mathrm{Gal}(\mathbb{F}_{p^n}/\mathbb{F}_p) \cong \mathbb{Z}/n\mathbb{Z}$ *generated by Frobenius. Since* $\overline{\mathbb{F}}_p = \varinjlim \mathbb{F}_{p^n}$ *is an algebraic closure of* $\mathbb{F}_p$*, we get*

$$\mathrm{Gal}(\overline{\mathbb{F}}_p/\mathbb{F}_p) = \varprojlim \mathbb{Z}/n\mathbb{Z} =: \widehat{\mathbb{Z}}.$$

Example 1.15.18. *Let* $\mathbb{Q}(\xi_n; n \geq 1) = \cup_{n\geq 1}\mathbb{Q}(\xi_n)$*. Since* $\mathrm{Gal}(\mathbb{Q}(\xi_n)/\mathbb{Q}) \cong (\mathbb{Z}/n\mathbb{Z})^\times$*, we get*

$$\mathrm{Gal}(\mathbb{Q}(\xi_n; n \geq 1)/\mathbb{Q}) \cong \varprojlim_n (\mathbb{Z}/n\mathbb{Z})^\times = \widehat{\mathbb{Z}}^\times.$$

Remark 1.15.19. *In a profinite group, a (normal) subgroup of finite index need not be closed. In our situation, there exists* $H < G = \mathrm{Gal}(K/F)$ *of finite index such that* $\overline{H} = G$*, hence* $K^H = F = K^G$*. This does not contradict Artin's Lemma (since* $|H| = +\infty$*).*

***Example**: Let* $K = \mathbb{Q}(\sqrt{2}, \sqrt{3}, \sqrt{5}, ...)$ *be the extension obtained by adjoining* $\sqrt{p}$ *to* $\mathbb{Q}$*, for all prime numbers* p*. If* p_n *denotes the* n*-th prime number, then*

$$G := \mathrm{Gal}(K/\mathbb{Q}) \cong \varprojlim_{n\geq 1} \mathrm{Gal}(\mathbb{Q}(\sqrt{p_1}, \ldots, \sqrt{p_n})/\mathbb{Q}) \cong \varprojlim_{n\geq 1} \prod_{k=1}^{n} \mathbb{Z}/2\mathbb{Z} \cong \prod_{n\geq 1} \mathbb{Z}/2\mathbb{Z}.$$

This group admits a subgroup $H := \bigoplus_{n\geq 1} \mathbb{Z}/2\mathbb{Z}$ *which is proper and dense (hence not closed). By taking the pre-image of a proper subgroup of finite index in the quotient group* G/H*, we obtain a (proper) subgroup of finite index in* G *which is not closed (because it contains* H*).*

1.16 Norm and trace

Let K/F be a finite extension.

Definition 1.16.1. *For any* $\alpha \in K$*, let* $m_\alpha : K \to K$ *be the* F*-linear map defined by* $m_\alpha(x) = \alpha x$*. Set*

$$\mathrm{N}_{K/F}(\alpha) = \det(m_\alpha) \in F$$
$$\mathrm{Tr}_{K/F}(\alpha) := \mathrm{Tr}(m_\alpha) \in F.$$

By linear algebra, we have the following facts.

- $\mathrm{N}_{K/F}(-)$ is multiplicative: $\mathrm{N}_{K/F}(\alpha\beta) = \mathrm{N}_{K/F}(\alpha)\mathrm{N}_{K/F}(\beta)$. If $a \in F$, then $\mathrm{N}_{K/F}(a\alpha) = a^{[K:F]}\mathrm{N}_{K/F}(\alpha)$.
- $\mathrm{Tr}_{K/F}(-)$ is additive and F-linear: $\mathrm{Tr}_{K/F}(\alpha + \beta) = \mathrm{Tr}_{K/F}(\alpha) + \mathrm{Tr}_{K/F}(\beta)$; if $a \in F$ then $\mathrm{Tr}_{K/F}(a\alpha) = a\mathrm{Tr}_{K/F}(\alpha)$.
- In particular, $\alpha \in F$ implies $\mathrm{N}_{K/F}(\alpha) = \alpha^{[K:F]}$ and $\mathrm{Tr}_{K/F}(\alpha) = [K : F]\alpha$.

Corollary 1.16.2. *The trace map* $\mathrm{Tr}_{K/F} : K \to F$ *is either surjective or identically zero.*

Lemma 1.16.3 (Transitivity)**.** *Let $F \subset E \subset K$ be a tower of finite extensions. Then*

$$\mathrm{N}_{E/F} \circ \mathrm{N}_{K/E} = \mathrm{N}_{K/F}, \quad \mathrm{Tr}_{E/F} \circ \mathrm{Tr}_{K/E} = \mathrm{Tr}_{K/F}.$$

Proof. An easy exercise. $\square$

We have the following explicit description.

Lemma 1.16.4. *Let $K = F(\alpha)$ be a simple extension of F. Let P be the minimal polynomial, say $P(X) = X^n + a_{n-1}X^{n-1} + \cdots + a_0$. Then*

$$\mathrm{N}_{F(\alpha)/F}(\alpha) = (-1)^n a_0, \quad \mathrm{Tr}_{F(\alpha)/F} = -a_{n-1}.$$

Proof. Use the F-basis $\{1, \alpha, ..., \alpha^{n-1}\}$ to represent the linear transformation m_α. $\square$

Proposition 1.16.5. *If K/F is finite extension, then*

$$\mathrm{N}_{K/F}(\alpha) = \prod_{\sigma \in \mathrm{Hom}_F(K, \overline{F})} \sigma(\alpha)^{[K:F]_i}, \quad \mathrm{Tr}_{K/F}(\alpha) = [K:F]_i \sum_{\sigma \in \mathrm{Hom}_F(K,\overline{F})} \sigma(\alpha).$$

Proof. Consider the tower $F \subset F(\alpha) \subset K$. We saw that

$$\mathrm{N}_{K/F}(\alpha) = \mathrm{N}_{F(\alpha)/F}(\alpha)^{[K:F(\alpha)]}. \tag{6}$$

On the other hand, recall that $\mathrm{Hom}_F(K, \overline{F}) = \mathrm{Hom}_F(K_s, \overline{F})$ and similarly for $F(\alpha)$. Consider the following surjective map

$$\mathrm{Hom}_F(K, \overline{F}) \to \mathrm{Hom}_F(F(\alpha), \overline{F}), \quad \sigma \mapsto \sigma|_{F(\alpha)}.$$

For each $\tau \in \mathrm{Hom}_F(F(\alpha), \overline{F})$, there are exactly $[K_s : F(\alpha)_s]$ elements in the pre-image of τ. Thus, we have

$$\Big(\prod_{\sigma \in \mathrm{Hom}_F(K,\overline{F})} \sigma(\alpha)\Big)^{[K:F]_i} = \prod_{\sigma \in \mathrm{Hom}_F(F(\alpha),\overline{F})} \sigma(\alpha)^{[K:F(\alpha)]_s \cdot [K:F]_i}.$$

Note that

$$[K : F(\alpha)]_s [K : F]_i = [K : F(\alpha)][F(\alpha) : F]_i.$$

Comparing this with (6), we are left to show

$$\mathrm{N}_{F(\alpha)/F}(\alpha) = \Big(\prod_{\sigma \in \mathrm{Hom}_F(F(\alpha),\overline{F})} \sigma(\alpha)\Big)^{[F(\alpha):F]_i}.$$

Now, if P is the minimal polynomial of α, then $P = P_e(X^{p^e})$ where $p^e = [F(\alpha) : F]_i$ and $\deg P_e = [F(\alpha) : F]_s$. Write $P(X) = X^n + a_{n-1}X^{n-1} + \cdots + a_0$, then $\mathrm{N}_{F(\alpha)/F}(\alpha) = (-1)^n a_0$ and the above equality can be directly checked. $\square$

Corollary 1.16.6. *Let K/F be a finite extension. Then K/F is separable iff $\mathrm{Tr}_{K/F}$ is surjective iff $\mathrm{Tr}_{K/F}$ is non-zero.*

Proof. If K/F is inseparable, then F has positive characteristic, say p. Proposition 1.16.5 implies that $\mathrm{Tr}_{K/F}$ is identically zero.

If K/F is separable, we need to prove that $\mathrm{Tr}_{K/F}$ is non-zero. By Proposition 1.16.5, it is enough to find some α such that $\sum_\sigma \sigma(\alpha) \neq 0$. We conclude by the next theorem. $\square$

Theorem 1.16.7 (Dedekind)**.** *Let $\chi_1, ..., \chi_n$ be distinct characters $G \to L^\times$. They are linearly independent: if $c_1, ..., c_n \in L$ satisfy*

$$c_1\chi_1(g) + \cdots + c_n\chi_n(g) = 0, \quad \forall g \in G$$

then $c_i = 0$ for $1 \leq i \leq n$.

Proof. The proof is analogue to the technique used in Artin's lemma 1.5.4. The assumption that χ_i are distinct is necessary. $\square$

Example 1.16.8. *Let K/F be an extension of finite fields. Then $\mathrm{N}_{K/F}$ and $\mathrm{Tr}_{K/F}$ are both surjective.*

Indeed, since K/F is separable, $\mathrm{Tr}_{K/F}$ is surjective by Corollary 1.16.6. For the surjectivity of $\mathrm{N}_{K/F}$, we write $F = \mathbb{F}_q$ and $K = \mathbb{F}_{q^d}$. We know that $\mathbb{F}_{q^d}^\times$ is a cyclic group of order $q^d - 1$; let x be a generator. Then

$$\mathrm{N}_{K/F}(x) = \prod_{\sigma \in \mathrm{Gal}(K/F)} \sigma(x) = \prod_{i=0}^{d-1} x^{q^i} = x^{\frac{q^d-1}{q-1}} =: a \in \mathbb{F}_q^\times.$$

It is clear that a has order $q-1$, hence is a generator of the cyclic group $F^\times = \mathbb{F}_q^\times$. Since $\mathrm{N}_{K/F}$ is multiplicative, the result follows.

1.17 Galois cohomology

Let G be a finite group. A G-module is an abelian group A together with an action of G, i.e. a map $G \times A \to A$, $(g, a) \mapsto g(a)$ such that

(a) $(gg')a = g(g'a)$;

(b) $1a = a$;

(c) $g(a + a') = ga + ga'$.

A morphism between two G-modules A_1, A_2 is a group homomorphism $f : A_1 \to A_2$ which is G-equivariant, i.e. $f(ga_1) = gf(a_1)$ for any $g \in G$ and $a_1 \in A_1$.

Let $\mathbb{Z}[G]$ denote the group algebra of G. Precisely,

$$\mathbb{Z}[G] = \{\sum_{g \in G} a_g g, \ a_g \in \mathbb{Z}\}$$

which is a free $\mathbb{Z}$-module with a natural basis indexed by elements of G. The multiplication in $\mathbb{Z}[G]$ is given by

$$(\sum_{g \in G} a_g g)(\sum_{h \in G} b_h h) = \sum_{k \in G} c_k k$$

where $c_k = \sum_{gh=k} a_g b_h$. It is direct to check that a G-module is the same as a (left) $\mathbb{Z}[G]$-module.

Example 1.17.1. *Let K be a Galois extension of F with galois group G. Then $(K, +)$ and $(K^\times, \times)$ are G-modules.*

Given a G-module A, we define cohomology groups $H^n(G, A)$ for $n \geq 0$ as follows.

- If $n = 0$, $C^0(G, A) := A$; if $n \geq 1$,
$$C^n(G, A) := \{\text{functions } f : G^n \to A\}.$$
- Define $d^n : C^n(G, A) \to C^{n+1}(G, A)$ as follows:
$$\begin{aligned}(d^n)f(g_1, \ldots, g_{n+1}) &= g_1 f(g_2, \ldots, g_{n+1}) + \textstyle\sum_{i=1}^n (-1)^i f(g_1, \ldots, g_i g_{i+1}, \ldots, g_{n+1}) \\ &\quad + (-1)^{n+1} f(g_1, \ldots, g_n).\end{aligned}$$
- One checks $d^n \circ d^{n-1} = 0$, so we obtain a complex
$$0 \to C^0(G, A) \to C^1(G, A) \to \cdots \to C^n(G, A) \to \cdots.$$

Definition 1.17.2. *For $n \geq 0$, define $Z^n(G, A) := \mathrm{Ker}(d^n)$ and call its elements n-cocycles, $B^n(G, A) := \mathrm{Im}(d^{n-1})$ and call its elements n-coboundaries; define the cohomology groups*
$$H^n(G, A) := \frac{Z^n(G, A)}{B^n(G, A)}.$$

Example 1.17.3. *For $n = 0$, $H^0(G, A) = \mathrm{Ker}(d^0)$, where $d^0 : A \to C^1(G, A)$ is given by $a \mapsto (g \mapsto ga - a)$. It is clear that $H^0(G, A) = A^G$.*

Example 1.17.4. *For $n = 1$, $d^1 : C^1(G, A) \to C^2(G, A)$ is given by*
$$(d^1 f)(g_1, g_2) = g_1 f(g_2) + (-1) f(g_1 g_2) + f(g_1).$$
Thus $Z^1(G, A)$ is exactly the set of so-called crossed homomorphisms, *i.e. $f : G \to A$ satisfying*
$$f(g_1 g_2) = f(g_1) + g_1 \cdot f(g_2).$$
By Example 1.17.3, $B^1(G, A)$ consists of those crossed homomorphisms of the form $ga - a$ for $a \in A$, called principal crossed homomorphisms. *We then have*
$$H^1(G, A) := \frac{\{\textit{crossed homomorphisms}\}}{\{\textit{principal crossed homomorphisms }\}}.$$

Remark 1.17.5. *When the action of G on A is trivial, i.e. $g \cdot a = a$ for any $g \in G$ and $a \in A$, a crossed homomorphism is just a usual group homomorphism. Moreover, as $ga - a = 0$, we have $Z^1(G, A) = 0$ and so*
$$H^1(G, A) = \mathrm{Hom}(G, A),$$
where $\mathrm{Hom}(G, A)$ denotes the set of group homomorphisms from G to A.

Remark 1.17.6. *Note that if f is a crossed homomorphism, then we always have $f(1) = 0$ as $f(1) = f(1 \cdot 1) = f(1) + f(1)$.*

Assume that $G = \langle \sigma \rangle$ is a cyclic group of order n. Then a crossed homomorphism $f : G \to A$ is determined by $\alpha := f(\sigma)$ and α satisfies

$$(1 + \sigma + \cdots + \sigma^{n-1})\alpha = f(\sigma^n) = f(1) = 0. \tag{7}$$

Conversely, if $\alpha \in A$ satisfies (7), then the formula

$$f(\sigma^i) := (1 + \sigma + \cdots + \sigma^{i-1})\alpha$$

defines a crossed homomorphism $f : G \to A$. Thus there is a one-to-one correspondence between the set of crossed homomorphisms and $\alpha \in A$ satisfying (7).

Theorem 1.17.7. *Let K be a Galois extension of F with Galois group G. Then $H^1(G, K^\times) = 0$ and $H^1(G, K) = 0$.*

Proof. Let f be a crossed homomorphism $G \to K^\times$. In multiplicative notation, this means

$$f(\sigma\tau) = f(\sigma) \cdot \sigma(f(\tau))$$

and we want to find $x \in K^\times$ such that $f(\sigma) = \frac{\sigma x}{x}$ for all $\sigma \in G$. Since $f(\tau) \neq 0$ for any $\tau \in G$, the map

$$\sum_{\tau \in G} f(\tau)\tau : K^\times \to K^\times$$

is not identically zero by Theorem 1.16.7; choose $\gamma \in K^\times$ such that $\beta := \sum_{\tau \in G} f(\tau)\tau(\gamma) \neq 0$. Then

$$\sigma(\beta) = \sum_{\tau \in G} \sigma(f(\tau))\sigma\tau(\gamma) = \sum f(\sigma)^{-1} f(\sigma\tau) \cdot \sigma\tau(\gamma) = f(\sigma)^{-1}\beta.$$

Therefore, $f(\sigma) = \frac{\beta}{\sigma(\beta)} = \frac{\sigma(x)}{x}$ with $x = \beta^{-1}$ (recall $\beta \neq 0$).

For the additive version, we need to use the fact that the trace map $\mathrm{Tr}_{K/F} : K \to F$ is not identically zero (as K/F is separable). Take $\gamma \in K$ with $\mathrm{Tr}_{K/F}(\gamma) \neq 0$ and let $\beta = \sum_{\tau \in G} f(\tau)\tau(\gamma)$. Then

$$\sigma(\beta) = \sum_{\tau \in G} [\sigma f(\tau) \cdot \sigma\tau(\gamma)] = \sum_{\tau \in G} (f(\sigma\tau) - f(\sigma)) \cdot \sigma\tau(\gamma) = \beta - f(\sigma) \cdot \sum_{\tau} \tau(\gamma).$$

Since $\sum_{\tau \in G} \tau(\gamma) = \mathrm{Tr}_{K/F}(\gamma) \neq 0$, we get $f(\sigma) = \sigma(x) - x$ for $x = \beta/(-\mathrm{Tr}(\gamma))$. $\square$

Next we prove the famous Hilbert's Theorem 90.

Theorem 1.17.8 (Hilbert's Theorem 90). [17] *Let K be a finite extension of F with cyclic Galois group $G = \langle \sigma \rangle$.*

(i) If $\alpha \in K^\times$ with $\mathbb{N}_{K/F}(\alpha) = 1$ then $\alpha = \sigma(\beta)/\beta$ for some $\beta \in K^\times$.

(ii) If $\mathrm{Tr}_{K/F}(\alpha) = 0$, then $\alpha = \sigma(\beta) - \beta$ for some $\beta \in K$.

[17] See Jacobson, Basic algebra I, §4.15 for a treatment without using cohomology of groups.

Proof. (ii) Let $n = [K : F]$. Since $G = \langle\sigma\rangle$ is cyclic, the condition $\mathrm{Tr}_{K/F}(\alpha) = 0$ means $\sum_{i=0}^{n-1} \sigma^i(\alpha) = 0$. By Remark 1.17.6, there is a crossed homomorphism $f : G \to K$ with $f(\sigma) = \alpha$. Then apply Theorem 1.17.7. (i) is proved in a similar way. $\square$

Example 1.17.9. *Here is an application of Hilbert's Theorem 90. Consider the extension $\mathbb{Q}(i)/\mathbb{Q}$. Let $a + bi \in \mathbb{Q}(i)$ be an element of norm 1, i.e. $N(a + bi) = a^2 + b^2 = 1$. Note that $\mathrm{Gal}(\mathbb{Q}(i)/\mathbb{Q}) = \{1, \sigma\}$ is a cyclic group of order 2, where c denotes the complex conjugation. Then there exists $c + di$ such that*

$$a + bi = \frac{\sigma(c + di)}{c + di} = \frac{(c^2 - d^2) - 2icd}{c^2 + d^2}.$$

We deduce that solutions of $a^2 + b^2 = 1$ in $\mathbb{Q}$ has the form

$$\left(\frac{c^2 - d^2}{c^2 + d^2}, \frac{2cd}{c^2 + d^2}\right), \quad c, d \in \mathbb{Z}.$$

1.18 Kummer theory

Recall that we have considered the following situation: K is the splitting field of $X^n - a$ for some $a \in F$; we know that $\mathrm{Gal}(K/F)$ is solvable, with both $\mathrm{Gal}(F(\zeta_n)/F)$ and $\mathrm{Gal}(K/F(\zeta_n))$ being abelian. Now we study in detail Galois extensions of type $K/F(\zeta_n)$ in a more general context.

In this subsection, we fix a field F and $n \in \mathbb{N}$ satisfying the following condition:

$$F \text{ contains a primitive } n\text{-th root of unity.}$$

This is equivalent to requiring that F contains n distinct roots of unity. Remark that this implies that p does not divide n when $\mathrm{char}(F) = p$.

Denote by μ_n the subgroup of n-th roots of unity in F. Note that μ_n is in fact cyclic.

Lemma 1.18.1. *Let $a \in F^\times$ and m be the order of a in $F^\times/(F^\times)^n$. Then every irreducible factor of $X^n - a \in F[X]$ has the form $X^m - b$ for some $b \in F^\times$.*

Proof. Let $\alpha \in \overline{F}$ be a root of $P(X) = X^n - a$. It suffices to prove that the minimal polynomial of α, say Q, has the form $X^m - b$.

By definition of m, $a^m \in (F^\times)^n$, so there exists $b \in F^\times$ such that $a^m = b^n$. Since $\alpha^n = a$, we get $\alpha^{nm} = b^n$, i.e. $\alpha^m/b \in \mu_n \subset F^\times$, so $\alpha^m \in F^\times$. As a consequence, Q divides $X^m - b$ for some $b \in F^\times$. In particular $d := \deg Q \leq m$.

We need to prove $m = d$. Since $P = \prod_{i=0}^{n-1}(X - \alpha\xi_n^i)$, there exists a subset S of $\mathbb{Z}/n\mathbb{Z}$ such that $Q = \prod_{i\in S}(X - \alpha\xi_n^i)$ (thus $|S| = d$). Expanding it, we see that $\alpha^d \xi_n^{\sum_{i\in S} i} \in F$. Since $\xi_n \in F$, we deduce $\alpha^d \in F^\times$, so $a^d \in (F^\times)^n$ (as $\alpha^n = a$). But m is the order of a in $F^\times/(F^\times)^n$, so $m|d$ and finally $m = d$. $\square$

Remark 1.18.2. *In Lemma 1.18.1, we see that any irreducible factor of $X^n - a$ has degree dividing n. This is not true if F does not contain μ_n; for example, $X^3 - 1 \in \mathbb{Q}[X]$ has an irreducible factor of degree 2.*

Proposition 1.18.3. *Let $K = F(\sqrt[n]{a})$ for some $a \in F^\times$. Then K/F is a cyclic Galois extension of degree m, where m is the order of a in $F^\times/(F^\times)^n$.*

Here $\sqrt[n]{a}$ denotes a fixed n-th root of a.

Proof. Since F contains μ_n, $F(\sqrt[n]{a})$ is Galois over F. It has degree m by Lemma 1.18.1.

Write $\alpha = \sqrt[n]{a}$. We define a map as follows:

$$\phi : \mathrm{Gal}(K/F) \to \mu_n, \quad \sigma \mapsto \frac{\sigma(\alpha)}{\alpha}.$$

This is a group homomorphism (noting that $\mathrm{Gal}(K/F)$ fixes $\mu_n \subset F$):

$$\phi(\sigma\tau) = \frac{\sigma\tau(\alpha)}{\alpha} = \frac{\sigma(\phi(\tau)\cdot\alpha)}{\alpha} = \frac{\sigma(\alpha)\cdot\phi(\tau)}{\alpha} = \phi(\sigma)\phi(\tau). \tag{8}$$

It is injective since α generates the simple extension $F(\alpha)$ over F. Since μ_n is cyclic, so is $\mathrm{Gal}(K/F)$. □

Remark 1.18.4. *The homomorphism ϕ in* (8) *does not depend on the choice of $\sqrt[n]{a}$. If α' is another n-th root of a, then $\alpha' = \xi_n^i\alpha$ for some i and*

$$\frac{\sigma(\alpha')}{\alpha'} = \frac{\sigma(\xi_n^i)\sigma(\alpha)}{\xi_n^i\alpha} = \frac{\sigma(\alpha)}{\alpha}.$$

Definition 1.18.5. *A Kummer extension is an abelian extension K/F of exponent dividing n.*

Theorem 1.18.6. *Then K/F is a Kummer extension if and only if K takes the form $F(\sqrt[n]{a_1}, \dots, \sqrt[n]{a_r})$ for some $a_i \in F^\times$.*

Proof. $\Longleftarrow$ This is because each $F(\sqrt[n]{a_i})/F$ is abelian with exponent n and $\mathrm{Gal}(K/F) \hookrightarrow \prod_i \mathrm{Gal}(F(\sqrt[n]{a_i})/F)$.

$\Longrightarrow$ First assume K/F is a cyclic extension of order m (dividing n). By assumption F contains a primitive m-th root of unity, say ξ_m (just take $\xi_n^{n/m}$). Since $N_{K/F}(\xi_m) = (\xi_m)^{[K:F]} = (\xi_m)^m = 1$, we apply Hilbert's Theorem 90 to find some α with $\xi_m = \sigma(\alpha)/\alpha$, i.e. $\sigma(\alpha) = \alpha\xi_m$. Therefore $\sigma^i(\alpha) = \alpha\xi_m^i$, and α has m distinct conjugates in K, hence $[F(\alpha) : F] \geq m$ using Proposition 1.4.17 and finally $F(\alpha) = K$. It is left to check $a := \alpha^n \in F$ (in fact $\alpha^m \in F$), or equivalently $\sigma(\alpha^n) = \alpha^n$, which is clear.

Now we treat the general case. Since $G = \mathrm{Gal}(K/F)$ is abelian, it is a direct product of cyclic groups of exponent n, i.e.

$$G = C_1 \times \cdots \times C_r$$

with each C_i being cyclic, of order dividing n. Let $H_j = \prod_{i\neq j} C_i \times \{1\}$ and $K_j = K^{H_j}$. Then $\mathrm{Gal}(K_j/F) \cong C_j$ is cyclic of order dividing n, hence of the form $F(\sqrt[n]{a_j})$ by the above discussion. Lemma 1.18.7 below shows that $K = K_1 \cdots K_r$ and finishes the proof. □

Lemma 1.18.7. *If K/F is a finite Galois extension with Galois group $G = C_1 \times \cdots C_r$,* [18] *then $K = K_1 \cdots K_r$, where K_j is the fixed field of $H_j := \prod_{i \neq j} C_j \times \{1\}$.*

Proof. By induction, we only need prove it for $r = 2$. Observe that $K_1 \cap K_2 = F$: let $x \in K_1 \cap K_2$, then x is fixed by H_1 and H_2, hence by G, hence belongs to F. Since K_1/F and K_2/F are Galois extensions,[19] we know by Proposition 1.6.3(ii)

$$\mathrm{Gal}(K_1K_2/K_2) = \mathrm{Gal}(K_1/(K_1 \cap K_2)) = \mathrm{Gal}(K_1/F).$$

In particular, we have

$$[K_1K_2 : F] = [K_1K_2 : K_2][K_2 : F] = [K_1 : F][K_2 : F] = |C_1||C_2| = |G|,$$

hence $K_1K_2 = K$. □

Remark 1.18.8. *The proof of Theorem 1.18.6 gives the following: If K/F is a cyclic Galois extension such that $\mathrm{Gal}(K/F) \cong \mathbb{Z}/n\mathbb{Z} = \langle\sigma\rangle$, then $K = F(\sqrt[n]{a})$ for some $a \in F^\times$.*

Reformulation using Galois cohomology

We have defined $H^n(G, A)$ for a G-module A, with $H^0(G, A) = A^G$. We admit the following fact from homological algebra: if $0 \to A_1 \to A \to A_2 \to 0$ is a short exact sequence of G-modules, then there is a long exact sequence

$$0 \to A_1^G \to A^G \to A_2^G \xrightarrow{\partial} H^1(G, A_1) \to H^1(G, A) \to H^1(G, A_2) \to \cdots$$

where ∂ is called the connecting morphism.

Let K/F be a finite Galois extension with $G := \mathrm{Gal}(K/F)$. We have the following exact sequence of G-modules:

$$1 \to \mu_n \to K^\times \xrightarrow{x \mapsto x^n} (K^\times)^n \to 1$$

where μ_n carries the trivial G-action. We obtain a long exact sequence

$$1 \to \mu_n \to F^\times \to (K^\times)^n \cap F^\times \to H^1(G, \mu_n) \to H^1(G, K^\times) = 0$$

where the last equality follows from Hilbert's Theorem 90. Moreover, note that $H^1(G, \mu_n) \cong \mathrm{Hom}(G, \mu_n)$, see Remark 1.17.5. Therefore we obtain an isomorphism of groups, called *Kummer isomorphism*

$$\begin{array}{rcl} ((K^\times)^n \cap F^\times)/(F^\times)^n & \cong & \mathrm{Hom}(G, \mu_n) \\ a & \mapsto & (\chi_a : \sigma \mapsto \frac{\sigma(\alpha)}{\alpha}). \end{array} \tag{9}$$

[18] C_i not necessarily cyclic, or even abelian

[19] For general field extensions, we don't have $[K_1K_2 : K_2] = [K_1 : K_1 \cap K_2]$. Example, let $K_1 = \mathbb{Q}(\sqrt[3]{2})$, $K_2 = \mathbb{Q}(\sqrt[3]{2}\xi_3)$, then K_1K_2 is the splitting field of $X^3 - 2$ over $\mathbb{Q}$, hence $[K_1K_2 : K_2] = 2$, while $K_1 \cap K_2 = \mathbb{Q}$ and $[K_1 : \mathbb{Q}] = 3$.

Here we need to choose an n-th root α of a, but χ_a is independent of the choice as explained in Remark 1.18.4 .

Given $\chi \in \mathrm{Hom}(G, \mu_n)$, let $H_\chi < G$ be the kernel of χ and $K_\chi := K^{H_\chi}$. Then Kummer theory implies that

$$K_\chi = F(\sqrt[n]{a})$$

where $a \in (K^\times)^n \cap F^\times$ is chosen so that its class in $(K^\times \cap F^\times)/(F^\times)^n$ corresponds to χ.

Moreover, (9) induces a bi-multiplicative pairing

$$G \times (K^\times \cap F^\times)/(F^\times)^n \to \mu_n, \quad (\sigma, a) \mapsto \chi_a(\sigma)$$

which is *perfect* when K/F is a Kummer extension.

1.19 Artin-Schreier extensions

Let F be a field of characteristic p.

Lemma 1.19.1. *For any $a \in F$, the polynomial $X^p - X - a \in F[X]$ is either irreducible or completely reducible (i.e. every irreducible factor has degree 1).*

Proof. Write $f = X^p - X - a$. First observe that if $\alpha \in \overline{F}$ is a root of f, then $\alpha + j$ (for $j \in \mathbb{F}_p$) is also a root of f. Hence f decomposes in $\overline{F}[X]$ as

$$f = \prod_{j \in \mathbb{F}_p} (X - (\alpha + j)).$$

In particular, if $f(X)$ has one root in F, then f is completely reducible over F.

We assume f is reducible in $F[X]$, say $f = gh$, with $1 \leq d := \deg g \leq p - 1$. Then g has the form

$$g = \prod_{j \in S} (X - (\alpha + j))$$

for some subset $S \subset \mathbb{F}_p\}$ with $|S| = d$. Expanding the expression we get $\sum_{j \in S} (\alpha + j) = d\alpha + \sum_{j \in S} j \in F$, and so $d\alpha \in F$. As $1 < d < p - 1$, $d \neq 0$ in F, so $\alpha \in F$. As explained in the first paragraph, this implies that f is completely reducible. $\square$

Proposition 1.19.2. *A field extension K/F is cyclic of degree p if and only if $K = F(\alpha)$ where α is a root of $X^p - X - a \in F[X]$ and a is not of the form $c^p - c$ for any $c \in F$.*

Proof. $\Longleftarrow$ The condition that $a \neq c^p - c$ for any $c \in F$ means that $X^p - X - a \in F[X]$ is irreducible. As in the proof of Lemma 1.19.1, the conjugates of α are just $\alpha + j$ for $j \in \mathbb{F}_p$. Since these elements are distinct, $F(\alpha)$ is a Galois extension over F of degree p.

$\Longrightarrow$ Assume K/F is cyclic of degree p, say $\mathrm{Gal}(K/F) = \langle \sigma \rangle$. Since $\mathrm{Tr}_{K/F}(1) = p = 0$, Hilbert Theorem 90 implies that $1 = \sigma(\alpha) - \alpha$ for some $\alpha \in K$, i.e. $\sigma(\alpha) = \alpha + 1$. Consequently, $\sigma^i(\alpha) = \alpha + i$ for $1 \leq i \leq p - 1$, and in this way we get *all* the conjugates of α (p conjugates in total). This shows $[F(\alpha) : F] = p$, thus $K = F(\alpha)$. Finally, since $\mathrm{Gal}(K/F) = \langle\sigma\rangle$ and

$$\sigma(\alpha^p - \alpha) = (\alpha+1)^p - (\alpha+1) = \alpha^p - \alpha,$$

by Galois correspondence we deduce $a := \alpha^p - \alpha \in F$, so the minimal polynomial divides $X^p - X - a$. They must be equal comparing their degrees. $\square$

Definition 1.19.3. *An Artin-Schreier extension is a finite Galois extension whose group is an elementary abelian p-group.*

Theorem 1.19.4. *An extension is Artin-Schreier if and only if $K = F(\alpha_1, \dots, \alpha_r)$ where α_j is a root of $X^p - X - a_j$ for suitable $a_j \in F$.*

Proof. Analogue to the proof of Theorem 1.18.6. $\square$

Reformulation via Galois cohomology

Let F be as above, and K/F be a finite Galois extension with $G = \mathrm{Gal}(K/F)$. It is clear the map $f : x \mapsto x^p - x$ is a homomorphism of G-modules $(K, +) \to (K, +)$; let $\mathcal{P}(K)$ denote the image of f. Consider the short exact sequence of G-modules

$$0 \to \mathbb{F}_p \to K \xrightarrow{f} \mathcal{P}(K) \to 0$$

where $\mathcal{P}(K) := \{x^p - x | x \in K\} \subset K$ is an additive subgroup of K.

We obtain a long exact sequence

$$0 \to \mathbb{F}_p \to F \to \mathcal{P}(K) \cap F \to H^1(G, \mathbb{F}_p) \to H^1(G, K) = 0$$

where the last equality follows from Hilbert Theorem 90. Note that $H^1(G, \mathbb{F}_p) = \mathrm{Hom}(G, \mathbb{F}_p)$, we deduce an isomorphism

$$(\mathcal{P}(K) \cap F)/\mathcal{P}(F) \cong \mathrm{Hom}(G, \mathbb{F}_p), \quad a \mapsto \theta_a \tag{10}$$

where the homomorphism $\theta_a : G \to \mathbb{F}_p$ is given by $\theta_a(\sigma) = \sigma(\alpha) - \alpha$ by choosing a root α of $X^p - X - a$. It is direct to check that θ_a is well-defined (i.e. independent of the choice of α) and is indeed a group homomorphism.

Let $H_\theta < G$ be the kernel of $\theta \in \mathrm{Hom}(G, \mathbb{F}_p)$ and $K_\theta = K^{H_\theta}$, then $K_\theta = F(\alpha)$ if θ corresponds to $a \in (\mathcal{P}(K) \cap F)/\mathcal{P}(F)$ and α is a root of $X^p - X - a$.

The isomorphism (10) also implies a pairing

$$\begin{array}{ccc} G \times (\mathcal{P}(K) \cap F)/\mathcal{P}(F) & \longrightarrow & \mathbb{F}_p \\ (\sigma, a) & \longmapsto & \theta_a(\sigma) \end{array}$$

which is *perfect* when K/F is Artin-Schreier.

1.20 Transcendental extensions

Let F be a field and Ω/F an extension. Given $\alpha_1, \ldots, \alpha_n \in \Omega$, there exists an F-homomorphism

$$\iota : F[X_1, \ldots, X_n] \to \Omega, \quad X_i \mapsto \alpha_i.$$

Definition 1.20.1. *The α_i are said to be* algebraically independent *over F if ι is injective; otherwise they are* algebraically dependent *over F.*

That is, they are algebraically dependent over F if there exists a non-zero polynomial $f(X_1, \ldots, X_n)$ such that $f(\alpha_1, \ldots, \alpha_n) = 0$.

An infinite set $A \subset \Omega$ is said to be *algebraically independent* over F if every finite subset of A is algebraically independent; otherwise, it is algebraically dependent over F.

Lemma 1.20.2. *Let $A \subset \Omega$ and $\beta \in \Omega$. Suppose that A is algebraically independent over F, but that $A \cup \{\beta\}$ is not. Then β is algebraic over $F(A)$.*

Proof. The assumption means that there exist $\alpha_1, \ldots, \alpha_n \in A$ and a non-zero polynomial $f(X_1, \ldots, X_n, Y) \in F[X_1, \ldots, X_n, Y]$ such that $f(\alpha_1, \ldots, \alpha_n, \beta) = 0$. Then we can write

$$f = g_m Y^m + g_{m-1} Y^{m-1} + \cdots + g_0.$$

Since A is algebraically independent, the g_i's (for $1 \leq i \leq m$) are not all zero, from which the result follows. $\square$

Definition 1.20.3. *A* transcendence basis *for Ω over F is an algebraically independent set $A \subset \Omega$ such that Ω is algebraic over $F(A)$.*

Theorem 1.20.4. *Every algebraically independent subset of Ω is contained in a transcendence basis for Ω over F. In particular, transcendence bases exist.*

Conversely, if $A \subset \Omega$ is a subset such that Ω is algebraic over $F(A)$, then any maximal algebraically independent subset of A is a transcendence basis.

Proof. Let S be the set of algebraically independent subsets of Ω containing the given one. We partially order it by inclusion. Let T be a totally ordered subset of S and let $B = \cup_{A \in T} A$. We see that $B \in S$, i.e. B is still algebraically independent, because every algebraic relation only involves finitely many elements. So, by Zorn's lemma, there exists a maximal algebraically independent subset A in S and Lemma 1.20.2 shows that Ω is algebraic over $F(A)$. The last assertion is clear. $\square$

Similar to the fact that two bases of a finite dimensional F-vector space have the same cardinality, it can be shown that any two (possibly infinite) transcendence bases for Ω have the same cardinality. We first prove a key lemma.

Lemma 1.20.5. *(The exchange property) Let $A := \{\alpha_1, \ldots, \alpha_n\}$ be a subset of Ω. Let $\beta \in \Omega$ be an element which is algebraic over $F(\alpha_1, \ldots, \alpha_n)$ but not algebraic over $F(\alpha_1, \ldots, \alpha_{n-1})$. The following statements hold.*

(i) α_n is algebraic over $F(A')$, where $A' := \{\alpha_1, \ldots, \alpha_{n-1}, \beta\}$.
(ii) If A is algebraically independent, then so is A'.
(iii) If A is a transcendence basis of Ω, then so is A'.

Proof. (i) Since β is algebraic over $F(\alpha_1, \ldots, \alpha_n)$, there exist $u_i \in F(\alpha_1, \ldots, \alpha_n)$, not all 0, such that

$$u_r\beta^r + \cdots + u_1\beta + u_0 = 0.$$

We may further assume $u_i \in F[\alpha_1, \ldots, \alpha_n]$. Write $u_i = f_i(\alpha_1, \ldots, \alpha_n)$ for $f_i \in F[X_1, \ldots, X_n]$ and put $f = \sum_{i=0}^r f_i Y^i$, we get a non-zero polynomial $f(X_1, \ldots, X_n, Y) \in F[X_1, \ldots, X_n, Y]$ such that

$$f(\alpha_1, \ldots, \alpha_n, Y) \neq 0, \quad f(\alpha_1, \ldots, \alpha_n, \beta) = 0.$$

Now write f as a polynomial in X_n,

$$f = \sum_{i=0}^{d} g_i(X_1, \ldots, X_{n-1}, Y) X_n^i \tag{11}$$

and note that at least one of the polynomials $g_i(\alpha_1, \ldots, \alpha_{n-1}, Y) \neq 0$, say g_{i_0}; this is because $f(\alpha_1, \ldots, \alpha_n, Y) \neq 0$. Since β is not algebraic over $F(\alpha_1, \ldots, \alpha_{n-1})$, we have $g_{i_0}(\alpha_1, \ldots, \alpha_{n-1}, \beta) \neq 0$ (recall the definition !). Therefore $f(\alpha_1, \ldots, \alpha_{n-1}, X_n, \beta) \neq 0$ (as a polynomial in X_n) by (11), and gives an algebraic relation of α_n over $F(\alpha_1, \ldots, \alpha_{n-1}, \beta)$.

(ii) Assume that A' is algebraically dependent. Then, since $A' \backslash \{\beta\}$ is algebraically independent, by Lemma 1.20.2, β is algebraic over $F(\alpha_1, \ldots, \alpha_{n-1})$ which contradicts the assumption.

(iii) We need to show that Ω is algebraic over $F(A')$. It suffices to show that $F(A)$ is algebraic over $F(A')$, and again reduces to showing α_n is algebraic over $F(A')$, which is true by (i). $\square$

Theorem 1.20.6. *If A and B are two transcendence bases for Ω/F, then A and B have the same cardinality. The cardinality is called the* transcendence degree *of Ω/F.*

Example 1.20.7. *The transcendence degree of $\mathbb{Q}(\pi, e)$ over $\mathbb{Q}$ is either 1 or 2; the precise answer is unknown because it is not known whether π and e are algebraically independent.*

We recall some basic facts about the cardinality of a set A, measuring the size of A. It is denoted by $|A|$ or $\#A$.

(i) When A is finite, $|A|$ denotes the number of its elements.

(ii) Two sets A, B are said to have the same cardinality if there exists a bijection between A and B. We write $|A| \leq |B|$ if there is an injective map of A into B. The Cantor-Schröder-Bernstein theorem says that $|A| = |B|$ if and only if $|A| \leq |B|$ and $|B| \leq |A|$.

(iii) Assume *Axiom of choice*, the cardinalities of infinite sets are denoted by

$$\aleph_0 < \aleph_1 < \aleph_2 < \cdots$$

For each ordinal α, $\aleph_{\alpha+1}$ is the least cardinal number strictly greater than $\aleph_\alpha$.

(iv) The cardinality of the natural numbers is denoted $\aleph_0$, while the cardinality of the real numbers by "$\mathfrak{c}$". The diagonal argument shows that $\aleph_0 < \mathfrak{c}$. The **continuum hypothesis**[20][21] says that $\mathfrak{c} = \aleph_1$.

(v) If B is infinite and $|B| \geq |A|$, then $|A \dot\cup B| = |A| + |B| = |B|$.

(vi) If $A \neq \emptyset$ and B is infinite and $|B| \geq |A|$, then $|A \times B| = |B|$. It suffices to treat the case $|A| = |B|$, i.e. to prove $|B \times B| = |B|$. (Exercise.)

(vii) If $|A|$ is infinite, let $S = \{B \subset A, B \text{ finite}\}$. Then $|S| = |A|$. Proof: S is a disjoint union of S_n, where $S_n = \{B \subset A : |B| = n\}$. It is clear that $|S_n| \leq |A^n| = |A|$ (since A is infinite). Hence $|S| \leq |A \times \mathbb{N}| = |A|$.[22]

(viii) If B is infinite and if $f : A \to B$ is such that for any $b \in B$, $f^{-1}(b)$ is a finite set, then $|A| \leq |B|$. Indeed, consider $B_n \subset B$, the subset of $b \in B$ such that $|f^{-1}(b)| = n$. Then

$$f : A_n := f^{-1}(B_n) \to B_n$$

is an n-to-1 map, so $|A_n| \leq |(B_n)^n| \leq |B^n| = |B|$ (as B is infinite). Therefore $A \hookrightarrow B \times \mathbb{N}$ as in (vii), hence the result.

Proof of Theorem 1.20.6. First treat the case when one of A, B is finite, and we must show that the other is also finite and $|A| = |B|$. We may assume A is finite and write $A = \{\alpha_1, \ldots, \alpha_n\}$. We do induction on n. If $n = 0$, i.e. $A = \emptyset$, then Ω is algebraic over F, hence B is also empty. Suppose $n \geq 1$ and the result is true for $n-1$ case. Consider $A \cap B$. If it is not empty, then Ω has a transcendence basis over $F(A \cap B)$ of cardinal

$$A \backslash (A \cap B) < n,$$

and by inductive hypothesis, $B \backslash (A \cap B)$ has the same cardinality, hence $|A| = |B|$.

Assume $A \cap B = \emptyset$. Put $K = F(\alpha_1, \ldots, \alpha_{n-1})$. Since Ω is not algebraic over K, there must exist some $\beta \in B$ which is not algebraic over K. Since β is algebraic over $F(\alpha_1, \ldots, \alpha_n)$, Lemma 1.20.5 shows that $A' := \{\alpha_1, \ldots, \alpha_{n-1}, \beta\}$ is also a transcendence basis. But $A' \cap B \neq \emptyset$, the above argument then implies that $|B| = |A'| = |A|$.

Now we prove the case when A and B are both infinite. It suffices to prove the inequality $|A| \geq |B|$ by symmetry. For any $x \in A$, it is algebraic over $F(B)$, hence

[20] It is independent of the ZF axiomatization of set theory.

[21] The generalized continuum hypothesis (GCH) states that for every infinite set A, there are no cardinals strictly between $|A|$ and $2^{|A|}$.

[22] Here for each n, we choose an injection $\iota_n : S_n \to A$, then $S \to A \times \mathbb{N}$, $s \mapsto (\iota_n(s), n)$ if $s \in S_n$, is injective.

algebraic over $F(S_x)$ for some *finite* subset $S_x \subset B$. Put $S = \cup_{x\in A} S_x \subset B$. Then $|S| \le |A|$ by property (viii) (which implies $|\dot{\cup}_{x\in A} S_x| \le A$). On the other hand, any element of A is algebraic over $F(S)$ and Ω is algebraic over $F(A)$, so is algebraic over $F(S)$. Hence $S \subset B$ must be a maximal algebraically independent subset, so $S = B$ and $|B| \le |A|$. $\square$

Proposition 1.20.8. *Let $F \subset E \subset K$ be field extensions. Then*

$$\operatorname{trdeg}(K/F) = \operatorname{trdeg}(K/E) + \operatorname{trdeg}(E/F).$$

Proof. If A, B are respectively a transcendence basis of E/F and K/E, then $A\dot{\cup}B$ is a transcendence basis for K/F. $\square$

Proposition 1.20.9. *Any two algebraically closed fields with the same transcendence degree over F are F-isomorphic.*

Proof. Choose transcendence bases A and A' for the two fields Ω and Ω'. By assumption, there exists a bijection $A \to A'$, which extends uniquely to an F-isomorphism $F[A] \to F[A']$, and hence to an an F-isomorphism of the fields $F(A) \to F(A')$. Identifying $F(A)$ and $F(A')$, Ω and Ω' are algebraic closures of the same field, hence are isomorphic. $\square$

Proposition 1.20.10. *Let F be an infinite field. We have the following statements.*

(i) $|F[X]| = |F|$.

(ii) *If K/F is an algebraic extension, then $|K| = |F|$.*

(iii) *If $K = F(A)$, where $A = \{\alpha_i\}$ is an arbitrary set of indeterminates, then $|F(A)| = \max(|A|, |F|)$.*

Proof. (i) Let $F[X]_n$ denote the subset of polynomials of degree n. Then $F[X]_n$ is bijective to F^n, so that $|F[X]_n| = |F|$ since F is infinite. Therefore there exists an injective map $F[X] \hookrightarrow F \times \mathbb{N}$ and the result follows.

(ii) We may assume K is an algebraic closure of F. Then K is a disjoint union of finite sets, each of which is the set of roots of a monic irreducible polynomial. We get a map $K \to F[X]$ (with image being the set of monic irreducible polynomials), with finite fibers. So $|K| \le |F[X]| = |F|$, giving the result.

(iii) Since $F(A)$ embeds into $F[A] \times F[A]$, we have $F(A) = F[A]$. If A is finite, then we may do induction, hence reduce to show $|F[X]| = |F|$, which is proved in (i). For infinite A, we have

$$F[A] = \bigcup_{A' \subset A,\ A' \text{ finite}} F[A'] \hookrightarrow H \times F,$$

where $H = \{A' \subset A, A' \text{ finite}\}$. Since $|H| = |A|$, we get the result. $\square$

Corollary 1.20.11. *Let F be an infinite field and K/F be any extension. Then*

$$|K| = \max(|F|, \operatorname{trdeg}(K/F)).$$

Proof. Let A be a set of transcendence basis of K over F, then K is an algebraic extension of $F(A)$. So $|K| = |F(A)| = \max(|F|, |A|)$ by Proposition 1.20.10. $\square$

Example 1.20.12. *(i) We know $|\mathbb{C}| = |\mathbb{R}| = \mathfrak{c} > |\mathbb{Q}|$. Hence $\operatorname{trdeg}(\mathbb{R}/\mathbb{Q}) = |\mathbb{R}|$.*

(ii) There is an isomorphism of fields $\mathbb{C} \cong \overline{\mathbb{Q}_p}$, where $\mathbb{Q}_p$ is the field of p-adic numbers. To see this, we only need to check $\operatorname{trdeg}(\mathbb{Q}_p/\mathbb{Q}) = |\mathbb{R}|$. By Corollary 1.20.11, it is enough to show $|\mathbb{Q}_p| = |\mathbb{R}|$. This follows from the construction of $\mathbb{Q}_p$.

Remark 1.20.13. *If we have an integral domain R containing a field F, then we define the transcendence degree of R over F to be that of $\operatorname{Frac}(R)$. In commutative algebra, when R is noetherian, this is equal to the Krull dimension of R.*

Definition 1.20.14. *An extension Ω/F is called* purely transcendental *if it has a transcendence basis A such that $\Omega = F(A)$.*

In general, subextensions of a purely transcendental extension need not be purely transcendental. However, this is true when the transcendence degree is 1.

Theorem 1.20.15 (Lüroth)**.** *Let X be an indeterminante. If $F \subset K \subset F(X)$, then $K = F(\alpha)$ for some $\alpha \in F(X)$. In particular, every nontrivial extension of F contained in $F(X)$ is purely transcendental over F.*

2 Module theory

2.1 Modules and homomorphisms

We fix a ring R with unit.

Definition 2.1.1. *A left R-module is an abelian group $(M,+)$ equipped with a map*

$$R \times M \to M, \quad (r,m) \mapsto rm$$

called scalar multiplication*, satisfying*

- $r(r'm) = (rr')m$
- $r(m+m') = rm + rm'$
- $(r+r')m = rm + r'm$
- $1m = m$

for all $r, r' \in R$ and $m, m' \in M$.

If the scalar multiplication is written on the right $(m,r) \mapsto mr$, we get the notion of right modules, e.g. $m(rr') = (mr)r'$. [23]

Example 2.1.2. *(1) For $R = \mathbb{Z}$, a $\mathbb{Z}$-module is just an abelian group.*

(2) For $R = F$ a field, an R-modules is an F-vector space.

(3) If F is a field, then an $F[X]$-module M is an F-module with an additional action of X on M which is F-linear. In other words, an $F[X]$-module is an F-vector space M combined with a linear map from M to M.

(4) Let $R = M_n(F)$ where F is a field, and let $V = F^n$ (viewed as column vectors). Then R acts on V by matrix multiplication, making V a left R-module. If we view V as row vectors, then it becomes a right R-module.

(5) If M is a left R-module and $S \to R$ is a ring homomorphism, then M becomes an S-module.

If S is another ring, an (R,S)-bimodule is an abelian group which is a left R-module and a right S-module such that

$$r(ms) = (rm)s, \quad \forall r \in R, s \in S.$$

Example 2.1.3. *(1) Every ring R is an (R,R)-bimodule.*

(2) When R is commutative, every left R-module is naturally an (R,R)-bimodule by setting $rmr' := rr'm$.

Definition 2.1.4. *A subset $N \subset M$ is said a left* sub-R-module *of M if N is a commutative subgroup of M and $rn \in N$ if $n \in N$.*

Given a submodule N of M, we can form the quotient module $M/N := \{x + M\}$.

[23] A left R-module is the same thing as a right R^{op}-module, where R^{op} denotes the opposite ring: as a set $R^{\mathrm{op}} = R$, and with the same addition law, but $a * b := b \cdot a$.

Example 2.1.5. *R itself is a left (or right) R-module; left (right) submodules of R are called left (right) ideals.*

The set of submodules of a given module M, together with the two binary operations $+$ and $\cap$,[24] forms a lattice which satisfies the modular law: Given submodules U, N_1, N_2 of M such that $N_1 \subset N_2$, then the following two submodules are equal:

$$(N_1 + U) \cap N_2 = N_1 + (U \cap N_2).$$

If $\{m_i\}$ is a family of elements in M, there exists a minimal submodule of N containing $\{m_i\}$, called the submodule generated by them, denoted by $\langle m_i \rangle$.

Definition 2.1.6. *A map $\phi : M \to M'$ between two R-modules is called a homomorphism if*

- *ϕ is a homomorphism for the underlying abelian groups;*
- *$\phi(rm) = r\phi(m)$ for $r \in R$ and $m \in M$.*

Let $\ker(\phi) := \{m \in M | \phi(m) = 0\}$ be the *kernel* of ϕ, $\mathrm{im}(\phi) := \{m' \in M' | \exists m \in M,\ m' = \phi(m)\}$ be the *image* of ϕ. One checks that $\ker(\phi)$ is a submodule of M and $\mathrm{im}(\phi)$ is a submodule of M'. Recall the following facts:

(1) if $\phi : M \to M'$ is a homomorphism, then $M/\ker(\phi) \cong \mathrm{im}(\phi)$.

(2) if ϕ is surjective then $M/\ker(\phi) \cong M'$.

(3) if M_1, M_2 are submodules of M, then

$$(M_1 + M_2)/M_1 \cong M_2/(M_1 \cap M_2).$$

(4) Consider $\mathrm{Hom}_R(M, M')$ which is an additive group:

$$(\phi + \phi')(m) = \phi(m) + \phi'(m).$$

But note that in general there is no R-module structure on $\mathrm{Hom}_R(M, M')$.

(5) When $M = M'$, we get $\mathrm{End}_R(M)$, which becomes a ring: if $\phi : M \to M$ and $\phi' : M \to M$, then the composition $\phi \circ \phi'$ is still an R-module homomorphism from M to M. Setting multiplication $\phi \cdot \phi' := \phi \circ \phi'$ defines a multiplication operation on $\mathrm{End}_R(M)$, which is in general not commutative ! For example, when $M = F^n$, $\mathrm{End}_F(M) = M_n(F)$ is not commutative.

[24] Arbitrary intersection of submodules is still a submodule

Direct sum and direct product

Let $(M_i)_{i\in I}$ be a family of left R-modules. We set

$$\prod_{i\in I} M_i = \big\{(m_i)_{i\in I} : m_i \in M_i\big\}$$

and call it the *direct product* of M_i. Also set

$$\bigoplus_{i\in I} M_i := \big\{(m_i) : m_i \in M_i,\ m_i = 0 \text{ for almost all i}\big\},$$

and call it the *direct sum* of M_i. Clearly $\bigoplus_{i\in I} M_i$ is a submodule of $\prod_{i\in I} M_i$, and they are equal when I is a finite set.

For each $i \in I$, we have a natural inclusion

$$\iota_i : M_i \to \bigoplus_i M_i, \quad m \mapsto (m_i)_{i\in I},\ m_i = m \text{ and } m_j = 0, \forall j \neq i.$$

The universal property of $\bigoplus_i M_i$ is the following: given an R-module N and a family of R-homomorphisms $\phi_i : M_i \to N$, then there exists a unique homomorphism $\phi : \bigoplus_{i\in I} M_i \to N$ making the following diagram commute:

$$\begin{tikzcd} M_i \arrow[r, "\phi_i"] \arrow[d] & M . \\ \bigoplus_{i\in I} M_i \arrow[ur, "\phi"'] & \end{tikzcd}$$

In other words, we have

$$\operatorname{Hom}_R\big(\bigoplus_{i\in I} M_i, -\big) = \prod_{i\in I} \operatorname{Hom}_R(M_i, -).$$

Similarly we have the universal property for $\prod_{i\in I} M_i$, that is,

$$\operatorname{Hom}_R\big(-, \prod_{i\in I} M_i\big) \cong \prod_{i\in I} \operatorname{Hom}_R(-, M_i).$$

2.2 Free modules

Definition 2.2.1. *An R-module is called* free *if it is isomorphic to a* direct sum *of copies of R, i.e.*

$$M \cong \bigoplus_{i\in I} R_i$$

for some indexing set I, where each R_i is a copy of R. We write $R^{(I)}$ for the direct sum above (and reserve the notation R^I for the direct product).

An R-module M is free iff it has a basis, i.e. a set $\{e_i : i \in I\}$ such that any element of M is a *unique* finite linear combination of the e_i's. Unlike vector spaces over fields, submodules of free modules need not be free.

Remark 2.2.2. *By convention, the zero module is free with $\emptyset$ as basis.*

Recall that our fixed ring R is non-zero.

Proposition 2.2.3. *If $R^{(I)} \cong R^{(J)}$ as left R-modules where I is infinite, then $|I| = |J|$.*

Proof. We fix an isomorphism $\iota : R^{(J)} \xrightarrow{\sim} R^{(I)}$. Consider the canonical basis $\{e_i : i \in I\}$ for $R^{(I)}$ and the basis $\{f_j : j \in J\}$ for $R^{(J)}$. Then each $\iota(f_j)$ is contained in the sub-R-module $\bigoplus_{i \in S_j} Re_i$ for some finite subset $S_j \in I$. Let $I_0 := \bigcup_{j \in J} S_j \subset I$. We must have $I_0 = I$, because $\bigoplus_{i \in I_0} Re_i$ contains the whole $R^{(J)}$. If J is finite, then $I_0 = I$ is also finite which is impossible. Hence J is infinite, and

$$|I| \leq |J \times \mathbb{N}| = |J|.$$

Exchanging I and J, we also get $|J| \leq |I|$. $\square$

However, Proposition 2.2.3 is no longer true when both I and J are finite.

Example 2.2.4. *Let F be a field and $V = F^{(\mathbb{N})}$ which is an infinite dimensional F-vector space. Let $R = \mathrm{End}_F(V)$. Then by choosing a bijection $\mathbb{N} \to \mathbb{N} \dot{\cup} \mathbb{N}$, we get an isomorphism $V \to V \oplus V$. Applying $\mathrm{Hom}_F(V, -)$, we get $R \cong R \oplus R$.*

Definition 2.2.5. *A ring R is said to have IBN (Invariant Basis Number) if for any $n, m \in \mathbb{N}$, $R^n \cong R^m$ implies $n = m$. In this case, we define n to be the rank of R^n.*

Proposition 2.2.6. (i) *If R is a field, then R has IBN.*

(ii) *If R is commutative, then R has IBN.*

(iii) *If R is a finite ring, then R has IBN.*

Proof. (i) This is a standard fact in linear algebra.

(ii) Take a maximal ideal $\mathfrak{m}$ of R. If $R^n \cong R^m$, then module $\mathfrak{m}$ we get $(R/\mathfrak{m})^n \cong (R/\mathfrak{m})^n$ and the result follows from (i).

(iii) If $R^n \cong R^m$, then $|R^n| = |R^m|$, hence $n = m$. $\square$

Theorem 2.2.7. *Every R-module is a quotient of some free R-module.*

Proof. Take a set of generators of M, say $\{a_\lambda\}_{\lambda \in I}$; it does exist since we can take M itself. Then by universal property, there is a homomorphism of R-modules $R^{(I)} \to M$, which is obviously surjective. $\square$

2.3 Finitely generated modules

Definition 2.3.1. *An R-module M is called* finitely generated *if it is isomorphic to a quotient of a finite free module R^n, equivalently, if there exists $m_1, \dots, m_r$ such that*

$$M = Rm_1 + \cdots + Rm_r.$$

It is also called a finite R-module.

An R-module M is called cyclic *if it is a quotient of R, i.e. it can be generated by one element. In other words, a cyclic R-module is of the form $R/\mathfrak{a}$ for some left ideal $\mathfrak{a}$.*

Example 2.3.2. *The left module $V = F^n$ over $M_n(F)$ is finitely generated. In fact it is a cyclic module generated by $m_1 = (1, 0, \dots, 0)^T$.*

Recall that in a ring R with unity, Zorn's lemma implies that R admits left maximal ideals and right maximal ideals. We have an analogue for finitely generated modules.

Proposition 2.3.3. *Let M be a non-zero finitely generated R-module. Then every proper submodule of M is contained in a maximal proper submodule of M.*

Proof. By assumption, M has a finite set of generators $\{m_i; i = 1, \dots, r\}$. A submodule of M is proper if and only if it does not contain all the m_i. Using this, we see that the union of an increasing chain of proper submodules of M is still proper. Thus we may conclude by Zorn's lemma. □

Definition 2.3.4. *An R-module M is called* simple *if $M \neq 0$ and M has no submodules other than M and $\{0\}$.*

Corollary 2.3.5. *Any non-zero finitely generated R-module admits (at least) one simple quotient module.*

Example 2.3.6. *(i) As a $\mathbb{Z}$-module, $\mathbb{Q}$ is not finitely generated (looking at denominators of a finite set of generators) and has no maximal proper submodule.*

(ii) $\mathbb{Q}$ is not a free $\mathbb{Z}$-module. If it were, then since any $r \neq 0, r' \neq 0$ are $\mathbb{Z}$-linear dependent, it must be a cyclic module. But this is not true by (i).

Annihilators

For any $m \in M$, Rm is a cyclic submodule of M. Denote

$$\mathrm{Ann}_R(m) := \{r \in R \mid rm = 0\}$$

which is a left ideal of R; we call it the *annihilator* of m. Denote

$$\mathrm{Ann}_R(M) := \{r \in R \mid rM = 0\}$$

and call it the *annihilator* of M. Note that $\mathrm{Ann}_R(M)$ is a *two-sided* ideal of R.

Definition 2.3.7. *(1) An element $m \in M$ is called* torsion *if* $\mathrm{Ann}_R(m) \neq 0$. *The module M is called* torsion *if any element of M is torsion; called* torsion-free *if the only torsion element in M is zero.*

(2) M is called a faithful *module if* $\mathrm{Ann}_R(M) = 0$.

Remark 2.3.8. *(i) In general, an element $m \in M$ and the submodule generated by m can have different annihilators. For example, if $R = M_n(F)$, $M = F^n$ and $m_1 = (1, 0, \dots, 0)^T$ as in Example 2.3.2, then $Rm_1 = M$ and*

$$\mathrm{Ann}_R(m_1) = \big\{(a_{ij}) \in R \mid a_{i1} = 0\big\}, \quad \mathrm{Ann}(M) = 0.$$

(ii) The above example also shows that there exists a torsion (cyclic) module which is faithful.

Modules over PID

Let R be a commutative PID (principal ideal domain). Typical examples are $R = \mathbb{Z}$ or $R = F[X]$ where F is a field.

Theorem 2.3.9. *Let M be a finitely generated R-module.*

(i) *There is a R-module decomposition*

$$M = R^m \bigoplus \big(\bigoplus_{i=1}^{s} R/(p_i^{n_i})\big)$$

where $p_1, \dots, p_s$ are prime elements and $n_i \geq 1$. Up to reordering, the ideals $(p_i^{n_i})$ are uniquely determined by M.

(ii) *There is a unique increasing sequence of proper ideals*

$$(d_1) \supseteq (d_2) \supseteq \cdots \supseteq (d_n)$$

such that M is isomorphic to the sum of cyclic modules:

$$M \cong R^m \bigoplus \big(\bigoplus_i R/(d_i)\big).$$

The ideals (d_i) are called invariant factors *of M.*

Remark that a finite abelian group can be viewed as a finitely generated (torsion) $\mathbb{Z}$-module, so Theorem 2.3.9 implies the structure theorem for finite abelian groups.

Proposition 2.3.10. *Let M be a free R-module of rank n and N be a non-zero submodule of rank $m \leq n$. Then N is also free. Moreover, there exists a basis $\{e_1, \dots, e_n\}$ for M and elements $a_1 | a_2 | \cdots | a_m$ of R such that $\{a_1 e_1, \dots, a_m e_m\}$ is a basis for N.*

2.4 Exact sequences

A sequence of R-module homomorphisms (finite or run to infinity in either direction)

$$C^\bullet : \quad \cdots \to M^{-1} \to M^0 \xrightarrow{f^0} M^1 \xrightarrow{f^1} M^2 \xrightarrow{f^2} M^3 \to \cdots$$

is called a *cochain complex* if $f^i \circ f^{i-1} = 0$ for any i. It is called exact if $\mathrm{im}(f^i) = \ker(f^{i-1})$ for any i. We set

$$H^i(C^\bullet) := \ker(f^i)/\mathrm{im}(f^{i-1}).$$

Then the complex $C^\bullet$ is exact if and only if $H^i(C^\bullet) = 0$ for any i.

In particular, we have the following basic facts.

(1) $0 \to M \xrightarrow{f} N$ is exact if and only if f is injective; also written as $M \hookrightarrow N$.

(2) $M \xrightarrow{g} N \to 0$ is exact if and only if g is surjective, also written as $M \twoheadrightarrow N$.

(3) An exact sequence of the form

$$0 \to M \xrightarrow{f} N \xrightarrow{g} P \to 0$$

is called a *short exact sequence*, which means that f is injective, g is surjective, and $\ker(g) = \mathrm{im}(f)$.

(4) Let $f : M \to N$ be a homomorphism of R-modules. Putting

$$\mathrm{coker}(f) := N/\mathrm{im}(f),$$

we obtain an exact sequence

$$0 \to \ker(f) \to M \xrightarrow{f} N \to \mathrm{coker}(f) \to 0.$$

The next famous result is referred as *Snake lemma*.

Lemma 2.4.1. *Consider a commutative diagram of R-modules with exact rows:*

$$\begin{tikzcd}
 & A \arrow[r, "f"] \arrow[d, "\alpha"] & B \arrow[r, "g"] \arrow[d, "\beta"] & C \arrow[r] \arrow[d, "\gamma"] & 0 \\
0 \arrow[r] & A' \arrow[r, "f'"] & B' \arrow[r, "g'"] & C' &
\end{tikzcd}$$

Then there exists an exact sequence

$$\ker(\alpha) \xrightarrow{f} \ker(\beta) \xrightarrow{g} \ker(\gamma) \xrightarrow{\delta} \mathrm{coker}(\alpha) \xrightarrow{\bar{f'}} \mathrm{coker}(\beta) \xrightarrow{\bar{g'}} \mathrm{coker}(\gamma).$$

Moreover, if f is injective, then f is also injective; if g' is surjective, then $\bar{g}'$ is also surjective.

Proof. We only give the construction of δ. Given $c \in \ker(\gamma) \subset C$, we can lift it to $b \in B$ as g is surjective. Since $\gamma(g(b)) = \gamma(c) = 0$, we have $g'(\beta(b)) = 0$, i.e. $\beta(b) \in \ker(g') = \operatorname{im}(g')$. Thus there exists $a' \in A'$ such that $\beta(b) = f'(a')$. We then define

$$\delta(c) := \text{image of } a' \text{ in } \operatorname{coker}(\alpha).$$

The definition does not depend on the choice of b, because different choices would differ by an element in A, which maps to 0 in $\operatorname{coker}(\alpha)$. □

2.5 Chain conditions

Definition 2.5.1. *An R-module M is said to be noetherian if it satisfies the following ascending chain condition (ACC): every chain*

$$M_1 \subset M_2 \subset M_3 \subset \cdots$$

of submodules of M stabilizes, that is, there exists n such that $M_{n'} = M_n$ for all $n' \geq n$.

Definition 2.5.2. *An R-module M is said to be* artinian *if it satisfies the following descending chain condition (DCC): every chain*

$$M_1 \supset M_2 \supset M_3 \supset \cdots$$

of submodules of M stabilizes, that is, there exists n such that $M_{n'} = M_n$ for all $n' \geq n$.

Lemma 2.5.3. *The following characterization are equivalent:*

(1) M is noetherian;

(2) Every non-empty family $\mathcal{S}$ of submodules of M contains a maximal element with respect to $\subset$.

(3) Every submodule is finitely generated.

Proof. (1)$\Longrightarrow$(2) Take any element M_1 in $\mathcal{S}$. If it is already maximal, then we are done; if not, we can find another element $M_2 \in \mathcal{S}$ such that $M_1 \subsetneq M_2$. In this way, we get a sequence of strictly ascending chain. The noetherian property implies that the sequence ends after finite steps.

(2)$\Longrightarrow$(3) Fix a submodule M' of M. Consider the set $\mathcal{S}$ of all finitely generated submodules of M'. It is non-empty since $(0) \in \mathcal{S}$. By (2), $\mathcal{S}$ has a maximal element, say N. We claim that $M' = N$. Otherwise, choose $m \in M' \backslash N$, then $N + Rm$ is still finitely generated, a contradiction to the maximality of N. Hence $M' = N$ is finitely generated.

(3)$\Longrightarrow$(1) Consider an ascending chain $M_1 \subset M_2 \subset \cdots$, and let M' denote the union of the chain, which is still a submodule of M. By (3), M' is finitely generated, say by $\{m_1, \ldots, m_r\}$. Then there exists $n \gg 1$ such that $m_1, \ldots, m_r \in M_n$, so that $M_n = M'$ and the chain is stable. □

Similarly, being artinian is equivalent to that every non-empty family $\mathcal{S}$ of submodules contains a minimal element. However, there is no counterpart (3) for artinian modules.

Definition 2.5.4. *A ring R is said to be left noetherian (resp. left artinian) if it is noetherian (resp. artinian) as a left R-module. Note that left submodules of R are just left ideals of R.*

Remark 2.5.5. *There exist rings which are left noetherian but not right noetherian. For example:*

$$R = \left\{ \begin{pmatrix} \alpha & 0 \\ \beta & \gamma \end{pmatrix},\ \alpha \in \mathbb{Z}, \beta, \gamma \in \mathbb{Q} \right\}.$$

Example 2.5.6. (i) *$\mathbb{Z}$ is noetherian, but not artinian.*

(ii) *Consider the submodule $\mathbb{Z}[1/p]/\mathbb{Z}$ of $\mathbb{Q}/\mathbb{Z}$ (as a $\mathbb{Z}$-module). It is not noetherian, because the ascending chain*

$$\mathbb{Z}\frac{1}{p} \subset \mathbb{Z}\frac{1}{p^2} \subset \cdots$$

does not stabilize. But it is artinian, because every descending chain has the form

$$\mathbb{Z}\frac{1}{n_1} \supset \mathbb{Z}\frac{1}{n_2} \supset \cdots$$

for suitable $n_i \in \mathbb{N}$ satisfying $n_{i+1}|n_i$, hence stabilizes.

(iii) *Modules of finite cardinality are both noetherian and artinian.*

Proposition 2.5.7. *Let $0 \to N \to M \to P \to 0$ be a short exact sequence. Then M is noetherian (resp. artinian) if and only if N and P are both noetherian (resp. artinian).*

Proof. We first prove the following claim: let N, M_1, M_2 be submodules of M such that

$$M_1 \subset M_2, \quad N + M_1 = N + M_2, \quad N \cap M_1 = N \cap M_2,$$

then $M_1 = M_2$. Let $z \in M_2$. Then $z \in N + M_2 = N + M_1$, $z = n + m_1$. Since $M_1 \subset M_2$, we also have $n \in M_2$, hence $n \in N \cap M_2 = N \cap M_1$, i.e. $z \in M_1$.

Now prove the assertion. Let $M_1 \subset M_2 \subset \cdots$ be an ascending chain of submodules of M. Since N is noetherian, the chain

$$N \cap M_1 \subset N \cap M_2 \subset \cdots$$

stabilizes. Since $P \cong M/N$ is noetherian, the chain $(N + M_1)/N \subset (N + M_2)/N \subset \cdots$ stabilizes, hence $N + M_1 \subset N + M_2 \subset \cdots$ also stabilizes. The claim allows to show that $M_1 \subset M_2 \subset \cdots$ stabilizes. □

Corollary 2.5.8. *Let N, M be submodules of some R-module. If M and N are both noetherian (resp. artinian), then $M + N$ is noetherian (resp. artinian) as well.*

Proof. Alternatively, use $(M+N)/M \cong N/(M \cap N)$, which is a homomorphic image of N, hence is noetherian (resp. artinian). $\square$

Proposition 2.5.9. *If R is left noetherian (resp. artinian), then every finitely generated left R-module is noetherian (resp. artinian).*

Proof. This is because M is a quotient of R^n for some $n > 0$. $\square$

Definition 2.5.10. *An R-module M is said to be* finitely presented *if there exists an exact sequence*

$$R^m \to R^n \to M \to 0.$$

Corollary 2.5.11. *If R is left noetherian, then every finitely generated left R-module is finitely presented.*

Remark 2.5.12. *We will see later that a left artinian ring is also left noetherian. But this is wrong, as we already see, for an artinian module.*

We say a left ideal I of R is *nil* if each element of I is nilpotent.

Proposition 2.5.13. *Let R be a left artinian ring, and $I \subset R$ be a left ideal. If I is nil, then it is nilpotent.*

Proof. We have a descending chain $I \supset I^2 \supset \cdots$ which must stabilize, say $I^k = J$ whenever $k \gg 0$. Suppose that $J \neq 0$.

Since $J^2 = J \neq 0$, the set of left ideals of R such that $J \cdot \mathfrak{a} \neq 0$ is nonempty, hence admits a minimal one $\mathfrak{a}_0$. Choose $a \in \mathfrak{a}_0$ such that $Ja \neq 0$. Note that Ja is also a left ideal, and satisfies $J \cdot (Ja) = J^2 a = Ja \neq 0$. However, $Ja \subset Ra \subset \mathfrak{a}_0$ (since $\mathfrak{a}_0$ is a left ideal), we must have $Ja = \mathfrak{a}_0$. We can therefore choose $y \in J$ such that $ya = a$, hence $(1-y)a = 0$. But, y is nilpotent, $1-y$ is invertible with inverse being $\sum_{i \geq 0} y^i$ (finite sum), so $a = 0$, contradiction. $\square$

Remark 2.5.14. *When R is commutative, a nil ideal in a noetherian ring is nilpotent: write $I = (x_1, \ldots, x_r)$.*

Proposition 2.5.15. *Let R be left noetherian (resp. artinian) ring. Then $M_n(R)$ is also noetherian (resp. artinian).*

Proof. Since $M_n(R)$ is finite free over R, it is a noetherian (resp. artinian) R-module whenever R is. On the other hand, we have a natural ring morphism $R \to M_n(R)$, sending r to the matrix $r\mathrm{I}_n$. This allows to view $M_n(R)$-modules as R-modules. The result then follows. $\square$

2.6 Hilbert basis theorem

We prove the following famous theorem.

Theorem 2.6.1. *If R is left noetherian, so is the polynomial ring $R[X]$.*

For a polynomial $f = \sum_{k=0}^{n} a_k X^k \in R[X]$, with $a_n \neq 0$, we call $L(f) := a_n$ the *initial* or *leading* coefficient of f. For a left ideal $\mathfrak{a} \subset R[x]$ and $k \geq 0$, we let

$$L_k(\mathfrak{a}) := \{L(f) : \deg(f) = k, f \in \mathfrak{a}\} \cup \{0\}.$$

Lemma 2.6.2. *(1) $L_k(\mathfrak{a})$ is a left ideal of R and $L_k(\mathfrak{a}) \subset L_{k+1}(\mathfrak{a})$ for all k.*

(2) If $\mathfrak{a}'$ is another ideal of R such that $\mathfrak{a} \subset \mathfrak{a}'$ and $L_k(\mathfrak{a}) = L_k(\mathfrak{a}')$, for all $k \geq 0$, then $\mathfrak{a} = \mathfrak{a}'$.

Proof. (1) If $f(X), g(X) \in \mathfrak{a}$ then $f(X) + g(X)$ and $rg(X)$ $(r \in R)$ also in $\mathfrak{a}$. The second inclusion holds because $Xf(X) \in \mathfrak{a}$.

(2) Let $g(X) \in \mathfrak{a}'$ and $k = \deg g$. Since $L_k(\mathfrak{a}) = L_k(\mathfrak{a}')$, there exists $f_k(X) \in \mathfrak{a}$ such that $g(X) - f_k(X)$ has degree $\leq k-1$. But this new element belongs to $\mathfrak{a}'$ since $\mathfrak{a} \subset \mathfrak{a}'$, we can continue this procedure which terminates after finite steps and shows that $g \in \mathfrak{a}$. □

Proof of Theorem 2.6.1. Let $(\mathfrak{a}_j)_{j \geq 1}$ be an ascending chain of left ideals in $R[X]$. We consider the family of left ideals of R:

$$\{L_i(\mathfrak{a}_j), \quad i \geq 0, j \geq 1\}.$$

Note the following facts:

- if i is fixed, the chain $\{L_i(\mathfrak{a}_j),\ j \in \mathbb{N}\}$ is ascending;
- if j is fixed, the chain $\{L_i(\mathfrak{a}_j),\ i \in \mathbb{N}\}$ is also ascending.

We will prove that there exists $k \gg 0$ such that $L_i(\mathfrak{a}_j) = L_i(\mathfrak{a}_k)$ for all $j \geq k$ and all $i \geq 0$. Hence $\mathfrak{a}_j = \mathfrak{a}_k$ for $j \geq k$ by Lemma 2.6.2.

Let $L_p(\mathfrak{a}_q)$ be a maximal element of the family $\{L_i(\mathfrak{a}_j),\ i \geq 0, j \geq 1\}$, that is $L_p(\mathfrak{a}_q) = L_i(\mathfrak{a}_j)$ if $i \geq p$ and $j \geq q$. For $0 \leq i \leq p-1$ fixed, the chain $L_i(\mathfrak{a}_j)$ stabilizes, there exists $q(i)$ sufficiently large such that $L_i(\mathfrak{a}_{j(i)}) = L_i(\mathfrak{a}_j)$ for all $j \geq q(i)$. If we put

$$k := \max\{q, q(0), \dots, q(p-1)\},$$

then $L_i(\mathfrak{a}_j) = L_i(\mathfrak{a}_k)$ for all $i \geq 0$ and $j \geq k$, as desired. □

Corollary 2.6.3. *If R is left noetherian, so is $R[X_1, \dots, X_n]$. More generally, if S is a finitely generated R-algebra, then S is left noetherian.*

2.7 Schur's lemma

Recall that an R-module is *simple* if $M \neq 0$ and M has no submodules other than M and $\{0\}$

Example 2.7.1. (i) *M is simple if and only if $M \cong R/\mathfrak{a}$ for some maximal left ideal.*

(ii) *Over a field, a simple R-module is just a vector space of dimension 1.*

(iii) *We view $M := R^n$ as a left module over $M_n(R)$. Then submodules of M are of the form $\mathfrak{a}^n$ for some left ideal $\mathfrak{a}$ (Exercise). If $R = D$ is a division ring (for example, $R = F$ is a field), then M is a simple module.*

Lemma 2.7.2 (Schur's Lemma). *Let M_1, M_2 be simple R-modules and $\varphi : M_1 \to M_2$ be a homomorphism. Then either $\varphi = 0$ or φ is an isomorphism. In particular, the endomorphism ring of a simple R-module is a division ring.*

Proof. Easy but important. □

Corollary 2.7.3. *Let M be a simple module, then $\mathrm{End}_R(M^{\oplus n}) \cong M_n(D)$ where $D := \mathrm{End}_R(M)$ is a division ring by Schur's lemma.*

Proof. We construct a morphism $\mathrm{End}_R(M^{\oplus n}) \to M_n(D)$: if $\varphi \in \mathrm{Hom}_R(M^{\oplus n}, M^{\oplus n})$, we get an element in $\mathrm{End}_R(M) \cong D$:

$$\alpha_{jk} : M \xrightarrow{\iota_k} M^{\oplus n} \xrightarrow{\varphi} M^{\oplus n} \xrightarrow{\mathrm{pr}_j} M,$$

then we send φ to $(\alpha_{jk})_{j,k} \in M_n(D)$. One checks that the ring structure of $\mathrm{End}_R(M^{\oplus n})$ corresponds to matrix multiplication in $M_n(D)$. (Left as Exercise.) □

More generally, if $M = M_1^{\oplus n_1} \oplus \cdots \oplus M_r^{\oplus n_r}$, where $M_1, \dots, M_r$ are distinct simple R-modules, then

$$\mathrm{End}_R(M) \cong \prod_{i=1}^{r} M_{n_i}(D_i), \quad D_i := \mathrm{End}_R(M_i).$$

2.8 The Jordan-Hölder theorem

Let M be a left R-module. A *normal series* of M is a finite chain of submodules

$$M = M_0 \supseteq M_1 \supseteq \cdots \supseteq M_r = (0). \tag{12}$$

If $M_i \supsetneq M_{i+1}$, we say the normal series has no repetitions. A *refinement* of a normal series (12) is a new normal series obtained by inserting additional terms in (12).

Definition 2.8.1. *(1) A* composition series *of M is a normal series such that $M_i \supsetneq M_{i+1}$ for all i and M_i/M_{i+1} is a simple module. The corresponding set (with multiplicities)*

$$\{M_0/M_1, M_1/M_2, \dots, M_{r-1}/M_r\}$$

is called the set of composition factors *(or* Jordan-Hölder factors*) and denoted by* JH(M).

(2) Two composition series are called equivalent if $r = s$ and their composition factors are isomorphic up to permutation, taking into account of multiplicities.

Note that, not every module has a composition series. For example, $\mathbb{Z}$ (over itself) has no composition series.

Theorem 2.8.2. *(1) (Jordan) If an R-module M has one composition series of length r, then every composition series of M has the same length r. Moreover, every normal series without repetitions can be refined to a composition series.*

(2) (Hölder) If an R-module M has a composition series, then any two composition series are equivalent.

Proof. (1) The theorem is true for $r = 1$. We proceed by induction on r, assuming the result holds for modules having a composition series of length $\leq r - 1$. Let

$$M = M_0 \supsetneq M_1 \supsetneq \cdots \supsetneq M_r = (0), \tag{13}$$

be a composition series of length r. Fix another composition series

$$M = N_0 \supsetneq N_1 \supsetneq \cdots \supsetneq N_s = (0);$$

we need to show $s = r$.

By (13), M_1 has a composition series of length $r - 1$. If $N_1 \subseteq M_1$, then we must have $N_1 = M_1$, as M/N_1 is simple by definition while M_1/N_1 is a proper submodule of M/N_1. Therefore $r - 1 = s - 1$ by the inductive hypothesis, hence $r = s$. So we may assume $N_1 \nsubseteq M_1$. Since M/M_1 is simple, we get

$$M = M_1 + N_1.$$

From the isomorphisms

$$M/N_1 \cong M_1/M_1 \cap N_1, \quad M/M_1 \cong N_1/M_1 \cap N_1,$$

we see that $M_1/(M_1 \cap N_1)$ and $N_1/(M_1 \cap N_1)$ are both simple. Consider the diagram

$$M = M_1 + N_1 \supsetneq \begin{matrix} M_1 \\ N_1 \end{matrix} \supsetneq M_1 \cap N_1 \supseteq M_1 \cap N_2 \supseteq \cdots \supseteq M_1 \cap N_s = (0).$$

We first observe that the series $M_1 \cap N_1 \supseteq \cdots \supseteq M_1 \cap N_s = (0)$, although has repetitions, has all subquotients either 0 or simple modules: this is because

$$M_1 \cap N_i / M_1 \cap N_{i+1} \hookrightarrow N_i/N_{i+1}.$$

In particular, $M_1 \cap N_1$ admits one composition series (by deleting the repetitions). If the length is ℓ, then $\ell + 1 = r - 1$ (by induction), hence $\ell = r - 2$. Hence, N_1 has one composition series of length $r-1$, and by the inductive hypothesis again, any composition series of N_1 has length $r-1$, so $s-1=r-1$ and $s=r$.

(2) We know that any two composition series have the same length:

$$(*) \quad M = M_0 \supsetneq M_1 \supsetneq \cdots \supsetneq M_r = (0)$$

$$(**) \quad M = N_0 \supsetneq N_1 \supsetneq \cdots \supsetneq N_r = (0).$$

We proceed by induction on r. The theorem being trivial if $r=1$, we assume $r \geq 2$ and the result holds for modules of length $< r$. If $M_1 = N_1$, then we have two composition series of M_1 which are equivalent by induction, and the two series of M are also equivalent.

Assume $M_1 \neq N_1$, so that $M = M_1 + N_1$. By taking a fixed composition series for $M_1 \cap N_1$, we obtain as above two composition series of M:

$$(*') \quad M = M_1 + N_1 \supsetneq M_1 \supsetneq M_1 \cap N_1 \supsetneq \cdots \supsetneq M_r = (0)$$

$$(**') \quad M = M_1 + N_1 \supsetneq N_1 \supsetneq M_1 \cap N_1 \supsetneq \cdots \supsetneq M_r = (0).$$

They are indeed composition series because $M_1/M_1 \cap N_1$ and $N_1/M_1/N_1$ are both simple. The induction hypothesis implies that the two composition series of $M_1 \cap N_1$ are equivalent. We conclude by noting that the extra terms are just

$$\{M/M_1, M_1/(M_1 \cap N_1)\} = \{M/N_1, N_1/(M_1 \cap N_1)\}.$$

Hence $(*) \sim (*') \sim (**') \sim (**)$. □

Example 2.8.3. *Consider $C_{12} = \mathbb{Z}/12\mathbb{Z}$. It has several composition series:*

$$C_1 < C_2 < C_6 < C_{12}$$

$$C_1 < C_2 < C_4 < C_{12}$$

$$C_1 < C_3 < C_6 < C_{12}$$

but they all have the same Jordan-Hölder factors: C_2, C_2, C_3.

However, note that C_4 and $C_2 \times C_2$ have the same Jordan-Hölder factors, but they are not isomorphic.

Theorem 2.8.4. *An R-module M has a composition series if and only if M satisfies both chain conditions, i.e. M is noetherian and artinian.*

Proof. $\Longrightarrow$ is clear.

$\Longleftarrow$ Let $M_0 = M$. If $M_0 \neq (0)$, then let M_1 be a maximal proper submodule of M (use ascending chain condition!). If $M_1 \neq (0)$, we similarly define M_2 and continue if $M_2 \neq (0)$. In this way, we get

$$M = M_0 \supsetneq M_1 \supsetneq \cdots \supsetneq .$$

The descending chain condition assures the sequence stabilizes, hence $M_r = 0$ for some $r \gg 0$. We get a composition series of length r since M_i/M_{i+1} is simple. □

Corollary 2.8.5. *Consider a short exact sequence* $0 \to M_1 \to M \to M_2 \to 0$. *Then* M *has a composition series if and only if both* M_1 *and* M_2 *have one. Moreover,* $\mathrm{JH}(M) = \mathrm{JH}(M_1) \cup \mathrm{JH}(M_2)$.

Proof. Because the analogous statement hold for noetherian (resp. artinian) modules. $\square$

Definition 2.8.6. *If* M *has a composition series of length* r, *then we define the length of* M *to be* r*:*

$$\ell(M) = r.$$

When M *has no composition series, we set* $\ell(M) = \infty$.

Proposition 2.8.7. *(1) If* $0 \to M_1 \to M \to M_2 \to 0$ *is exact, then* $\ell(M) = \ell(M_1) + \ell(M_2)$.

(2) If M_1, M_2 *are two submodules of* M, *then*

$$\ell(M_1) + \ell(M_2) = \ell(M_1 + M_2) + \ell(M_1 \cap M_2).$$

Proof. Obvious. $\square$

Corollary 2.8.8. *Let* D *be a division ring. Then, up to isomorphism,* D^n *is the unique simple* $M_n(D)$*-module.*

Proof. Write $R = M_n(D)$. By Example 2.7.1(iii), D^n is a simple R-module.

Let M be a simple R-module. Then M is a quotient of R. We know that R has a composition series $R = \bigoplus_{i=1}^n D^n$, with composition factors $\{D^n\}$. So M is isomorphic to D^n by Jordan-Hölder theorem. $\square$

2.9 Semi-simple modules

Definition 2.9.1. *A module* M *is said to be* semi-simple *if it is a* direct sum *of simple modules.*

We view the zero module 0 as a direct sum over $\emptyset$, so it is semi-simple.

Remark 2.9.2. *A direct product of simple modules is in general not semi-simple. For example, a simple* $\mathbb{Z}$*-module has the form* $\mathbb{Z}/p\mathbb{Z}$ *for some prime* p. *Thus, any semi-simple* $\mathbb{Z}$*-module is* torsion. *However, the direct product* $\prod_{p \text{ prime}} \mathbb{Z}/p\mathbb{Z}$ *is not torsion (it contains* $\mathbb{Z}$ *as a submodule), so it is not semi-simple.*

Proposition 2.9.3. *The following are equivalent for an* R*-module* M.

(1) M *is semi-simple.*

(2) For every submodule $N \subset M$, *there exists another submodule* N' *such that* $M = N \oplus N'$. *(*N' *is usually called a* complement *of* N*.)*

(3) M *is equal to the sum of its simple submodules.*

Proof. (1)$\Longrightarrow$(2). Write $M = \bigoplus_{i \in I} S_i$. We may assume $N \subsetneq M$. Let

$$\mathfrak{F} = \big\{ J \subset I : N \cap (\bigoplus_{j \in J} S_j) = 0 \big\}.$$

Clearly $\mathfrak{F}$ is non-empty (since $N \subsetneq M$) and is inductive. By Zorn's lemma there is a maximal element $J \in \mathfrak{F}$. We want to show that $M = N \oplus (\bigoplus_{j \in J} S_j)$; denote the latter module by P. For any $i \in I$, $P \cap S_i$ is either 0 or S_i because S_i is simple. If $P \cap S_i = 0$, then $N \cap (\bigoplus_{j \in J \cup \{i\}} S_j) = 0$, [25] a contradiction to the choice of J. Hence $P \cap S_i = S_i$ for any $i \in I$, so $P = M$.

(2)$\Longrightarrow$(3) Let N be the sum of all the simple submodules of M; we will show that $N = M$. If this is not the case, then $M = N \oplus N'$ where $N' \neq 0$ by (2). Let T be a non-zero cyclic submodule of N'. Then T contains a maximal proper submodule, say T_1, as T is finitely generated. Now T_1 has a complement in M, it also has a complement in T by Lemma 2.9.4 below, say $T = T_1 \oplus T_2$. Since $T_2 \cong T/T_1$ (and T_1 is a maximal proper submodule in T), T_2 is simple, thus $T_2 \subset N \cap T$, which is a contradiction since the sum is direct. This shows that $T = 0$ and so $N = M$, as required.

(3)$\Longrightarrow$(1) We may assume $M \neq 0$ (and so $I \neq \emptyset$). Suppose $M = \sum_{i \in I} S_i$ with each S_i simple. Consider

$$\mathfrak{F} := \big\{ J \subset I : \bigoplus_{j \in J} S_j = \sum_{j \in J} S_j \big\}.$$

Here the condition $\bigoplus_{j \in J} S_j = \sum_{j \in J} S_j$ means that the natural surjective map $\bigoplus_{j \in J} S_j \twoheadrightarrow \sum_{j \in J} S_j$ is an isomorphism. Then $\mathfrak{F}$ is non-empty as any singleton of I belongs to $\mathfrak{F}$; it is also inductive with respect to the partial order defined by inclusion. By Zorn's lemma we can find a maximal element $J \in \mathfrak{F}$. We claim that $M = \bigoplus_{j \in J} S_j$. Indeed, since M is the sum of all S_i, it suffices to prove $S_i \subset \oplus_{j \in J} S_j$ for any $i \in I$. But, if this is not the case, then $S_i \cap (\bigoplus_{j \in J} S_j) = 0$ because S_i is simple. Therefore $J \cup \{i\} \in \mathfrak{F}$, a contradiction to the choice of J. $\square$

Lemma 2.9.4. *If N, N' are submodules of M such that $M = N \oplus N'$, then*

$$P = N \oplus (N' \cap P)$$

for any $N \subset P \subset M$.

Proof. Let $x \in P \subset M = N \oplus N'$, we may write $x = y + y'$ with $y \in N$, $y' \in N'$. Since $y \in N \subset P$, we have $y' \in P$, so $y' \in N' \cap P$. This proves $P = N + (N' \cap P)$; this is clearly a direct sum. $\square$

The proof above in fact gives more:

[25] let $x \in N \cap (\oplus_{j \in J \cup \{i\}} S_j)$, and write $x = x_i + \sum_{j \in J} x_j$ (almost all x_j are zero in the sum $\sum_j \in J$), then $x_i \neq 0$ by assumption, hence $x_i = x - \sum_j x_j$ showing that $N \cap P \neq 0$, contradiction.

Corollary 2.9.5. *Let M be a semi-simple module: $M = \bigoplus_{i\in I} S_i$, with each S_i simple. For any submodule $N \subset M$, there is $J \subset I$ such that*

$$M = N \oplus \big(\bigoplus_{i\in J} S_i\big).$$

In particular, if M is semi-simple, then so is every submodule and every quotient.

Proof. The first assertion follows from the proof of Proposition 2.9.3. Consequently, modulo $\bigoplus_{i\in J} S_i$, we obtain an isomorphism $N \cong \bigoplus_{I\setminus J} S_i$, so N is semi-simple. □

Remark 2.9.6. *Although N is isomorphic to $\bigoplus_{I\setminus J} S_i$ (in Corollary 2.9.5), it is not always true that N is equal to $\oplus_K S_i$ for suitable $K \subset I$. (As an example, consider the subspace inside $M = F \oplus F$:*

$$N = \{(x, x) : x \in F\},$$

where F is a field.)

Definition 2.9.7. *We say two simple R-modules have the same type if they are isomorphic. A semi-simple module is called* isotypic *if it is a sum of simple modules all of the same type; also called σ-isotypic if the type is σ.*

If M is a semi-simple module, then $M = \bigoplus_\sigma M_\sigma$, namely M decomposes as the direct sum of its σ-isotypic summands for distinct σ. In general, for any R-module M, we define its socle to be the sum of all its simple submodules. Hence $\mathrm{soc}(M)$ is always a semi-simple module, but it could be zero; it is nonzero for a module satisfying the descending chain condition; this is extremely important when studying modular representation theory. It is easy to see that if M semi-simple and σ-isotypic, then so are its submodules and quotient modules.

Corollary 2.9.8. *If $M = \bigoplus_\sigma M_\sigma$, then $\mathrm{End}(M) = \prod_\sigma \mathrm{End}(M_\sigma)$.*

2.10 Indecomposable modules

Definition 2.10.1. *An R-module M is called decomposable if there exists a decomposition $M = M_1 \oplus M_2$ with M_i both nonzero; otherwise we say M is indecomposable.*

A module M is called strongly indecomposable if $M \neq 0$ and $\mathrm{End}(M)$ is local.

We recall the definition of a local ring:

Definition 2.10.2. *Let A be a (unital) ring. It is called* local *if the set of non-units of A is an (two-sided) ideal (hence automatically maximal ideal).*

By a *unit* in A we mean an element which has both a left inverse and a right inverse (which must be equal).

Lemma 2.10.3. *The following statements are equivalent:*

(i) A has a unique maximal left ideal;

(i') A has a unique maximal right ideal;

(ii) $A/\mathrm{rad}(A)$ *where* $\mathrm{rad}(A)$ *denotes the Jacobson ideal of* A*;*

(iii) A is local.

Proof. By definition, $\mathrm{rad}(A)$ is defined to be the intersection of all maximal left ideals in A. We will prove later on that it is also equal to the intersection of all maximal right ideals in A, thus $\mathrm{rad}(A)$ is a (two-sided) ideal.

(i) $\Rightarrow$ (ii) Admitting the above fact, $J(A)$ is the unique maximal left ideal in A. We claim that this implies $A/J(A)$ is a division ring. More generally, we claim that a ring which only admits two left ideals, 0 and D iteself, is a division ring. Indeed, if $a \in D\backslash\{0\}$, then $D \cdot a = D$ (being a nonzero left ideal of D), thus there exists $b \in D\backslash\{0\}$ such that $ba = 1$. By the same reason there exists $a' \in D\backslash\{0\}$ such that $a' = 1$. We deduce that b is a unit with inverse $a = a'$, so a is also a unit.

(i') $\Rightarrow$ (ii) Similar.

(ii) $\Rightarrow$ (i) + (i') Clear.

(i) + (i') $\Rightarrow$ (iii) If $x \in A$ is a non-unit, then it does not have a left inverse (or a right inverse), so must be contained in $\mathrm{rad}(A)$ which is the unique maximal left ideal (resp. right ideal). Thus $\mathrm{rad}(R)$ is equal to the set of non-units in A and A is local.

(iii) $\Rightarrow$ (i) + (i') Clear. □

Definition 2.10.4. *Let A be a (unital) ring. An element $e \in A$ is called an* idempotent *if $e^2 = e$. In this case, $1 - e$ is also an idempotent since $(1-e)^2 = 1 - 2e + e^2 = 1 - s$, and we have $e(1-e) = 0$.*

If A is local, it contains no idempotents except 0 and 1. Indeed, if e is such that $e^2 = e$, then $e(1-e) = 0$ so $e, 1-e$ are non-units; hence $e, 1-e \in \mathfrak{m}$, and also $1 \in \mathfrak{m}$.

Remark 2.10.5. *There exists indecomposable modules but not simple. Usually, two simple modules can have "extensions". For example, $\mathbb{Z}/p^2\mathbb{Z}$ is indecomposable but not simple; in fact it is a non-split extension of the simple module $\mathbb{Z}/p\mathbb{Z}$ by itself.*

Another example: $\mathbb{Z}$ itself is an indecomposable module of infinite length. Indeed, any submodule of $\mathbb{Z}$ is an ideal, of the form $n\mathbb{Z}$. If $\mathbb{Z} = n\mathbb{Z} + m\mathbb{Z}$, then $(n,m) = 1$; but they always have non-zero intersection, i.e. the sum is not a direct sum.

Proposition 2.10.6. *An R-module M is indecomposable if and only if the only idempotents in $\mathrm{End}_R(M)$ are 0 and 1. In particular, strongly indecomposable implies indecomposable.*

Proof. If $M = M_1 \oplus \cdots \oplus M_n$ where $M_j \neq 0$ and $n > 1$, then the projections $e_j : M \to M_j$ is an idempotents and $e_j \neq 0, 1$. Conversely, suppose $\mathrm{End}_R(M)$ contains an idempotent $e \neq 0, 1$. Then $M = eM \oplus (1-e)M$ where $eM, (1-e)M \subset M$ are both nonzero. To see this, it is enough to check the sum is direct: if $ex = (1-e)y$, then $ex = e(ex) = e(1-e)y = 0$. □

Lemma 2.10.7 (Fitting). *Let M be an R-module, which is both noetherian and artinian. For every $f \in \mathrm{End}_R(M)$, there exists a canonical decomposition*

$$M = \mathrm{im}(f^\infty) \oplus \ker(f^\infty)$$

such that each summand is f-stable, and

(1) $f|_{\mathrm{im}(f^\infty)}$ is an isomorphism

(2) $f|_{\ker(f^\infty)}$ is nilpotent.

Proof. Consider the chains

$$\mathrm{im}(f) \supset \mathrm{im}(f^2) \supset \mathrm{im}(f^3) \supset \cdots,$$
$$\ker(f) \subset \ker(f^2) \subset \ker(f^3) \subset \cdots.$$

By our hypothesis, they are stable and we denote their limit by $\mathrm{im}(f^\infty)$ and $\ker(f^\infty)$, respectively.

Let us show $\mathrm{im}(f^\infty) \cap \ker(f^\infty) = 0$. Take $n \gg 0$ so that $\ker(f^n) = \ker(f^\infty)$ and $\mathrm{im}(f^n) = \mathrm{im}(f^\infty)$. Write $x \in \mathrm{im}(f^n)$ as $x = f^n(y_n)$ for some $y_n \in M$. If $x \in \ker(f^n)$, then $f^n(x) = f^{2n}(y_n) = 0$, so $y_n \in \ker(f^{2n}) = \ker(f^\infty) = \ker(f^n)$, thus $x = 0$.

Next we show $M = \mathrm{im}(f^\infty) + \ker(f^\infty)$, which will imply the direct sum decomposition. Retain the assumption that $n \gg 0$ so that everything stabilizes at n. Given $x \in M$, choose $y \in M$ with $f^n(x) = f^{2n}(y)$ (always possible). Then

$$x = x - f^n(y) + f^n(y)$$

from which we get $M = \mathrm{im}(f^\infty) + \ker(f^\infty)$. □

Theorem 2.10.8. *Let M be an indecomposable module satisfying both chain conditions. Then any endomorphism of M is either nilpotent or an automorphism. In particular, M is strongly indecomposable.*

Proof. By Fitting's lemma, we have either $M = \mathrm{im}(f^\infty)$ or $M = \ker(f^\infty)$, hence the first assertion. Next, we need to show $\mathrm{End}(M)$ is local, i.e. the ideal I of all non-automorphisms is an ideal; this follows from Lemma 2.10.9 below. □

Lemma 2.10.9. *If for any $x \in A$, it is either nilpotent or invertible, then A is a local ring with the maximal ideal consisting of all nilpotent elements.*

Proof. Let $\mathfrak{m} = \{x \in A : x \text{ nilpotent}\}$. We need to check that $\mathfrak{m}$ is a two-sided ideal of A. Let $x \in \mathfrak{m}$ and $r \in A$ arbitrary, to show $xr, rx \in \mathfrak{m}$, it is equivalent to show they are not invertible. We may assume $x \neq 0$ and, since x is nilpotent, let $n \geq 2$ be minimal such that $x^n = 0$ and $x^{n-1} \neq 0$. Since $x^{n-1}(xr) = 0$ and $(rx)x^{n-1} = 0$, we get rx, xr are not invertible. Finally let $x, y \in \mathfrak{m}$, we need to check $x + y \in \mathfrak{m}$: if not, $x + y =: u$ is invertible, then $x = u - y$ is also invertible: $[\sum_{i\geq 0}(u^{-1}y)^i] \cdot u^{-1}$ (resp. $u^{-1} \cdot \sum_{i\geq 0}(yu^{-1})^i$) provides the left inverse (resp. right inverse), as $u^{-1}y$ (resp. yu^{-1}) is nilpotent.[26] This gives a contradiction with $x \in \mathfrak{m}$. □

[26] here note the subtlety: since A is not necessarily commutative, that y is nilpotent does not imply that $u^{-1}y$ is nilpotent in general. However, we certainly have $u^{-1}y$ is not invertible, hence nilpotent by assumption

2.11 Krull-Schmidt theorem

Lemma 2.11.1. *Let M, N be R-modules such that $M \neq 0$ and N is indecomposable. Let $f : M \to N$, $g : N \to M$ such that $gf : M \to M$ is an isomorphism. Then f and g are both isomorphisms.*

Proof. Since gf is an isomorphism, g is surjective and f is injective. We need to show that f is surjective and g is injective. We first assume $gf = \mathrm{Id}_M$. Consider $\mathrm{im}(f)$ and $\ker(g)$, both are submodules of N. It is easy to check that $N = \mathrm{im} f \oplus \ker(g)$:

- if $x \in \mathrm{im}(f) \cap \ker(g)$ then $x = f(y)$ and $0 = g(x) = gf(y) = y$, hence $x = 0$.

- for any $x \in N$, let $y = g(x) \in M$, then $x = (x - f(y)) + f(y) \in \ker(g) + \mathrm{im}(f)$.

But N is indecomposable, so either $\ker g = 0$ and $\mathrm{im} f = N$, or $\ker g = N$ and $\mathrm{im} f = 0$ (but in this case $gf = 0$, i.e. $M = 0$). In general case, let $h : M \to M$ be the inverse of fg, then $(gf)h = g(fh) = \mathrm{Id}_M$, so that g, fh are both isomorphisms by the argument above. $\square$

Theorem 2.11.2. *Let $M \neq 0$ be an R-module that is noetherian and artinian. There exists a decomposition*

$$M = M_1 \oplus \cdots \oplus M_r, \quad r \in \mathbb{Z}_{\geq 1}$$

as a direct sum of indecomposable submodules. Moreover, the integer r is unique and the indecomposable summands M_i are unique up to isomorphism and permutation.

Proof. We do induction on $\ell(M)$, which is finite by Theorem 2.8.4. If M is already indecomposable, we are done. If $M = M_1 \oplus M_2$ with $M_1, M_2 \neq 0$, then $\ell(M_1), \ell(M_2) < \ell(M)$, and by induction we have decompositions for M_1, M_2, taking direct sums gives one for M.

Suppose $M = M_1 \oplus \cdots \oplus M_r$ and $M = N_1 \oplus \cdots \oplus N_s$ are two decompositions. We shall argue by induction on $\max\{r, s\}$. Without loss of generality, we assume $s \geq r$. Fixing M_1, we want to find some $1 \leq j \leq s$ such that $M_1 \cong N_j$. For $j = 1, \ldots, s$, set

$$u_j : N_j \hookrightarrow M \twoheadrightarrow M_1, \quad v_j : M_1 \hookrightarrow M \twoheadrightarrow N_j.$$

Then $u_j v_j \in \mathrm{End}_R(M_1)$, and $\sum_{j=1}^{s} u_j v_j = \mathrm{Id}_{M_1}$. In fact, we have

$$\sum_{j=1}^{s} u_j v_j : M_1 \hookrightarrow M \to \bigoplus_{j=1}^{s} N_j \to M \twoheadrightarrow M_1,$$

and the composition of the middle two morphisms is just Id_M. Since M_1 is indecomposable and satisfies both chain conditions, $\mathrm{End}_R(M_1)$ is a local ring by Theorem 2.10.8. This implies that at least one of $u_j v_j$ is an isomorphism: otherwise Id_{M_1} would be also in the maximal ideal. Therefore by Lemma 2.11.1, u_j, v_j define isomorphisms $M_1 \cong N_j$.

Up to permutation, we may assume $j = 1$. The next step is to "cancel out" M_1 and N_1. Since $u_1 : N_1 \to M \xrightarrow{\mathrm{pr}_1} M_1$ is an isomorphism, and $\mathrm{pr}_1(\bigoplus_{i\geq 2} M_i) = 0$, we have

$$N_1 \cap \big(\bigoplus_{i\geq 2} M_i\big) = 0.$$

It follows easily that $M = N_1 \oplus (\bigoplus_{i\geq 2} M_i)$, from which we obtain that M/N_1 is isomorphic to both $\bigoplus_{2\leq i\leq r} M_i$ and $\bigoplus_{2\leq j\leq s} N_j$, and we can conclude by induction. □

Remark 2.11.3. *The Krull-Schmidt theorem fails if M is only assumed to be noetherian or artinian.*

2.12 Tensor product of modules

We first recall the notion of bimodules. Let R, S be two rings with unit.

Definition 2.12.1. *An (R, S)-bimodule is an additive group which is a left R-module and a right S-module such that*

$$r(ms) = (rm)s, \quad \forall r \in R, s \in S.$$

Homomorphisms between (R, S)-bimodules are just homomorphisms which are simultaneously homomorphisms between left R-modules and right S-modules.

Example 2.12.2.
- *Every ring R is an (R, R)-bimodule.*
- *If R is commutative, then every left R-module is naturally an (R, R)-bimodule: $rmr' := rr'm$.*

Convention: ${}_RM$ denotes left R-module, and M_S for right S-module, ${}_RM_S$ for (R, S)-bimodule. It is clear that $\mathrm{Hom}_R({}_RM_S, {}_RN)$ is a left S-module, while $\mathrm{Hom}_S({}_RM_S, N_S)$ is a right R-module.

Definition 2.12.3. *Consider M_R, ${}_RN$ and an additive group A. A map*

$$\phi : M \times N \to A$$

is called a balanced product *if it satisfies:*
(i) $\phi(x + x', y) = \phi(x, y) + \phi(x' + y)$
(ii) $\phi(x, y + y') = \phi(x, y) + \phi(x, y')$
(iii) $\phi(xr, y) = \phi(x, ry)$
for any $x, x' \in M$, $y, y' \in N$ and $r \in R$. The set of balanced products over R from $M \times N$ to A is denoted by $Bil(M, N; A)$. It forms an additive group.

Remark 2.12.4. *(1) Every ring R is an (R, R)-bimodule. So the ring multiplication $(r, r') \mapsto rr'$ is an R-balanced product $R \times R \to R$.*

(2) We have a natural identification $Bil(M, N; A) = \mathrm{Hom}_R(M, \mathrm{Hom}_{\mathbb{Z}}(N, A))$, here $\mathrm{Hom}_{\mathbb{Z}}(N, A)$ is viewed as a right R-module.

Definition 2.12.5. *The tensor product of M and N over R is an abelian group $M \otimes_R N$ together with a balanced product*

$$\otimes : M \times N \to M \otimes_R N$$

which is universal in the following sense: for every abelian group A and every balanced product $\phi : M \times N \to A$, there is a unique group homomorphism $\tilde{\phi} : M \otimes_R N \to A$ such that

$$\begin{tikzcd} M \times N \arrow[r, "\otimes"] \arrow[dr, "\phi"'] & M \otimes_R N \arrow[d, "\tilde{\phi}"] \\ & A \end{tikzcd}$$

Remark 2.12.6. *We have a natural isomorphism*

$$\mathrm{Hom}_{\mathbb{Z}}(M \otimes_R N, A) \cong \mathrm{Hom}_R(M, \mathrm{Hom}_{\mathbb{Z}}(N, A)).$$

Proposition 2.12.7. *Tensor products of M and N, if they exist, are unique up to a unique isomorphism.*

Proof. It follows from the definition of $M \otimes_R N$: in fact follows from the universal property. This is a general fact: If T, T' are two tensor products, then we let $T' = A$ and use the universal property of T, to get a morphism $T \to T'$; on the other hand let $T = A$ and use the universal property of T' to get $T' \to T$. They have to be inverse to each other by uniqueness: indeed, the uniqueness implies that if $\alpha \circ \otimes = \otimes$, then $\alpha = \mathrm{Id}$. $\square$

Theorem 2.12.8. *Tensor product $M \otimes_R N$ exists.*

Proof. Consider the free $\mathbb{Z}$-module F with the set $M \times N$ as its basis, i.e F is a free module with basis $e_{(x,y)}$ for every $(x, y) \in M \times N$. We will define $M \otimes_R N$ as a quotient of F. Define I to be the submodule *generated* by elements

$$(x + x', y) - (x, y) - (x', y)$$

$$(x, y + y') - (x, y) - (x, y')$$

$$(xr, y) - (x, ry).$$

Put $M \otimes_R N := F/I$ and $\otimes : M \times N \to M \otimes_R N$ is given by $(x, y) \mapsto$ (image of $e_{(x,y)}$). Then this gives the tensor product: for this we check the universal property. For any balanced product $\phi : M \times N \to A$, we get a morphism of $\mathbb{Z}$-modules

$$\tilde{\phi} : F \to A, \quad e_{(x,y)} \mapsto \phi((x, y))$$

then by construction $I \subset \ker \tilde{\phi}$, that is, $\tilde{\phi}$ induces a morphism $M \otimes_R N \to A$. $\square$

Corollary 2.12.9. *Every element of $M \otimes_R N$ can be written (non-uniquely) as a finite sum*

$$\sum_i x_i \otimes y_i, \quad x_i \in M, y_i \in N,$$

where $x \otimes y$ denotes the image of (x, y) under $\otimes$.

Example 2.12.10. *When $R = k$ is a field, we have the usual notion of tensor products of k-vector spaces $V \otimes_k W$; this is again a k-vector space. Moreover, if $\{v_i\}$, $\{w_j\}$ are basis of V, W respectively, then $\{v_i \otimes w_j\}$ is a basis of $V \otimes W$.*

Remark 2.12.11. *Attention on $M \times N$ (direct sum; abelian group structure) and the map $\otimes : M \times N \to M \otimes_R N$ (this map is not a group homomorphism).*

Lemma 2.12.12. *If M is an (R, S)-bimodule, and N is (S, T)-bimodule, then $M \otimes_R N$ is naturally an (R, T)-bimodule. The module structure is given by*

$$r(x \otimes y) = (rx) \otimes y$$

$$(x \otimes y)t = x \otimes (yt).$$

Proof. Obvious check. □

Lemma 2.12.13. *Let $\varphi : {}_RM_S \to {}_RM'_S$ and $\psi : {}_SN_T \to {}_SN'_T$ be homomorphisms. There is a unique homomorphism of (R, T)-bimodules such that*

$$\varphi \otimes \psi : M \otimes_S N \to M' \otimes_S N', \quad x \otimes y \mapsto \varphi(x) \otimes \psi(y).$$

Such hommorphisms behave well under composition: $(\varphi \otimes \psi) \circ (\alpha \otimes \beta) = (\varphi\alpha) \otimes (\psi\beta)$.

Proof. First define a balanced product $M \times N \to M' \otimes_R N'$ as the composition

$$M \times N \to M' \times N' \xrightarrow{\otimes} M' \otimes_R N'.$$

Then use the universal property of $M \otimes_R N$. □

Lemma 2.12.14. *(1) We have a natural isomorphism $M \otimes_R R \cong M$. Here we view R as an (R, R)-bimodule. similarly we have $R \otimes_R M \cong M$.*

(2) We have $(M \otimes_R M') \otimes_S M'' \cong M \otimes_R (M' \otimes_S M'')$, for all $M_R, {}_RM'_S, {}_SM''$.

(3) When R is commutative, then there are canonical isomorphism $M \otimes_R N \cong N \otimes_R M$ between R-modules, sending $x \otimes y$ to $y \otimes x$.

Proof. An easy exercise. □

Lemma 2.12.15. *We have $(\bigoplus_{i\in I} M_i) \otimes_R N \cong \bigoplus_{i\in I}(M_i \otimes_R N)$ and $M \otimes_R (\bigoplus_{i\in I} N_i) \cong \bigoplus_{i\in I}(M \otimes_R N_i)$.*

Proof. First, the maps $M_i \times N \to (\bigoplus_{i\in I} M_i) \otimes N$ induce a map $M_i \otimes_R N \to (\bigoplus_I M_i) \otimes N$; then use the universal property of direct sum. □

Remark 2.12.16. *Tensor product does not distribute direct products. For example, let $M = \mathbb{Q}$, $N_p = \mathbb{Z}/p\mathbb{Z}$, then $\mathbb{Q} \otimes_{\mathbb{Z}} \mathbb{Z}/p\mathbb{Z} = 0$. Since $\prod_p \mathbb{Z}/p\mathbb{Z}$ contains elements which are not torsion (for example $\mathbb{Z}$ embeds in it), $\mathbb{Q} \otimes_{\mathbb{Z}} (\prod_p \mathbb{Z}/p\mathbb{Z})$ is non-zero.*

As a generalization of the isomorphisms

$$\mathrm{Hom}_{\mathbb{Z}}(M \otimes_R N, A) \cong Bil(M, N; A) \cong \mathrm{Hom}_R(M, \mathrm{Hom}_{\mathbb{Z}}(N, A))$$

We have the following property.

Proposition 2.12.17. *(Adjunction isomorphism) Let $M_R, {}_RN_S, P_S$, then there exists a canonical isomorphism:*

$$\mathrm{Hom}_S(M \otimes_R N, P) \cong \mathrm{Hom}_R(M, \mathrm{Hom}_S(N, P)).$$

Similarly, for ${}_SM_R, {}_RN, {}_SP$ there exists a canonical isomorphism

$$\mathrm{Hom}_S(M \otimes_R N, P) \cong \mathrm{Hom}_R(N, \mathrm{Hom}_S(M, P)).$$

2.13 Tensor product of algebras

In this part, R is a commutative algebra.

Definition 2.13.1. *An (associative) R-algebra is a ring A (with unit) which is equipped with an R-module structure, such that the multiplication $A \times A \to A$ is balanced, in particular:*

$$x(ry) = (xr)y = (rx)y = r(xy).$$

A homomorphism between R-algebras is a ring homomorphism which is also R-linear.

Equivalently, the R-algebra structure is equivalent to give a ring homomorphism $R \to A$ with image in the center of A.

Let A and B be two R-algebras. Since A and B are both R-modules, we can form $A \otimes_R B$ the tensor product; this is also an R-module. This can further be made to an R-algebra: define first the product on elements of the form $a \otimes b$ by

$$(a \otimes b)(a' \otimes b') := (aa') \otimes (bb')$$

and extend by linearity to all of $A \otimes B$. This makes $A \otimes B$ to be an R-algebra, with identity element $1_A \otimes 1_B$. If A and B are both commutative, so is $A \otimes_R B$.

In the case $A = B$, by universal property of tensor product, we get a homomorphism $\mu : A \otimes_R A \to A$ between R-modules, sending $x \otimes y$ to xy.

Linear disjointness

Now we consider cases where A, B are both fields. Let F be field, and $\overline{F}$ a fixed algebraic closure of F. Let E, K be two extensions of F contained in $\overline{F}$. In general $E \otimes_F K$ need not be a field, even need not be a domain. For example, $\mathbb{C} \otimes_{\mathbb{R}} \mathbb{C}$ is not a domain. Let $a = 1 \otimes i$, $b = i \otimes 1$, then $a^2 = b^2 = -(1 \otimes 1)$, so that $(a + b)(a - b) = 0$. But a, b are linearly independent over $\mathbb{R}$.

Definition 2.13.2. *We say that E and K are* linearly disjoint *over F if the following condition is satisfied: whenever $x_1, \dots, x_m$ are linearly independent elements of E over F, they are also linearly independent over K.*

Proposition 2.13.3. *The following are equivelent:*

(i) *E and K are linearly disjoint over F;*

(ii) *the natural map $m : E \otimes_F K \to E.K$ is injective;*

(iii) *$E \otimes_F K$ is a field.*

Proof. (i)$\Longleftrightarrow$(ii) By definition of linear disjointness.

(ii)$\Longrightarrow$(iii) We claim that m is an isomorphism. The image of m is the subring $E[K]$ generated by E and K. But, by Proposition 1.2.2 it is a field. Namely, any element in $E[K]$ is contained in $E[S]$ for some finite subset S of K. So (iii) follows from (ii).

(iii)$\Longrightarrow$(ii) Trivial. $\square$

Remark 2.13.4. *If E and K are linearly disjoint over F, then $E \cap K = F$. Indeed, take any $x \in E \cap K$ and consider $\{1, x\}$ which are linearly dependent over K. Hence they are also linearly dependent over F, thus $x \in F$.*

Example 2.13.5. *$\mathbb{Q}(\sqrt[3]{2})$ and $\mathbb{Q}(\sqrt[3]{2}i)$ are not linearly disjoint over $\mathbb{Q}$, although their intersection gives $\mathbb{Q}$. Otherwise, $K \otimes_{\mathbb{Q}} L \cong K.L$ would have degree 6 over $\mathbb{Q}$, while $K \otimes_{\mathbb{Q}} L$ has degree 9 over $\mathbb{Q}$.*

Example 2.13.6. *If E/F is finite Galois, then E and K are linearly disjoint over $E \cap K$. Indeed, we may assume $E \cap K = F$. By Proposition 1.6.3, we have $[EK : K] = [E : F]$, which is equal to $[E \otimes_F K : K]$, hence the result.*

Galois descent

Let K/F be a finite Galois extension with Galois group G. Let V be a K-vector space. We say V has a *G-action*: if $G \times V \to V$ is a G-action satisfying the extra condition:

$$g(xv) = g(x)g(v), \quad x \in K, v \in V.$$

Proposition 2.13.7 (Galois descent)**.** *Let V be a K-vector space with a G-action. Then we have an isomorphism of K-spaces*

$$K \otimes_F V^G \xrightarrow{\sim} V, \quad x \otimes v \mapsto xv.$$

Moreover, the morphism is G-equivariant in the sense that:

$$g(x \otimes v) = (gx) \otimes v \mapsto g(xv).$$

Proof. The map $K \times V^G \to V$ gives a balanced product, so the morphism in Lemma is well-defined.

Injectivity: let $v_1, \ldots, v_n \in V^G$ be a set of linearly independent elements over F, we show that they are still linearly independent over K. If not, we choose a relation $\sum_{i=1}^r a_i v_i = 0$ with least r, where $a_i \in K$ and we may assume $a_1 = 1$. Using the G-action, we get

$$v_1 + \sum_{i \geq 2} g(a_i) v_i = 0$$

hence $g(a_i) = a_i$ for all $g \in G$ and $i \geq 2$, so that $a_i \in K^G = F$. This implies that the morphism is injective.

Surjectivity: Let V' be the image in V and $V'' = V/V'$. We define the trace map: $V \to V$ by $\mathrm{Tr}(v) := \sum_{\sigma \in G} \sigma(v)$. For any $v \in V$, we have $\mathrm{Tr}(v) \in V^G \subset V'$, hence $\mathrm{Tr} : V'' \to V''$ is identically zero, so that $V'' = 0$ by Lemma 2.13.8 below. ☐

Lemma 2.13.8. *Assume $V \neq 0$ has G-action and let $0 \neq v \in V$. Then there exists $a \in K$ such that $\mathrm{Tr}(av) \neq 0$. In particular, Tr is not identically zero.*

Proof. This is a form of the linear independence of distinct characters. We enumerate the elements of G as $\{\sigma_1, \ldots, \sigma_n\}$. We show inductively on r the following statement: for any $c_1, \ldots, c_r \in K$, not all zero, there is some $a \in K$ such that $\sum c_i \sigma_i(av) \neq 0$. ☐

Corollary 2.13.9. *Let K be a finite Galois extension of F with Galois group G. For any field extension L/K (not necessarily finite), we have*

$$K \otimes_F L \cong \prod_{\sigma \in G} L_\sigma$$

where L_σ is a K-algebra which is L when viewed as a ring, but with a K-module structure given by $k.x = \sigma(k)x$.

Proof. Let $\mathcal{A} = \prod_{\sigma \in G} L_\sigma$; we define a G-action on $\mathcal{A}$ by

$$\tau(x_\sigma)_{\sigma \in G} := (x_{\sigma\tau})_\sigma.$$

One easily checks that this is well-defined. Then (x_σ) is fixed by G if and only if $x_\sigma = x_{\sigma'}$ for any $\sigma, \sigma' \in G$. That is $\mathcal{A}^G \cong L$. By Proposition 2.13.7, we get

$$\mathcal{A} \cong K \otimes_F \mathcal{A}^G \cong K \otimes_F L.$$

☐

3 Ring theory

3.1 Semi-simple rings

Just as in module theory, we start by studying semi-simple rings and simple rings.

Definition 3.1.1. *A ring is R called left (resp. right)* semi-simple *if R is semi-simple as a left (resp. right) R-module.*

Lemma 3.1.2. *Let $R_1, \ldots, R_m$ be left semi-simple rings, then so is $R = R_1 \times \cdots \times R_m$.*

Proof. Obvious. $\square$

Example 3.1.3. *Let D be a division ring, $M_n(D)$ is a left (and right) semi-simple ring: since*

$$M_n(D) = D^n \oplus \cdots \oplus D^n$$

where D^n is simple. Moreover, D^n is the unique simple module over $M_n(D)$ (by Jordan-Hölder theorem).

Proposition 3.1.4. *A left semi-simple ring is left noetherian and left artinian. Hence it is of finite length and has composition series.*

Proof. Since R is semi-simple, write $R = \bigoplus_{i \in I} \mathfrak{a}_i$, where each $\mathfrak{a}_i$ is a simple left submodule of R. This means that $\mathfrak{a}_i$ is a minimal left ideal of R. We show that I is a finite index set. Indeed, there exists a finite subset $I_0 \subset I$ and $e_i \in \mathfrak{a}_i$ for $i \in I_0$ such that $1 = \sum_{i \in I_0} e_i \in \bigoplus_{i \in I_0} \mathfrak{a}_i$. Then for any $x \in R$,

$$x = x.1 = x\big(\sum_{i \in I_i} e_i\big) = \sum_{i \in I_0} (xe_i) \in \sum_{i \in I_0} \mathfrak{a}_i.$$

Thus $I = I_0$ is finite. $\square$

Proposition 3.1.5. *Let D be a division ring and $R = M_n(D)$. Then*

(i) There exists a unique simple left R-module V up to isomorphism; moreover, R acts faithfully on V.

(ii) We have $\mathrm{End}_R({}_RV) \cong D^{\mathrm{op}}$ as rings.

Proof. (i) See Corollary 2.8.8 for the first assertion and Remark 2.3.8 for the second.

(ii) We saw that $V = D^n$. The homomorphism of rings[27]

$$\Delta : D^{\mathrm{op}} \to \mathrm{End}_R({}_RV), \quad d \mapsto [\rho_d : v \mapsto vd]$$

is well-defined, noting that $\rho_{dd'} = \rho_{d'} \circ \rho_d$:

$$dd' \mapsto (V \xrightarrow{d'} V \xrightarrow{d} V) = d \circ d'.$$

[27] In $\mathrm{End}_R(M)$, the multiplication is given by: $\phi \cdot \phi' := \phi \circ \phi'$.

We show that it is an isomorphism. The injectivity is clear. Let $f \in \mathrm{End}_R({}_RV)$ and write

$$f(1,0,\dots,0)^{\mathrm{T}} = (d,*,\dots,*)^{\mathrm{T}}.$$

Since f is an R-module homomorphism and

$$\begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix} = \begin{pmatrix} a_1 & 0 & \dots & 0 \\ \vdots & & \vdots & \\ a_n & 0 & \cdots & 0 \end{pmatrix} \begin{pmatrix} 1 \\ \vdots \\ 0 \end{pmatrix},$$

we see that f sends $(a_1,\dots,a_n)^{\mathrm{T}}$ to $(a_1d,\dots,a_nd)^T$, i.e. $f = \Delta(d)$. □

Wedderburn-Artin theorem

Theorem 3.1.6 (Wedderburn-Artin)**.** *Let R be a left semi-simple ring. There exists a direct sum decomposition*

$$R = M_{n_1}(D_1) \times \cdots \times M_{n_r}(D_r) \tag{14}$$

where $r \geq 1$, $n_i \in \mathbb{Z}_{\geq 1}$ and D_i are division rings. Such a decomposition is unique up to isomorphism and permutation of (n_i, D_i).

Up to isomorphism there are exactly r left simple modules of R.

Proof. Since R is semi-simple, we can decompose it as

$${}_RR = \bigoplus_{i=1}^{r} V_i^{\oplus n_i}$$

where $V_1,\dots,V_r$ are distinct simple left R-modules.

Set $D_i = (\mathrm{End}_R(V_i))^{\mathrm{op}}$ for $i = 1,\dots,r$, which is a division ring by Schur's lemma. We have

$$\mathrm{End}_R({}_RR) = \bigoplus_{i=1}^{r} \mathrm{End}_R(V_i^{\oplus n_i}) = \bigoplus_{i=1}^{r} M_{n_i}(\mathrm{End}_R(V_i)) = \bigoplus_{i=1}^{r} M_{n_i}(D_i^{\mathrm{op}}).$$

The key is that we have $R^{\mathrm{op}} \cong \mathrm{End}({}_RR)$, sending x to $[r \mapsto rx]$, so that

$$R^{\mathrm{op}} \cong \bigoplus_{i=1}^{r} M_{n_i}(D_i^{\mathrm{op}}).$$

But taking transpose $A \mapsto {}^tA$ of matrices gives a ring isomorphism[28]

$$M_{n_i}(D_i^{\mathrm{op}}) \cong M_{n_i}(D_i)^{\mathrm{op}},$$

so $R \cong \prod_i M_{n_i}(D_i)$. The last assertion follows from the Jordan-Hölder theorem and the fact that any simple left module appears as a composition factor of R. □

[28] For example, $\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)\left(\begin{smallmatrix} a' & b' \\ c' & d' \end{smallmatrix}\right) = \left(\begin{smallmatrix} aa'+bc' & * \\ * & * \end{smallmatrix}\right)$, while $\left(\begin{smallmatrix} a' & c' \\ b' & d' \end{smallmatrix}\right)\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) = \left(\begin{smallmatrix} a'a+c'b & * \\ * & * \end{smallmatrix}\right)$.

Corollary 3.1.7. *A ring R is left semi-simple if and only if it is right semi-simple.*

Proof. Assume R is left semi-simple. Then it has the form (14) in Wedderburn-Artin theorem, which is clearly right semi-simple. $\square$

Proposition 3.1.8. *A ring R is semi-simple if and only if every left (or right) R-module is semi-simple.*

Proof. Every left R-module M is a quotient module of a free R-module. Since R is semi-simple, $R^{(I)}$ is also semi-simple R-module, so is M by Corollary 2.9.5. The converse is trivial taking $M = R$. $\square$

3.2 Simple rings

Definition 3.2.1. *A ring R is called* simple *if it has no two-sided ideals except* $\{0\}$ *and itself.*

Remark 3.2.2. *Be attention that a simple ring need not be a simple (left) module over itself.*

Examples of simple rings are fields, division rings.

Lemma 3.2.3. *For any two-sided ideal $\mathfrak{a}$ of R, the set $M_n(\mathfrak{a})$ is a two-sided ideal of $M_n(R)$. Moreover, $\mathfrak{a} \mapsto M_n(\mathfrak{a})$ gives a bijection between the set of two-sided ideals of R and the set of two-sided ideals of $M_n(R)$.*

Proof. It is easy to check that $M_n(\mathfrak{a})$ is a two-sided ideal of $M_n(R)$.

Conversely: if $I \subset M_n(R)$ is a *two-sided* ideal, we show that $I = M_n(\mathfrak{a})$ with

$$\mathfrak{a} := \{x_{11} \in R : \exists (x_{ij}) \in I\}.$$

Indeed, given $(x_{ij}) \in I$, for each (i, j), there exists another $(y_{ij}) \in I$ (using permutation matrices) such that $y_{11} = x_{ij}$, hence $x_{ij} \in \mathfrak{a}$ by definition of $\mathfrak{a}$. $\square$

Corollary 3.2.4. *If D is a division ring, then $M_n(D)$ is a simple ring.*

Remark that simple rings need not be semi-simple. Below is an example.

Example 3.2.5. *Let V be a F-vector space of countably infinite dimension and $R =$* $\mathrm{End}_F(V)$*. Then R has a unique proper non-zero two-sided ideal I,* [29] *consisting of those k-linear maps with finite rank. So $\overline{R} := R/I$ is a simple ring. Next we show that it is not semi-simple. Indeed, we already see that R does not satisfy IBN (invariant basis number), so R/I does not satisfy IBN.*[30] *However, semi-simple rings satisfy IBN. (If S is semi-simple, S has a composition series, so $\ell(S)$ is finite. If $S^m \cong S^n$, then $m = n$ by Krull-Schmidt theorem.)*

[29] Proof: let $f \notin I$, hence $\mathrm{im} f$ is of infinite dimension. Choose W to be a complement of $\ker f$ in V. Then $f|_W$ is injective (as $W \cap \ker f = 0$) and induces an isomorphism $f : W \xrightarrow{\sim} \mathrm{im} f$; as a consequence W is of infinite dimension. Since V, W, $\mathrm{im}(f)$ are all of countably infinite dimension, there exist two bijections $g : V \to W$ and $h : \mathrm{im} f \to V$ (both viewed as an element in R in obvious way), so that the composition $h \circ f \circ g : V \to V$ is an isomorphism.

[30] see Lam, Lectures on modules and rings, §1.1, Remark 1.5.

Weyl algebras

Let F be a field of characteristic 0. The polynomial ring $F[x]$ carries the usual F-linear derivation $\frac{d}{dx}$ Define the *Weyl algebra* (in one variable) to be the ring of F-linear operators on $F[x]$ generated by $\frac{d}{dx}$ together with the multiplication operator defined by the polynomials in $F[x]$, namely

$$A(F) := F\langle x, \frac{d}{dx}\rangle.$$

Note that this is a non-commutative ring; in fact we have

$$\frac{d}{dx} \circ x - x \circ \frac{d}{dx} = 1.$$

Below we will write y for $\frac{d}{dx}$, so that

$$A(F) = F\langle x, y : yx - xy = 1\rangle.$$

By induction, it is to check the following relations

$$yx^i - x^i y = ix^{i-1}, \quad y^i x - xy^i = iy^{i-1}, \quad \forall i \geq 1. \tag{15}$$

For example, by induction $yx^{i-1} - x^{i-1}y = (i-1)x^{i-2}$ so

$$yx^i = (xy+1)x^{i-1} = xyx^{i-1} + x^{i-1} = x(x^{i-1}y + (i-1)x^{i-2}) + x^{i-1} = x^i y + ix^{i-1}.$$

As a consequence, every element in $A(F)$ can be written in a unique way as a finite sum $\sum_{i,j\geq 0} c_{ij}x^i y^j$, $c_{ij} \in F$. In other words, $\{x^i y^j,\ i,j \geq 0\}$ forms a basis of $A(F)$ as a vector space over F.

Proposition 3.2.6. *$A(F)$ is a simple ring.*

Proof. Let J be a non-zero two-sided ideal in $A(F)$ and fix $0 \neq f \in J$. Write $f = \sum_{j=0}^n f_j(x)y^j$ where $f_j(x) \in F[x]$ and $f_n(x) \neq 0$; here $n = \deg_y(f)$. Since J is a two-sided ideal, $fx - xf \in J$. Using (15) we have

$$fx - xf = \sum_{j=1}^{n} jf_j(x)y^{j-1}.$$

In particular, $\deg_y(fx - xf) < \deg_y(f)$. If $n \geq 1$ then after n steps we see that J contains the non-zero element $n!f_n(x) \in F[x]$. Write this new element as $\sum_{i=0}^m c_i x^i$, where $c_i \in F$ and $c_m \neq 0$. Using (15) again, we see that (after m steps) J contains the non-zero element $m!c_m \in F$. Hence $J = A(F)$. ∎

Remark 3.2.7. *It can be shown that $A(F)$ is a left (and right) noetherian ring. Indeed, it is equipped with a* good *filtration such that the associated graded ring is commutative and noetherian. On the other hand, one can show that $A(F)$ is not Artinian (Exercise).*

The ring $A(F)$ has very interesting properties and play an important role in representation theory. Below we give an example.

Proposition 3.2.8. *If M is a left $A(F)$-module such that $\dim_F M < \infty$, then $M = 0$.*

Proof. Now x and $y = \frac{d}{dx}$ are two F-linear operators on M. A well-known fact from linear algebra says that the trace of the commutator of two linear operators on a finite dimensional vector space is zero. Thus the trace of $yx - xy$ is 0. But we have $yx - xy = 1$ on M, so $\mathrm{Tr}(yx - xy) = \dim_F M$, hence $M = 0$. $\square$

Remark 3.2.9. *More generally, for $n \geq 1$ the Weyl algebra in n variables is defined as*

$$A_n(F) := F\langle x_1, \ldots, x_n, \frac{\partial}{\partial x_1}, \ldots, \frac{\partial}{\partial x_n}\rangle.$$

An easy induction on n shows that $A_n(F)$ is a simple ring.

Double centralizer theorem for simple rings

Let M be a left R-module and $R' := \mathrm{End}_R(M)$.

We can view M as a left R'-module, via $(\phi, m) \mapsto \phi(m)$. Let $R'' := \mathrm{End}_{R'}(M)$, then $R'' \hookrightarrow E$. There is a natural morphism

$$\lambda : R \to R'' \tag{16}$$

sending r to $[m \mapsto rm]$. Indeed, one checks that for all $f \in R'$ and $m \in M$,

$$f \cdot rm = f(rm) = r(fm) = r(f \cdot m)$$

by definition of the action of f and the assumption that f is a left R-module homomorphism.

Example 3.2.10. *Let $R = M_n(D)$ where D is a division algebra. Let $M = D^n$ be the (unique) simple left R-module. Then $\mathrm{End}_R(M) \cong D^{\mathrm{op}}$ by the proof of Proposition 3.1.5 (i.e. every element of $\mathrm{End}_R(M)$ has the form $\rho_d : x \mapsto xd$ for some $d \in D$). Now view M as a left D^{op}-module, then $R \xrightarrow{\sim} \mathrm{End}_{D^{\mathrm{op}}}(M)$, and the isomorphism is given by $A \mapsto (l_A : x \mapsto Ax)$.*

Theorem 3.2.11. *(Rieffel) Let R be a simple ring and $\mathfrak{a}$ be a non-zero left ideal of R, $R' = \mathrm{End}_R(\mathfrak{a})$ and $R'' = \mathrm{End}_{R'}(\mathfrak{a})$. Then the natural map $\lambda : R \to R''$ is an isomorphism.*

Proof. The kernel of λ is a two-sided ideal, so λ is injective (as R is simple). Since $\mathfrak{a}R$ is a two-sided ideal, we have $\mathfrak{a}R = R$ and so $\lambda(\mathfrak{a})\lambda(R) = \lambda(R)$.

Claim: $\lambda(\mathfrak{a})$ is a left ideal of R''.

Indeed, let $a \in \mathfrak{a}$ and $h \in R''$, it suffices to check that $h \cdot \lambda(a) = \lambda(h(a))$. For any $x \in \mathfrak{a}$, the LHS is $x \mapsto h(ax)$, while the RHS is $x \mapsto h(a) \cdot x$. They are equal, because h is a morphism of R'-modules and $\rho_x : a \mapsto ax$ belongs to R', i.e. $h(\rho_x \cdot a) = \rho_x \cdot h(a)$.

The claim implies $\lambda(\mathfrak{a}) = R''\lambda(\mathfrak{a})$, so

$$R'' = R''\lambda(R) = R''\lambda(\mathfrak{a})\lambda(R) = \lambda(\mathfrak{a})\lambda(R) = \lambda(R),$$

i.e. λ is surjective. $\square$

Proposition 3.2.12. *Let R be a simple ring. Then R is semi-simple if and only if it is artinian. If this is the case, then $R \cong M_n(D)$ for some division ring D.*

Proof. We already see that semi-simple rings are artinian. Now assume R is simple and artinian. Let $\mathfrak{a}$ be a minimal (non-zero) left ideal in R, which exists as R is artinian. In particular, $\mathfrak{a}$ is simple and $D := R' = \mathrm{End}_R(\mathfrak{a})$ is a division ring by Schur's lemma. Rieffel's theorem says that $R \cong \mathrm{End}_D(\mathfrak{a})$, so we are left to show $\dim_D(\mathfrak{a}) = n$ is finite, so that

$$R \cong M_n(\mathrm{End}_D(D)) = M_n(D^{\mathrm{op}}).$$

Indeed, if $\dim_D(\mathfrak{a})$ were infinite, then $R = \mathrm{End}_D(\mathfrak{a})$ would contain a proper two-sided ideal consisting of D-linear endomorphisms of $\mathfrak{a}$ which are of finite rank. This is impossible as R is simple. $\square$

Rieffel'stheorem can be stated as a form of the *double centralizer property.*

Definition 3.2.13. *Let E be a ring and $R \subset E$ be a subring. The centralizer of R in E is given by (or $Z_E(R)$)*

$$C_E(R) = \{z \in E : rz = zr \ \forall r \in R\}.$$

Clearly $C_E(C_E(R)) \supseteq R$; but the equality does not hold in general.

Now we specialize to the case $E = \mathrm{End}_{\mathbb{Z}}(M)$: the endomorphism ring of the abelian group M. There is a natural ring morphism $R \to E$ induced by the module structure $R \times M \to M$. In general it need not be injective (in fact it is injective if and only if M is a faithful R-module); let $\bar{R}$ be its image in E. We also have $R' \hookrightarrow E$.

Lemma 3.2.14. *We have $C_E(\bar{R}) = R'$ and $C_E(R') = R''$.*

Proof. Almost by definition: let $f : M \to M$ be a homomorphism as abelian groups; it commutes with $\bar{R}$ if and only if f is an R-module homomorphism. The second statement is proved similarly. $\square$

An R-module M is said to have the *double centralizer property* or to be a *balanced module* if $C_E(C_E(\overline{R})) = \overline{R}$ (equivalently $\overline{R} = R''$). Thus Rieffel's theorem says that a non-zero left ideal $\mathfrak{a}$ in a simple ring is balanced and faithful.

3.3 Jacobson radicals

The study of the Jacobson radical comes from the study of the semi-simplicity. E. Cartan defined the radical of a Lie algebra $\mathfrak{g}$ as the largest solvable ideal of $\mathfrak{g}$, and used it to define and classify finite-dimensional semi-simple Lie algebras. In the associative case, Wedderburn and Artin developed a parallel theory by defining the radical as the largest nilpotent ideal, but this approach breaks down for general rings because there might not be a largest nilpotent ideal. After decades of attempts to generalise the notion, Jacobson resolved this problem by defining the radical as the intersection of all maximal left ideals.

Definition 3.3.1. *We define*

$$\mathrm{rad}(R) = \bigcap_{\text{maximal left ideal } \mathfrak{m}} \mathfrak{m}$$

and call it the left Jacobson radical *of R. It is also denoted by $J(R)$.*

We define similarly the right Jacobson radical of R. We will see later that they coincide.

Lemma 3.3.2. *Let $y \in R$, the following are equivalent:*

(1) $y \in \mathrm{rad}(R)$.

(2) $yM = 0$ for every simple left R-module M. (That means, the Jacobson radical consists of those "bad" elements which cannot be detected by the action on semi-simple modules.)

(3) $1 - xy$ has a left inverse for every $x \in R$.

(3') $1 - xyz$ has an inverse for any $x, z \in R$. Here we mean that it has both a left and right inverse; note that the left and right inverse must coincide: $ax = 1, xb = 1$ implies $a = b$.[31]

Proof. (1)$\Longrightarrow$(3) If not, Zorn's lemma shows that $1 - xy$ is contained in a left maximal left ideal $\mathfrak{m}$, then $1 \in \mathfrak{m}$ as $y \in \mathfrak{m}$.

(3)$\Longrightarrow$(2) If $ym \neq 0$ for some $m \in M$, then $M = R(ym)$ (as M is simple), therefore $m = x(ym)$ for some $x \in R$. But $1 - xy$ has a left inverse, so $m = 0$.

(2)$\Longrightarrow$(1) Every simple module corresponds to a maximal left ideal: $M = R/\mathfrak{m}$.

(3')$\Longrightarrow$(3) is trivial. Next we combine (2) and (3) to imply (3'). Fix $x, z \in R$. Since $yM = 0$ for any simple M, $(yz)M = y(zM) \subset yM = 0$, so by (3) $1 - x(yz)$ has a left inverse, say u. Rather that to show $1 - xyz$ has a right inverse, we show that u has a left inverse (hence u is invertible with inverse $1 - xyz$).

[31] In general, it could happen that $x \in R$ has only left inverses, or even has infinitely many left inverses. For example, in $R = \mathrm{End}_F(V)$ with $V = F^{(\mathbb{N})}$, choose $\{e_n; n \geq 0\}$ a basis of V. Let $g \in R$ sending e_n to e_{n+1}, and for any $v \in V$, let $f_v \in R$ sending e_0 to v and e_{n+1} to e_n. Then $f_v \circ g = \mathrm{Id}_V$. This also implies that g does not have right inverse; in fact g is not surjective.

Indeed, the equality $u(1-xyz)=1$ implies that

$$u = 1 + uxyz = 1 + (ux)yz,$$

which has a left inverse again by (3) (as seen in last paragraph). $\square$

Corollary 3.3.3. (i) *We have* $\mathrm{rad}(R) = \bigcap_{M \text{ simple}} \mathrm{Ann}_R(M)$.

(ii) $\mathrm{rad}(R)$ *is a two-sided ideal.*

Proof. (i) If $y \in \mathrm{rad}(R)$, then $yM = 0$ for any simple left module M, so that $y \in \mathrm{Ann}(M)$ for any simple M. The converse is also clear.

(ii) Note that $\mathrm{Ann}_R(M)$ is a two-sided ideal and apply (i). $\square$

Corollary 3.3.4. *The left radical and right radical coincide.*

Proof. The property in Lemma 3.3.2 is symmetric. $\square$

Example 3.3.5. *(1)* $\mathrm{rad}(\mathbb{Z}) = 0$*;* $\mathrm{rad}(\mathbb{Z}/12\mathbb{Z}) = 6\mathbb{Z}/12\mathbb{Z}$.

(2) If $(R, \mathfrak{m})$ *is local, then* $\mathrm{rad}(R) = \mathfrak{m}$.

(3) If R is semi-simple, then $\mathrm{rad}(R) = 0$. *Indeed, decompose R as $\mathfrak{a}_1 \oplus \cdots \oplus \mathfrak{a}_n$, then the maximal ideals of R are exactly $\bigoplus_{j \neq i} \mathfrak{a}_j$ and their intersection is zero.*

(4) If $R \to S$ is surjective, then $f(\mathrm{rad}(R)) \subset \mathrm{rad}(S)$.

(5) For any ideal $\mathfrak{a}$ contained in $\mathrm{rad}(R)$, $\mathrm{rad}(R/\mathfrak{a}) = \mathrm{rad}(R)/\mathfrak{a}$.

(6) If R is a non-commutative ring, $\mathrm{rad}(R)$ *is not necessarily equal to the intersection of all maximal* two-sided *ideals of R. For example, if $V = F^{(\mathbb{N})}$ and $R = \mathrm{End}_F(V)$, then* $\mathrm{rad}(R) = 0$*, while it has a unique non-zero two-sided ideal I.*[32]

(7) For $R = \mathbb{C}[x_1, x_2, \cdots, x_n]/I$, the Jacobson radical is $\sqrt{I}$, the radical of I. (To get this, you need Hilbert's Nullstellensatz.)*

Lemma 3.3.6. *(Nakayama's lemma) Let M be a finitely generated non-zero left R-module. Then* $\mathrm{rad}(R).M \neq M$.

[32] Let $\{e_k, k \in \mathbb{N}\}$ be a basis of V. For each k, let I_k be the set of linear maps with images contained in $V_k := \oplus_{i \neq k} Fe_i$. This is a right ideal of R; we claim that it is maximal. Let $f \notin I_k$, then $\mathrm{im}(f) + V_k = V$. Let $v \in \mathrm{im}(f) \setminus V_k$ and let $f^{-1}(Fv)$ be the preimage. Choose a decomposition

$$V = f^{-1}(Fv) \oplus W.$$

Sending W to 0, we get an endomorphism $\tilde{f} \in fR : V \xrightarrow{\mathrm{pr}} f^{-1}(Fv) \xrightarrow{f} V$ whose image is exactly Fv. Now $\ker(\tilde{f})$ is of countably infinite dimension, there exists a linear bijection $g : \ker \tilde{f} \xrightarrow{\sim} V_k$. One checks that $\tilde{f} + g : V \to V$ is bijective. This shows that $I_k + fR$ contains an invertible element, so $I_k + fR = R$.

Proof. Since M is finitely generated and non-zero, it has a maximal proper submodule M', so that M/M' is simple. By Lemma 3.3.2, $a(M/M') = 0$ for all $a \in \mathrm{rad}(R)$, hence $\mathrm{rad}(R).M \subset M' \subsetneq M$.

A second proof. Let $x_1, \ldots, x_n$ be a set of generators of M, with n taken to be minimal. Then every element in M has the form $\sum_{i=1}^n r_i x_i$ where $r_i \in R$. Since $\mathrm{rad}(R)$ is a two-sided ideal, every element of $\mathrm{rad}(R)M$ can be written as the form $\sum_{i=1}^n r_i x_i$, with $r_i \in \mathrm{rad}(R)$. Consider $\sum_{i=1}^n x_i \in M$. If $\mathrm{rad}(R)M = M$, we get

$$\sum_{i=1}^n x_i = \sum_{i=1}^n r_i x_i$$

hence $\sum_{i=1}^n (1 - r_i)x_i = 0$. Since $1 - r_1$ is invertible by Lemma 3.3.2, we get

$$x_1 = -\sum_{i \geq 2} (1 - r_1)^{-1}(1 - r_i)x_i,$$

so M can be generated by $x_2, \ldots, x_n$. This contradicts the minimality of n.

A third proof when R is commutative. Let $x_1, \ldots, x_n$ be a set of generators of M. Assuming $\mathrm{rad}(R)M = M$, we can represent x_i as $\sum_j a_{ij}x_j$, where $a_{ij} \in \mathrm{rad}(R)$. Letting $A = (a_{ij})$, we get $\mathbf{x} = A\mathbf{x}$ where $\mathbf{x} = (x_1, \ldots, x_n)^{\mathrm{T}}$, i.e. $(1 - A)\mathbf{x} = 0$. Let A^* be the adjoint matrix of $1 - A$, i.e. $A^*(1 - A) = \det(1 - A)$, then we obtain $\det(1 - A)\mathbf{x} = 0$. Since $\det(1 - A)$ is of the form $1 - a$ with $a \in \mathrm{rad}(R)$, it is invertible, hence $\mathbf{x} = 0$. $\square$

Remark 3.3.7. *For Nakayama's lemma, we will often use the following statements for a local ring $(R, \mathfrak{m})$. (Check that they are equivalent.)*

- *If M is a finitely generated non-zero left R-module, then $\mathfrak{m} \cdot M \neq M$;*
- *If M is a finitely generated non-zero left R-module, and $x_1, x_2, \cdots, x_n \in M$, their images in $M/\mathfrak{m}M$ generate $M/\mathfrak{m}M$ as $R/\mathfrak{m}$-module, then $x_1, x_2, \cdots, x_n$ generate M as R-module;*
- *If N is a finitely generated non-zero left R-module, and $f\colon M \to N$ is a left R-module homomorphism and $\bar{f}\colon M/\mathfrak{m}M \to N/\mathfrak{m}N$ is surjective, then f is surjective.*

A geometric picture: for a vector bundle $E \to X$, and a point $P \in X$, then any basis of the fiber of E over P can be extended to a basis of sections of $E|_U \to U \subset X$ for some neighborhood $P \in U \subset X$. Here, $E|_U$ corresponds to M and the fiber of E over P corresponds to $M/\mathfrak{m}M$.

3.4 Nil ideals and Nilpotent ideals

Definition 3.4.1. *Let $\mathfrak{a}$ be a left (or right) ideal of R. We say*

- *$\mathfrak{a}$ is nil, if every $x \in \mathfrak{a}$ is nilpotent, i.e. $x^{n(x)} = 0$ for some $n \in \mathbb{N}$;*

- $\mathfrak{a}$ *is nilpotent, if* $\mathfrak{a}^n = 0$ *for some* $n \in \mathbb{N}$.

Of course, nilpotent ideals are nil. The converse need not be true however. For example, $R = \mathbb{Z}[x_1, x_2, \cdots]/(x_1^2, x_2^3, \cdots)$, $\mathfrak{a} = (x_1, x_2, \cdots)$ is nil, but not nilpotent.

Remark 3.4.2. (i) *A finite sum of nilpotent left ideals of R is still nilpotent. (Check this!). But for nil ideals, it is an open question. We have*[(33)]

Köthe's conjecture*: in any ring, the sum of two nil left ideals is nil.*

(ii) *Let x be a nilpotent element, Rx need not be a nil ideal. For example, in $M_2(F)$, $x := \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$ is nilpotent, but in Rx, we have $\left(\begin{smallmatrix} 0 & 0 \\ 1 & 0 \end{smallmatrix}\right)\left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right) = \left(\begin{smallmatrix} 0 & 0 \\ 0 & 1 \end{smallmatrix}\right)$, which is not nilpotent.*

(iii) *A nil ideal is nilpotent in the following two cases: R is left artinian (see Proposition 2.5.13), or R is left noetherian. The result in the noetherian case is named* Levitzki's theorem. *Note that left artinian ring is always left noetherian.*[(34)]

Proposition 3.4.3. *If $\mathfrak{a}$ is a nil left ideal of R, then $\mathfrak{a} \subset \mathrm{rad}(R)$.*

Proof. Let $y \in \mathfrak{a}$, need to show $1 - xy$ is invertible for any $x \in R$. Since $xy \in \mathfrak{a}$, xy is nilpotent, hence the inverse of $1 - xy$ is given by $\sum_{i \geq 0} (xy)^i$. □

Remark that this does not mean that any nilpotent element belongs to $\mathrm{rad}(R)$, compare Remark 3.4.2(ii). Also, it is not true that any element in $\mathrm{rad}(R)$ is nilpotent even if we assume that there exists a largest nilpotent ideal or R is commutative. For example, take $R = F[[t]]$, then $\mathrm{rad}(R) = tF[[t]]$ is the maximal ideal, but any non-zero element in $\mathrm{rad}(R)$ is not nil.

However, in the artinian case, we have

Theorem 3.4.4. *Let R be a left artinian ring. Then, the largest left (also, right, two-sided) nilpotent ideal exists and equals* $\mathrm{rad}(R)$.

Proof. By the former property, we only have to prove that $J = \mathrm{rad}(R)$ is nilpotent; the proof is analogue to Proposition 2.5.13. We have a descending chain $J \supset J^2 \supset \cdots$, which stablizes since R is artinian; say $J^k = I$ for $k \gg 0$. We want to show $I = 0$. Suppose it is not the case: $I \neq 0$. Again the descending condition gives a minimal element $\mathfrak{a}_0$ in the non-empty set

$$\{\mathfrak{a} : \text{left ideal such that } I\mathfrak{a} \neq 0\}.$$

Choose $a \in \mathfrak{a}_0$ such that $Ia \neq 0$ (of course we get $Ra \subset \mathfrak{a}_0$ hence $Ra = \mathfrak{a}_0$, but this is not enough to conclude). Since $I^2 = I$ by the construction of I, we have

$$I(Ia) = I^2 a = Ia \neq 0,$$

thus $Ia = \mathfrak{a}_0$ by minimality.

In sum, there exists $y \in I \subset J$ such that $a = ya$, namely $(1 - y)a = 0$. However $1 - y \in R^\times$ since $y \in J$, thus $a = 0$, a contradiction. □

(33) A very important conjecture in non-commutative ring theory, posed in 1930's; see Agata Smoktunowicz's ICM talk (2006).

(34) See Wiki's term on "nil ideal" or "Levitzki's theorem".

Remark 3.4.5. *The above theorem explains that why the Jacobson radical is a generalisation of Wedderburn's (resp. Artin's) radical for the finite-dimensional (one-sided artinian) associative ring to a general setting.*

Remark 3.4.6. *When R is commutative, the nilradical of R is defined as the intersection of all prime ideals of R: $\bigcap_{\mathfrak{p}\text{ prime}} \mathfrak{p}$, which equals the set of nilpotent elements of R. According to the properties above, it is contained in* $\mathrm{rad}(R)$ *and equals* $\mathrm{rad}(R)$ *if we assume R is artinian.*

3.5 Semi-primitive and semi-primary rings

The notion of Jacobson radical leads to a slight generalisation of semi-simple rings.

Definition 3.5.1. *A ring is called* semi-primitive *if* $\mathrm{rad}(R) = 0$. *(This property is symmetric for left and right.) Sometimes, it is also called Jacobson-semisimple.*

By definition, if $\mathfrak{a}$ is a two-sided ideal contained in $\mathrm{rad}(R)$, then $\mathrm{rad}(R/\mathfrak{a}) = \mathrm{rad}(R)/\mathfrak{a}$. As a consequence, $R/\mathrm{rad}(R)$ is semi-primitive.

As examples, semi-simple rings are semi-primitive. Also, $\mathrm{End}_F(V)$ with $V = F^{(\mathbb{N})}$ is semi-primitive (but not semi-simple by Example 3.2.5).

Theorem 3.5.2. *The following are equivalent:*

(i) *R is semi-simple.*

(ii) *R is semi-primitive and left artinian.*

Remark that since R is not known to be noetherian (although this is indeed the case), we don't *a priori* have a composition series of R.

Proof. (i)$\Longrightarrow$(ii) It follows from Proposition 3.1.4.

(ii)$\Longrightarrow$(i) Assume R is semi-primitive and left artinian. We first note the following two facts.

(a) Since R is left artinian, every non-zero left ideal contains a minimal left ideal. Note that it must be principal and simple as a left R-module.

(b) Every minimal left ideal I is a direct summand of R. In fact, since $\cap_{\mathfrak{m}}\mathfrak{m} = \mathrm{rad}(R) = 0$, there exists a maximal left ideal $\mathfrak{m}$ which does not contain I. Hence $R = \mathfrak{m} + I$; but $\mathfrak{m} \cap I = 0$ since I is minimal. We deduce that $R = I \oplus \mathfrak{m}$.

Now we prove that R is a semi-simple ring. Take a minimal left ideal $\mathfrak{a}_1 \subset R$; if $\mathfrak{a}_1 = R$, then we are done. If not, write $R = \mathfrak{a}_1 \oplus \mathfrak{b}_1$. Since $\mathfrak{b}_1 \neq 0$, it contains a minimal left ideal $\mathfrak{a}_2$; if $\mathfrak{a}_2 = \mathfrak{b}_1$, then we are done. Recall that $\mathfrak{a}_2$ has a complementary in R, so it has a complementary in any submodule of R which contains $\mathfrak{a}_2$, see Lemma 2.9.4. In particular, if $\mathfrak{a}_2 \neq \mathfrak{b}_1$, then $\mathfrak{b}_1 = \mathfrak{a}_2 \oplus \mathfrak{b}_2$ for some $\mathfrak{b}_2 \neq 0$. Continuing in this way we get

$$R := \mathfrak{b}_0 \supsetneq \mathfrak{b}_1 \supsetneq \mathfrak{b}_2 \supset \cdots \tag{17}$$

and minimal left ideals $\mathfrak{a}_1, \mathfrak{a}_2, \ldots$, such that

$$\mathfrak{a}_i \oplus \mathfrak{b}_i = \mathfrak{b}_{i-1}.$$

But R is artinian, so the chain (17) is finite and we get a finite direct sum decomposition $R = \bigoplus_{1 \leq i \leq n} \mathfrak{a}_i$. Since each $\mathfrak{a}_i$ is a simple R-module, R is semi-simple. □

Remark 3.5.3. *R and $R/\mathrm{rad}(R)$ have the same simple left modules. Also, a ring is semi-primitive if and only if it has a faithful semi-simple module. (check these).*

Definition 3.5.4. *A ring is called semi-primary if $\mathrm{rad}(R)$ is a nilpotent ideal and $R/\mathrm{rad}(R)$ is semi-simple.*

If R is semi-primary, then every left or right nil ideal is nilpotent, since it is contained in $\mathrm{rad}(R)$. So $\mathrm{rad}(R)$ is the largest nilpotent left ideal; it is also the largest nilpotent right ideal.

Proposition 3.5.5. *Left artinian rings are semi-primary.*

Proof. Denote $J = \mathrm{rad}(R)$ in the proof. According to Theorem 3.4.4, it is enough to prove that R/J is semi-simple. But $\mathrm{rad}(R/J) = J/J = 0$, so that R/J is semi-primitive. Moreover, R/J is artinian since R is, therefore R/J is semi-simple by Theorem 3.5.2. □

Theorem 3.5.6 (Hopkins-Levitzki)**.** *Let R ba semi-primary ring and M be a left module. The following statements are equivalent:*

(i) *M is left noetherian;*

(ii) *M is left artinian;*

(iii) *M has a composition series, or equivalently, M has finite length.*

Proof. It is obvious that (iii) implies (i) and (ii). We show that if M is either noetherian or artinian, then M has a composition series. Let $J = \mathrm{rad}(R)$ be the Jacobson radical of R. Consider the series

$$M \supset \cdots \supset J^i M \supset J^{i+1} M \supset \cdots$$

This is finite since $J^n = 0$ for $n \gg 0$. Study $F_i := J^{i-1}M/J^iM$, each may be viewed as an R/J-module. Since R/J is a semi-simple ring, each F_i is a semi-simple R/J-module. If M is artinian (or noetherian), then F_i has finite length, since it is a direct sum of simple modules. The additivity of $\ell(\cdot)$ shows that M has finite length. □

Corollary 3.5.7. *A left artinian ring is automatically left noetherian.*

Proof. Combine Proposition 3.5.5 and Theorem 3.5.6. □

In summary:

$$\begin{tikzcd}
\text{simple} \arrow[r, Rightarrow, dashed, "\text{(artinian)}"] & \text{semi-simple} \arrow[r, Rightarrow] \arrow[d, Rightarrow, shift left] & \text{left artinian} \arrow[r, Rightarrow] \arrow[d, Rightarrow] & \text{semi-primary} \\
& \text{semi-primitive} \arrow[u, Rightarrow, dashed, shift left, "\text{(artinian)}"] & \text{left noetherian} &
\end{tikzcd}$$

3.6 Density theorem for semi-simple modules

Let M be a left R-module. Let $R' = \mathrm{End}_R(M)$. We can also view M as a left R'-module:

$$R' \times M \to M; \quad (f, m) \mapsto f(m).$$

Recall that the multiplication in R' is defined by $f \cdot g = f \circ g$.

We have seen that there is a natural ring homomorphism in (16)

$$\lambda : R \to R'' := \mathrm{End}_{R'}(M).$$

We want to know how large the image of λ is. The density theorem, due to Jacobson, says that any element in $\mathrm{End}_{R'}(M)$ can be approximated by an element in the image when M is semi-simple.

Example 3.6.1. *Let's consider a simple case: $R = F$ is a field, $M = F^n$. Then $R' = M_n(F)$ acts on F^n. We have the morphism $F \to \mathrm{End}_{R'}(F^n)$; it is surjective. Indeed, F^n is a simple R'-module generated by $\mathbf{e} := (1, 0, \ldots, 0)^{\mathrm{T}}$: any $\mathbf{x}$ is of the form $A\mathbf{e}$, where $A = (\mathbf{x}|\mathbf{0} \cdots \mathbf{0})$ (the matrix whose first column is $\mathbf{x}$ and the other columns are $\mathbf{0}$). So for $\varphi \in \mathrm{End}_{R'}(F^n)$,*

$$\varphi(\mathbf{x}) = A \cdot \varphi(\mathbf{e}) = a\mathbf{x}$$

where a is the first coefficient of $\varphi(\mathbf{e})$. In particular, taking $\mathbf{x} = \mathbf{e}$, we see that $\varphi(\mathbf{e}) = a\mathbf{e} = (a, 0, \ldots, 0)^{\mathrm{T}}$.

Definition 3.6.2. *We say that $R \to R'' := \mathrm{End}_{R'}(M)$ has a dense image, or R acts densely on ${}_{R'}M$, if for every $\varphi \in R''$, any $x_1, \ldots, x_n \in M$, there exists $r \in R$ such that*

$$rx_i = \varphi(x_i), \quad \forall i = 1, \ldots, n.$$

Remark 3.6.3. *If we equip M with the discrete topology and R'' with the coarsest topology such that for every $m \in M$, the morphism $f \mapsto f(m) \in M$ is continuous. Then Definition 3.6.2 means that the image of R is a dense subset of R''.*

Theorem 3.6.4 (Jacobson density theorem)**.** *Suppose M is a semi-simple left R-module, and let $R' = \mathrm{End}_R(M)$. Then $R \to R'' = \mathrm{End}_{R'}(M)$ has dense image. Furthermore, if M is finitely generated over R', then the homomorphism $R \to \mathrm{End}_{R'}(M)$ is surjective.*

We first prove a lemma, which is just the special case of the theorem when $n = 1$.

Lemma 3.6.5. *Let M, R' be as above. Then for any $\varphi \in R''$ and any $x_1 \in M$, there exists $r \in R$ such that $rx_1 = \varphi(x_1)$.*

Proof. We need check $\varphi(x_1) \in Rx_1$. For this we may assume $x_1 \neq 0$. If M is simple, this is trivial. Since M is semi-simple, we can write

$$M = Rx_1 \oplus M'$$

for some submodule M' (since every submodule is a direct summand). If $\mathrm{pr} : M \to Rx_1$ denotes the projection, then $\mathrm{pr} \in R'$, so

$$\varphi(x_1) = \varphi(\mathrm{pr}(x_1)) = \mathrm{pr}(\varphi(x_1)).$$

But $\{y \in M : \mathrm{pr}(y) = y\}$ is just Rx_1, thus $\varphi(x_1) \in Rx_1$ as desired. □

Lemma 3.6.6. *Given $n > 1$ and $\varphi \in \mathrm{End}_{R'}(M)$, we have $\varphi^{\oplus n} \in \mathrm{End}_{R'_n}(M^{\oplus n})$, where $R'_n = \mathrm{End}_R(M^{\oplus n})$.*

Proof. We know that $R'_n = M_n(R')$. Given $g \in R'_n$, we represent g by $(g_{i,j})_{1\leq i,j\leq n}$, with $g_{i,j} \in R'$. Given $(m_1, \dots, m_n) \in M^{\oplus n}$, since $\varphi \in \mathrm{End}_{R'}(M)$, we have (the check is as in $n = 1$ case)

$$\begin{aligned}
\varphi^{\oplus n} g(m_1, \dots, m_n) &= \varphi^{\oplus n}\left(\textstyle\sum_{j=1}^n g_{1,j}(m_j), \dots, \sum_{j=1}^n g_{n,j}(m_j)\right) \\
&= \left(\textstyle\sum_{j=1}^n \varphi g_{1,j}(m_j), \dots, \sum_{j=1}^n \varphi g_{n,j}(m_j)\right) \\
&= \left(\textstyle\sum_{j=1}^n g_{1,j}(\varphi(m_j)), \dots, \sum_{j=1}^n g_{n,j}(\varphi(m_j))\right) \\
&= g(\varphi(m_1), \dots, \varphi(m_n)) \\
&= g\varphi^{\oplus n}(m_1, \dots, m_n).
\end{aligned}$$

□

Proof of Theorem 3.6.4. We want to find some $r \in R$ such that $\varphi(x_i) = rx_i$, i.e.

$$(\varphi(x_1), \dots, \varphi(x_n)) = (rx_1, \dots, rx_n).$$

The trick is to apply Lemma 3.6.5 to the induced endomorphism

$$\varphi^{\oplus n} : M^{\oplus n} \to M^{\oplus n}.$$

Indeed, by Lemma 3.6.6 we have that $\varphi^{\oplus n}$ is a homomorphism of $\mathrm{End}_R(M^{\oplus n})$-modules. Then we can apply Lemma 3.6.5 to $\varphi^{\oplus n}$ and the element $(x_1, \dots, x_n) \in M^{\oplus n}$, and obtain an $r \in R$ such that

$$(rx_1, \dots, rx_n) = r(x_1, \dots, x_n) = (\varphi(x_1), \dots, \varphi(x_n)),$$

giving the desired result.

The surjectivity assertion is clear under the assumption, since in this case φ is determined by its values on a finite generating set for M (over R'). □

Example 3.6.7. *Let F be a field and $V = \mathbb{F}^{(\mathbb{N})}$. Let I be the set of all finite rank F-linear morphisms on V. Let R be the subring of $\mathrm{End}_F(V)$ generated by I and the identity 1. One can show that: (i) V is a simple R-module; (ii) $F = \mathrm{End}_R(V)$. However, the morphism $R \to R'' = \mathrm{End}_F(V)$ is just the natural inclusion, which is not surjective.*

Let us now discuss the condition under which M is a finitely generated R'-module.

Proposition 3.6.8. *Assume M is a semi-simple left R-module. Then M is also semi-simple as a left R'-module. If moreover R is left artinian, then M is finitely generated over R'.*

Proof. (i) We claim that: if $x \in M$ generates a simple left R-submodule, then $R'x$ is a simple R'-module. Let $a' \in R'$ and $a'x \neq 0$, we need to show that $a'x$ generates $R'x$, i.e. $x = b'a'x$ for some $b' \in R'$. Define a morphism $\ell : Rx \to M$ by sending u to $a'u$; this is an R-module homomorphism since $a' \in R' = \mathrm{End}_R(M)$. Since Rx is simple, ℓ induces an isomorphism $\ell : Rx \xrightarrow{\sim} Ra'x$. Take a decomposition of R-modules $M = Ra'x \oplus N$, and consider

$$M \xrightarrow{\mathrm{pr}} Ra'x \xrightarrow{\ell^{-1}} Rx \hookrightarrow M.$$

This is an R-module homomorphism sending $a'x$ to x, in other words, it defines an element $b' \in R'$ such that $b'a'x = x$.

The claim implies the result, because M being semi-simple, thus decomposes as $\oplus_{i\in I} N_i$ where N_i are simple. Hence every element $x \in M$ can be uniquely written as $x = \sum_{i\in I} x_i$, with $x_i = 0$ for almost all i. If $x_i \neq 0$, then $Rx_i = N_i$ (as N_i is simple), so $R'x_i$ is simple by the claim. Thus $x \in \sum_{i\in I} R'x_i$, a finite sum of simple R'-submodules of M. This shows that, as an R'-module, M is the sum of its simple submodules, hence is semi-simple by Proposition 2.9.3.

(ii) Consider the following set

$$\mathcal{S} = \{\mathfrak{a} \text{ left ideal of } R : \ \mathfrak{a} = \mathrm{Ann}_R(X), \text{ finite } X \subset M\}.$$

Since R is artinian, $\mathcal{S}$ contains a minimal element $\mathfrak{a}$ and let X be the finite set associated to $\mathfrak{a}$. Note that $\mathfrak{a}$ could be zero, i.e. $\mathrm{Ann}_R(X) = 0$ for some finite X. The minimality implies that, for any $y \in M$,

$$\mathrm{Ann}_R(X \cup \{y\}) = \mathfrak{a} \cap \mathrm{Ann}_R(y) = \mathfrak{a}$$

i.e. $\mathfrak{a} \subset \mathrm{Ann}_R(y)$. In other words, if $r \in R$ annihilates X, then r annihilates M.

We claim that $M = R'X$, i.e. generated by X as an R'-module. If not, since M is a semi-simple R'-module by (i), it decomposes as $M = R'X \oplus N$. Let

$$\mathrm{pr} : M \twoheadrightarrow N \hookrightarrow M$$

denote the projection to N. Then it is an R'-endomorphism of M, i.e. $\mathrm{pr} \in R''$. By Jacobson density, for $X \cup \{y\}$ (here $y \in N$ is non-zero) there exists $r \in R$ such that $\mathrm{pr} = r$ on $X \cup \{y\}$. In particular, $rX = 0$, which implies $r \in \mathfrak{a}$; but $\mathrm{pr}(y) = y$, hence $ry = y \neq 0$; this contradicts what we have remarked: $\mathfrak{a} \subset \mathrm{Ann}_R(y)$. $\square$

As a consequence of Jacobson density theorem, we prove the following double centralizer theorem for semi-simple modules. As in §3.2 let $E := \mathrm{End}_{\mathbb{Z}}(M)$ and $\bar{R}$ be the image of $R \to E$. (This map is injective if and only if M is a faithful R-module.)

Theorem 3.6.9. *Assume R is left artinian and let M be a semi-simple left R-module. Then*

$$C_E(C_E(\bar{R})) = \bar{R}.$$

The theorem says that semi-simple modules over a (left) artinian ring are balanced.

Proof. By Proposition 3.6.8, M' is a finitely generated semi-simple R'-module. Then we apply Jacobson Density theorem. □

Burnside's theorem

We present now some applications of Jaconson density theorem to representation theory.

Theorem 3.6.10. *Let F be an algebraically closed field, and R be a finite dimensional F-algebra.*(35)

(i) *Let V be a simple R-module, then $\mathrm{End}_R(V) \cong F$ and $\lambda : R \to \mathrm{End}_F(V)$ is surjective.*

(ii) *As a left R-module, $\mathrm{End}_F(V)$ decomposes as $V^{\oplus \dim V}$.*

(iii) *Let $V_1, \ldots, V_r$ be a set of pairwise non-isomorphic simple R-modules. Then the map*

$$\bigoplus_{i=1}^{r} \lambda_i : R \to \bigoplus_{i=1}^{r} \mathrm{End}_F(V_i)$$

is surjective. In particular, $\dim_F R \geq \sum_{i=1}^{r} n_i^2$, where $n_i = \dim_F V_i$.

Proof. (i) Since F lies in the center of R, $D := \mathrm{End}_R(V)$ (which is a division algebra) is an F-algebra. Since V is simple, it is a quotient of R. In particular, $\dim_F V$ is finite. The first assertion follows from Lemma 3.6.11 below and the second follows from Jacobson density theorem.

(ii) Let $n := \dim_F V$ and choose $e_1, \ldots, e_n$ a basis of V over F, say the canonical one. Consider the map

$$\mathrm{End}_F(V) \to V \oplus \cdots \oplus V, \quad A \mapsto (Ae_1, \ldots, Ae_n).$$

It is a morphism of R-modules, and is a bijection by identifying $\mathrm{End}_F(V)$ with $M_n(F)$ using the basis.

(iii) By (i), each $\mathrm{End}_F(V_i)$ can be viewed as an R-module, and decomposes as $V_i^{\oplus n_i}$. Theorem 3.6.4 says that the projection of R to each factor is surjective; since the V_i are pairwise non-isomorphic, $\bigoplus_i \lambda_i$ is also surjective.(36) Indeed, being a submodule of $\bigoplus_{i=1}^r V_i^{\oplus n_i}$, the image is isomorphic to $\bigoplus_{i=1}^r V_i^{s_i}$ for $s_i \leq n_i$. If $s_i \neq n_i$ for some i, then the projection to the i-th component is not surjective. □

(35) One important example for such R is a group algebra over a finite group G.

(36) Be attention that: a morphism $R \to \oplus_i S_i$ is not necessarily surjective even if each $R \to S_i$ is surjective.

Lemma 3.6.11. *Let D be a finite dimensional division F-algebra and F be algebraically closed. Then $D = F$.*

Proof. Let $x \in D$ and consider the subring $F[x] \subset D$. This is a (commutative) domain as D is a division ring. Since $\dim_F F[x] \leq \dim_F D < \infty$, the natural surjective ring homomorphism $F[X] \to F[x]$ is not an isomorphism. Its kernel is a non-zero prime (hence maximal) ideal, so $F[x]$ is a finite field extension of F. But F is algebraically closed, so $F[x] = F$ and $x \in F$. □

Example 3.6.12. *If $R = M_n(F)$, it has only one simple module F^n; the relation* $\dim_F R = (\dim V)^2$ *holds.*

3.7 Primitive rings

Recall that a ring is semiprimitive if and only if its Jacobson radical is zero. Also recall that a left R-module M is called faithful if $\mathrm{Ann}_R(M) = 0$. This is equivalent to the injectivity of the natural map $R \to \mathrm{End}_{\mathbb{Z}}(M)$ (endomorphism ring as an abelian group). As a consequence, if M is faithful, the following morphism is also injective:

$$R \to R'' = \mathrm{End}_{R'}(M), \quad R' := \mathrm{End}_R(M).$$

Proposition 3.7.1. *A ring R is semi-primitive if and only if there exists a faithful semi-simple left R-module.*

Proof. Let M be a faithful semi-simple R-module, then $\mathrm{Ann}_R(M) = 0$; but we have $\mathrm{rad}(R) \subset \mathrm{Ann}_R(M)$, hence $\mathrm{rad}(R) = 0$. Conversely, assume $\mathrm{rad}(R) = 0$ and set

$$M = \bigoplus_{N \text{ simple}} N$$

where the sum runs over all simple R-modules N. Then M is semi-simple and

$$\mathrm{Ann}_R(M) = \bigcap_N \mathrm{Ann}_R(N) = \mathrm{rad}(R) = 0,$$

where the middle equality follows from Lemma 3.3.2. Hence M is faithful. □

Proposition 3.7.1 gives another characterization of semi-primitive rings and motivates the following definition.

Definition 3.7.2. *A ring R is called left (resp. right)* primitive *if there exists a faithful simple left (resp. right) R-module.*

Obviously left (or right) primitive rings are semi-primitive.

Remark 3.7.3. *The notion of primitivity is not left-right symmetric (although semi-primitivity is).*

Lemma 3.7.4. *A simple ring R is left (resp. right) primitive.*

Proof. By taking a maximal left ideal of R, we obtain a simple left R-module, say M. Since $\mathrm{Ann}_R(M)$ is a two-sided ideal in R and since R is simple, we must have $\mathrm{Ann}_R(M) = 0$. $\square$

Example 3.7.5. (i) *For any division ring D, $M_n(D)$ is simple, hence is primitive.*

(ii) *Let $V = F^{(\mathbb{N})}$, then $\mathrm{End}_F(V)$ is a primitive ring because V becomes a faithful simple $\mathrm{End}_F(V)$-module. Moreover, the subring defined in Example 3.6.7 is also primitive. However, $\mathrm{End}_R(V)$ is not a simple ring because it contains a non-zero proper two-sided ideal (cf. Example 3.2.5).*

Proposition 3.7.6. *Let R be a left artinian ring. Then*

(i) *R is semi-simple if and only if R is semi-primitive;*

(ii) *R is simple if and only if R is left primitive.*

Proof. (i) is proved in Theorem 3.5.2.

(ii) Assume R is left primitive. Then R is semiprimitive since $\mathrm{rad}(R) = 0$. By (i) it follows that R is semi-simple. Now use the Wedderburn-Artin theorem to get

$$R \cong M_{n_1}(D_1) \times \cdots \times M_{n_r}(D_r)$$

and that the simple R-modules are $D_i^{n_i}$. Need to show $r = 1$. If $r \geq 2$, then R can not act faithfully on any simple module, so R can not be left primitive. $\square$

Theorem 3.7.7. *Let R be a left primitive ring, V be a faithful simple left R-module. Set $D = \mathrm{End}_R(V)$, which is a division ring by Schur's lemma. Then $\lambda : R \to \mathrm{End}_D(V)$ has dense image. Moreover,*

(i) if R is left artinian, then $n := \dim_D(V)$ is finite and $R \cong M_n(D)$.

(ii) if R is not left artinian, then $\dim_D(V)$ is infinite and for every $n \in \mathbb{Z}_{\geq 1}$ there exists a subring R_n of R together with a surjective ring homomrophism $R_n \twoheadrightarrow M_n(D)$.

Proof. Note that λ is injective by the faithfulness of V. The density of $\lambda(R)$ follows from Jacobson's theorem.

- Assume that $n := \dim_D V$ is finite. Then λ is surjective and thus $R \cong M_n(D)$, and is a simple ring. In particular R is left artinian.

- Assume that $\dim_D(V)$ is infinite. There exists a family $\{v_i\}_{i \in \mathbb{Z}_{\geq 1}}$ of D-linearly independent vectors in V. For every n, let V_n be the linear span of $\{v_1, \ldots, v_n\}$ and set

$$R_n := \{r \in R : r(V_n) \subset V_n\}$$
$$\mathfrak{a}_n := \{r \in R : r(V_n) = 0\}.$$

Then R_n is a subring of R and $\mathfrak{a}_n$ is a two-sided ideal of R_n. We have the natural ring homomorphism $\lambda_n : R_n/\mathfrak{a}_n \hookrightarrow \mathrm{End}_D(V_n)$. Furthermore, Jacobson density theorem implies that λ_n is actually surjective, hence $R_n \twoheadrightarrow R_n/\mathfrak{a}_n \cong M_n(D)$.

Note that $\mathfrak{a}_n$ is a left ideal of R and $\mathfrak{a}_{n+1} \subset \mathfrak{a}_n$ for all n, thus R is not left artinian.

The above shows that $\dim_D(V) < \infty$ if and only if R is left artinian. The finishes the proof. $\square$

3.8 Central simple algebras

From now on let F be a field and consider F-algebras.

Definition 3.8.1. *Let A be an F-algebra, the center of A is denoted by $Z(A)$*[(37)]*, i.e. $Z(A) = C_A(A)$. We say that A is central if $Z(A) = F$.*

Definition 3.8.2. *Let A be an F-algebra. We say that A is* central simple *if*
- *A is a simple ring,*
- *A is a central F-algebra.*

Remark 3.8.3. *If A is a simple ring, $Z(A)$ is always a field. Indeed, let $a \in Z(A)$ be non-zero, then $Aa = aA$ is a two-sided ideal of A, hence equals to A, i.e. a is invertible. So $Z(A)$ is a commutative division algebra, namely a field.*

We will only consider central simple F-algebras which are *finite dimensional* over F (hence are artinian). By Proposition 3.2.12 they are of the form $M_n(D)$, where D is a central division F-algebra with $Z(D) = F$. Recall (by Wedderburn-Artin theorem) that D and n can be recovered by the decomposition ${}_AA = M^{\oplus n}$ where M is the unique simple left A-module, and $D \cong \mathrm{End}_A(M)$. Thus we are reduced to studying finite dimensional central division F-algebras.

Example 3.8.4. *(i) If $F = \mathbb{C}$, then there is no non-trivial central division algebra by Lemma 3.6.11.*

(ii) Over $\mathbb{R}$, except $\mathbb{R}$ itself, we have Hamilton's quaternion algebra: $\mathbb{H} = \mathbb{R} \oplus \mathbb{R}i \oplus \mathbb{R}j \oplus \mathbb{R}k$, with the identities

$$i^2 = j^2 = k^2 = ijk = -1,$$

which determine all the possible products of i, j, k. This is the unique non-trivial one by a theorem of Frobenius.

(iii) Over a finite field $\mathbb{F}$, a theorem of Wedderburn says that any central division algebra is a field, hence equals to $\mathbb{F}$ itself.

(iv) Let E/F be a finite cyclic extension of degree n, and let $\mathrm{Gal}(E/F) = \langle s \rangle$. Let $A = A(E, s, \gamma)$ be the subalgebra of $M_n(E)$ generated by all of the diagonal matrices

$$\mathrm{diag}(b, sb, \ldots, s^{n-1}b), \quad b \in E$$

and the matrix

$$z = \begin{pmatrix} 0 & 1 & 0 & \cdots \\ 0 & 0 & 1 & \cdots \\ 0 & 0 & 0 & 1 \\ \gamma & 0 & 0 & 0 \end{pmatrix}$$

(37) For the German "Zentrum"

where γ is a non-zero element of F. Note that $b \mapsto \operatorname{diag}(b, sb, \ldots, s^{n-1}b)$ allows us to embed E into A. We check that

$$zb = s(b)z, \quad z^n = \gamma$$

and every element of A can be written uniquely as the form

$$b_0 + b_1 z + \cdots + b_{n-1} z^{n-1}$$

where the $b_i \in E$. One checks that A is central simple and $\dim_F A = n^2$; moreover, $A \cong M_n(F)$ if and only if γ is the norm of an element of E.

Let A, B be two F-algebras. Then $A \otimes_F B$ is also an F-algebra. Moreover,

$$A \to A \otimes_F B, \quad a \mapsto a \otimes 1$$

is an injective ring homomorphism so we can view A as a subalgebra of $A \otimes_F B$. Similarly for $B \hookrightarrow A \otimes_F B$. Moreover, inside $A \otimes_F B$, we have $AB = BA$ (commutativity). If we assume both A, B finite dimensional over F, then $\dim_F A \otimes_F B = \dim_F A \cdot \dim_F B$.

Proposition 3.8.5. *If A, B are subalgebras of an F-algebra S, then $S \cong A \otimes_F B$ if and only if*

(1) $ab = ba$ for all $a \in A, b \in B$;

(2) there exists a basis $\{a_\alpha\}$ of A/F such that every element of S can be written in a unique way as $\sum_\alpha a_\alpha b_\alpha$, where $b_\alpha \in B$ are almost all 0.

If S is finite dimensional over F, then condition (2) can be replaced by

(2') $S = AB$ and $\dim_F S = \dim_F A \cdot \dim_F B$.

Proof. We construct an algebra homomorphism $A \otimes_F B \to S$ given by $a \otimes b \mapsto ab$; this is well-defined thanks to (1). Condition (2) means that this is bijective. If S is finite dimensional, then (2') implies it is bijective. □

Lemma 3.8.6. *(i) For any F-algebra B, there is an isomorphism of F-algebras*

$$M_n(F) \otimes_F B \cong M_n(B).$$

(ii) For any division F-algebra D, we have as F-algebras:

$$M_n(F) \otimes_F M_m(D) \cong M_{nm}(D).$$

Proof. (i) We identify $b \in B$ with $\operatorname{diag}(b, \ldots, b) \in M_n(B)$ and write $A = M_n(F)$. Then $AB = BA$. To check condition (2), we let $E_{ij} \in A$ be the canonical basis over F (only (i,j)-entry is non-zero and equals 1). It is clear that that any matrix $(b_{ij}) \in M_n(B)$ can be written uniquely as $\sum_{i,j} b_{ij} E_{ij}$.

(ii) Left as an exercise. □

If A is an F-algebra, so is A^{op}.

Definition 3.8.7. *Let $A^e := A \otimes_F A^{\text{op}}$, and call it the* enveloping algebra *of A.*

If A is a subalgebra of some F-algebra B, then we can view B as an A^e-module as follows:

$$(a \otimes a') \cdot b := aba'.$$

This is well-defined as B is an F-algebra. In particular, A itself is an A^e-module. We have:

- A^e-submodules of A as A^e-module are just two-sided ideals of A. Thus, if A is simple ring, then A is a simple A^e-module.

- $\text{End}_{A^e}(A) = Z(A)$. Indeed, let $f : A \to A$ be an endomorphism of A^e-modules. Then as left A-module, f has the form $x \mapsto xd$ for some $d \in A$; as left A^{op}-module (i.e. right A-module), f has the form $x \mapsto cx$ for some $c \in A$. Taking $x = 1$, we have $c = f(1) = d$ and $cx = xc$ for any $x \in A$. Therefore $c \in Z(A)$. In particular, if A is central, then $\text{End}_{A^e}(A) = F$.

  More generally, if A is a subalgebra of some F-algebra B, then

  $$\text{Hom}_{A^e}(A, B) \cong C_B(A).$$

Lemma 3.8.8. *If A is a finite dimensional central F-algebra, then A is simple if and only if the homomorphism*

$$\begin{array}{rcl} \lambda : A^e = A \otimes_F A^{\text{op}} & \to & \text{End}_F(A) \\ a \otimes b & \mapsto & [x \mapsto axb]. \end{array}$$

is an isomorphism. If this holds, we have $A^e \cong M_n(F)$ where $n := \dim_F A$.

Proof. We view A as an A^e-module as above. Assume first A is a simple ring. Then A is simple as A^e-module and $\text{End}_{A^e}(A) = F$ since A is central. Apply Jacobson density theorem to $R = A^e$ and $M = A$, $R' = F$, we get a surjection $\lambda : A^e \twoheadrightarrow \text{End}_F(A)$ since A is artinian. Comparing dimensions, it is in fact an isomorphism. We also have $\text{End}_F(A) \cong M_n(F)$ if $n = \dim_F A$.

Conversely, assume A is not simple and let $I \subset A$ be a proper non-zero two-sided ideal. The image of λ must send I to I, hence λ can not be surjective. ☐

Let A be a finite dimensional central simple F-algebra, and C be any F-algebra. Let $B := A \otimes_F C$.

Proposition 3.8.9. *We have $C_B(A) = C$.*

Proof. Let c_α be a basis of C/F, then any element $b \in B$ can be written uniquely as $b = \sum_\alpha a_\alpha \otimes c_\alpha$. Let $a \in A$, then the condition $ab = ba$ (for any $a \in A$) is equivalent to

$$\sum_\alpha aa_\alpha \otimes c_\alpha = \sum_\alpha a_\alpha a \otimes c_\alpha$$

hence equivalent to $aa_\alpha = a_\alpha a$ for any $a \in A$. But A is central, meaning that $a_\alpha \in F$, so that $b \in C$. □

Note that we only use the fact that A is central. We have the converse of the above result.

Theorem 3.8.10. *Let B be an F-algebra, not necessarily of finite dimensional over F, and A be a finite dimensional F-subalgebra which is central simple. Set $C := C_B(A)$, then*

(i) the natural homomorphism $A \otimes_F C \to B$ given by $a \otimes c \mapsto ac$ between F-algebras is an isomorphism.

(ii) there is a bijection

$$\{\textit{two-sided ideals of } C\} \to \{\textit{two-sided ideals of } B\}, \quad I \mapsto AI = A \otimes_F I$$

whose inverse is given by $I = AI \cap C$.

(iii) $Z(B) = Z(C)$.

We first prove a lemma.

Lemma 3.8.11. *Let A be as above, and M be an A^e-module. Then the morphism*

$$\Theta : A \otimes_F \mathrm{Hom}_{A^e}(A, M) \to M, \quad a \otimes \phi \to \phi(a)$$

is an isomorphism.

Proof. Prove the injectivity. Let $\{a_1, \ldots, a_n\}$ be a basis of A/F (where $n = \dim_F A$). Let $x = \sum_{i=1}^n a_i \otimes \phi_i$ be an element in $\mathrm{Ker}(\Theta)$, i.e. $\sum_{i=1}^n \phi_i(a_i) = 0$. Since A is CSA, $A^e \cong \mathrm{End}_F(A)$ by Lemma 3.8.8. Choose $r \in A^e$ such that $r(a_1) = a_1$, and $r(a_i) = 0$ for all $2 \leq i \leq n$. Then since ϕ_i are A^e-module homomorphisms, we obtain

$$0 = r0 = \sum_i \phi_i(ra_i) = \phi_1(a_1).$$

Now, since A is a simple A^e-module, ϕ_1 is either an injection or identically zero, so we deduce $\phi_1 = 0$, so $a_1 \otimes \phi_1 = 0$. Similarly we obtain $x = 0$.

Prove the surjectivity. Since A^e is a simple and artinian ring, any A^e-module, in particular M, is semi-simple. Moreover, $M \cong A^{(S)} = \oplus_{\alpha \in S} A_\alpha$ with $A_\alpha \cong A$, since A is the unique simple A^e-module (up to isomorphism). We fix an isomorphism $\phi_\alpha : A \to A_\alpha$ for each component A_α (indeed $\mathrm{Hom}_{A^e}(A, A_\alpha) \cong F$). Let $x \in M$ and write it as a finite sum $\sum_\alpha x_\alpha$ with $x_\alpha \in A_\alpha$, then $x = \sum_\alpha a_\alpha \otimes \phi_\alpha$, where a_α is chosen so that $\phi_\alpha(a_\alpha) = x_\alpha$. □

Proof of Theorem 3.8.10. (i) Since $\mathrm{Hom}_{A^e}(A, B) = C_B(A) = C$, the result follows from Lemma 3.8.11 with $M = B$.

(ii) Let I be a two-sided ideal in C, then AI is a two-sided ideal in $B = AC$. Claim: $AI \cap C = I$. We fix a basis $\{a_1 = 1, a_2, \ldots, a_n\}$ of A/F, then any element $b \in B$ can be written uniquely as $\sum_{i=1}^n a_i c_i$, with $c_i \in C$ (don't confuse with c_α above). Then $b \in AI$ if and only if $c_i \in I$, and $b \in C$ if and only if, by unicity, $c_1 = b$ and $c_i = 0$ for $2 \leq i \leq n$. This shows $b \in I$, and hence the association $I \mapsto AI$ is injective.

Now let I' be a two-sided ideal of B, we need to show $I' = AI$ for some two-sided ideal I in C; then we must have $I = I' \cap C$ the last paragraph. Since I' is a submodule of B as A^e-module, we have an isomorphism by Lemma 3.8.11

$$A \otimes_F \mathrm{Hom}_{A^e}(A, I') = I'.$$

The inclusion $I' \subset B$ allows us to identify $I := \mathrm{Hom}_{A^e}(A, I')$ with a subspace of C, and one checks that I is a two-sided ideal of C.

(iii) We must have $Z(B) \subset C$ as $C_B(A) = C$. This implies $Z(B) \subset Z(C)$. Conversely, if $c \in Z(C)$, c also commutes with A, so it commutes with the whole $B = AC$, so $c \in Z(B)$. $\square$

Corollary 3.8.12. *Let A be a finite dimensional central simple algebra. For every F-algebra C, the F-algebra $A \otimes_F C$ is simple if and only if C is; $A \otimes_F C$ is central (over $Z(C)$) if and only if C is central.*

Proof. Theorem 3.8.10(ii) implies the equivalence for simplicity, and (iii) implies the equivalence for centrality. $\square$

In particular, we get the following consequence when $C = K$ is a field extension of F.

Corollary 3.8.13. *Let A be a finite dimensional F-algebra and K/F be a field extension, then A is central simple over F if and only if $A_K := A \otimes_F K$ is central simple over K.*

Example 3.8.14. *We have $\mathbb{H} \otimes_{\mathbb{R}} \mathbb{C}$ is still a CSA over $\mathbb{C}$, so is of the form $M_n(D)$ for some division algebra D over $\mathbb{C}$. We already see that $D = \mathbb{C}$, and looking at dimensions, we get $n = 2$, i.e.*

$$\mathbb{H} \otimes_{\mathbb{R}} \mathbb{C} \cong M_2(\mathbb{C}).$$

(Find an explicit isomorphism!)

Corollary 3.8.15. *If $A_1, \ldots, A_n$ are all finite dimensional central simple F-algebra, so is $A_1 \otimes_F \cdots \otimes_F A_n$.*

Although the above proof relies on the finiteness of A_i, however, the statememt is also true for infinite dimensional case.

Proposition 3.8.16. *If $A_1, \ldots, A_n$ are all central simple F-algebra, so is $A_1 \otimes_F \cdots \otimes_F A_n$.*

Proof. The centrality is easy. Let us show that if A, B are central simple F-algebras, then $A \otimes_F B$ is still simple.

Choose a F-basis $\{b_1, b_2, \cdots\}$ of B, and assume that I is a two-sided ideal of the tensor product. Then, we choose a non-zero element u, such it can be a finite sum $\sum a_i \otimes b_i$, $a_i \in A$ and the number of non-zero terms is minimal. Without loss of generality, we assume $a_1 \neq 0$.

Since $Aa_1A = A$, one can even choose a such element u in the ideal, such that the first term $a_1 = 1$.

Thus, take any element $x \in A$ and consider the commutator of u and x, then, $[u, x] \in I$ and it has less number of non-zero terms. Thus, $[u, x] = 0$. The central property of A implies that $x \in B$. By assumption, B is simple, so x generates the whole B. Now, the proposition follows by $1 \otimes B$ generates $A \otimes_F B$ as a two sided ideal. $\square$

Splitting fields

Definition 3.8.17. *The matrix algebra $M_n(F)$ is called a* split *central simple algebra over F. If A is finite dimensional central simple over F, then an extension field K of F is called a splitting field for A if $A_K := A \otimes_F K$ is split over K. We also say K splits A.*

We have the following simple properties:

- Any K/F is a splitting field of $M_n(F)$.
- If K is a splitting field of A and B, then it is a splitting field of $A \otimes_F B$,(38) and of A^{op}, hence of A^{e}.
- If $A = M_n(D)$ for some central division F-algebra D, then K is a splitting field of A if and only if K is a splitting field of D. As a consequence, we only need to consider splitting fields of a central division F-algebra.

  Proof. In fact, since $M_n(D) \cong M_n(F) \otimes_F D$, we always have

  $$M_n(D) \otimes_F K \cong (M_n(F) \otimes_F K) \otimes_K (D \otimes_F K) \cong M_n(K) \otimes_K M_m(\Delta) \cong M_{mn}(\Delta),$$

  here we used that, since $D \otimes_F K$ is a central simple algebra over K (because D is over F), we have $D \otimes_F K \cong M_m(\Delta)$ for some central division K-algebra Δ. Now if K splits D, then $\Delta = K$, hence K splits A.

  Conversely, if K splits A, then $M_n(D) \otimes_F K \cong M_r(K)$ for some r. The isomorphism $M_{mn}(\Delta) \cong M_r(K)$ then shows that $\Delta = K$ and $mn = r$, so K also splits D. Indeed, in Wedderburn-Artin theorem, if $A \cong M_n(D)$ is a simple artinian ring (where D is a division ring), then D and n are uniquely determined by A. $\square$

(38) Because $(A \otimes_F B) \otimes_F K \cong (A \otimes_F K) \otimes_K (K \otimes_F B) = M_m(K) \otimes_K M_n(K) \cong M_{mn}(K)$.

- Splitting fields exist. Indeed, if $K = \overline{F}$ is an algebraical closure of F, then K splits D.

The next result is less obvious.

Proposition 3.8.18. *Let D be a finite dimensional central division F-algebra, then every maximal subfield K of D contains F and is a splitting field of D.*

Lemma 3.8.19. *Let A be a subalgebra of some F-algebra B, and view B as an $A \otimes_k B^{\mathrm{op}}$-module. Then $\mathrm{End}_{A\otimes_F B^{\mathrm{op}}}(B) \cong C_B(A)$.*

Proof. We have $\mathrm{End}_{A\otimes_F B^{\mathrm{op}}}(B) \hookrightarrow \mathrm{End}_{B^{\mathrm{op}}}(B) \cong B$, so we can identify $\mathrm{End}_{A\otimes_F B^{\mathrm{op}}}(B)$ with a subalgebra in B, and it makes sense to compare it with $C_B(A)$. For $y \in B$, let $\varphi_y : B \to B$, sending x to yx (which is B^{op}-equivariant). It is $A \otimes_F B^{\mathrm{op}}$-equivariant if and only if

$$\varphi_y((a \otimes b)x) = (a \otimes b)\varphi_y(x)$$

i.e.

$$y(axb) = a(yx)b,$$

if and only if $ya = ay$ for any $a \in A$, if and only if $y \in C_B(A)$. □

Proof of Proposition 3.8.18. We first prove that $C_D(K) = K$. Let $K' = C_D(K)$. Then $K' \supset K$ and if $K' \neq K$, we can choose $x \in K' \backslash K$. As in the proof of Lemma 3.6.11, the subalgebra generated by K and x is a subfield of D strictly containing K, which contradicts the maximality of K. Thus $C_D(K) = K$. In particular, K contains F (as $F = Z(D) \subseteq C_D(K)$).

Write $R := K \otimes_F D^{\mathrm{op}}$. We make D into a left R-module by $(k \otimes d)x := kxd$. Then D is a simple faithful R-module, because R is also a simple ring. Lemma 3.8.19 implies that $\mathrm{End}_R(D) \cong C_D(K)$, thus $\mathrm{End}_R(D) = K$ by the last paragraph. By Jacobson density theorem we deduce that $R \cong \mathrm{End}_K(D) = M_n(K)$ where $n = \dim_K D$. In other words, K splits D^{op}, hence also splits D. □

The proof gives the following result.

Corollary 3.8.20. *Let K be the maximal subfield of a finite dimensional central division F-algebra D, then $C_D(K) = K$. Moreover, $(\dim_F K)^2 = \dim_F D$.*

Proof. By the isomorphism $K \otimes_F D^{\mathrm{op}} \cong \mathrm{End}_K(D) \cong M_n(K)$, where $n = \dim_K D$, we obtain

$$\dim_F D = \dim_K(K \otimes_F D^{\mathrm{op}}) = n^2.$$

Since $\dim_F D = \dim_F K \cdot \dim_K D$, we also get $\dim_F K = n$. □

Remark 3.8.21. *It can be shown that every central division F-algebra D contains a maximal subfield which is separable over F. Thus any central simple F-algebra splits over a finite separable extension of F.*

Example 3.8.22. *In general, maximal subfields of D are not unique. If $\mathbb{H} = \mathbb{R} \oplus \mathbb{R}i \oplus \mathbb{R}j \oplus \mathbb{R}k$, with $i^2 = j^2 = k^2 = ijk = -1$, denotes the usual Hamilton 4-dimensional division algebra over R, then $\mathbb{H}$ contains $\mathbb{C} = \mathbb{R}[i]$ as a maximal subfield. Also $\mathbb{R}[j], \mathbb{R}[k]$ are maximal subfields, both isomorphic to $\mathbb{C}$. Of course, $\mathbb{C}$ is separable over $\mathbb{R}$.*

Skolem-Noether theorem

Theorem 3.8.23. *Let B be a finite dimensional central simple F-algebra and A be a simple subalgebra of B. For any homomorphisms $f, g : A \to B$ of F-algebras, there exists $y \in B^\times$ such that*

$$g(a) = y^{-1} f(a) y, \quad \forall a \in A.$$

In particular, any $f : A \to B$ can be extended to an automorphism of B which is inner, that is, of the form $a \mapsto y^{-1} a y$ for some $y \in B^\times$.

Note that since A is simple, any F-algebra homomorphism is non-zero, hence injective.

Proof. Set $R := A \otimes_F B^{\mathrm{op}}$. It is a finite-dimensional simple F-algebra by Theorem 3.8.12. Any module for R is semi-simple and any two simple R-modules are isomorphic, by combining Proposition 3.1.5 and Proposition 3.2.12. Hence any two R-modules of the *same* finite dimension over F are isomorphic.

There are two left R-modules structures on B. One is defined using the usual left and right multiplication by A and B, i.e. $(a \otimes b)x = f(a)xb$. The other one is: $(a \otimes b)x = g(a)xb$. These two R-module structures must be intertwined by some $\ell \in \mathrm{Aut}_R(B)$, namely

$$\ell(f(a)xb) = g(a)\ell(x)b \tag{18}$$

for all $a \in A$, $x, b \in B$. On the other hand, ℓ is a right B-module endomorphism of B, it has the form $x \mapsto yx$ with $y = \ell(1)$. Note that $y \in B^\times$, because ℓ is an automorphism and ℓ^{-1} also has the form $x \mapsto y'x$ with $y' = \ell^{-1}(1)$, thus $yy' = y'y = 1$.

Taking $x = 1$ and $b = 1$ in (18), we obtain

$$yf(a) = g(a)y,$$

i.e. $g(a) = yf(a)y^{-1}$ for $a \in A$.

For the last statement, take g to be the natural embedding $A \hookrightarrow B$. Then f can be extended to the inner automorphism $a \mapsto y^{-1}ay$. □

Corollary 3.8.24 (Skolem-Noether)**.** *Any automorphism of a finite dimensional central simple algebra is inner.*

Corollary 3.8.25. *Let A be a finite dimensional central simple F-algebra and K splits A. If $f : A \otimes_F K \cong M_n(K)$ and $g : A \otimes_F K \to M_n(K)$ are two K-algebra isomorphisms. Then there exists $P \in \mathrm{GL}_n(K)$ such that for all $a \in A$:*

$$g(a \otimes 1) = P^{-1} f(a \otimes 1) P.$$

Proof. Apply Theorem 3.8.23 to $B = M_n(K)$ and $A := A \otimes_F K$ here. Then there exists $P \in \mathrm{GL}_n(K)$ such that $g = P^{-1} f P$ on A. ☐

Corollary 3.8.26. *If K_1, K_2 are subfields of A and if $K_1 \cong K_2$ as F-algebras, then K_1 and K_2 are conjugate, i.e. there exists $y \in A^\times$ such that $K_2 = y^{-1} K_1 y$.*

Proof. In fact, fix an isomorphism $\alpha : K_1 \xrightarrow{\sim} K_2$. Then we get two embeddings $K_1 \hookrightarrow A$, one is the canonical inclusion, the other is via inclusion $K_1 \overset{\alpha}{\xrightarrow{\sim}} K_2 \subset A$. Then we apply Theorem 3.8.23. ☐

Theorem 3.8.27 (Wedderburn)**.** *Any finite division ring is a field.*

Proof. Assume D is a finite division simple F-algebra (this implies that F is a finite field). Then any maximal subfield of D is an extension of F with degree n (determined by D). Such a finite field is unique up to isomorphism ! Therefore we can apply Corollary 3.8.26 to deduce that two maximal subfields are conjugate. Now consider $D^\times$ which is a multiplicative group, and $K^\times$ (where K is a fixed maximal subfield). Every element of $D^\times$ is contained in some maximal subfield. Hence

$$D^\times = \bigcup_{a \in D^\times} a^{-1} K^\times a.$$

We have the following result from finite group theory. From which we deduce that $K = D$ and D is commutative.

Lemma 3.8.28. *Let H be a proper subgroup of a finite group G, then $G \neq \bigcup_{g \in G} g^{-1} H g$.*

Proof. If $g' = hg$ for $h \in H$, then $g'^{-1} H g' = g^{-1} H g$, so there are at most $[G : H]$ distinct sets $g^{-1} H g$. But each one of them has the same cardinality as H, and they have non-trivial intersection (as they all contain the identity element), the result easily follows. ☐

Theorem 3.8.29 (Frobenius)**.** *The only finite dimensional division $\mathbb{R}$-algebras are $\mathbb{R}$, $\mathbb{C}$ and $\mathbb{H}$.*

Proof. Let D be such an $\mathbb{R}$-algebra. If the center of D is $\mathbb{C}$, then $D \cong \mathbb{C}$ by Lemma 3.6.11. So we assume $Z(D) = \mathbb{R}$ and $D \neq \mathbb{R}$ from now on. Let K be a maximal subfield of D. Since $[D : \mathbb{R}] > 1$, so $[K : F] > 1$ and K must be isomorphic to $\mathbb{C}$. As $[K : \mathbb{R}] = 2$, this implies $[D : \mathbb{R}] = 4$ by Corollary 3.8.20. We will construct a subalgebra of D which is isomorphic to $\mathbb{H}$, hence $D \cong \mathbb{H}$ comparing their dimensions.

Since $K \cong \mathbb{C}$, we can find $i \in K$ such that $i^2 = -1$. By Theorem 3.8.23 the $\mathbb{R}$-embedding $K \hookrightarrow D$, sending i to $-i$, is given by a conjugation: there exists $j \in D^\times$ such that $j^{-1} i j = -i$. We see that $j \notin \mathbb{R} = Z(D)$ and we want to prove that $j^2 \in -\mathbb{R}_{\geq 0}$, hence $j^2 = -1$ up to multiplication by an element of $\mathbb{R}$. The equality $j^{-1} i j = -i$ is just $i^{-1} j i = -j$. Let K' be the field generated by $\mathbb{R}$ and j; then $K' \cong \mathbb{C}$ and

$$K' \to K', \quad x \mapsto i^{-1} x i$$

is a well-defined field isomorphism which fixes $\mathbb{R}$. This corresponds to the unique non-trivial element σ in $\mathrm{Gal}(\mathbb{C}/\mathbb{R})$, hence is just the complex conjugation. Since in $\mathbb{C}$, $\sigma(x) = -x$ if and only if $x^2 \in -\mathbb{R}_{\geq 0}$, we get $j^2 = -r$ with $r \in \mathbb{R}_{\geq 0}$. Replacing j by $j/\sqrt{r}$ we obtain $j^2 = -1$, and the equality $j^{-1}ij = -i$ still holds true. Let $k := ij$, then clearly $k^2 = -1$ and $ijk = -1$. It is easy to check that $\{1, i, j, k\}$ are linearly independent over $\mathbb{R}$,[39] hence the subalgebra generated by i, j, k is isomorphic to $\mathbb{H}$, and the proof is finished. $\square$

Double centralizer theorem for central simple algebras

Theorem 3.8.30. *Let B be a central simple F-algebra.*

(i) For every semi-simple F-subalgebra A of B, we have $C_B(C_B(A)) = A$.

(ii) If A is simple, then $C_B(A)$ is simple. Furthermore, $\dim_F B = (\dim_F A) \cdot (\dim_F C_B(A))$.

Remark 3.8.31. *(1) If A is central simple, then this is obvious from Theorem 3.8.10. Indeed, it implies that $B \cong A \otimes_F C_B(A)$, and since B is central simple, so is $C_B(A)$. Apply the theorem again we get $A = C_B(C_B(A))$, giving (i). But it is crucial to not restrict to* central *subalgebras, since maximal subfields of a division algebra are typically not central.*

(2) If we apply Theorem 3.8.30 to $B = D$ is division central F-algebra, and $A = K$ a maximal subfield of D, we get $C_D(K) = K$, thus $\dim_F D = (\dim_F K)^2$ *which recovers Corollary 3.8.20.*

(3) Although B is central simple, it may contain a semi-simple F-subalgebra: if $B = M_n(F)$, then $\{\mathrm{diag}(b_1, \ldots, b_n)\}$ is isomorphic to $F \times F \times \cdots \times F$ which is a semi-simple subalgebra. We check that $C_B(A) = A$, hence $C_B(C_B(A)) = A$. We could also consider another semi-simple sub-algebra: $A' = \{\mathrm{diag}(b_1, \ldots, b_{n-2}, b, b)\}$, which is isomorphic to $F \times \cdots \times F$ ($n-1$ copies). Then $C_B(A')$ consists of matrices of the form

$$\begin{pmatrix} b_1 & 0 & 0 & 0 \\ 0 & b_2 & 0 & 0 \\ 0 & 0 & \ldots & 0 \\ 0 & 0 & 0 & M_2(F) \end{pmatrix}.$$

Taking $C_B(\cdot)$ again, we get back A'.

Proof. (i) Set $R := A \otimes_F B^{\mathrm{op}}$; it is a semi-simple F-algebra. This is because A is semi-simple, hence is a direct product of simple rings by Wedderburn-Artin theorem, and we apply Corollary 3.8.12. As consequence, any module over R is semi-simple; in particular B, viewed as R-module, is semi-simple.

Lemma 3.8.19 implies that $R' := \mathrm{End}_R(B) \cong C_B(A)$. By Jacobson density theorem, the morphism $R \to \mathrm{End}_{R'}(B)$ is surjective. Given $x \in C_B(C_B(A))$, the endomorphism

[39] Indeed, this follows from the equality $(a + bi + cj + dk)(a - bi - cj - dk) = a^2 + b^2 + c^2 + d^2$.

$y \mapsto xy$ of B commutes with the action of R', hence comes from R (by the surjection $E \twoheadrightarrow \mathrm{End}_{R'}(B)$).[40]

We need to show this implies $x \in A$. Since B is central simple, the homomorphism $B \otimes_F B^{\mathrm{op}} \to \mathrm{End}_F(A)$, $a \otimes b \mapsto [y \mapsto ayb]$ is an isomorphism by Lemma 3.8.8, hence $A \otimes_F B^{\mathrm{op}} \to \mathrm{End}_F(B)$ is injective. Since

$$x \otimes 1 \in (A \otimes B^{\mathrm{op}}) \cap (B \otimes 1) = A \otimes 1$$

we obtain $x \in A$.

(ii) Assume A is simple. Then R is also simple. Write R as $M_s(D)$ by Wedderburn-Artin theorem with D a division F-algebra, and let M be the unique simple R-module. Then $M \cong D^{\oplus s}$ and $D^{\mathrm{op}} := \mathrm{End}_R(M)$. Write ${}_RB = M^{\oplus r}$ viewed as R-modules. It follows that

$$C_B(A) \cong R' = \mathrm{End}_R(B) \cong M_r(D)$$

is simple. The second assertion follows from the following equalities:

$$\dim B = r \cdot \dim M = rs \dim D$$

$$\dim A \cdot \dim B = \dim R = s^2 \dim D$$

$$\dim C_B(A) = r^2 \cdot \dim D.$$

Indeed,

$$(\dim B)^2 = r^2 s^2 (\dim D)^2 = (\dim R) \cdot \dim C_B(A) = \dim A \cdot \dim B \cdot \dim C_B(A).$$

□

3.9 Brauer groups

Definition 3.9.1. *Let A and B be central simple F-algebras. We say that A is similar to B, written as $A \sim B$, if there exists $n, m \geq 1$ such that $M_n(A) \cong M_m(B)$.*

Lemma 3.9.2. *This defines an equivalence relation. Moreover, if $A \sim A'$ and $B \sim B'$, then $A \otimes_F A' \sim B \otimes_F B'$.*

Proof. Exercise. □

Trivially, if $A = M_n(D)$ for some central division F-algebra D, then $A \sim D$. If D, D' are central division F-algebras, then $D \sim D'$ if and only if $D \cong D'$ by Wedderburn-Artin theorem. For any A we write $[A]$ for its equivalent class. Let $\mathrm{Br}(F)$ be the monoid formed by the equivalence classes of central simple F-algebras, the multiplication being given by $\otimes$ and the unit being the class of $M_n(F)$, or F.

[40]Remark: it is in fact an isomorphism, since $R' \hookrightarrow B \otimes_F B^{\mathrm{op}} \cong \mathrm{End}_F(B)$, while $\mathrm{End}_{R'}(B)$ also embeds in $\mathrm{End}_F(B)$.

Proposition 3.9.3. $\mathrm{Br}(F)$ *is an abelian group, called the Brauer group of* F*.*

Proof. Since A is central simple, we have $A \otimes_F A^{\mathrm{op}} \cong M_n(F) \sim F$, so $[A]$ has an inverse and $\mathrm{Br}(F)$ becomes a group. Commutativity follows from the fact $A \otimes_F B \cong B \otimes_F A$ $\square$

Example 3.9.4. *1.* $\mathrm{Br}(\mathbb{R}) \cong \mathbb{Z}/2\mathbb{Z}$*, the nontrivial element being represented by Hamilton's quaternion algebra. As a consequence,* $\mathbb{H}^{\mathrm{op}} \cong \mathbb{H}$*.*

2. If F *is a finite field, then any finite division algebra is a field (Wedderburn's theorem).*

3. If F *is separably closed, then* $\mathrm{Br}(F)$ *is trivial.*

4. $\mathrm{Br}(F)$ *is always torsion as a* $\mathbb{Z}$*-module. (This uses a cohomological interpretation of* $\mathrm{Br}(F)$*.)*

4 Representation theory

4.1 Representations

We fix a group G (usually will be a finite group) and a field F.

Definition 4.1.1. *By a linear representation* (V, ρ) *of a group* G*, we mean a homomorphism of groups* $G \to \mathrm{Aut}_F(V)$ *(bijective linear transformations). The degree of* ρ *is the dimension of* V *over* F*.*

This is equivalent to giving an F-linear action of G on V: $G \times V \to V$.

A morphism between two representations (V_1, ρ_1) and (V_2, ρ_2) is an F-linear map $f : V_1 \to V_2$ such that the diagram

$$\begin{tikzcd} V_1 \arrow[r, "f"] \arrow[d, "\rho_1(g)"'] & V_2 \arrow[d, "\rho_2(g)"] \\ V_1 \arrow[r, "f"'] & V_2 \end{tikzcd}$$

commutes for every $g \in G$. We write $\mathrm{Hom}_G(V_1, V_2)$ for the set of morphisms from V_1 to V_2. It is an F-vector space.

Two representations are said to be *isomorphic* or *equivalent* if there exists a bijective morphism between them.

In this course, we will only consider finite dimensional representations. Let $\underline{\mathrm{Rep}}_F(G)$ denote the category of finite dimensional representations of G over F.

Matrix representation: Let $B = \{e_1, ..., e_n\}$ be a basis for V/F. If $a \in \mathrm{End}_F(V)$, we write

$$ae_i = \sum_{j=1}^{n} \alpha_{ij} u_j, \quad 1 \leq i \leq n$$

and obtain the matrix (α_{ij}). In this way we get a group homomorphism $\rho_B : G \to \mathrm{GL}_n(F)$. Such a homomorphism is called a *matrix representation of* G *of degree* n. A change of basis from B to $B' = \{e'_1, \cdots, e'_n\}$ replaces ρ_B by $\rho_{B'} = P\rho_B P^{-1}$, where P is the transition matrix.

(1) Trivial representation: $V = F$, $\rho(g)v = v$ for any $g \in G$ and $v \in V$. The matrix representation is $g \mapsto \mathrm{Id}$.

(2) A representation of degree 1 is a homomorphism $G \to \mathrm{GL}_1(F) = F^\times$, also called *character*. When G is finite, $\rho(g)$ are roots of unity. For example, for $\rho : G = \mathbb{Z}/4\mathbb{Z} \to \mathbb{C}^\times$, its image must lie in $\{1, -1, i, -i\}$.

(3) Permutation representation. Let X be a finite set on which G acts. Define a vector space V of dimension $n := |X|$, with basis $\{e_x\}_{x\in X}$. Let G act on e_x via $g \cdot e_x := e_{g.x}$. This gives a linear representation of G on V. In particular,

   • If $X = \{1, \ldots, n\}$, $G = S_n$, we get a representation of S_n to $\mathrm{GL}_n(F)$, each σ with value being a permutation matrix. More generally, if there exists a group homomorphism $G \to S_n$, we get a representation of G.

   • Let $X = G$ and G acts on it via $g.x = gx$. The resulting representation of G on V is called the *left regular representation.* A given linear representation $\rho : G \to \mathrm{Aut}_F(V)$ is isomorphic to the left regular representation, if and only if there exists $v \in V$, such that $(\rho(g)v)_{g\in G}$ is a basis of V.

(4) Let V be the F-vector space spanned by homogenous polynomials in x, y of degree n, so that $\dim_F V = n + 1$. Let $G = \mathrm{SL}_2(F)$ act on V by:

$$\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) x^i y^{n-i} = (ax + cy)^i (bx + dy)^{n-i}$$

   One checks that this defines a representation of G.

4.2 Group algebras

Definition 4.2.1. *The group algebra $F[G]$ is the F-algebra having the set G as a basis: every $a \in F[G]$ is written uniquely as $a = \sum_{g\in G} a_g g$ (finite sum); the multiplication is given*

$$(\sum_{g\in G} a_g g)(\sum_{h\in G} b_h h) = \sum_{g,h} (a_g b_h)(gh) = \sum_{\gamma\in G} c_\gamma \gamma$$

where $c_\gamma = \sum_{gh=\gamma} a_g b_h$.

Remark 4.2.2. *In general, $F[G]^\times$ contains G but is larger than G. $F[G]$ is commutative if and only if G is abelian.*

Example 4.2.3. *Let $G = C_4$ be the cyclic group of order 4 and let s be a generator. Then $G = \{1, s, s^2, s^3\}$, and $F[G] = F \oplus Fs \oplus Fs^2 \oplus Fs^3$ as a vector space over F. As an algebra, we have an isomorphism*

$$F[X]/(X^4 - 1) \xrightarrow{\sim} F[G], \quad X \mapsto s.$$

(i) *If $F = \mathbb{C}$, then this implies $F[G] \cong \mathbb{C} \times \mathbb{C} \times \mathbb{C} \times \mathbb{C}$, using Chinese Remainder theorem and the decomposition in $F[X]$: $X^4 - 1 = (X+1)(X-1)(X+i)(X-i)$.*

(ii) *If $F = \mathbb{R}$, then $F[G] \cong \mathbb{R} \times \mathbb{R} \times \mathbb{C}$;*

(iii) *If $F = \mathbb{F}_2$, then $F[G] \cong F[X]/(X-1)^4$.*

Remark 4.2.4. *It can happen that $G_1 \ncong G_1$ but $F[G_1] \cong F[G_2]$. For example, if G is any abelian group of order n, then $\mathbb{C}[G]$ is semi-simple (see Theorem 4.2.9 below) and*

$$C[G] \cong \mathbb{C} \times \mathbb{C} \times \cdots \times \mathbb{C} \text{ (n factors)}.$$

However, Perlis and Walker proved that if G_1, G_2 are finite abelian groups such that $\mathbb{Q}[G_1] \cong \mathbb{Q}[G_2]$ as $\mathbb{Q}$-algebras, then $G_1 \cong G_2$ as groups.

Universal property of $F[G]$: for any F-algebra R, we have a natural isomorphism

$$\mathrm{Hom}_{F\text{-Alg}}(F[G], R) \cong \mathrm{Hom}_{\mathrm{Gr}}(G, R^\times).$$

Proposition 4.2.5. *There is an equivalence between the categories $\underline{\mathrm{Rep}}_F(G)$ and (left) $F[G]$-Mod, given as follows. Let (V, ρ) be a representation of G over k, we make V into an $F[G]$-module by*

$$\sum_{g \in G} a_g g \cdot v := \sum_{g \in G} a_g \rho(g) v.$$

Proof. We use

$$\mathrm{Hom}_{F\text{-Alg}}(F[G], \mathrm{End}_k(V)) \cong \mathrm{Hom}_G(G, \mathrm{Aut}_F(V)).$$

□

We can define many concepts in representation theory using the corresponding ones in module theory. Examples are as follows. Let (V, ρ) be a representation of G over F.

- A submodule U of V is the same thing as a subspace U which is stable by G. We call it a *sub-representation*. Its quotient V/U is still an $F[G]$-module, called *quotient representation*. This gives a short exact sequence

$$0 \to U \to V \to U/V \to 0.$$

- V is said *irreducible* if it is simple as an $F[G]$-module; *absolutely irreducible* if it is absolutely simple as an $F[G]$-module, i.e. $V \otimes_F E$ is still simple as an $E[G]$-module for any field extension E/F; *reducible* if not irreducible. V is called *completely reducible* (also called semi-simple) if it is semi-simple as an $F[G]$-module. V is called *indecomposable* if it is indecomposable as an $F[G]$-module.

Lemma 4.2.6. *Let M be simple left R-module, where R is a finite dimensional F-algebra. The following are equivalent:*

(1) M is absolutely simple;

(2) $M \otimes_F E$ is simple for every finite extension E of F;

(3) $\mathrm{End}_R(M) = F$;

(4) the homomorphism $R \to \mathrm{End}_F(M)$ is surjective.

Proof. (1)$\Longrightarrow$(2) is trivial.

(2)$\Longrightarrow$(3) Let $\varphi \in \mathrm{End}_R(M)$. In particular, it is an element in $\mathrm{End}_F(M)$. Choose E/F large enough to contain an eigenvalue of φ, say $\lambda \in E$. Set $\varphi_E := E \otimes_F \varphi \in \mathrm{End}_E(M \otimes_F E)$. Then $\varphi_E - \lambda$ is not a bijection, i.e. not invertible in $\mathrm{End}_E(M \otimes_F E)$. But $M \otimes_F E$ is simple by assumption, so $\varphi_E = \lambda$, hence $\varphi = \lambda$ too (and $\lambda \in F$).

(3)$\Longrightarrow$(4) follows from Jacobson density theorem.

(4)$\Longrightarrow$(1) For every field extension E/F, the homomorphism $R \otimes_F E \to \mathrm{End}_E(M \otimes_F E)$ is again surjective. By linear algebra, one sees that $M \otimes_F E$ is a simple $R \otimes_F E$-module. Indeed, if $N \neq 0$ is a proper $R \otimes_F E$-submodule of $M \otimes_F E$, we may choose $v \in N$ and $w \notin N$, and define an E-linear map $M \otimes_F E \to M \otimes_F E$ sending v to w; clearly such a map can not lie in the image of $R \otimes_F E$. $\square$

Definition 4.2.7. *A field F is called a* splitting field *of G if any simple $F[G]$-module is absolutely simple. We also say that G* splits *over F.*

Corollary 4.2.8. *Given F and G, there exists a finite extension E/F over which G splits.*

Basic operations on representations:

Change of fields Let (ρ, V) be a representation of G over F. Let E/F be a field extension. Then $(\rho \otimes 1, V \otimes_F E)$ is a representation of G over E.

Direct sum if $V = V_1 \oplus V_2$ with $\rho = (\rho_1, \rho_2)$.

External tensor product Let (V_i, ρ_i) be a representation of G_i over F, for $i = 1, 2$. Then $\rho_1 \boxtimes \rho_2$ becomes a representation of $G_1 \times G_2$.

Tensor product Let (V_1, ρ_1), (V_2, ρ_2) be representations of G over F, then $V_1 \otimes_F V_2$ becomes a representation by letting

$$(\rho_1 \otimes \rho_2)(g) = \rho_1(g) \otimes \rho_2(g).$$

As $F[G]$-modules, we get $V_1 \otimes_F V_2$, via $g(v_1 \otimes v_2) = (gv_1) \otimes (gv_2)$.

Hom-spaces Let (V_1, ρ_1), (V_2, ρ_2) be representations of G over F. The space $\mathrm{Hom}_F(V_1, V_2)$ is equipped with a representation of G:

$$g \cdot f := \rho_2(g) \circ f \circ \rho_1(g)^{-1} \in \mathrm{Hom}_F(V_1, V_2). \tag{19}$$

We have $\mathrm{Hom}_G(V_1, V_2)$ is just $\mathrm{Hom}_F(V_1, V_2)^G$.

Contragredient representation Given (V_1, ρ_1), take $V_2 = F$ the trivial representation, we get the *contragredient* representation $\mathrm{Hom}_F(V_1, F)$, denoted by V_1^*. Choose $B = \{e_i\}$ a base of V_1, and $B^* = \{e_i^*\}$ the dual base of V_1^*.

Pull-back/restriction/inflation Let $\varphi : H \to G$ be a group homomorphism. then we deduce a representation of H from a representation of G. In fact, φ induces an F-algebra homomorphism $F[H] \to F[G]$. When $H \to G$ is an inclusion, the pull-back is called the restriction; when it is a quotient, it is called the inflation from G to H.

Reduction Let H be a *normal* subgroup of G and (V, ρ) be a representation of G. Set $V^N := \{v \in V : hv = v\}$. Then V^N becomes a representation of G/H.

Regular representation: $F[G]$ viewed as a left $F[G]$-module, corresponds to the left regular representation of G.

Theorem 4.2.9. *(Maschke's theorem) Assume G is finite. Then $F[G]$ is a semi-simple ring iff* $\mathrm{char}(F)$ *does not divide $|G|$. In particular, if* $\mathrm{char}(F) = 0$*, $F[G]$ is semi-simple.*

Proof. $\Longleftarrow$ Let V be any $F[G]$-module. We show that V is semi-simple, i.e. if W is an $F[G]$-submodule, then it has a complement in V. This implies in particular $F[G]$ is a semi-simple module over itself, hence is a semi-simple ring. Take an F-vector space W' such that $V = W \oplus W'$ as vector spaces, and let $\pi : V \twoheadrightarrow W$ be the projection map. Set(41)

$$\pi' := \frac{1}{|G|} \sum_{g \in G} \rho(g)^{-1} \circ \pi \circ \rho(g).$$

We check that $\pi' \in \mathrm{Hom}_{F[G]}(V, W)$: if $g' \in G$, then

$$\begin{aligned} \rho(g')^{-1} \circ \pi' \circ \rho(g') &= \tfrac{1}{|G|} \textstyle\sum_{g \in G} \rho(g')^{-1} \rho(g)^{-1} \pi \rho(g) \rho(g') \\ &= \tfrac{1}{|G|} \textstyle\sum_{g \in G} \rho(gg')^{-1} \pi \rho(gg') \\ &= \tfrac{1}{|G|} \textstyle\sum_{g \in G} \rho(g)^{-1} \pi \rho(g) \\ &= \pi. \end{aligned}$$

Moreover, since $\pi|_W = \mathrm{id}$, we still have $\pi'|_W = \mathrm{id}$. It follows that π' is a projection of $F[G]$-modules, thus

$$V = W \oplus \ker(\pi').$$

$\Longrightarrow$ Put $z = \sum_{g \in G} g$. Then $g'z = zg' = z$ for all $g' \in G$, i.e. z lies in the center of $F[G]$, and

$$z^2 = \sum_{g' \in G} g'z = |G|z = 0.$$

It follows that $F \cdot z$ is a nil ideal of $F[G]$, thus $\mathrm{rad}(F[G]) \neq 0$ by Proposition 3.4.3, and $F[G]$ is not semi-simple. $\square$

(41) this construction is called *averaging* π

Let $Z(F[G])$ denote the center of $F[G]$.

Proposition 4.2.10. *Let $G = C_1 \sqcup C_2 \sqcup \cdots \sqcup C_s$ be the decomposition of G into conjugacy classes, where s denotes the number of conjugacy classes. Put*

$$c_i = \sum_{g_i \in C_i} g_i.$$

Then $\{c_1, ..., c_s\}$ forms a basis for $Z(F[G])$. In particular, $\dim_F Z(F[G]) = s$.

Proof. If $g \in G$, then $g^{-1}c_i g = \sum_{g_i \in C_i} g^{-1}g_i g = c_i$ by definition of a conjugacy class. Hence $c_i \in Z(F[G])$. Now let $c = \sum_g \gamma_g g \in Z(F[G])$. If $h \in G$,

$$h^{-1}ch = \sum_g \gamma_g h^{-1}gh = \sum_g \gamma_{hgh^{-1}} g$$

so the condition $h^{-1}ch = c$ gives $\gamma_{hgh^{-1}} = \gamma_g$, $\forall h \in G$. Thus any two elements g and g' in the same conjugacy class have the same coefficient in the expression $c = \sum_g \gamma_g g$, so c is a linear combination of the c_i's. It is clear that $c_1, \ldots, c_s$ are linearly independent. □

Theorem 4.2.11. *Assume* $\mathrm{char}(F) \nmid |G|$ *so that $F[G]$ is semi-simple. We write*

$$F[G] = A_1 \oplus \cdots \oplus A_r = M_{n_1}(D_1) \oplus \cdots \oplus M_{n_r}(D_r),$$

where D_i are division F-algebras. Let $d_i := [D_i : F]$.

(i) *There are exactly r isomorphism classes of irreducible representations $\{V_i : 1 \le i \le r\}$ of G, of dimension respectively $d_i n_i$ over F. Moreover,*

$$\sum_{i=1}^{r} n_i^2 d_i = |G|.$$

(ii) *The multiplicity of V_i in the regular representation is n_i.*

(iii) *If F is algebraically closed (e.g. $F = \mathbb{C}$), then $D_i \cong F$ and*

$$\sum_{i=1}^{r} n_i^2 = |G|.$$

Proof. Obvious from Wedderburn-Artin theorem. □

Remark 4.2.12. *Without the assumption* $\mathrm{char}(F) \nmid |G|$, *we have the following variant of Theorem 4.2.11. Let $J := \mathrm{rad}(F[G])$ denote the Jacobson radical of $F[G]$, then the quotient ring $F[G]/J$ is semisimple, so we may apply Wedderburn-Artin theorem to $F[G]/J$. Moreover, any simple $F[G]$-module is annihilated by J (see Lemma 3.3.2), so $F[G]$ and $F[G]/J$ have the same simple modules. Since $|G| = \dim_F(F[G]) > \dim_F(F[G]/J)$, as above we get $\sum_{i=1}^{r} n_i^2 d_i < |G|$.*

Since $Z(A_i) = Z(D_i)$ (but note that D_i is not necessarily central), in particular,

$$\dim_F Z(F[G]) = \sum_{i=1}^{r} [Z(D_i) : F].$$

Again, if F is algebraically closed, then $\dim_F Z(F[G]) = r$.

Putting together, we get

$$s = \sum_{i=1}^{r} [Z(D_i) : F] \geq r$$

and the equality holds if and only if $Z(D_i) = F$ for all i, i.e. each D_i is central. This is the case if F is algebraically closed.

Corollary 4.2.13. *With the notation in Theorem 4.2.11, F is a splitting field of G if and only if $D_i = F$ for any $1 \leq i \leq r$. If this is the case, we have $s = r$, i.e. the number of isomorphism classes of irreducible representations of G is equal to the number of conjugacy classes in G.*

Proof. This is a combination of Proposition 3.1.5 and Lemma 4.2.6. □

Example 4.2.14. *We saw that $\mathbb{R}[C_4] \cong \mathbb{R} \oplus \mathbb{R} \oplus \mathbb{C}$. This example shows that, when F is not algebraically closed, D_i need not be central over F.*

Example 4.2.15. *We have $\mathbb{C}[S_3] \cong \mathbb{C} \oplus \mathbb{C} \oplus M_2(\mathbb{C})$. This is because S_3 has 3 conjugacy classes, hence $r = 3$ and*

$$n_1^2 + n_2^2 + n_3^2 = 6,$$

which forces (up to renumeration) $n_1 = n_2 = 1$, $n_3 = 2$.

Example 4.2.16. *Let $Q_8 = \{\pm 1, \pm i, \pm j, \pm k\}$ be the quaternion group, a subgroup of $\mathbb{H}$. It has 5 conjugacy classes:*

$$\{1\},\ \{-1\},\ \{i, -i\},\ \{j, -j\},\ \{k, -k\}.$$

The irreducible $\mathbb{C}$-representations of Q_8 have dimension $1, 1, 1, 1, 2$ respectively. Precisely, the quotient group $G/\{\pm 1\}$ is abelian of order 4, isomorphic to $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$, inducing 4 1-dim representations. Fixing an isomorphism $\mathbb{H} \otimes_{\mathbb{R}} \mathbb{C} \cong M_2(\mathbb{C})$, we obtain one 2-dim representation of Q_8 via $Q_8 \hookrightarrow \mathbb{H} \otimes_{\mathbb{R}} \mathbb{C} \cong M_2(\mathbb{C})$. However, this last representation can not be defined over $\mathbb{R}$.

For irreducible $\mathbb{R}$-representations of $\mathbb{H}$, again 4 of them are 1-dimensional and one of them is 4-dimensional which is induced by the natural action of Q_8 on $\mathbb{H}$. This corresponds to the fact that

$$\mathbb{R}[Q_8] = \mathbb{R} \oplus \mathbb{R} \oplus \mathbb{R} \oplus \mathbb{R} \oplus \mathbb{H}.$$

Corollary 4.2.17. *If G is abelian, then any absolutely irreducible representation of G is of degree 1.*

Proof. We may assume F is a splitting field of G. Hence there are $d = |G|$ conjugacy classes and $\sum_{i=1}^{d} n_i^2 = |G| = d$, showing $n_i = 1$.

This is true even without the assumption $\mathrm{char}(F) \nmid |G|$. Indeed, $\mathrm{rad}(F[G])$ annihilates any simple module, so we may assume $\mathrm{rad}(F[G]) = 0$, thus $F[G]$ is semi-simple (as it is artinian). Therefore $F[G]$ decomposes as a direct sum of simple matrix rings $M_{n_i}(D_i)$. Since G is abelian, $F[G]$ is commutative. But $M_{n_i}(D_i)$ is commutative only when $n_i = 1$. $\square$

Remark 4.2.18. *If the field F is algebraically closed and characteristic 0, then the converse statement is still true. However, without the algebraically closed condition, the above example shows that the converse statement is wrong: all absolutely irreducible representations of G are of degree 1 cannot imply that G is abelian.*

4.3 Character of representations

From now on, we consider linear representations over F, such that $\mathrm{char}(F) \nmid |G|$ and G splits over F. This is equivalent to saying that every irreducible representation of G is absolutely irreducible.

Let V be a finite dimensional F-vector space. For $a \in \mathrm{End}_F(V)$ we let $\mathrm{Tr}(a)$ denote the trace of a. It is also the sum of all eigenvalues of a.

Definition 4.3.1. *Let (V, ρ) be a linear representation of G. We define the map $\chi_\rho = \chi_V : G \to F$ by*

$$\chi_\rho(g) = \mathrm{Tr}(\rho(g)).$$

The F-valued function χ_ρ is called the character *of the representation ρ. We extend χ to a function on $F[G]$ by linearity.*

If ρ is irreducible, then χ_ρ is called irreducible. *If $F = \mathbb{C}$, χ_ρ is called a* complex character. *The degree of ρ is called the* degree *of χ_ρ.*

Basic properties:

- Equivalent representations have the same characters.
- If E/F is a field extension, then $\chi_{\rho\otimes E} = \chi_\rho$. Indeed, choose $\{e_i\}$ a basis of V/F, then $\{e_i \otimes 1\}$ is a basis of $V \otimes_F E$ over E.
- Any character is a *class function*, i.e. it is constant on every conjugacy class: $\chi_\rho(h^{-1}gh) = \chi_\rho(g)$.
- $\deg \rho = \chi_\rho(1)$.
- If $U \subset V$ is a sub-representation, then $\chi_V = \chi_U + \chi_{V/U}$. Indeed, choose a basis of U over F, and extend it to a basis of V, then with respect to this basis the matrix of $\rho(g)$ is upper triangular, of the form $\left(\begin{smallmatrix} \rho_U & * \\ 0 & \rho_{V/U} \end{smallmatrix}\right)$.

- If ρ_1, ρ_2 are representations of G, then $\chi_{\rho_1 \oplus \rho_2} = \chi_{\rho_1} + \chi_{\rho_2}$, $\chi_{\rho_1 \otimes \rho_2} = \chi_{\rho_1}\chi_{\rho_2}$.
- $\chi_{\rho^*}(g) = \chi_\rho(g^{-1})$: this is because $\rho^*_{B^*} = ({}^t\rho_B)^{-1}$ where B^* is the dual basis of B.
- If $F \subset \mathbb{C}$, then $\chi_\rho(g^{-1}) = \overline{\chi_\rho(g)}$. To see this, we may assume $F = \mathbb{C}$. Any eigenvalue of $\chi_\rho(g)$ is a root of unity, and eigenvalues of $\rho(g^{-1})$ are just the inverse of the ones of $\rho(g)$. As $\epsilon^{-1} = \overline{\epsilon}$ for any root of unity ϵ, the result follows.

Example 4.3.2. (i) *If ρ is of degree 1, then its character is given by $\chi = \rho$.*

(ii) *Let V be the permutation representation of G acting on a finite set X. Then $\chi_V(g) = \{x \in X : gx = x\}$.*

(iii) *The character of the regular representation is given by*

$$\chi_{\mathrm{reg}}(g) = \begin{cases} 0 & \text{if } g \neq 1 \\ |G| & \text{if } g = 1. \end{cases}$$

So for any $a = \sum_{x\in G} a_x x \in F[G]$, we can determine a_x by computing

$$\chi_{\mathrm{reg}}(ag^{-1}) = \sum_{x\in G} a_x \chi_{\mathrm{reg}}(ag^{-1}) = a_g|G|. \tag{20}$$

Denote by $\mathrm{cf}(G)$ the F-vector space of class functions on G. We define an inner product $\langle\cdot,\cdot\rangle$ on $\mathrm{cf}(G)$ by setting

$$\langle \alpha, \beta\rangle = \frac{1}{|G|}\sum_{g\in G} \alpha(g)\beta(g^{-1}). \tag{21}$$

We have the following facts:

- $\langle \alpha, \beta\rangle = \langle \beta, \alpha\rangle$;
- $\langle\cdot,\cdot\rangle$ is bilinear on $\mathrm{cf}(G) \times \mathrm{cf}(G)$.

Remark 4.3.3. *Be attention with the definition using (when $F \subset \mathbb{C}$)*

$$\langle \alpha, \beta\rangle' = \frac{1}{|G|}\sum_{g\in G} \alpha(g)\overline{\beta(g)}.$$

For this one we have $\langle \beta, \alpha\rangle' = \overline{\langle \alpha, \beta\rangle'}$, and if α, β are both characters, it coincides with (21). *In particular, if α, β are characters, then $\langle \alpha, \beta\rangle \in \mathbb{R}$ and $\langle \alpha, \alpha\rangle \geq 0$.*

Recall that $F[G] = A_1 \oplus \cdots \oplus A_r$, each A_i corresponds to an irreducible representation of G, say V_i. Viewed as a left $F[G]$-module, we have a direct sum decomposition

$$F[G] \cong \bigoplus_{i=1}^{r} V_i^{\oplus n_i}$$

where $n_i = \dim_F V_i$. We write e_i for the idempotent corresponding to A_i, so that

$$1 = \sum_i e_i.$$

We have $e_i|_{V_j} = \delta_{ij}$.

Proposition 4.3.4. *With the above notation and write $\chi_i = \chi_{V_i}$, we have*

$$e_i = \frac{n_i}{|G|} \cdot \sum_{g \in G} \chi_i(g^{-1}) g.$$

In particular, $n_i \neq 0$ in F.(42)

Proof. Denote by χ_{reg} the (left) regular character. Write $e_i = \sum_{x \in G} a_x x$. We will use (20) to determine a_x:

$$\chi_{\mathrm{reg}}(e_i g^{-1}) = \sum_{x \in G} a_x \chi_{\mathrm{reg}}(x g^{-1}) = a_g |G|, \quad \forall g \in G.$$

On the other hand, we know $\chi_{\mathrm{reg}} = \sum_{i=1}^r n_i \chi_i$ (here we use that n_i is exactly the multiplicity of V_i in the regular representation), and that $\chi_i(e_j g^{-1}) = 0$ if $i \neq j$ and $\chi_i(e_i g^{-1}) = \chi_i(g^{-1})$ (as $g^{-1} = \sum_j e_j g^{-1}$), thus

$$\chi_{\mathrm{reg}}(e_i g^{-1}) = n_i \cdot \chi_i(g^{-1}), \quad \forall g \in G.$$

Hence $a_g = \frac{n_i}{|G|} \chi_i(g^{-1})$ as required. ☐

Theorem 4.3.5. *(Schur's first orthogonality relation) With the above notation, $\langle \chi_i, \chi_j \rangle = \delta_{ij}$.*

Proof. Recall that e_i acts as identity on V_i, so $\chi_i(e_i) = \dim V_i = n_i$. By Proposition 4.3.4, we get

$$n_i = \chi_i(e_i) = \frac{n_i}{|G|} \sum_{g \in G} \chi_i(g^{-1}) \chi_i(g).$$

Since n_i is invertible in F, we get the result by definition of $\langle -, - \rangle$.

Similarly, we have $\chi_i(e_j) = 0$ if $i \neq j$, which gives the result. ☐

Theorem 4.3.6. *$\{\chi_i : 1 \leq i \leq r\}$ is an orthonormal basis for $\mathrm{cf}(G)$.*

Proof. We already see that χ_i are orthonormal; so they must be linearly independent. In fact, if $\chi := \sum_i \alpha_i \chi_i = 0$, then take $\langle \chi, \chi_i \rangle$ for any i, we get $\alpha_i = 0$.

It forms a basis because the number of conjugacy classes in G is equal to the number of irreducible representations of G. ☐

Corollary 4.3.7. *The space $\mathrm{cf}(G)$ admits two bases: the irreducible characters χ_i and the characteristic functions $\mathbb{1}_C$ of the conjugacy classes $C \subset G$.*

(42) We will see later that $n_i \mid |G|$ for all i, so this result is a consequence of the assumption $\mathrm{char}(F) \nmid |G|$.

Corollary 4.3.8. *Furthermore, let us assume* $\mathrm{char}(F) = 0$.

(i) *Let* W *be a representation of* G *and* V *be an irreducible representation of* G. *Then the multiplicity of* V *in* W *is equal to* $\langle \chi_W, \chi_V \rangle$.

(ii) *Two representations of* G *with the same characters are isomorphic.*

(iii) *Let* W *be a representation of* G. *Then* W *is irreducible if and only if* $\langle \chi_W, \chi_W \rangle = 1$.

Proof. Clear. □

Remark 4.3.9. *Let* V, W *be two* F*-representations of* G, *and* $\mathrm{char}(F)$ *does not devide* $|G|$, *then* $\dim_F \mathrm{Hom}_G(V, W) = \langle \chi_V, \chi_W \rangle$. *(check it.)*

Theorem 4.3.10. *(Second orthogonality relation) For* $g \in G$, *let* C_g *be the conjugacy class containing* g. *We have*

$$\sum_{i=1}^{r} \chi_i(g)\chi_i(h^{-1}) = \begin{cases} |G|/|C_g| & g \sim h \\ 0, & \text{otherwise} \end{cases}$$

Remark 4.3.11. *If we set* $Z_G(g) := \{x \in G : x^{-1}gx = g\}$, *then this is a subgroup of* G, *and* $|G|/|C_g| = |Z_G(g)|$. *In particular, this relation allows us to determine the size of* $Z_G(g)$.

Proof. Let $\mathbb{1}_{C_g} \in \mathrm{cf}(G)$ be the characteristic function on C_g. Since $\{\chi_i,\ 1 \leq i \leq r\}$ forms a basis of $\mathrm{cf}(G)$, we can write

$$\mathbb{1}_{C_g} = \sum_{i=1}^{r} \lambda_i \chi_i, \quad \lambda_i \in F.$$

The coefficient λ_i is equal to

$$\lambda_i = \langle \mathbb{1}_{C_g}, \chi_i \rangle = \frac{1}{|G|} \sum_{h \in G} \mathbb{1}_{C_g}(h)\chi_i(h^{-1}) = \frac{1}{|G|} \sum_{h \in C_g} \chi_i(h^{-1}) = \frac{|C_g|}{|G|} \chi_i(g^{-1})$$

where the last equality holds because χ_i is a class function. We get

$$\mathbb{1}_{C_g}(h) = \frac{|C_g|}{|G|} \sum_i \chi_i(g^{-1})\chi_i(h).$$

To obtain the result, it suffices to note that $\mathbb{1}_{C_g}(h) = 1$ if and only if $g \sim h$. □

4.4 Examples

We always take $F = \mathbb{C}$ in this subsection.

4.4.1 Cyclic groups

Let $G = \langle g \rangle$ of order n, i.e. $G \cong \mathbb{Z}/n\mathbb{Z}$. It is abelian, so there are n isomorphism classes of irreducible representations. Let $\zeta_n \in \mathbb{C}$ be a primitive n-th root of unity. For $0 \le i \le n-1$, define ρ_i to be

$$\rho_i(g^k) = \zeta_n^{ik} \in \mathbb{C}^\times$$

which is a 1-dimensional representation of G.

4.4.2 Character table of S_3

Let $G = S_3$. It has order 6, and three conjugacy classes:

$$C_1 = \{\mathrm{id}\}, \quad C_2 = \{(12),(13),(23)\}, \quad C_3 = \{(123),(132)\}.$$

By the relation $n_1^2 + n_2^2 + n_3^2 = 6$, we must have $n_1 = 1, n_2 = 1, n_3 = 2$, i.e. there are two irreducible representations of degree 1, and one of degree 2. It is easy to construct the irreducible representations of degree 1: the trivial one and the character sgn. We know already

	C_1	C_2	C_3
$\sharp$	1	3	2
χ_1	1	1	1
χ_2	1	-1	1
χ_3	2	α	β

The orthogonal relations $\langle \chi_3, \chi_1 \rangle = \langle \chi_3, \chi_2 \rangle = 0$ imply that

$$\frac{1}{6}(2 + 3\alpha + 2\beta) = 0$$

$$\frac{1}{6}(2 - 3\alpha + 2\beta) = 0$$

we get $\beta = -1$ and $\alpha = 0$. In fact, since $\chi_2\chi_3$ is also an irreducible character of G, it must be equal to χ_3, hence $\alpha = 0$.

Let us construct the representation ρ_3. Since S_3 acts naturally on $X = \{1,2,3\}$, we obtain a permutation representation of degree 3, say π. Its character is given by

$$\chi_\pi(C_1) = 3, \quad \chi_\pi(C_2) = 1, \quad \chi_\pi(C_3) = 0.$$

Such a representation is not irreducible (always), since $F(e_1 + e_2 + e_3)$ is stable under S_3. The quotient is then isomorphic to ρ_3, because the character is $\chi_\pi - \chi_1$, which is equal to χ_3.

As a consequence, we have a decomposition $\mathbb{C}[S_3] \cong \mathbb{C} \oplus \mathbb{C} \oplus M_2(\mathbb{C})$.

4.4.3 Character table of A_4

Let $G = A_4$. It has order 12. Its conjugacy classes are

$$C_1 = \{\mathrm{id}\}, \quad C_2 = \{(12)(34), (13)(24), (14)(23)\}$$

$$C_3 = \{(123), (142), (134), (243)\}, \quad C_4 = \{(132), (124), (143), (234)\}.$$

So there are in total 4 non-isomorphic irreducible representations. On the other hand, A_4 contains a normal subgroup of order 4, i.e. $V_4 = C_1 \cup C_2$. The quotient A_4/V_4 is abelian of order 3. There are three non-isomorphic irreducible representations of A_4/V_4, all of degree 1. Inflation to A_4 gives us three non-isomorphic representations of A_4 of degree 1. This implies that the last irreducible representation has degree 3.

Let $\omega = e^{2\pi i/3}$ be a primitive 3-th root of unity. Note that the class of (123) in A_4/V_4 is a generator, and $(123)^2 = (132)$. We obtain the character table as follows:

	C_1	C_2	C_3	C_4
$\sharp$	1	3	4	4
χ_1	1	1	1	1
χ_2	1	1	ω	ω^2
χ_3	1	1	ω^2	ω
χ_4	3	α	β	γ

We must have $\beta = \gamma = 0$ as $\chi_2\chi_4$ is still an irreducible character of G of degree 3, hence must be equal to χ_4. To determine α, we use the orthogonal relation for rows $\langle \chi_4, \chi_1 \rangle = 0$, i.e. $3 + 3\alpha + 0 + 0 = 0$, hence $\alpha = -1$.

Let us construct the representation ρ_4. Since A_4 acts on $X = \{1, 2, 3, 4\}$, we get as above a permutation of degree 4, say π. Its character is given by

$$\chi_\pi(C_1) = 4, \quad \chi_\pi(C_2) = 0, \quad \chi_\pi(C_3) = 1, \quad \chi_\pi(C_4) = 1.$$

We have $\rho_1 \subset \pi$ as sub-representation, whose quotient has character $\chi_\pi - \chi_1$, which is χ_4.

Next, we consider the tensor product $\rho_4 \otimes \rho_4$, which has degree 9. It has character χ_4^2. If we write $\rho_4 \otimes \rho_4 = \bigoplus_{i=1}^4 \rho_i^{d_i}$, then

$$d_1 = \langle \chi_1, \chi_4^2 \rangle = \frac{1}{12}(1 \times 1 \times 9 + 3 \times 1 \times 1) = 1$$

$$d_2 = \langle \chi_2, \chi_4^2 \rangle = 1, \quad d_3 = \langle \chi_3, \chi_4^2 \rangle = 1$$

$$d_4 = \langle \chi_4, \chi_4^2 \rangle = \frac{1}{12}(27 - 3) = 2.$$

We have a decomposition

$$\mathbb{C}[A_4] \cong \mathbb{C} \oplus \mathbb{C} \oplus \mathbb{C} \oplus M_3(\mathbb{C}).$$

However, as χ_2, χ_3 are not defined over $\mathbb{R}$, we have

$$\mathbb{R}[A_4] \cong \mathbb{R} \oplus \mathbb{C} \oplus M_3(\mathbb{R}).$$

4.4.4 Character table of D_n

Recall that $D_n = \langle r, s : r^n = 1, s^2 = 1, srs = r^{-1}\rangle$. It has order $2n$. We first determine its conjugacy classes. Every element of D_n has the form r^k or sr^k, where $0 \le k \le n-1$. We have the following:

- since $sr^k s = r^{-k}$, we get $(sr^k)^2 = 1$, and $\{r^k, r^{-k}\}$ are in the same conjugacy class. If n is even, $\{r^0 = 1\}$ and $\{r^{n/2}\}$ form single classes, so $\{r^k\}$ splits into $\frac{n}{2}+1$ conjugacy classes. When n is odd, they split into $\frac{n+1}{2}$ conjugacy classes.
- r^k and $sr^{k'}$ can not be in one conjugacy class.
- $r^{-i}sr^i = sr^{2i}$, so that $s \sim sr^2 \sim sr^4 \sim \cdots$. So when n is even $\{sr^k; 0 \le k \le n-1\}$ splits into two conjugacy classes; when n is odd, it forms a single conjugacy class.

In summary, if n is even, there are $\frac{n}{2} + 1 + 2$ conjugacy classes; if n is odd, there are $\frac{n+1}{2} + 1$ conjugacy classes.

Below we assume n is even (and leave the case n is odd as an exercise).

- Degree 1 representations, i.e. $\rho : D_n \to \mathbb{C}^\times$. Since sr^k has order 2, they take images in $\{\pm 1\}$, this implies $\rho(r^k) \in \{\pm 1\}$ too. We have four such representations ρ_i such that

	r^k	sr^k
ρ_1	1	1
ρ_2	1	-1
ρ_3	$(-1)^k$	$(-1)^k$
ρ_4	$(-1)^k$	$(-1)^{k+1}$

- Degree 2 representations. Let $\zeta = e^{2\pi i/n}$ be a primitive n-th root of unity. For $h \ge 0$, define

$$\rho^h(r) = \begin{pmatrix} \zeta^h & 0 \\ 0 & \zeta^{-h} \end{pmatrix}, \quad \rho^h(s) = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}. \tag{22}$$

  It only depends on $h \pmod n$, so we assume $0 \le h \le n-1$. Moreover ρ^h and ρ^{n-h} are isomorphic (with the same character), so we assume $0 \le h \le n/2$.

  If $h = 0$, then $\rho^0 = \rho_1 + \rho_2$. If $h = \frac{n}{2}$, then $\rho^h = \rho_3 + \rho_4$. If $0 < h < n/2$, we show that ρ^h are irreducible and non-isomorphic. Let $V = \mathbb{C}^2 = \mathbb{C}e_1 \oplus \mathbb{C}e_2$, and $W \subset V$ be a sub-representation. Then $\dim W = 1$, say spanned by $v = \alpha_1 e_1 + \alpha_2 e_2$. Since $\rho^h(s)v = \alpha_2 e_1 + \alpha_1 e_2$, we must have $\alpha_1 = \alpha_2$, and we may assume $v = e_1 + e_2$. But this vector is not stable under $\rho^h(r)$ as $\zeta^h \ne \zeta^{-h}$. This shows the irreducibility. They are non-isomorphic, by looking at their associated characters:

$$\chi_h(r^k) = \zeta^{hk} + \zeta^{-hk} = 2\cos\frac{2\pi hk}{n}.$$

- We already get $4 \times 1 + (\frac{n}{2} - 1) \times 2^2 = 2n$, so this gives all irreducible representations.

Remark 4.4.1. *The group D_n can be characterized as the group of symmetries of a regular n-gon in the plane. This allows us to construct the representations ρ^h geometrically.*

As an example, when $n = 4$, there are 5 conjugacy classes,

$$C_1 = \{1\}, \quad C_2 = \{r^2\}, \quad C_3 = \{r, r^3\}$$

$$C_4 = \{s, sr^2\}, \quad C_5 = \{sr, sr^3\}.$$

the character table is

	C_1	C_2	C_3	C_4	C_5
$\sharp$	1	1	2	2	2
χ_1	1	1	1	1	1
χ_2	1	1	1	-1	-1
χ_3	1	1	-1	1	-1
χ_4	1	1	-1	-1	1
χ_5	2	α	0	0	0

4.4.5 Character table of Q_8

Let $Q = Q_8$ be the quarternion group. By definition, Q is of order 8 defined by

$$Q = \{1, i, j, k, -1, -i, -j, -k\}, \quad i^2 = j^2 = k^2 = ijk = -1$$

or equivalently (setting $a = i, b = j$)

$$Q = \{1, a, a^2, a^3, b, ab, a^2b, a^3b | a^4 = 1, b^2 = a^2, bab^{-1} = a^{-1}\}.$$

It has 5 conjugacy classes:

$$C_1 = \{1\}, \quad C_2 = \{-1\}, \quad C_3 = \{i, -i\},$$

$$C_4 = \{j, -j\}, \quad C_5 = \{k, -k\}.$$

Since $\sum_i n_i^2 = 8$, we know the degree of any irreducible representation is ≤ 2. Moreover, we see that four out of the five irreducible representations have degree 1 and only one has degree 2: $1 + 1 + 1 + 1 + 2^2 = 8$. As $H = \{1, -1\}$ is a normal subgroup of Q, we first study the quotient group Q/H, which is isomorphic to $\mathbb{Z}_2 \times \mathbb{Z}_2 = \{1, \bar{i}, \bar{j}, \bar{k}\}$. Since this is an abelian group, it has 4 irreducible representations, all of degree 1. Inflation gives us all the irreducible representations of degree 1 of Q. In all we obtain

	C_1	C_2	C_3	C_4	C_5
$\sharp$	1	1	2	2	2
χ_1	1	1	1	1	1
χ_2	1	1	1	-1	-1
χ_3	1	1	-1	1	-1
χ_4	1	1	-1	-1	1
χ_5	2	α	0	0	0

Then we use the orthogonal relation to show $\alpha = -2$ as in the case of D_4. *In particular, we see that D_4 and Q_8 have the same character table.*

We want to construct the representation ρ_5. Recall that Q is a subgroup of $\mathbb{H}^\times$, and $\mathbb{H}$ can be embedded in $\mathrm{GL}_2(\mathbb{C})$ by sending $a = i \mapsto \left(\begin{smallmatrix} i & 0 \\ 0 & -i \end{smallmatrix}\right)$, $b = j \mapsto \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$, $ab = k \mapsto \left(\begin{smallmatrix} 0 & i \\ i & 0 \end{smallmatrix}\right)$. So ρ_5 is simply the embedding $\rho_5 : Q \hookrightarrow \mathrm{GL}_2(\mathbb{C})$.

Corollary 4.4.2. *The representation ρ_5 is a faithful representation of Q.*

We deduce that

$$\mathbb{C}[Q_8] \cong \mathbb{C} \oplus \mathbb{C} \oplus \mathbb{C} \oplus \mathbb{C} \oplus M_2(\mathbb{C}).$$

Note that χ_i for $1 \leq i \leq 4$ are defined over $\mathbb{R}$, but ρ_5 is not. The natural action of Q_8 on $\mathbb{H}$ gives an irreducible $\mathbb{R}$-representation of degree 4. As a consequence,

$$\mathbb{R}[Q_8] \cong \mathbb{R} \oplus \mathbb{R} \oplus \mathbb{R} \oplus \mathbb{R} \oplus \mathbb{H}.$$

4.5 Induced representations and Frobenius reciprocity

In this subsection, G is a finite group and F can be a field of arbitrary characteristic, and not necessarily a splitting field of G.

Definition 4.5.1. *Let $H \subset G$ be a subgroup, W be a linear representation of H. We call*

$$\mathrm{Ind}_H^G W := F[G] \otimes_{F[H]} W$$

the induced representation of W from H to G.

Remark 4.5.2. *Sometimes $\mathrm{Ind}_H^G W$, for (σ, W) representation of H, is defined as follows:*

$$\mathrm{Ind}_H^G W := \{f : G \to W, f(hg) = \sigma(h) f(g), \forall h \in H, g \in G\} = \mathrm{Hom}_{F[H]}(F[G], W),$$

with the action of G given by $(g'f)(g) := f(gg')$. This is isomorphic to $k[G] \otimes_{k[H]} W$ check!.

By definition, if $K \subset H \subset G$ are subgroups, then we have $\mathrm{Ind}_H^G \mathrm{Ind}_K^H \cong \mathrm{Ind}_K^G$.

Proposition 4.5.3. *$F[G]$ is a (right) free $F[H]$-module, with G/H as a basis.*

Proof. If T is a set of representatives for G/H, then

$$F[G] = \bigoplus_{g \in T} gF[H].$$

Sometimes, we write simply $\bigoplus_{\bar{g} \in G/H} gF[H]$, with $g \in G$ any representative of $\bar{g}$. $\square$

Corollary 4.5.4. *(i) $\dim_F \mathrm{Ind}_H^G W = [G : H] \dim_F W$;*

(ii) Every element of $\mathrm{Ind}_H^G W$ can be written as

$$\sum_{\bar{g} \in G/H} g \otimes w_g.$$

(iii) As $F[H]$-module, $\mathrm{Ind}_H^G W$ has a natural decomposition as $W \oplus W^c$ for some submodule W^c.

Proof. We have a decomposition

$$\mathrm{Ind}_H^G W \cong \bigoplus_{\bar g\in G/H} g(F[H])\otimes_{F[H]} W = \bigoplus_{g\in T} g\otimes W. \tag{23}$$

Here given $\bar g \in G/H$, the left coset $\bar g(F[H])$ becomes a right $F[H]$-module. Note that $g\otimes W$ is well-defined, not depend on the choice of $g\mapsto \bar g$.

(iii) We note that $1\otimes W$ and $\bigoplus_{\bar g\neq 1} g\otimes W$ are both stable under H-action. □

Theorem 4.5.5 (Frobenius reciprocity)**.** *Let W be a representation of H and V a representation of G. We have*

(1) $\mathrm{Hom}_G(\mathrm{Ind}_H^G W, V) \cong \mathrm{Hom}_H(W, \mathrm{Res}_H^G V)$*;*

(2) $\mathrm{Hom}_G(V, \mathrm{Ind}_H^G W) \cong \mathrm{Hom}_H(\mathrm{Res}_H^G V, W)$.

In some books, $\mathrm{Res}_H^G V$ is also simply written as $V|_H$. For the proof, we first recall the adjoint property of tensor product. If A, B are two rings, ${}_AN_B, {}_BM, {}_AS$ are modules, we have

$$\mathrm{Hom}_A(N\otimes_B M, S) \xrightarrow{\sim} \mathrm{Hom}_B(M, \mathrm{Hom}_A(N, S)), \tag{24}$$

given by $f\mapsto [m\mapsto [n\mapsto f(n\otimes m)]]$ and the inverse given by $\psi\mapsto [n\otimes m\mapsto \psi(m)(n)]$.

Proof. (1) Since $\mathrm{Ind}_H^G W = F[G]\otimes_{F[H]} W$, it is a special case of (24). More precisely,

- if $f\in \mathrm{Hom}_G(\mathrm{Ind}_H^G W, V)$, it is sent to $w\mapsto f(1\otimes w)\in V$.
- if $\psi\in \mathrm{Hom}_H(W, \mathrm{Res}_H^G V)$, it is sent to $\sum_{g\in T} g\otimes w_g \mapsto \sum_{g\in T} g\cdot \psi(w_g)$.

(2) Fix a set of representatives T of G/H containing 1. Note that the natural projection $\mathrm{pr}: \mathrm{Ind}_H^G W \twoheadrightarrow W = 1\otimes W$ sending $\sum_{g\in T} g\otimes w_g$ to w_1 is a morphism of $k[H]$-modules. Indeed, $hg = g'h'$ for some $g'\in T$ and $h'\in H$. Although in general $g\neq g'$, we have $g'=1$ if and only if $g=1$. Hence

$$h\cdot \big(\sum_{g\in T} g\otimes w_g\big) = 1\otimes hw_g + \sum_{1\neq g'\in T} g'\otimes h'w_g.$$

Let $f\in \mathrm{Hom}_G(V, \mathrm{Ind}_H^G W)$, we send it to $\mathrm{pr}\circ f\in \mathrm{Hom}_H(\mathrm{Res}_H^G V, W)$. Conversely, let $\psi\in \mathrm{Hom}_H(\mathrm{Res}_H^G V, W)$, we sent it to

$$\tilde\psi : v\mapsto \sum_{g\in T} g\otimes \psi(g^{-1}v).$$

We need to check the following facts:

- The expression $\tilde\psi(v)$ does not depend on the choice of T. In fact, if T' is another one, then for any $g\in T$, there corresponds to a unique $g'\in T'$ such that $g = g'h\in g'H$, so that
$$g\otimes \psi(g^{-1}v) = g'h\otimes h^{-1}\psi(g'^{-1}v) = g'\otimes \psi(g'^{-1}v).$$

- $\tilde{\psi}$ is a morphism of G-representations. Let $g' \in G$, then
$$\tilde{\psi}(g'v) = \sum_{g\in T} g \otimes \psi(g^{-1}g'v) = \sum_{g\in T} g'(g'^{-1}g) \otimes \psi(g^{-1}g'v) = g' \cdot \sum_{g\in T} g'^{-1}g \otimes \psi(g^{-1}g'v).$$
It suffices to note that $\{g'^{-1}g : g \in T\}$ forms another set of representatives for G/H.

- It is clear that $\mathrm{pr} \circ \tilde{\psi} = \psi$. To check $\widetilde{\mathrm{pr} \circ f} = f$, we write $f = \sum_{g\in T} g \otimes f_g$, then $\mathrm{pr} \circ f = f_1$ and we need to check $f_g(v) = f_1(g^{-1}v)$ for any $g \in T$. In fact, this is clear since f itself is a morphism of $F[G]$-modules, so
$$f(g^{-1}v) = g^{-1}f(v)$$
which is just
$$\sum_{t\in T} t \otimes f_t(g^{-1}v) = \sum_{t\in T} g^{-1}t \otimes f_t(v).$$
The unicity of the expression (for fixed T) implies $1 \otimes f_1(g^{-1}v) = 1 \otimes f_g(v)$, and the result follows.

Another proof of (2): use the fact that $\mathrm{Ind}_H^G W \simeq \mathrm{Hom}_{F[H]}(F[G], W)$ for any finite-dimensional representation W and G is finite, then apply the 'tensor-hom' adjunction. ☐

Corollary 4.5.6. *If V is irreducible, and if $\mathrm{Res}_H^G V$ contains a sub-representation isomorphic to W, then V is a quotient of $\mathrm{Ind}_H^G W$.*

Proof. The injection $W \hookrightarrow \mathrm{Res}_H^G V$ (in particular non-zero) induces a non-zero map $\mathrm{Ind}_H^G W \to V$. Since V is irreducible, this map must be surjective. ☐

Corollary 4.5.7. *Assume H is an abelian subgroup of G, let $d = [G : H]$. Assume F is algebraically closed. Then every irreducible representation V of G has degree $\leq d$.*

Proof. Consider $\mathrm{Res}_H^G V$. It contains an irreducible sub-representation, since it is of finite length as $F[H]$-module. Since H is abelian and F is algebraically closed, it must be of dimension 1, say χ by Corollary 4.2.17. Therefore V is a quotient of $\mathrm{Ind}_H^G \chi$. ☐

Example 4.5.8. *Every irreducible $\mathbb{C}$-representation of D_n has dimension ≤ 2, because $H := \langle r \rangle$ is an abelian subgroup of index 2. Moreover, assume n is even and let ρ^h, $0 < h < \frac{n}{2}$, be the representation of D_n of degree 2 defined in (22). Then*
$$\rho^h = \mathrm{Ind}_H^G \chi^h$$
where $\chi^h : H \to \mathbb{C}^\times$ sending r to ζ_n^h. In fact, this induced representation has basis $1 \otimes w$ and $s \otimes w$, so the matrix of s is $\left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$, and the matrix of r is $\left(\begin{smallmatrix} \zeta^h & 0 \\ 0 & \zeta^{-h} \end{smallmatrix}\right)$ because $rs = sr^{-1}$.

Corollary 4.5.9. *Let V be a representation of G and W be a subrepresentation of $\mathrm{Res}_H^G V$. Then $V \cong \mathrm{Ind}_H^G W$ if and only if V is generated by W as an $F[G]$-module and*
$$\dim V = [G : H] \dim W.$$

Proof. The condition is clearly necessary. It is also sufficient by Corollary 4.5.6 and by counting dimensions. $\square$

Proposition 4.5.10. *We have a canonical isomorphism* $\mathrm{Ind}_H^G(W^*) \cong (\mathrm{Ind}_H^G W)^*$.

Proof. Write $V = \mathrm{Ind}_H^G W$ for simplicity. Since $V = W \oplus W^c$ as $F[H]$-modules, define

$$\widehat{W} = \{f \in V^*, \mathrm{supp}(f) \subset W\}.$$

which is a submodule of V^* isomorphic to W^*.

Similarly, we define $\widehat{g \otimes W}$ as the functionals in V^* supported on $g \otimes W$. Then $V^* = \bigoplus_{g \in G/H} \widehat{g \otimes W}$ and it is direct to check that $g\widehat{W} = \widehat{g \otimes W}$.[43] In particular, V^* is generated by W^*, so we conclude by Corollary 4.5.9. $\square$

Remark 4.5.11. *This gives another proof for Theorem 4.5.5(2)* $\mathrm{Hom}_G(V, \mathrm{Ind}_H^G W) \cong \mathrm{Hom}_H(\mathrm{Res}_H^G V, W)$. *Indeed,* $\mathrm{Hom}_G(V, V') \cong \mathrm{Hom}_G(V'^*, V^*)$ *(sending f to f^*) since $V^{**} \cong V$ as G-representations*[44]*. Hence*

$$\mathrm{Hom}_G(V, \mathrm{Ind}_H^G W) \cong \mathrm{Hom}_G((\mathrm{Ind}_H^G W^*), V^*) \overset{(1)}{\cong} \mathrm{Hom}_H(W^*, \mathrm{Res}_H^G V^*) \cong \mathrm{Hom}_H(\mathrm{Res}_H^G V, W).$$

Proposition 4.5.12. *Let H be a finite subgroup of G. Let W be a representation of H and V a representation of G. There is a canonical isomorphism*

$$\mathrm{Ind}_H^G(W \otimes \mathrm{Res}_H^G V) \cong (\mathrm{Ind}_H^G W) \otimes V.$$

Proof. Using the decomposition of $F[H]$-modules $\mathrm{Ind}_H^G W = W \oplus W^c$, we obtain an embedding

$$W \otimes \mathrm{Res}_H^G V \hookrightarrow (\mathrm{Ind}_H^G W) \otimes V$$

which induces a morphism $\mathrm{Ind}_H^G(W \otimes \mathrm{Res}_H^G V) \to (\mathrm{Ind}_H^G W) \otimes V$. Then use Corollary 4.5.9 to show it is an isomorphism. $\square$

Induced characters

Now we assume $\mathrm{char}(F)$ does not divide $|G|$ and F is a splitting field of G.

Proposition 4.5.13. *Let W be a representation of H. For every $x \in G$, we have*

$$\chi_{\mathrm{Ind}_H^G W}(x) = \sum_{\bar{g} \in G/H, x\bar{g} = \bar{g}} \chi_W(g^{-1}xg)$$

where $g \in G$ is any representative of the coset $\bar{g}$.

[43] If $f \in \widehat{W}$, then $(gf)(g' \otimes w) := f(g^{-1} \cdot g' \otimes w)$ which vanishes unless $g' \in gH$.

[44] Since V is finite dimensional over F, we have an isomorphism as F-vector spaces $V^{**} \cong V$; then compare their matrix representations.

Proof. Write $F[G] \otimes_{F[H]} W = \bigoplus_{g \in G/H} g \otimes W$. The element x permutes the elements of G/H, only those $\bar{g} \in G/H$ with $x\bar{g} = \bar{g}$ contribute to $\chi_{\mathrm{Ind}_H^G W}(x)$.

Now take $\bar{g} \in G/H$ satisfying $x\bar{g} = \bar{g}$, i.e. $xgH = gH$, i.e. $g^{-1}xg \in H$. Then

$$x \cdot (g \otimes w) = g(g^{-1}xg) \otimes w = g \otimes (g^{-1}xgw)$$

so the matrix of x on $g \otimes W$ is the same as that of $g^{-1}xg$ on W. This gives the result. $\square$

In next corollary we further assume $\mathrm{char}(F) = 0$.

Corollary 4.5.14. *Let V be a representation of G and W a representation of H. Then*

$$\langle \chi_V, \mathrm{Ind}_H^G \chi_W \rangle_G = \langle \mathrm{Res}_H^G \chi_V, \chi_W \rangle_H$$

where we set $\mathrm{Ind}_H^G \chi_W := \chi_{\mathrm{Ind}_H^G W}$ *and* $\mathrm{Res}_H^G \chi_V := \chi_{\mathrm{Res}_H^G V}$.

Proof. This uses Frobenius reciprocity and the fact that

$$\langle \chi_V, \mathrm{Ind}_H^G \chi_W \rangle_G = \dim_F \mathrm{Hom}_G(V, \mathrm{Ind}_H^G W)$$

(and similarly for $\langle \mathrm{Res}_H^G \chi_V, \chi_W \rangle_H$). $\square$

Mackey decomposition theorem

Theorem 4.5.15. *Let H, L be subgroups of G and W be a representation of H. Then*

$$\mathrm{Res}_L^G \mathrm{Ind}_H^G W \cong \bigoplus_{s \in S} \mathrm{Ind}_{L \cap sHs^{-1}}^L \mathrm{Res}_{L \cap sHs^{-1}}^{sHs^{-1}} s \otimes W.$$

Here S is a set of representatives for $L \backslash G/H$, and for $s \in S$, $s \otimes W$ is the sHs^{-1}-representation given by

$$(shs^{-1}) \cdot (s \otimes w) := s \otimes hw.$$

Proof. We write $G = \coprod_{s \in S} LsH$. Then $F[G] = \bigoplus_{s \in S} Ls \cdot F[H]$ and

$$\mathrm{Ind}_H^G W = \bigoplus_{s \in S} Ls \otimes W.$$

Write V_s for the direct summand $Ls \otimes W$ for any fixed $s \in S$. Then V_s is an $F[L]$-submodule of $\mathrm{Ind}_H^G W$ and

$$\mathrm{Res}_L^G \mathrm{Ind}_H^G W = \bigoplus_{s \in S} V_s.$$

Let's study the structure of V_s. Let T_s be a set of representatives for $L/(L \cap sHs^{-1})$. Then

$$LsH = \coprod_{t \in T_s} tsH.$$

Indeed, (a) $tsH = t'sH$ if and only if $t^{-1}t' \in L \cap sHs^{-1}$, so the union is disjoint; (b) for any $l \in L$, let t be the representative such that $l \in t(L \cap sHs^{-1})$, then $lsH = tsH$.

We deduce that $V_s = Ls \otimes W = \bigoplus_{t \in T_s} ts \otimes W$. We are left to show $V_s \cong \mathrm{Ind}^L_{L \cap sHs^{-1}} s \otimes W$ as $F[L]$-modules. Take $t = 1$, we see that $s \otimes W$ is a submodule of V_s. Because the two spaces have the same dimension over F, it suffices to show V_s can be generated by $s \otimes W$ under L-action (by Corollary 4.5.9), but this is obvious. $\square$

Corollary 4.5.16. *Let N be a normal subgroup of G and W be a representation of N. Then*

$$\mathrm{Res}^G_N \mathrm{Ind}^G_N W \cong \bigoplus_{\bar{g} \in G/N} g \otimes W,$$

where the action is given by: $n \cdot (gw) = g(g^{-1}ng \cdot w)$.

Proof. Clear, since $N \backslash G/N = G/N$, hence $sNs^{-1} = N$. $\square$

Example 4.5.17. *Recall that $V_4 = \{1, (12)(34), (13)(24), (14)(23)\} \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$. It has 4 irreducible representations, all of dimension 1:*

	$(12)(34)$	$(13)(24)$
$\chi_{1,1}$	1	1
$\chi_{1,-1}$	1	-1
$\chi_{1,-1}$	-1	1
$\chi_{-1,-1}$	-1	-1

As an example, we consider $\mathrm{Ind}^{S_4}_{V_4} \chi_{1,-1}$. We first decompose $\mathrm{Res}^{S_4}_{V_4} \mathrm{Ind}^{S_4}_{V_4} \chi_{1,-1}$. Since V_4 is normal in S_4, we may apply Corollary 4.5.16. Note that S_4/V_4 is isomorphic to S_3. If we identify S_3 with the subgroup $\{\sigma : \sigma(4) = 4\}$ of S_4, then S_3 provides a set of representatives of S_4/V_4.

For each $s \in S_3$, we compute the action of V_4 on $s \otimes e$ where e is a basis element on which V_4 acts via $\chi_{1,-1}$. If $s = 1$, then we obtain $\chi_{1,-1}$; if $s = (12)$, we obtain $\chi_{1,-1}$ again; if $s = (13)$, then

$$(12)(34) \cdot (s \otimes e) = s \otimes \big((14)(23)e\big) = -s \otimes e$$

$$(13)(24) \cdot (s \otimes e) = -s \otimes e,$$

so we obtain the character $\chi_{-1,-1}$. Similarly if $s = (23)$, then $(12)(34)$ acts on $s \otimes e$ as -1 and $(13)(24)$ acts as 1, so we get $\chi_{-1,1}$. If $s = (123)$, we get $\chi_{-1,-1}$; if $s = (132)$, we get $\chi_{-1,1}$. In all, we obtain

$$\mathrm{Res}^{S_4}_{V_4} \mathrm{Ind}^{S_4}_{V_4} \chi_{1,-1} \cong \chi_{1,-1}^{\oplus 2} \oplus \chi_{-1,1}^{\oplus 2} \oplus \chi_{-1,-1}^{\oplus 2}.$$

Corollary 4.5.18. *Let H be a subgroup of G and W an irreducible representation of H. Assume $\mathrm{char}(F) \nmid |G|$ and F is a splitting field for G and H. Then*

(i) $\mathrm{Ind}^G_H W$ is irreducible if and only if

$$\mathrm{Hom}_{H \cap sHs^{-1}}(\mathrm{Res}^{sHs^{-1}}_{H \cap sHs^{-1}} s \otimes W, \mathrm{Res}^H_{H \cap sHs^{-1}} W) = 0$$

for any $s \in G$, $s \notin H$.

(ii) Assume H is a normal subgroup. Then $\mathrm{Ind}_H^G W$ is irreducible if and only if W and $s(W)$ are not isomorphic for any $s \in G \backslash H$.

Proof. Use twice Frobenous reciprocity. The assumption implies that $F[G]$ is semisimple, so $\mathrm{Ind}_H^G W$ is irreducible if and only if $\mathrm{Hom}_G(\mathrm{Ind}_H^G W, \mathrm{Ind}_H^G W) = F$. Applying Frobenius reciprocity we get

$$\begin{aligned}
\mathrm{Hom}_G(\mathrm{Ind}_H^G W, \mathrm{Ind}_H^G W) &\cong \mathrm{Hom}_H(\mathrm{Res}_H^G \mathrm{Ind}_H^G W, W) \\
&\cong \textstyle\bigoplus_{\bar{s} \in H\backslash G/H} \mathrm{Hom}_H(\mathrm{Ind}_{H\cap sHs^{-1}}^H \mathrm{Res}_{H\cap sHs^{-1}}^{sHs^{-1}} s\otimes W, W) \\
&\cong \textstyle\bigoplus_{\bar{s} \in H\backslash G/H} \mathrm{Hom}_{H\cap sHs^{-1}}(\mathrm{Res}_{H\cap sHs^{-1}}^{sHs^{-1}} s\otimes W, \mathrm{Res}_{H\cap sHs^{-1}}^H W),
\end{aligned}$$

which implies the result.

(ii) In this case, $sHs^{-1} = H$, so the result follows from (i). □

Example 4.5.19. *Let $G = \mathrm{GL}_2(\mathbb{F}_q)$, $B = \left(\begin{smallmatrix} * & * \\ 0 & * \end{smallmatrix}\right)$ the subgroup of upper triangular matrices. Let $\theta_1, \theta_2 : \mathbb{F}_q^\times \to \mathbb{C}^\times$ be two characters, and define*

$$\theta_1 \otimes \theta_2 : B \to \mathbb{C}^\times, \quad \left(\begin{smallmatrix} a & b \\ 0 & d \end{smallmatrix}\right) \mapsto \theta_1(a) \otimes \theta_2(d)$$

which is a character of B.

Proposition 4.5.20. *(i) We have $B\backslash G/B = \{1, s\}$, where $s = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$.*

(ii) $\mathrm{Ind}_B^G \theta_1 \otimes \theta_2$ is irreducible if and only if $\theta_1 \neq \theta_2$.

Proof. Exercise. □

4.6 Clifford's theorem

Theorem 4.6.1. *Let N be a normal subgroup of G. Let V be an irreducible representation of G.*

(i) $\mathrm{Res}_N^G V$ is semisimple and is a direct sum of irreducible representations of N of equal dimension.

(ii) Fix $W \hookrightarrow \mathrm{Res}_N^G V$ irreducible. Let $H := \{g \in G : gW \cong W\}$ [45] which is a subgroup of G containing N, then V is induced from some irreducible representation of H.

Remark 4.6.2. *In Jacobson's book, H is denoted by $T(W)$ and called* inertia subgroup *of W. It is not normal in general, and depends on W: in fact different choice of W yields conjugate of H.*

Proof. (i) First, we show that $\mathrm{Res}_N^G V$ always has an irreducible sub-representation, say W. Indeed, as an $F[N]$-module, V is artinian, hence contains a minimal sub-module, which must be simple.

[45] Here, N acts on gW via $n \cdot (gv) = g(g^{-1}ng \cdot v)$, and $\cong$ indicates isomorphic but not necessarily equal

For the chosen W, Frobenius reciprocity implies a non-zero G-morphism $\mathrm{Ind}_N^G W \to V$, which must be surjective because V is irreducible. Therefore, to show $\mathrm{Res}_N^G V$ is semi-simple, it is enough to show $\mathrm{Res}_N^G \mathrm{Ind}_N^G W$ is semi-simple. But by Mackey's formula, it is isomorphic to a direct sum of representation $s \otimes W$ for $s \in S$, a set of representatives of G/N. Since W is a simple $F[N]$-module, so is $s \otimes W$,[46] hence $\mathrm{Res}_N^G \mathrm{Ind}_N^G W$ is semi-simple.

(ii) It is easy to check that H is a subgroup of G, containing N (in fact $nW = W$ for $n \in N$). Let $W' := \sum_{\bar{h} \in H/N} hW$; then being a sum of simple modules W' is semi-simple, hence is isomorphic to $W^{\oplus m}$ for some $m \in \mathbb{N}$.

We prove that $V \cong \mathrm{Ind}_H^G W'$. The argument in (i) shows that $V = \sum_{\bar{g} \in G/H} gW'$ and we need to show it is a direct sum decomposition. The construction of H implies that, for g_1, g_2 such that $g_1 H \neq g_2 H$, $g_1 W'$ and $g_2 W'$ have no common Jordan-Hölder factors (as $F[N]$-modules). We conclude by Lemma 4.6.3 below. $\square$

Lemma 4.6.3. *Let R be a ring and M an R-module of finite length. Assume that $M_1, \ldots, M_r \subset M$ are submodules such that M_i and M_j have no common Jordan-Hölder factor for any $1 \leq i \neq j \leq r$. Then $\sum_{i=1}^r M_i = \bigoplus_{i=1}^r M_i$.*

Proof. Consider the natural surjective morphism $f : \bigoplus_{i=1}^r M_i \twoheadrightarrow \sum_{i=1}^r M_i$. Suppose f is not injective, and let $0 \neq S \subset \ker(f)$ be a simple submodule. For $1 \leq i \leq r$, let $S_i \subset M_i$ be the projection of S to M_i. As S is simple, we have either $S_i \cong S$ or $S_i = 0$. By assumption, M_i and M_j have no common Jordan-Hölder factor, hence $S_i = 0$ except for one i. This means that $S \subset M_i$ for some i. But $f|_{M_i}$ is injective, so $S = 0$, contradiction. $\square$

Corollary 4.6.4. *Let N be a normal subgroup of G and V be a semi-simple G-representation. Then $\mathrm{Res}_N^G V$ is still semi-simple.*

Example 4.6.5. *Consider the irreducible representation ρ_4 of S_4. We know*

$$\mathrm{Res}_{V_4}^{S_4} \rho_4 \cong \chi_{1,-1} \oplus \chi_{-1,1} \oplus \chi_{-1,-1}.$$

Take $W = \chi_{1,-1}$; we will determine H. Recall the construction of ρ_4: it is obtained as $V/(e_1 + e_2 + e_3 + e_4)$, where V denotes the permutation representation for G acting on $\{1,2,3,4\}$. Let $\mathfrak{e}$ denote the class of $(e_1 + e_2) - (e_3 + e_4)$, then it is easy to see that V_4 acts on $\mathbb{C} \cdot \mathfrak{e}$ via the character $\chi_{1,-1}$. Also, one checks that H is generated by V_4 and (12), which has order 8 and is isomorphic to D_4. That is, by Theorem 4.6.1, $\rho_4 \cong \mathrm{Ind}_H^{S_4} \chi_{1,-1}$.

4.7 Arithmetic considerations

4.7.1 Algebraic integers

If R is a commutative ring and $x \in R$. We say x is *integral* over $\mathbb{Z}$ if there exists $n \geq 1$ and $a_0, \ldots, a_{n-1} \in \mathbb{Z}$ such that

$$x^n + a_{n-1} x^{n-1} + \cdots a_0 = 0.$$

When $R = \mathbb{C}$, such x is called an *algebraic integer.*

[46] *Proof*: if $U \subsetneq s \otimes W$ were a proper submodule, then $s^{-1} U \subsetneq W$ would a proper submodule of W.

Example 4.7.1. *Consider $R = \mathbb{Q}$. Then $r \in \mathbb{Q}$ is integral over $\mathbb{Z}$ if and only if $r \in \mathbb{Z}$.*

Proposition 4.7.2. *Let $x \in R$. The following properties are equivalent:*

(i) *x is integral over $\mathbb{Z}$.*

(ii) *The submodule $\mathbb{Z}[x] \subset R$ is a finitely generated $\mathbb{Z}$-module.*

(iii) *There exists a finitely generated sub-$\mathbb{Z}$-module of R which contains $\mathbb{Z}[x]$.*

In particular, if R itself is finitely generated $\mathbb{Z}$-module, then any element of R is integral over $\mathbb{Z}$.

Proof. (i)$\Longrightarrow$(ii) The equation implies that $\mathbb{Z}[x]$ is spanned over $\mathbb{Z}$ by $\{1, x, \dots, x^{n-1}\}$.

(ii)$\Longrightarrow$(iii) trivial.

(iii)$\Longrightarrow$ (i) Since $\mathbb{Z}$ is noetherian, if $\mathbb{Z}[x]$ is contained in a finitely generated $\mathbb{Z}$-module, then $\mathbb{Z}[x]$ is a noetherian module. Consider the ascending chain

$$\mathbb{Z} \subset \mathbb{Z} + \mathbb{Z}x \subset \mathbb{Z} + \mathbb{Z}x + \mathbb{Z}x^2 \subset \cdots$$

which must stabilize. So for n large, $x^n \in \sum_{i<n} \mathbb{Z}x^i$. □

Corollary 4.7.3. *The set of integral elements in R form a subring of R.*

Proof. Easy. If x, y are integral, then $\mathbb{Z}[x], \mathbb{Z}[y]$ are finitely generated $\mathbb{Z}$-modules, so is $\mathbb{Z}[x, y]$. Since $x + y, xy \in \mathbb{Z}[x, y]$, we get the result. □

From now on, we consider $R := Z(\mathbb{C}[G])$, the center of $\mathbb{C}[G]$, which is a commutative ring.

Proposition 4.7.4. *Let $u = \sum_{g \in G} u(g)g \in R$ such that $u(g) \in \mathbb{C}$ are all algebraic integers, then u is integral over $\mathbb{Z}$.*

Here, we may view the coefficient $u(g)$ as a function $G \to \mathbb{C}$, which is moreover a class function. This explains the notation.

Proof. Let $C_1, \dots, C_r$ be the list of conjugacy classes in G and $\mathbb{1}_{C_i}$ be the characteristic function. Then the $\mathbb{1}_{C_i}$ form a basis of R over $\mathbb{C}$. We set

$$R' := \mathbb{Z}\mathbb{1}_{C_1} \oplus \cdots \oplus \mathbb{Z}\mathbb{1}_{C_r}.$$

Claim: this is a *subring*. In fact, we have

$$\mathbb{1}_{C_i}\mathbb{1}_{C_j} = \sum_{g \in C_i} g \sum_{h \in C_j} h = \sum_{k=1}^{r} \alpha_{ijk}\mathbb{1}_{C_k}$$

with α_{ijk} equals the number of (g, h) such that $l = gh$ for a fixed $l \in C_k$. In particular, the coefficients belong to $\mathbb{N}$. By Proposition 4.7.2, the $\mathbb{1}_{C_i}$ are integral over $\mathbb{Z}$.

Now, for $u \in R$ we can write $u = \sum_i \alpha_i \mathbb{1}_{C_i}$, with $\alpha_i = u(g_i)$ for any $g_i \in C_i$. Hence the α_i are algebraic integers. By Corollary 4.7.3, u is integral over $\mathbb{Z}$. □

Proposition 4.7.5. *Let V be an irreducible $\mathbb{C}$-representation of G. For any $u \in R$, it induces an element in* $\mathrm{End}_G(V)$. *Moreover,* $u|_V = \lambda \in \mathbb{C}$ *with*

$$\lambda = \frac{1}{\dim V} \sum_{g \in G} u(g) \chi_V(g)$$

where we write $u = \sum_{g \in G} u(g)g$.

(ii) If $u(g)$ are algebraic integers for all $g \in G$, then $\lambda \in \mathbb{C}$ is an algebraic integer.

Proof. (i) Since u is in the center of $\mathbb{C}[G]$, it induces an element in $\mathrm{End}_G(V)$. But V is irreducible so $\mathrm{End}_G(V) \cong \mathbb{C}$ by Schur's lemma, i.e. $u|_V = \lambda$ for some $\lambda \in \mathbb{C}$. To determine λ, we note that $\mathrm{Tr}(u) = (\dim V)\lambda$. Hence

$$\lambda = \frac{1}{\dim V} \sum_{g \in G} u(g) \chi_V(g).$$

(ii) In fact, by (i) we get a map $\omega_V : R \to \mathbb{C}$ sending u to λ which is a ring homomorphism. Since u is integral over $\mathbb{Z}$, so is λ. □

Corollary 4.7.6. *If we take $u = 1_C$ for some conjugacy class C, then $\frac{\chi_V(C)|C|}{\dim V}$ is an algebraic integer.*

4.7.2 Degrees of characters

Theorem 4.7.7. *Let V be an irreducible $\mathbb{C}$-representation of G, then* $(\dim V)||G|$.

Proof. Consider the element $u = \sum_{g \in G} \chi_V(g^{-1})g \in R$. Since $\chi_V(g^{-1})$ are algebraic integers (being sum of roots of unity), by Proposition 4.7.5 the number

$$\frac{1}{n} \sum_{g \in G} \chi_V(g^{-1})\chi_V(g) = \frac{|G|}{\dim V}\langle \chi_V, \chi_V \rangle = \frac{|G|}{\dim V}$$

is an algebraic integer. This implies $\frac{|G|}{\dim V} \in \mathbb{N}$, i.e. $(\dim V)||G|$. □

Theorem 4.7.8. *Let $Z(G)$ be the center of G. For any irreducible representation V of G,* $(\dim V)|[G : Z(G)]$.

Proof. Let $m \in \mathbb{N}$ and let $G_m := G \times G \times \cdots \times G$. Let (V, ρ) be an irreducible representation of G. We form the representation $(V^{\otimes m}, \rho^{\otimes m})$ of G_m:

$$V^{\otimes m} := \underbrace{V \otimes \cdots \otimes V}_{m \text{ times}}.$$

Then $V^{\otimes m}$ is irreducible for any m. If $c \in Z(G)$, the irreducibility of V implies that $\rho(c) = \gamma(c)\mathrm{Id}$ is scalar and $\gamma : Z(G) \to \mathbb{C}^\times$ is a group homomorphism. Hence for $c_j \in Z(G)$,

$$\rho^{\otimes m}(c_1, \ldots, c_m) = \gamma(c_1 \cdots c_m)\mathrm{Id}.$$

It follows that $D_m \subset \ker(\rho^{\otimes m})$, where

$$D_m := \{(c_1, \dots, c_m) \in Z(G) \times \cdots \times Z(G) \mid \prod_i c_i = 1\}.$$

We may thus regard $\rho^{\otimes m}$ as an irreducible representation of G_m/D_m It is clear that D_m is a normal subgroup of G_m of order $|Z(G)|^{m-1}$, so by Theorem 4.7.7 we see that $(\dim V)^m$ divides

$$|G|^m/|D_m| = |G|^m/|Z(G)|^{m-1} = (\frac{|G|}{|Z(G)|})^m \cdot |Z(G)|.$$

But this is true for all $m \in \mathbb{N}$, we deduce that $\dim V$ divides $\frac{|G|}{|Z(G)|}$. $\square$

Remark 4.7.9. *Theorem 4.7.8 can be strengthened as follows: If A is an abelian normal subgroup of G, then $(\dim V) | [G : A]$ for any irreducible $\mathbb{C}$-representation V of G.*

As an example, the group S_4 contains a normal abelian subgroup V_4. It has 5 irreducible characters, whose degrees are respectively 1, 1, 2, 3, 3, all dividing $6 = [S_4 : V_4]$.

4.7.3 Characters of dimension ≥ 2

Theorem 4.7.10. *Let V be an irreducible $\mathbb{C}$-representation of G and assume $\dim(V) \geq 2$. Then there exists $g \in G$ with $\chi_V(g) = 0$.*

Proof. Write $\chi = \chi_V$. There exists a finite Galois extension $L/\mathbb{Q}$ such that $\chi(g) \in L$ for all $g \in G$. Consider

$$B(\chi) := \prod_{\sigma \in \mathrm{Gal}(L/\mathbb{Q})} \prod_{g \in G} |\sigma(\chi(g))|^2.$$

It is clear that $B(\chi)$ is fixed by $\mathrm{Gal}(L/\mathbb{Q})$, hence belongs to $\mathbb{Q}$. However, all the values $\sigma(\chi(g))$ are sums of roots of unity, hence are integral elements. This implies that $B(\chi)$ is also an algebraic integer, thus $B(\chi) \in \mathbb{Z}_{\geq 0}$. Below we will prove $B(\chi) = 0$, which gives the result.

Since V is irreducible, one has

$$\frac{1}{|G|} \sum_{g \in G} \chi(g)\overline{\chi(g)} = \frac{1}{|G|} \sum_{g \in G} |\chi(g)|^2 = 1.$$

Moreover, this holds true for $\sigma \circ \chi$ (any σ) because it remains irreducible. Thus

$$\sum_{\sigma \in \mathrm{Gal}(L/\mathbb{Q})} \sum_{g \in G} |\sigma(\chi(g))|^2 = |G| \cdot [L : \mathbb{Q}].$$

By the arithmetic-geometric inequality, we obtain (setting $n = |G| \cdot [L : \mathbb{Q}]$)

$$B(\chi)^{1/n} \leq 1$$

with equality if and only if $|\chi(g)|^2 = |\chi(g')|^2$ for any $g, g' \in G$ (the action of σ does not change the value of $|\chi(g)|$). Since $\chi(1) = \dim(V) > 1$ by assumption and $\sum_{g \in G} |\chi(g)|^2 = |G|$, there must exist $g \in G$ such that $|\chi(g)|^2 < 1$. Hence $B(\chi) < 1$. However, it is an integer, so $B(\chi) = 0$. $\square$

Example 4.7.11. *Below is the character table of $G = S_4$:*

	C_1	C_2	C_3	C_4	C_5
χ_1	1	1	1	1	1
χ_2	1	-1	1	1	-1
χ_3	2	0	2	-1	0
χ_4	3	1	-1	0	-1
χ_5	3	-1	-1	0	1

from which we see that 0 *occurs in the last three rows.*

4.7.4 p-groups

We use representation theory to prove the following classical result in group theory.

Proposition 4.7.12. *Let p be a prime number. A group of order p^2 is abelian.*

Proof. We will prove that $\mathbb{C}[G]$ is commutative, this is equivalent to proving that all irreducible $\mathbb{C}$-representations of G are 1-dimensional. By Theorem 4.7.8, an irreducible representation can have dimension $1, p$, or p^2. But Theorem 4.2.11 implies that it can not be p^2 or p (because there always exists the 1-dimensional trivial representation). □

When G has order p^3, we can predict the degrees of its irreducible representations.

Proposition 4.7.13. *Let G be a non-abelian group of order p^3. Among the irreducible $\mathbb{C}$-representations of G, p^2 of them have dimension* 1 *and $p - 1$ of them have dimension p.*

Proof. We know the following facts:

- $Z(G)$ is non-trivial; moreover, $G/Z(G)$ can not be cyclic, otherwise G would be abelian. Thus $|Z(G)| = p$ and $G/Z(G) \cong \mathbb{Z}/p\mathbb{Z} \times \mathbb{Z}/p\mathbb{Z}$ (note that it is always abelian by Proposition 4.7.12).

- if $g \in G$, sending h to $h^{-1}gh$ induces a bijection between $G/Z_G(g)$ and C_g, so $|C_g|$ divides $|G| = p^3$. If $g \in Z(G)$, then $|C_g| = 1$. If $g \notin Z(G)$, then $|C_g| = p$ or p^2. But $Z_G(g)$ always contains $Z(G) \cup \{g\}$, so $|Z_G(g)| \geq p + 1$, consequently $|C_g| = p$. In all, there are $p^2 + p - 1$ conjugacy classes: p of them have size 1 and $p^2 - 1$ of them have size p.

On the other hand, if n_i denotes the dimension of irreducible representations, then $n_i \in \{1, p\}$ by Theorem 4.7.8 together with Theorem 4.2.11(iii). Since $G/Z(G)$ is abelian of order p^2, we obtain p^2 1-dimensional representations of $G/Z(G)$, hence of G by inflation. Using Theorem 4.2.11(iii), we easily deduce the result. □

Example 4.7.14. *Proposition 4.7.13 implies that a (non-abelian) group of order* 8*, like D_4 or Q_8, has* 4 *irreducible representations of dimension* 1 *and* 1 *irreducible representation of dimension* 2*.*

4.8 Supersolvable groups

Definition 4.8.1. *A finite group is called supersolvable if there exists an ascending chain*

$$\{1\} = G_0 \subsetneq G_1 \subsetneq \cdots \subsetneq G_n = G$$

of normal subgroups $G_i \lhd G$, such that G_i/G_{i-1} is cyclic for all i.

Example 4.8.2.

- *Abelian groups are supersolvable;*
- *D_n is supersolvable;*
- *subgroups and quotients of a supersolvable group are still supersolvable.*

Remark 4.8.3. *For a solvable group, we only require G_{i-1} to be normal in G_i and G_i/G_{i-1} to be* abelian. *As an example: A_4 is solvable but not supersolvable. Indeed, A_4 contains an abelian normal subgroup V_4 which is of order 4, isomorphic to $\mathbb{Z}/2\mathbb{Z}\times\mathbb{Z}/2\mathbb{Z}$ (in particular not cyclic). But A_4 has no (proper) cyclic normal subgroups.*

Lemma 4.8.4. *Let G be a non-abelian supersolvable group. There exists a normal abelian subgroup which is not contained in the center $Z(G)$ of G.*

Proof. The group $\bar{G} := G/Z(G)$ is again supersolvable. Take $\bar{G}_1$ to be the first subgroup in the ascending chain, which is a cyclic group. Let A be the inverse image in G of $\bar{G}_1$. Then A contains (properly) $Z(G)$, and is normal abelian (since $Z(G)$ commutes with a lifting of the generator of $\bar{G}_1$). □

Theorem 4.8.5. *Assume F is algebraically closed. Let G be a supersolvable group. Then every irreducible representation of G over F is induced from a 1-dimensional representation of some subgroup.*

Proof. Let (V,ρ) be an irreducible representation of G over F. If G is abelian, the result is obvious by Corollary 4.2.17. So we assume G is non-abelian in the rest of the proof.

First, we may assume $\ker(\rho)=\{1\}$, i.e. ρ is faithful.[47] Indeed, the result is true for $G/\ker(\rho)$, then there exists $H/\ker(\rho) \subset G/\ker(\rho)$ and $\chi : H/\ker(\rho) \to F^\times$ such that $\rho = \mathrm{Ind}_{H/\ker(\rho)}^{G/\ker(\rho)}\chi$. Then let $\widetilde{\chi}$ be the inflation of χ to H, we get

$$\rho = \mathrm{Ind}_H^G \widetilde{\chi}.$$

Since G is non-abelian, by Lemma 4.8.4 we may take a non-central abelian normal subgroup $A \lhd G$. Consider $\mathrm{Res}_A^G V$. It is semisimple by Theorem 4.6.1, hence is a direct sum of one-dimensional representations. We choose such an irreducible sub-representation η and let H be its inertia subgroup in Clifford's theorem. Then $V \cong \mathrm{Ind}_H^G W'$ for some irreducible representation W' by Theorem 4.6.1.

[47] A representation $\rho : G \to \mathrm{Aut}_F(V)$ is said to be *faithful* if $\ker(\rho)=\{1\}$. Note that it is *not* the same as saying V is a faithful $F[G]$-module.

We claim that $H \subsetneq G$. Otherwise, $V = W'$ is η-isotypic, namely is isomorphic to $\eta^{\oplus r}$ where $r = \dim V$. This implies $a \in A$ acts on V by the scalar $\eta(a)$. As a consequence, for any $g \in G$

$$\rho(ga) = \rho(g)\rho(a) = \rho(a)\rho(g) = \rho(ag).$$

As ρ is assumed to be faithful, we deduce $ga = ag$ for any $g \in G$, namely $A \subset Z(G)$, which contradicts the choice of A.

By induction on $|G|$, W' is induced from a 1-dimensional representation of some subgroup $H' \subset H$, say $W' \cong \mathrm{Ind}_{H'}^{H}\chi$. By transitivity of taking induction, we finally see that $V \cong \mathrm{Ind}_{H'}^{G}\chi$. $\square$

Remark 4.8.6. *A finite group is called* monomial*, if its complex irreducible representations are all monomial, i.e. induced from characters of degree 1. It is a theorem of Taketa (1930) that a monomial group is solvable; and supersolvable groups are monomial. But the converse is not true: S_4 is monomial but not supersolvable.*

Since A_n is not solvable for $n \geq 5$, so A_n, S_n are not monomial for $n \geq 5$ by Taketa's theorem.

Example 4.8.7. *Take $G = A_5$. It has 5 irreducible representations, and the character table is as follows:*

	C_1	C_2	C_3	C_4	C_5
χ_1	1	1	1	1	1
χ_2	3	-1	0	$\frac{1+\sqrt{5}}{2}$	$\frac{1-\sqrt{5}}{2}$
χ_3	3	-1	0	$\frac{1-\sqrt{5}}{2}$	$\frac{1+\sqrt{5}}{2}$
χ_4	4	0	1	-1	-1
χ_5	5	1	-1	0	0

where $C_1 = \{\mathrm{Id}\}$, $C_2 = \{(12)(34), ...\}$, $C_3 = \{(123), ...\}$, $C_4 = \{(12345), ...\}$, $C_5 = \{(21345), ...\}$.

Any proper subgroup of A_5 has order ≤ 12 (exercise!). Hence, χ_2, χ_3, χ_4 are not monomial representations. On the other hand, $A_4 < A_5$ is a subgroup of order 12, and χ_5 is induced from a 1-dimensional representation of A_4. In fact, take $\psi : A_4 \to \mathbb{C}^\times$ be a non-trivial character, and we claim that $\mathrm{Ind}_{A_4}^{A_5}\psi$ is irreducible, hence is isomorphic to ρ_5 (the unique irreducible representation of A_5 of degree 5). Indeed, by Frobenius reciprocity, $\rho_1 = \mathbb{1}$ does not occur in the decomposition of $\mathrm{Ind}_{A_4}^{A_5}\psi$ (as ψ is non-trivial). As $\mathrm{Ind}_{A_4}^{A_5}\psi$ has dimension 5, it must be isomorphic to ρ_5 for dimension reasons.

4.9 Induction theorems

We assume $F = \mathbb{C}$ in this section. Let $V_1, \ldots, V_r$ be the complete list of irreducible representations of G, with $\chi_i = \chi_{V_i}$ the characters. We may view χ_i as class functions on G. Let W be any representation of G, we know that $W = \bigoplus_{i=1}^r V_i^{m_i}$ with $m_i \in \mathbb{N}$ so that

$$\chi_W = \sum_{i=1}^{r} m_i\chi_i.$$

Recall that $W \otimes W'$ is also a representation of G, with character $\chi_W \chi_{W'}$.

Definition 4.9.1. *We let $R(G)$ be the* ring

$$R(G) := \mathbb{Z}\chi_1 \oplus \cdots \oplus \mathbb{Z}\chi_r,$$

with natural addition and multiplication, viewing $R(G)$ as a subring of $\mathrm{cf}(G)$*. To see that it is a unital ring, just note that $\chi_i \chi_j$ is the character corresponding to $V_i \otimes V_j$, hence can be written as an integral linear combination of χ_i; the trivial character χ_1 is the unit of $R(G)$.*

An element in $R(G)$ is called a generalized character *(or virtual character) of G.*

For $H \subset G$, we have natural maps

- the restriction $R(G) \to R(H)$;
- the induction $R(H) \to R(G)$: $\varphi \mapsto \mathrm{Ind}_H^G \varphi$, here $\mathrm{Ind}_H^G \varphi := \sum_{\bar{t} \in G/H, g\bar{t} = \bar{t}} \varphi(t^{-1}gt)$ (this generalizes the formula for characters of induced representations).

Moreover, the equality $\mathrm{Ind}(\varphi \mathrm{Res}\psi) = \mathrm{Ind}(\varphi)\psi$ (see Proposition 4.5.12) for $\varphi \in R(H)$ and $\psi \in R(G)$ shows that the image of $\mathrm{Ind} : R(H) \to R(G)$ is an *ideal* of $R(G)$.

Definition 4.9.2. *Let $\mathcal{H}$ be a class of subgroups in G. Let $R_{\mathcal{H}}(G)$ be the ideal of $R(G)$ generated by the image of Ind_H^G for $H \in \mathcal{H}$.*

We will consider two classes of subgroups of G: elementary subgroups and quasi-elementary subgroups.

4.9.1 Elementary groups and quasi-elementary groups

Definition 4.9.3. *Let p be a prime number. We call G a p-elementary group if it is of the form $G = C \times P$ where C is a cyclic group of order prime to p and P is a p-group (i.e. $|P| = p^n$ for some n).*

We call G elementary if it is p-elementary for some prime p.

Definition 4.9.4. *A group G is called* p-quasi-elementary *if it is a semi-direct product $C \rtimes P$ with C and P as above.*

We call G quasi-elementary *if it is p-quasi-elementary for some prime p.*

Of course, p-elementary groups are p-quasi-elementary groups.

Lemma 4.9.5. *In a p-quasi-elementary group G, the subgroup C is unique. Precisely, $C = G_{p'} := \{x \in G : o(x) \text{ is prime to } p\}$.*

Conversely, if $G_{p'}$ is a cyclic subgroup of G, then G is p-quasi-elementary.

Note that $G_{p'}$ need not be a subgroup in general.

Proof. We choose a presentation $G = C \rtimes P$. We claim that $G_{p'} = C$: $C \subset G_{p'}$ is clear; for the conserve, if $x \in G_{p'}$, the image of x in $G/C \cong P$ is still an element of order prime to p, but P is a p-group, so $\bar{x} = 1$.

Conversely, if $G_{p'}$ is a subgroup, then it must be normal as any element in gxg^{-1} also has order prime to p. Assume $G_{p'}$ is moreover cyclic, and let P be any Sylow p-subgroup of G, then $G = G_{p'} \rtimes P$ is quasi-elementary. □

Proposition 4.9.6. *(i) If G is p-elementary (resp. p-quasi-elementary), then so is any subgroup of G.*

(ii) If $G \subset G'$ with G p-elementary (resp. p-quasi-elementary), then so is any conjugate of H.

Proof. (i) Let H be a subgroup of G. We have $H_{p'} = H \cap G_{p'}$. If G is p-quasi-elementary, then $H_{p'}$ is also cyclic normal in H and $H/H_{p'} < G/G_{p'}$ is a p-group, thus H is p-quasi-elementary.

If G is p-elementary, then P is its unique Sylow p-subgroup. For any Sylow p-subgroup of H, it is contained in P, therefore $H_P := H \cap P$ is a Sylow p-subgroup of H. Since $H_{p'}$ commutes with H_P, one gets $H = H_{p'} \times H_P$ is p-elementary.

(ii) If $G = C \rtimes P$, then both xCx^{-1} and xPx^{-1} are subgroups of xGx^{-1}. We have xCx^{-1} is the set of elements in xGx^{-1} of order prime to p and it is cyclic since C is. This allows us to show xGx^{-1} is also p-quasi-elementary. If moreover $G = C \times P$, then $xGx^{-1} = (xCx^{-1}) \times (xPx^{-1})$. □

Remark 4.9.7. *Quasi-elementary groups are supersolvable. This is because a finite p-group is supersolvable. In fact, any finite p-group has a non-trivial center, by counting its orbits. Considering successively $G/Z(G)$, we get the result. Since C is normal cyclic, the pullback of a chain $P_i \subsetneq P_{i+1} \subsetneq \cdots$ of P (with $P_i \lhd P$) provides one for G.*

Example 4.9.8. *Given a p-quasi-elementary group $G = C \rtimes P$, we set $Z := \{g \in H \mid gy = yg, \forall y \in P\}$, then $Z \times P$ is a p-elementary subgroup of G.*

For example, for $G = S_3$, $C = A_3$, $P = \langle (12) \rangle$, we have $G = C \rtimes P$ is 2-quasi-elementary, and $Z = \{1\}$ so that $H = P$.

Lemma 4.9.9 (Isaacs)**.** *Let $G = C \rtimes P$ be a p-quasi-elementary group. Define Z as in Example 4.9.8, so that $H := Z \times P$ is a p-elementary subgroup of G. Suppose that $\lambda : G \to F^\times$ is a character which is trivial on H (i.e. $H \subset \ker \lambda$). Then λ itself is trivial.*

Example 4.9.10. *As an example, if $\lambda : S_3 \to F^\times$ is a character trivial on $P = \langle (12) \rangle$, then it is also trivial on any conjugate of P, hence trivial on the whole S_3.*

Proof. It suffices to show $C \subset \ker \lambda$. Let $K = C \cap \ker \lambda$ which is a normal subgroup of G. Then λ induces an injective homomorphism $C/K \hookrightarrow F^\times$. If $y \in P$, then

$$\lambda(ycy^{-1}) = \lambda(y) \cdot \lambda(c) \cdot \lambda(y)^{-1} = \lambda(c),$$

so $y(ck)y^{-1}$ lies in the coset cK. In particular, P acts on each coset cK.

Since P is a p-group and $|cK| = |K|$ is prime to p, by Lemma 4.9.11 below we have a fixed point $ck \in cK$ under this action, i.e. $ck \in Z$ for some $k \in K$. Since $\lambda|_Z = 1$ and $\lambda(k) = 1$, we get $\lambda(c) = 1$. The choice of c being arbitrary, the result follows. $\square$

Lemma 4.9.11. *If P is a finite p-group acting on is a finite set X. Then*

$$|X| \equiv \sharp\{\textit{fixed points}\} \ (\textit{mod } p).$$

In particular, if $|X|$ is prime to p, then there is a fixed point under this action.

Proof. For each $x \in X$, let O_x be the orbit of x under P-action. Then $X = \coprod_{\text{orbit}} O_x$. Each O_x has a p-power order: let H_x be the stablizer of x, then $|O_x| = [P : H_x]$. Moreover, x is a fixed point iff $O_x = \{x\}$ iff $H_x = P$. We get

$$|X| \equiv \sharp\{\text{fixed points}\} \mod p.$$

Since $|X|$ is prime to p, there exists at least one fixed point. $\square$

Theorem 4.9.12. *Let G be a quasi-elementary group. Any irreducible character of G is an integral linear combination of characters induced from linear characters of elementary subgroups of G.*

Here, by *linear character* we mean the character of a one-dimensional representation; it is just a group homomorphism $G \to F^\times$.

Proof. We do induction on $|G|$. Recall that G is then supersolvable, hence any irreducible representation V of G is induced from a one-dimensional representation W of some subgroup, say H. Moreover, $\dim V > 1$ if and only if $H \subsetneq G$. Since H is also quasi-elementary, we may apply the inductive hypothesis to (H, W) to express χ_W as an integral linear combination of characters induced linear characters of elementary subgroups of H. Since taking induction is transitive, we get an expression for χ_V.

We assume $\dim V = 1$ in the rest, i.e. $\chi = \chi_V$ is itself a linear character. Write $G = C \rtimes P$, and let $Z \subset C$ be the centralizer of P in C; i.e.

$$Z := \{g \in C \mid gy = yg, \forall\, y \in P\}.$$

Then $Z \times P =: H < G$ is elementary. If $G = H$, then G is itself elementary, and there is nothing to prove. Assume $H \subsetneq G$ and consider $\mathrm{Ind}_H^G \chi$ (here we view χ as a representation of H). **Claim**:

$$\mathrm{Ind}_H^G \chi = \chi \oplus (\oplus_i W_i)$$

with $\dim W_i > 1$. If not, let χ' be another one-dimensional sub-representation of $\mathrm{Ind}_H^G \chi$. Then Frobenius reciprocity shows $\mathrm{Hom}_H(\chi'|_H, \chi|_H) \neq 0$; but both χ and χ' are one-dimensional, hence $\chi'|_H = \chi|_H$. Then $\chi'\chi^{-1}$ is a character of G which is trivial on H. Lemma 4.9.9 implies that $\chi'\chi^{-1}$ is trivial on the whole G, i.e. $\chi' = \chi$.

As shown above, each W_i is induced from 1-dimensional representations of elementary subgroups of G. Since

$$\chi = \mathrm{Ind}_H^G(\chi) - \sum_i \chi_{W_i},$$

and noting that H is an elementary group, we get the result. $\square$

Example 4.9.13. *As an example, take $G = S_3$and $V = \mathbb{1}$ the trivial representation of G, . Then $H = \langle (12) \rangle$ (see Example 4.9.8) and*

$$\mathrm{Ind}_H^G(\mathbb{1}_H) \cong \mathbb{1} \oplus \rho_3,$$

where ρ_3 denotes the unique irreducible representation of G of degree 2. We may further write ρ_3 as $\mathrm{Ind}_{A_3}^G \omega$, where ω is a non-trivial character of A_3, and hence

$$\chi_{\mathbb{1}} = \chi_{\mathrm{Ind}_H^G(\mathbb{1}_H)} - \chi_{\mathrm{Ind}_{A_3}^G \omega}.$$

4.9.2 Solomon's induction theorem

Theorem 4.9.14 (Solomon)**.** *The trivial representation is an integral linear combination of characters induced from the trivial characters on quasi-elementary subgroups. In particular, $R_{\mathcal{H}'}(G) = R(G)$, where $\mathcal{H}'$ is the set of quasi-elementary subgroups of G.*

We start with several lemmas.

Lemma 4.9.15. *Let $\mathcal{H}$ be a class of subgroups in G closed under taking conjugates and intersections. Then the $\mathbb{Z}$-linear combinations of $\mathrm{Ind}_H^G 1$ form a possibly non-unital subring of $R(G)$.*

Proof. Indeed, by Mackey decomposition theorem, for $H, K \in \mathcal{H}$,

$$(\mathrm{Ind}_H^G \mathbb{1}) \otimes (\mathrm{Ind}_K^G \mathbb{1}) \cong \mathrm{Ind}_H^G(\mathrm{Res}_H^G \mathrm{Ind}_K^G \mathbb{1}) = \bigoplus_{g \in H \backslash G / K} \mathrm{Ind}_{H \cap gKg^{-1}}^G \mathbb{1}.$$

Remark that it is *not* an ideal in $R(G)$. $\square$

Lemma 4.9.16 (Banaschewski)**.** *Let A be a (possibly non-unital) subring of the ring of $\mathbb{Z}$-valued functions on a finite set X under pointwise addition and multiplication. If $1 \notin A$, then for some $x \in X$ and prime p, we have*

$$f(x) \equiv 0 \ (mod\ p) \quad \forall f \in A.$$

Remark 4.9.17. *The general form of this lemma says: if R is a commutative ring, and A is a possibly non-unital subring. Assume that A is not contained in any of a finite list of ideals $I_1, ..., I_n$, at most two of which are not prime, then A is not contained in their union.*

Proof. For each $x \in X$ we get a map $\mathrm{ev}_x : A \to \mathbb{Z}$; the image $\mathrm{ev}_x(A)$ is a subgroup of $\mathbb{Z}$, hence is of the form $n_x\mathbb{Z}$ for some $n_x \in \mathbb{Z}_{\geq 1}$. Suppose on the contrary that for every $x \in X$ we have $n_x = 1$. Let $f_x \in A$ be an element so that $f_x(x) = 1$; consider the product

$$\prod_{x \in X} (\mathbb{1}_X - f_x).$$

This vanishes at every $x \in X$, so vanishes identically. But expanding the product we see $\mathbb{1}_X \in A$. □

Proof of Solomon's theorem. We apply Lemma 4.9.16 to the ring A generated by the character of $\mathrm{Ind}_H^G \mathbb{1}$ for H running over quasi-elementary subgroups, viewed as functions on $X = G$. Assume $\mathbb{1}_G \notin A$, we get some $g \in G$ and some prime p such that $\chi_{\mathrm{Ind}_H^G \mathbb{1}}(g) \equiv 0 \mod p$ for every $H \in \mathcal{H}$. Hence, to show the result, it is enough to show that for each $g \in G$ and each prime p, there is a quasi-elementary subgroup $H_{g,p}$ such that $\chi_{\mathrm{Ind}_{H_{g,p}}^G \mathbb{1}}(g) \neq 0 \pmod p$.

We decompose $\langle g \rangle = C \times \langle g \rangle_p$ with $|C|$ prime to p. Let $N = N_G(C)$ be the normalizer of C in G and let $P \subset N$ be a Sylow p-subgroup. Then $H := C \rtimes P$ is a p-quasi-elementary subgroup.

Recall that

$$\chi_{\mathrm{Ind}_H^G \mathbb{1}}(g) = \#\{\bar{t} \in G/H : t^{-1}gt \in H\} = \#\{\bar{t} \in G/H : g(tH) = tH\}.$$

If $t^{-1}gt \in H$, then also $t^{-1}Ct \subset H$ as elements of C are all powers of g. But inside H, $t^{-1}Ct = H_{p'} = C$, so one gets $t \in N$ by definition. That is $\chi_{\mathrm{Ind}_H^G \mathbb{1}}(g) = \chi_{\mathrm{Ind}_H^N \mathbb{1}}(g)$, and more importantly, the values are equal to the number of fixed points for the natural action of $\langle g \rangle$ on N/H. Since $N = N_G(C)$, for any $t \in N$ we have $t^{-1}Ct = C \subset H$, so C acts trivially on N/H, so the action factors through $\langle g \rangle / C$ which is a p-group. By Lemma 4.9.11,

$$|N/H| \equiv \chi_{\mathrm{Ind}_H^N \mathbb{1}}(g) \pmod p.$$

Since N/H has order prime to p, we get $\chi_{\mathrm{Ind}_H^N \mathbb{1}}(g) \neq 0 \mod p$. □

4.9.3 Brauer's induction theorem

Theorem 4.9.18 (Brauer)**.** *Let $\mathcal{H}$ be the set of elementary subgroups of G. Then $R(G) = R_{\mathcal{H}}(G)$.*

Proof. First, since $R_{\mathcal{H}}(G)$ is an ideal, it suffices to show $\mathbb{1}_G \in R_{\mathcal{H}}(G)$. By Solomon's theorem, $\mathbb{1}_G = \sum_{H' \in \mathcal{H}'} c_{H'} \chi_{\mathrm{Ind}_{H'}^G \mathbb{1}}$. Then using Theorem 4.9.12, we can write each $\mathbb{1}_{H'}$ as an integral linear combination of characters induced from linear characters of elementary subgroups of H'. By transitivity of induction, we get an expression for 1_G. □

Corollary 4.9.19. *Every element in $R(G)$ can be expressed as*

$$\sum_{H \in \mathcal{H}} \sum_{\xi : \deg \xi = 1} n_{H,\xi} \chi_{\mathrm{Ind}_H^G(\xi)}, \quad n_{H,\xi} \in \mathbb{Z}.$$

Proof. First, Theorem 4.9.18 implies that any element in $R(G)$ can be written as

$$\sum_{H\in\mathcal{H}}\sum_{W\in\mathrm{Irr}(H)} n_{H,W}\chi_{\mathrm{Ind}_H^G W}, \quad n_{H,W}\in\mathbb{Z}.$$

Because elementary groups are supersolvable, we can express W as $\mathrm{Ind}_{H'}^{H}\xi'$ by Theorem 4.8.5, where H' is a subgroup of H (thus $H'\in\mathcal{H}$). We conclude by the transitivity of induction. □

Example 4.9.20. *In general, even for irreducible representation V of G, the coefficients c_ξ could be negative. To see this, just take a non-solvable group and an irreducible representation V of G which is not monomial. If $\chi_V=\sum_{i=1}^r d_i\xi_i$, $d_i\in\mathbb{N}$, then $r\geq 2$ or $r=1$ and $d_1\geq 2$ as V is not monomial, but this would contradict the irreducibility of V.*

4.10 Rationality

Let $F\subset\mathbb{C}$ be a subfield. If V is a linear F-representation of G, we denote by $V_{\mathbb{C}}$ the $\mathbb{C}$-representation $V\otimes_F\mathbb{C}$, obtained by extending scalars from F to $\mathbb{C}$. In general, even when V is irreducible, $V_{\mathbb{C}}$ can be reducible.

A linear $\mathbb{C}$-representation of G is said to be *realizable over F* (or *rational over F*) if it is isomorphic to $V\otimes_F\mathbb{C}$, where V is a representation of G defined over F.

Let F be a subfield of $\mathbb{C}$. If ρ is a $\mathbb{C}$-representation of G, a necessary condition for ρ to be realizable over F is that $\chi_\rho(g)\in F$ for all $g\in G$. In general, this is not sufficient.

Example 4.10.1. *Take $G=\mathbb{Q}_8$ and let ρ_5 be the unique irreducible 2-dimensional $\mathbb{C}$-representation of G, see §4.4.5. Then its character is given by*

	$\{1\}$	$\{-1\}$	$\{i,-i\}$	$\{j,-j\}$	$\{k,-k\}$
χ_5	2	-2	0	0	0

In particular, $\chi_5(g)\in\mathbb{Q}$ for all $g\in G$, but ρ_5 is not realizable over $\mathbb{Q}$ (or over $\mathbb{R}$).

We denote by $R_F(G)$ the group generated by the characters of the F-representations of G. It is a subring of $R(G)$ in Definition 4.9.1 (for $F=\mathbb{C}$). Moreover, if $V_1,\ldots,V_s$ are the irreducible representations of G over F, then the $\chi_i:=\chi_{V_i}$ form a $\mathbb{Z}$-basis of $R_F(G)$ and they are mutually orthogonal. However, in general, $\langle\chi_i,\chi_i\rangle$ need not be equal to 1.

Lemma 4.10.2. *Let V be a $\mathbb{C}$-representation of G. In order that V be realizable over F, it is necessary and sufficient that $\chi_V\in R_F(G)$.*

Proof. The condition is clearly necessary. Conversely, we can write $\chi_V=\sum_{i=1}^s n_i\chi_i$ with $n_i\in\mathbb{Z}$. We obtain:

$$\langle\chi_V,\chi_i\rangle=n_i\langle\chi_i,\chi_i\rangle.$$

Note that $\langle\chi_V,\chi_i\rangle\geq 0$ (this can be seen by taking base change to $\mathbb{C}$), so $n_i\geq 0$. Thus $V=\oplus_{i=1}^s V_i^{\oplus n_i}$ is realizable over F. □

Theorem 4.10.3 (Brauer). *If F contains the m-th roots of unity, then any $\mathbb{C}$-representation of G can be realized over F.*

Proof. By Brauer induction theorem, any $\chi_V = \sum_i n_i \chi_{\mathrm{Ind}_{H_i}^G \xi_i}$ where H_i are elementary subgroups, and ξ_i is a linear character of H_i. Since the exponent of H_i is a divisor of m, $\xi_i : H_i \to \mathbb{C}^\times$ takes values in $F^\times$ as F contains all m-th roots of unity. Thus, $\mathrm{Ind}_{H_i}^G \xi_i$ is realizable over F and $\chi_{\mathrm{Ind}_{H_i}^G \xi_i} \in R_F(G)$. We deduce $\chi_V \in R_F(G)$ and the result follows from Lemma 4.10.2. $\square$

4.11 Representations of symmetric groups

4.11.1 Review on symmetric groups

Let S_n be the symmetric group acting on $X = \{1, \dots, n\}$. Any $\sigma \in S_n$ can be represented as a product of disjoint *cycles*. A cycle is written in the form

$$(k_1 \cdots k_m)$$

where $k_i \in X$ (unique up to cyclic permutations). The integer m is called the *length* of the cycle. Recall some useful properties.

- Disjoint cycles commute in S_n.
- Every element $\sigma \in S_n$ can be written as a product $\sigma = \tau_1 \cdots \tau_r$, with τ_i disjoint cycles; if m_i denotes the length of τ_i, $\{m_1, \dots, m_r\}$ is called the *cycle shape* of σ.
- if $\sigma \in S_n$, $\tau = (k_1 \cdots k_m)$ is a cycle, then

  $$\sigma\tau\sigma^{-1} = (\sigma(k_1) \cdots \sigma(k_m))$$

  is again a cycle of length m.
- As a consequence, if $\sigma \sim \sigma'$ (conjugate) then they have the same cycle shape. The converse is also true. Hence the conjugacy classes in S_n are in bijection with the cycle shapes, also in bijection with the partitions of n. Here a *partition* λ of n is

  $$\lambda = (\lambda \geq \lambda_2 \geq \cdots \geq \lambda_r) = (\lambda_1, \lambda_2, \cdots, \lambda_r),\ \lambda_i \geq 1$$

  such that $\lambda_1 + \cdots + \lambda_r = n$. We writ $\lambda \vdash n$ to mean that λ is a partition of n.

Take $F = \mathbb{Q}$. For each $\lambda \vdash n$, we will construct an irreducible representation of S_n over $\mathbb{Q}$, called *Specht module* S^λ.

4.11.2 Young diagrams

Definition 4.11.1. *A* Young diagram *is an array of boxes with nonincreasing row length.*

Example 4.11.2.

$$3 \dashv (2,1) = \begin{array}{|c|c|}\hline & \\ \hline & \multicolumn{1}{c}{} \\ \cline{1-1} \end{array}$$

$$11 \dashv (5,4,2) = \begin{array}{|c|c|c|c|c|}\hline & & & & \\ \hline & & & & \multicolumn{1}{c}{} \\ \cline{1-4} & & \multicolumn{1}{c}{} & \multicolumn{1}{c}{} & \multicolumn{1}{c}{} \\ \cline{1-2} \end{array}$$

Definition 4.11.3. *Let $\lambda \vdash n$ be a partition. A Young tableau of shape λ consists of*

- *the Young diagram corresponding to λ and*
- *a way to fill the integers $1, \dots, n$ in the n boxes.*

The group S_n acts on the set of Young tableaux of shape λ pointwisely: action denoted by $\star$. The action is transitive: for any t, t', there exists $g \in S_n$ such that $t' = g \star t$.

Example 4.11.4.

$$(13) \star \begin{array}{|c|c|}\hline 2 & 1 \\ \hline 3 & \multicolumn{1}{c}{} \\ \cline{1-1} \end{array} = \begin{array}{|c|c|}\hline 2 & 3 \\ \hline 1 & \multicolumn{1}{c}{} \\ \cline{1-1} \end{array}$$

$$(1589) \star \begin{array}{|c|c|c|c|c|}\hline 2 & 3 & 5 & 10 & 7 \\ \hline 1 & 8 & 4 & 6 & \multicolumn{1}{c}{} \\ \cline{1-4} 9 & 11 & \multicolumn{1}{c}{} & \multicolumn{1}{c}{} & \multicolumn{1}{c}{} \\ \cline{1-2} \end{array} = \begin{array}{|c|c|c|c|c|}\hline 2 & 3 & 8 & 10 & 7 \\ \hline 5 & 9 & 4 & 6 & \multicolumn{1}{c}{} \\ \cline{1-4} 1 & 11 & \multicolumn{1}{c}{} & \multicolumn{1}{c}{} & \multicolumn{1}{c}{} \\ \cline{1-2} \end{array}.$$

Definition 4.11.5. *A tabloid is a (row) equivalence class of Young tableaux, where $t \sim t'$ if for each row, the set of values coincide. Denote by $\{t\}$ the equivalence class corresponding to t.*

Example 4.11.6.

$$\begin{array}{|c|c|c|}\hline 1 & 2 & 3 \\ \hline 4 & 5 & \multicolumn{1}{c}{} \\ \cline{1-2} \end{array} \sim \begin{array}{|c|c|c|}\hline 3 & 1 & 2 \\ \hline 5 & 4 & \multicolumn{1}{c}{} \\ \cline{1-2} \end{array} \sim \begin{array}{|c|c|c|}\hline 1 & 3 & 2 \\ \hline 4 & 5 & \multicolumn{1}{c}{} \\ \cline{1-2} \end{array} \sim \begin{array}{|c|c|c|}\hline 3 & 2 & 1 \\ \hline 5 & 4 & \multicolumn{1}{c}{} \\ \cline{1-2} \end{array}$$

In fact, there are $3!2! = 12$ tableaux in this equivalence class.

Lemma 4.11.7. *Given a shape $\lambda = (\lambda_1 \geq \lambda_2 \cdots \geq \lambda_r) \vdash n$.*

(i) let t be a Young tableau and $R_t := \{\sigma \in S_n : \sigma \star t \sim t\}$ (row stabilizer). Then R_t is isomorphic to $S_{\lambda_1} \times \cdots \times S_{\lambda_r}$.

(ii) The number of Young tableaux equivalent to a given one is $\prod_i \lambda_i!$.

Proof. Clear. $\square$

Now we also consider the column stabilizer, denoted by C_t (column stabilizer), which is the subgroup of S_n which fixes each column set-wise.

Example 4.11.8. *Consider* $t = \begin{array}{|c|c|c|}\hline 1 & 2 & 3 \\ \hline 4 & 5 \\ \cline{1-2} \end{array}$, *then*

$$C_t = \{1, (14), (25), (14)(25)\}.$$

Lemma 4.11.9. *Let* $\sigma \in S_n$. *Then*

$$C_{\sigma \star t} = \sigma C_t \sigma^{-1}, \quad R_{\sigma \star t} = \sigma R_t \sigma^{-1}.$$

Proof. Clear. □

Definition 4.11.10. *The permutation module* M^λ *is the (free)* F*-vector space generated by the tabloids of shape* λ*. The* S_n*-action* $\sigma\{t\} = \{\sigma t\}$ *on tabloids gives a left* $F[S_n]$*-module structure on* M^λ*.*

Lemma 4.11.11. M^λ *is a cyclic* $F[S_n]$*-module and has dimension* $\frac{n!}{\lambda_1! \cdots \lambda_r!}$*.*

Proof. Any $\{t\}$ is a generator of M^λ, so M^λ is cyclic. Moreover, since the action of G on tabloids is transitive, the number of tabloids of shape λ (which is also the dimension of M^λ) is equal to

$$\frac{n!}{\sharp R_t} = \frac{n!}{\prod_i \lambda_i!}.$$

□

In general, M^λ is not irreducible and could be very large.

Example 4.11.12. *There are 5 types for* $n = 4$*.*

- $\lambda = \begin{array}{|c|c|c|c|}\hline \ & \ & \ & \ \\ \hline \end{array}$: $\dim M^{(4)} = 1$.
- $\lambda = \begin{array}{|c|c|c|}\hline \ & \ & \ \\ \hline \ \\ \cline{1-1} \end{array}$: $\dim M^{(3,1)} = 4$.
- $\lambda = \begin{array}{|c|c|}\hline \ & \ \\ \hline \ & \ \\ \hline \end{array}$: $\dim M^{(2,2)} = 6$.
- $\lambda = \begin{array}{|c|c|}\hline \ & \ \\ \hline \ \\ \cline{1-1} \ \\ \cline{1-1} \end{array}$: $\dim M^{(2,1,1)} = 12$.
- $\lambda = \begin{array}{|c|}\hline \ \\ \hline \ \\ \hline \ \\ \hline \ \\ \hline \end{array}$: $\dim M^{(1,1,1,1)} = 24$.

Let us study $M^{(3,1)}$. It has a basis of tabloids given by the equivalence classes of

$$t_1 = \begin{array}{|c|c|c|}\hline 1 & 2 & 3 \\ \hline 4 \\ \cline{1-1} \end{array},\ t_2 = \begin{array}{|c|c|c|}\hline 2 & 3 & 4 \\ \hline 1 \\ \cline{1-1} \end{array},\ t_3 = \begin{array}{|c|c|c|}\hline 1 & 3 & 4 \\ \hline 2 \\ \cline{1-1} \end{array},\ t_4 = \begin{array}{|c|c|c|}\hline 1 & 2 & 4 \\ \hline 3 \\ \cline{1-1} \end{array}.$$

Since the action of S_4 permutes the $\{t_i\}$, and is isomorphic to the natural permutation representation of S_4 acting on $\{1,2,3,4\}$ (looking at the second row). So

$$f := \{t_1\} + \{t_2\} + \{t_3\} + \{t_4\}$$

is an $F[S_4]$-submodule of $M^{(3,1)}$ and its quotient (or complement) is just the Standard representation of S_4. We denote it by $S^{(3,1)}$. It is easy to see that $S^{(3,1)}$ is characterized by the three-dimensional subspace of vectors $\sum_{i=1}^4 c_i\{t_i\}$ with $\sum_{i=1}^4 c_i = 0$. It has a basis

$$\{t_1\} - \{t_2\},\ \{t_2\} - \{t_3\},\ \{t_3\} - \{t_4\}.$$

4.11.3 Specht modules

Definition 4.11.13. *Let t be a tableau, the* polytabloid *associated to t is*

$$e_t = \sum_{\sigma \in C_t} \operatorname{sgn}(\sigma)\sigma \star \{t\}. \tag{25}$$

*The Specht module S^λ associated to a cycle shape λ is the module spanned by polytabloids e_t where t is taken over all tableaux of shape λ (*but not direct sum*), i.e.*

$$S^\lambda = Fe_{t_1} + Fe_{t_2} + \cdots,\ t_i \text{ tableaux of shape } \lambda.$$

Example 4.11.14. *We continue the example with S_4. Even for $t \sim t'$, $e_t, e_{t'}$ need not be equal (as $C_t \neq C_{t'}$!). Take $t = t_1 = \begin{array}{|c|c|c|}\hline 1 & 2 & 3 \\ \hline 4 \\ \cline{1-1} \end{array}$, $t' = \begin{array}{|c|c|c|}\hline 2 & 1 & 3 \\ \hline 4 \\ \cline{1-1} \end{array}$, $t'' = \begin{array}{|c|c|c|}\hline 3 & 2 & 1 \\ \hline 4 \\ \cline{1-1} \end{array}$. Then*

$$e_t = \{t_1\} - \{\begin{array}{|c|c|c|}\hline 4 & 2 & 3 \\ \hline 1 \\ \cline{1-1} \end{array}\} = \{t_1\} - \{t_2\}$$

as $C_t = \{1, (14)\}$, and similarly $e_{t'} = \{t_1\} - \{t_3\}$, $e_{t''} = \{t_1\} - \{t_4\}$. This suggests that $S^{(3,1)}$ has dimension 3 and is the standard representation of S_4.

For convenience, we write

$$\kappa_t := \sum_{\sigma \in C_t} \operatorname{sgn}(\sigma)\sigma \in F[S_n].$$

so that $e_t = \kappa_t \star \{t\}$. Note that κ_t only depends on C_t. It has some nice properties.

Lemma 4.11.15. (i) *For $g \in S_n$, $g\kappa_t = \kappa_{g\star t} g$, equivalently $\kappa_{g\star t} = g\kappa_t g^{-1}$. As a consequence, $e_{g\star t} = g \star e_t$.*

(ii) *If* $g \in C_t$, *then* $g\kappa_t = \kappa_t g = \mathrm{sgn}(g^{-1})\kappa_t$.

(iii) $\kappa_t^2 = |C_t|\kappa_t$. *As a consequence,* $e_t = \frac{1}{|C_t|}\kappa_t \star e_t$ *(recall that* $\mathrm{char}(F) = 0$*).*

Proof. (i) It is a direct computation:

$$\begin{aligned} g \cdot \sum_{\sigma \in C_t} \mathrm{sgn}(\sigma)\sigma \cdot g^{-1} &= \sum_{\sigma \in C_t} \mathrm{sgn}(\sigma) g\sigma g^{-1} \\ &= \sum_{\sigma' \in C_{g\star t}} \mathrm{sgn}(\sigma')\sigma' \quad \text{with } \sigma' = g\sigma g^{-1} \\ &= \kappa_{g\star t} \cdot g. \end{aligned}$$

The last statement follows by definition: $e_{g\star t} = \kappa_{g\star t}(g \star t) = (g\kappa_t) \star t = g \star e_t$.

(ii) If $g \in C_t$, then

$$\begin{aligned} g \cdot \sum_{\sigma \in C_t} \mathrm{sgn}(\sigma)\sigma &= \mathrm{sgn}(g^{-1}) \sum_{\sigma \in C_t} \mathrm{sgn}(g\sigma) g\sigma \\ &= \mathrm{sgn}(g^{-1}) \sum_{\sigma' \in C_t} \mathrm{sgn}(\sigma')\sigma' \quad \text{with } \sigma' = g\sigma \\ &= \mathrm{sgn}(g^{-1})\kappa_t. \end{aligned}$$

The other assertion is proved in a similar way.

(iii) It is a consequence of (ii):

$$\kappa_t^2 = \sum_{g \in C_t} \mathrm{sgn}(g)\kappa_t g = \sum_{g \in C_t} \kappa_t = |C_t|\kappa_t.$$ □

Lemma 4.11.16. *The Specht module* S^λ *is cyclic. In fact, for any polytabloid* e_t*, we have* $S^\lambda = F[S_n] \cdot e_t$*.*

Proof. By definition S^λ is generated by all e_{t_i} for t_i of shape λ. But each t_i is of the form $g \star t$ for some $g \in S_n$, we conclude by Lemma 4.11.15(i). □

4.11.4 The dominance order

Let $\lambda, \mu \vdash n$. Let t, s be tableaux of shape λ and μ respectively.

Lemma 4.11.17. *If* C_t *contains a transposition* τ*, then* $\kappa_t \in F[S_n]\cdot(1-\tau)$*. If* $\tau \in C_t \cap R_s$*, then* $\kappa_t \star \{s\} = 0$*.*

Proof. Consider the coset decomposition $C_t = \coprod_i \sigma_i H$ where $H := \{1, \tau\}$ is viewed as a subgroup of C_t. Then

$$\kappa_t = \sum_{\sigma \in C_t} \mathrm{sgn}(\sigma)\sigma = \sum_i \mathrm{sgn}(\sigma_i)\sigma_i - \sum_i \mathrm{sgn}(\sigma_i)\sigma_i\tau = \sum_i \mathrm{sgn}(\sigma_i)\sigma_i \cdot (1 - \tau).$$

The condition $\tau \in R_s$ means that $(1 - \tau) \star \{s\} = 0$, hence the result. □

Definition 4.11.18. *Given $\lambda = (\lambda_1 \geq \cdots)$ and $\mu = (\mu_1 \geq \cdots)$ such that $\lambda, \mu \vdash n$, we say $\lambda \rhd \mu$ (called* dominance order*) if*

$$\sum_{i \leq k} \lambda_i \geq \sum_{i \leq k} \mu_i, \quad \forall k \geq 1.$$

Here we set $\lambda_i = \mu_i = 0$ for i sufficiently larege.

This defines a *partial order* on the set of partitions of n (not every pair (λ, μ) is comparable with respect to $\rhd$). If $\lambda \rhd \mu$ and $\mu \rhd \lambda$ then $\lambda = \mu$.

Example 4.11.19. *Take $n = 7$, $\lambda = (3, 1, 1, 1, 1)$ and $\mu = (2, 2, 2, 1)$. Then λ and μ are not comparable.*

Lemma 4.11.20. *Let $\lambda, \mu \vdash n$. Let t, s be tableaux of shape λ and μ respectively. If $\kappa_t \star \{s\} \neq 0$, then $\lambda \rhd \mu$.*

Proof. Since $\kappa_t \star \{s\} \neq 0$, for any b, c in the same row of s, they cannot lie in the same column of t. We claim that this implies $\lambda \rhd \mu$. Prove for example $\lambda_1 \geq \mu_1$. Write $\mu_1 = \boxed{b_1} \boxed{\cdots} \boxed{b_{\mu_1}}$, then b_i must lie in different column, so that λ_1, the number of columns of t, is bigger than μ_1. □

In the next lemma, we specialize to the case $\lambda = \mu$.

Lemma 4.11.21. *Let s, t be two tableaux of shape λ. Assume $\kappa_t \star \{s\} \neq 0$. Then*
(i) there exists $g \in C_t$ such that $\{s\} = g \star \{t\}$;
(ii) $\kappa_t \star \{s\} = \pm e_t$. In particular, κ_t defines an F-linear projection from M^λ to Fe_t.

Proof. (i) Since $\kappa_t \star \{s\} \neq 0$, $C_t \cap R_s$ does not contain any transposition (bc) by Lemma 4.11.17. So for any b, c in the same row of s, they can not lie in the same column of t. Consider the first row of s, say $\boxed{b_1} \boxed{b_2} \boxed{\cdots}$, then the b_i lie in different columns of t (although not necessarily the first row of t), we can then find an element $g \in C_t$ such that it sends b_i to the first row of t, i.e. the set of values appearing in the first row of s and of $g \star t$ coincide (here we use that λ, μ are of the same shape). Note that $C_{g \star t} = C_t$ (as $g \in C_t$), and $\kappa_{g \star t} \star \{s\} = \kappa_t \star \{s\} \neq 0$ (as $\kappa_{g \star t} = \kappa_t$), so we can continue the procedure row by row to get an element in C_t which sends $\{t\}$ to $\{s\}$.

(ii) Using (i), we may write $\{s\} = g \star \{t\}$ for some $g \in C_t$. By Lemma 4.11.15 we have

$$\kappa_t \star \{s\} = (\kappa_t \cdot g) \star \{t\} = \operatorname{sgn}(g^{-1}) \kappa_t \star \{t\} = \pm e_t.$$

Recall that $\{s : \text{shape } \lambda\}$ form a basis of M^λ. For this basis, we have either $\kappa_t \star \{s\} = 0$ or $\kappa_t \star \{s\} = \pm e_t$. The result follows from this. □

4.11.5 Irreducibility of S^λ

Theorem 4.11.22. *(i) The Specht module S^λ is absolutely irreducible for any $\lambda \vdash n$.*

(ii) $\mathrm{Hom}_{S_n}(S^\lambda, S^\mu) \neq 0$ if and only if $\lambda = \mu$.

(iii) The Specht modules S^λ (for $\lambda \vdash n$) form a complete set of isomorphism classes of irreducible representations of S_n over F.

Proof. (i) Since $\mathrm{char}(F) = 0$, S^λ is absolutely irreducible if and only if $\mathrm{End}_{S_n}(S^\lambda) = F$. Let $\varphi \in \mathrm{End}_{S_n}(S^\lambda)$ and t be a tableau of shape λ. By Lemma 4.11.21 and Lemma 4.11.15,

$$\varphi(e_t) = \varphi\big(\frac{1}{|C_t|}\kappa_t \star e_t\big) = \frac{1}{|C_t|}\kappa_t \star \varphi(e_t) \in Fe_t,$$

i.e. $\varphi(e_t) = ce_t$ for some $c \in F$. Since S^λ is a cyclic $F[S_n]$-module generated by e_t, we get $\varphi = c$.

(ii) Let $\varphi \in \mathrm{Hom}_{S_n}(S^\lambda, S^\mu)$ be non-zero. If t is a tableau of shape λ, then $\varphi(e_t) \neq 0$ since e_t generates S^λ. We then have

$$0 \neq \varphi(e_t) = \frac{1}{|C_t|}\kappa_t \star \varphi(e_t).$$

Write $\varphi(e_t) = \sum_i c_i\{s_i\}$ for s_i of shape μ. Then $\kappa_t \star \{s_i\} \neq 0$ for at least one s_i. This already implies $\lambda \trianglerighteq \mu$. On the other hand, since both S^λ and S^μ are irreducible, φ induces an isomorphism between S^λ and S^μ. Apply the same argument to φ^{-1} we obtain $\mu \trianglerighteq \lambda$, so $\lambda = \mu$.

(iii) Since the number of irreducible representations of S_n is equal to the number of conjugacy classes in S_n, which is again equal to the number of partitions of n. So we have constructed all absolutely irreducible representations, up to isomorphism. ☐

Theorem 4.11.23. *For $\mu \vdash n$, $M^\mu = \bigoplus_{\lambda \trianglerighteq \mu} (S^\lambda)^{\oplus \kappa_{\lambda,\mu}}$, with $\kappa_{\mu,\mu} = 1$.*

Proof. Given $\mu \vdash n$, we can decompose M^μ as a direct sum of S^λ, with multiplicities denoted by $\kappa_{\lambda,\mu}$. If S^λ appears in the decomposition, then

$$\mathrm{Hom}_{S_n}(S^\lambda, M^\mu) \neq 0.$$

The same argument as in Theorem 4.11.22(ii) shows that $\lambda \trianglerighteq \mu$. To prove $\kappa_{\mu,\mu} = 1$, it suffices to prove

$$\dim_F \mathrm{Hom}_{S_n}(S^\mu, M^\mu) = 1.$$

The proof is similar to that of Theorem 4.11.22(i). ☐

Now we explain, without proof, how to determine the numbers $\kappa_{\lambda,\mu}$, which are called *Kostka numbers*. Write $\mu = \{\mu_1, \ldots, \mu_s\}$. A Young tableau of shape λ and *weight* μ consists of a Young diagram and a way to fill $\{1, \ldots, 1, 2, \ldots, 2, \ldots, s, \ldots, s\}$ (with j appearing μ_j times). It is called *semi-standard*, if the values are non-decreasing for each row and strictly increasing for each column.

Theorem 4.11.24. *For $\lambda \rhd \mu$, $\kappa_{\lambda,\mu}$ is equal to the number of semi-standard tableaux of shape λ and weight μ.*

Example 4.11.25. *If $n = 5$ and $\lambda = (3,2) \rhd \mu = (2,1,1,1)$, there are three* semi-standard *ways to fill* $(1,1,2,3,4)$ *in the Young diagram of shape λ:*

$$\begin{array}{|c|c|c|}\hline 1 & 1 & 2 \\ \hline 3 & 4 \\ \cline{1-2}\end{array}, \quad \begin{array}{|c|c|c|}\hline 1 & 1 & 3 \\ \hline 2 & 4 \\ \cline{1-2}\end{array}, \quad \begin{array}{|c|c|c|}\hline 1 & 1 & 4 \\ \hline 2 & 3 \\ \cline{1-2}\end{array}$$

hence $\kappa_{\lambda,\mu} = 3$.

Remark 4.11.26. *In general, there are no nice formulas known for the Kostka numbers. However, some special cases are known. For example, if $\mu = (1,1,\dots,1)$ is the partition whose parts are all 1 then a semistandard Young tableau of weight μ is a standard Young tableau; the number of standard Young tableaux of a given shape λ is given by the hook-length formula. See Theorem 4.11.31 below.*

4.11.6 Dimension of S^λ

We fix a partition $\lambda \vdash n$.

Definition 4.11.27. *A Young tableau is called* standard *if for each row and each column, the values are ordered increasingly.*

Example of standard tableau:

$$\begin{array}{|c|c|c|}\hline 1 & 3 & 5 \\ \hline 2 & 4 \\ \cline{1-2}\end{array}.$$

Example of non-standard tableau:

$$\begin{array}{|c|c|c|}\hline 1 & 4 & 5 \\ \hline 2 & 3 \\ \cline{1-2}\end{array}.$$

Theorem 4.11.28. *The set $\{e_t : t \text{ standard}\}$ form a basis for S^λ. In particular, the dimension of S^λ is equal to the number of standard Young tableaux.*

Definition 4.11.29. *The* hook-length *of a given entry (i,j) in a Young diagram of shape λ, denoted $h_{i,j}$, is the number of entries (i',j) with $i' \geq i$, or (i,j') with $j' \geq j$.*

Example 4.11.30. *Let* $9 \dashv \lambda \leftrightarrow \begin{array}{|c|c|c|c|}\hline & & & \\ \hline & & \\ \cline{1-3} & \\ \cline{1-2}\end{array}$, *then* $h_{1,1} = 6$, $h_{1,2} = 5$.

Theorem 4.11.31 (Hook-length formula)**.** *The dimension of S^λ is given by*

$$\frac{n!}{\prod_{i,j} h_{i,j}}.$$

Example 4.11.32. *For $n = 4$,*

- $\lambda = \ydiagram{4}$: $\dim M^{(4)} = 1$, $\dim S^{(4)} = 1$.
- $\lambda = \ydiagram{3,1}$: $\dim M^{(3,1)} = 4$, $\dim S^{(3,1)} = 3$.
- $\lambda = \ydiagram{2,2}$: $\dim M^{(2,2)} = 6$, $\dim S^{(2,2)} = 2$.
- $\lambda = \ydiagram{2,1,1}$: $\dim M^{(2,1,1)} = 12$, $\dim S^{(2,1,1)} = 3$.
- $\lambda = \ydiagram{1,1,1,1}$: $\dim M^{(1,1,1,1)} = 24$, $\dim S^{(1,1,1,1)} = 1$.

Using the description of Kostka numbers, one checks that

$$M^{(4)} = S^{(4)}, \quad M^{(3,1)} \cong S^{(3,1)} \oplus S^{(4)},$$

$$M^{(2,2)} = S^{(2,2)} \oplus S^{(4)} \oplus S^{(3,1)}, \quad M^{(2,1,1)} \cong S^{(2,1,1)} \oplus S^{(4)} \oplus (S^{(3,1)})^{\oplus 2} \oplus S^{(2,2)}.$$

Finally, $M^{(1,1,1,1)}$ is just the regular representation of S_4, hence

$$M^{(1,1,1,1)} \cong \bigoplus_{\lambda} (S^\lambda)^{\dim S^\lambda}.$$

Example 4.11.33. *For $n = 5$, there are 7 partitions of 5, thus 7 isomorphic classes of irreducible representations of S_5.*

- $\lambda = \ydiagram{5}$: $\dim M^\lambda = 1$, $\dim S^\lambda = 1$.
- $\lambda = \ydiagram{4,1}$: $\dim M^\lambda = 5$, $\dim S^\lambda = 4$.
- $\lambda = \ydiagram{3,2}$: $\dim M^\lambda = 10$, $\dim S^\lambda = 5$.
- $\lambda = \ydiagram{3,1,1}$: $\dim M^\lambda = 20$, $\dim S^\lambda = 6$.

- $\lambda = \ydiagram{2,2,1}$: $\dim M^\lambda = 30$, $\dim S^\lambda = 5$.
- $\lambda = \ydiagram{2,1,1,1}$: $\dim M^\lambda = 60$, $\dim S^\lambda = 4$.
- $\lambda = \ydiagram{1,1,1,1,1}$: $\dim M^\lambda = 120$, $\dim S^\lambda = 1$.